MFT CHINA FUTURES

Research findings, all parts run so far

39 products · 2005-01 to 2021-10 · 5m / 15m / 30m / 60m / daily

130 figures · 235 cells recorded · 0 failed · built 2026-11-07

Every slide has a short explanation. Every graph carries a note on what it is and what it explains. Notes are written from the report text and figure titles.

EXECUTIVE SUMMARY

The answer in one page

- **No strategy clears every threshold.** Watch list: B volatility TS, B trend SEC, B carry XS, C3 EVT tail-event trades.
- **Volatility is forecastable, direction mostly is not.** HAR-RV is the best volatility model in every regime and time frame. Linear mean models explain 0.0000 of out-of-sample return.
- **Dependence:** Student-t copula (nu = 12) beats Gaussian; strategies fail together (lower-tail dependence 0.14).
- **Relationships:** 27 Granger links replicate out of sample; 0 of 10 fundamental chains do.
- **Only live candidate: the daily Stage 2 breakout (long).** About 83 / 72 / 71 bp net per trade in train / validation / test, test t 2.3. It still fails the full robustness test.
- **Everything faster than about 30 minutes fails cost** (round trip 6 bp). The newest step, RSI inside Stage 2 (18v), adds nothing.
- **Regime today:** HIGH volatility, BULL breadth.

METHOD

How the research is kept honest

- **One calendar split** for all frequencies: train to 2015-02-09, validation 2015-03-17 to 2018-06-11, test from 2018-07-11, 20-day embargo. The test period is 792 trading days and is read once.
- **Regime-balanced splits (train_R / val_R / test):** each volatility regime appears in train and validation as purged interleaved blocks; thresholds are learned per regime on train_R, never assumed.
- **"Supported" means:** same sign in all three periods, |t| above the Bonferroni bar for the number of searches, and for gated rules, better than the parent rule in train, validation and test.
- **Costs:** 6 bp round trip counted in every net number; overlapping trades are clustered into 10-day blocks so they are not counted as independent.
- **Causality:** 418 truncation checks on the real builders, 0 failed. Each structure is also run on the negated series and drift is removed.

PART A · DATA

What was done about the data

- 39 products have enough history; 2005-02-01 to 2021-10-18.
- Removed: 99 calendar-gap rows, 7,858 stale or zero-volume bars, 678 short-day rows, 8,040 duplicate timestamps (Step 1). Kept as real: 176 large daily moves, 214 one-minute moves over the daily limit.
- Contract rolls are back-adjusted; jump sizes are removed from returns.
- Every data-quality flag has a rule and no rule uses future data.

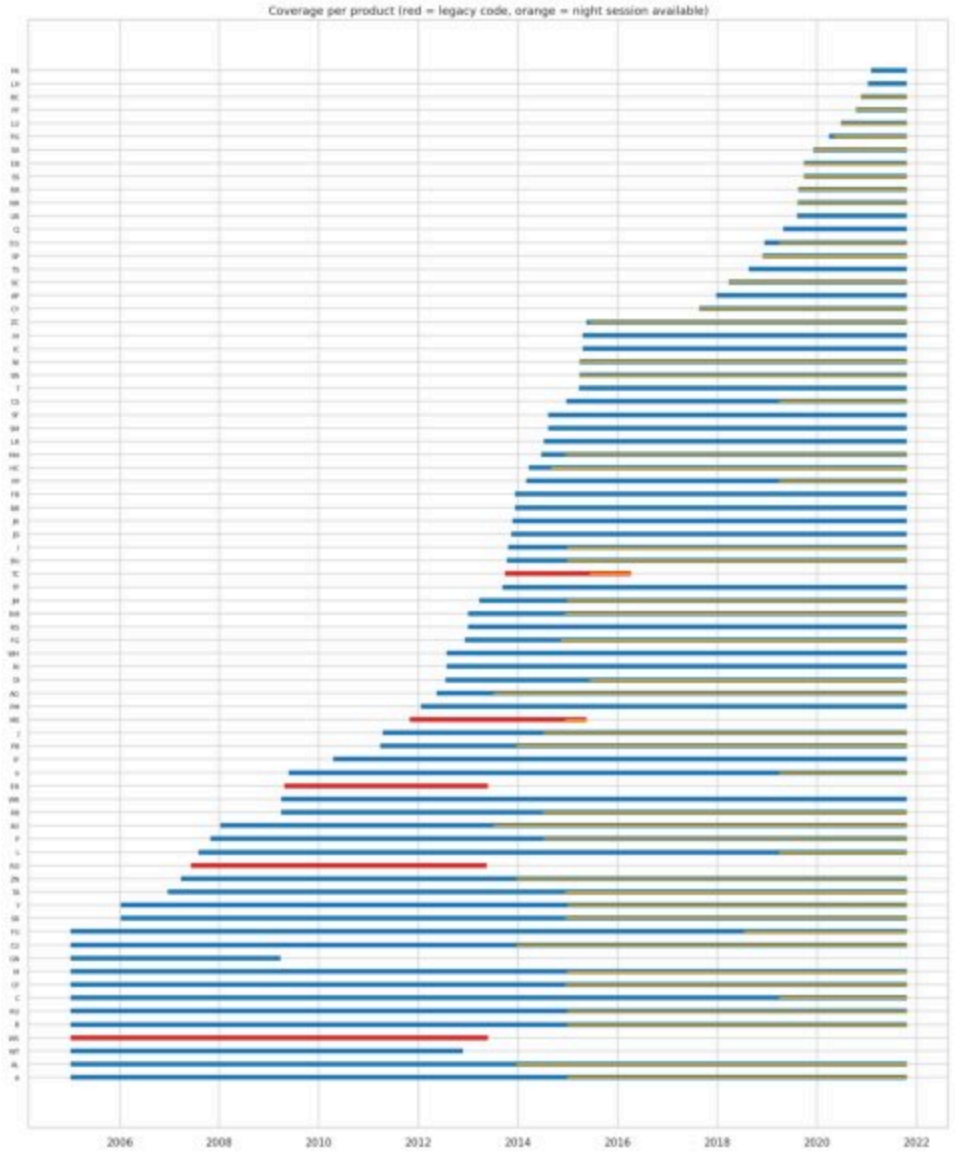
Next: the figures of Part A.

Part A

Data and EDA

Coverage, roll adjustment, data flags, volatility, intraday shape, correlation and first feature checks.

Coverage per product (red = legacy code, orange = night session available)



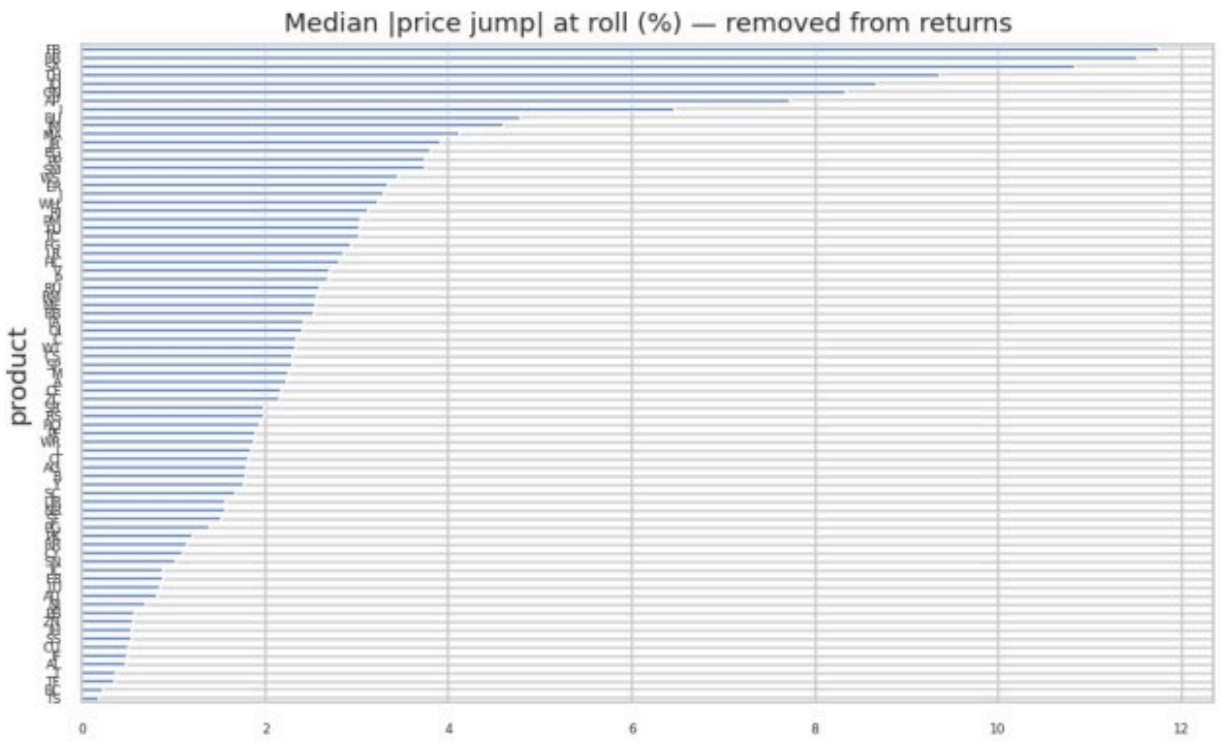
WHAT IT IS

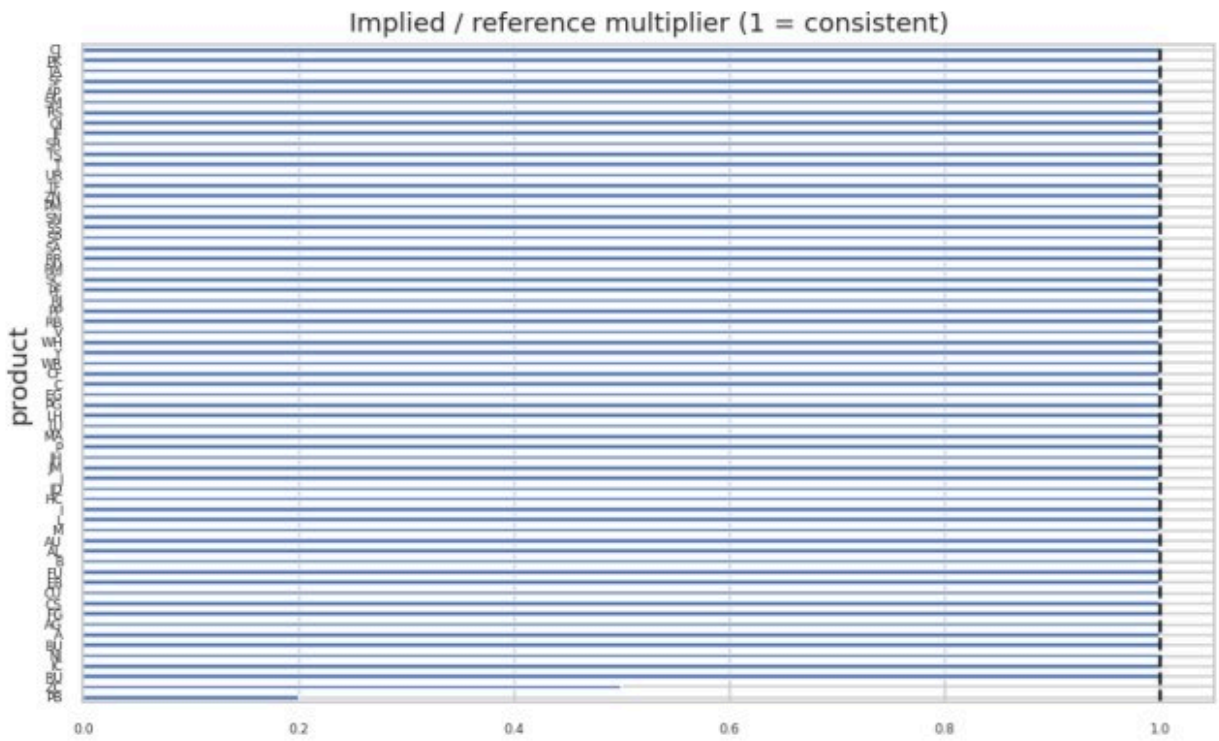
One bar per product showing the years of data available. Red = legacy contract code, orange = a night session exists.

WHAT IT EXPLAINS

How much history each of the 39 products has, and where intraday or night data start. Short-history products carry less weight in every later test; the usable sample is 2005-02 to 2021-10.

Median |price jump| at roll (%) | removed from returns





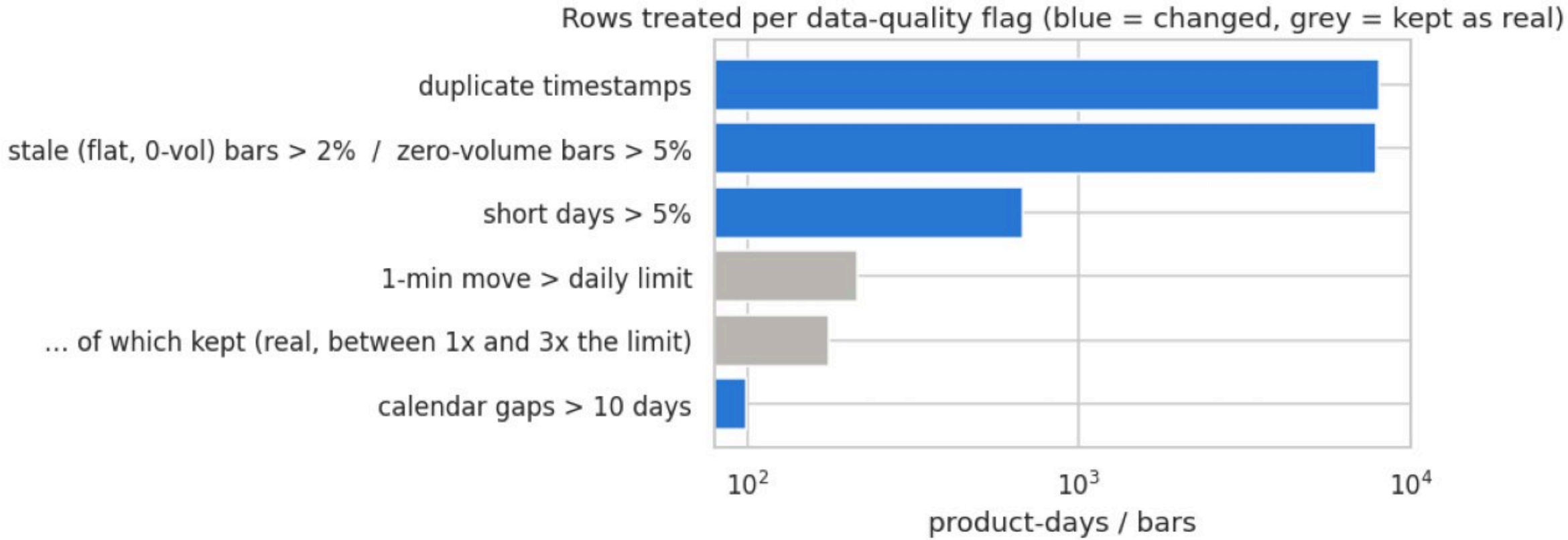
WHAT IT IS

Median size of the price jump on roll days (%), the amount removed so contract changes are not counted as returns.

WHAT IT EXPLAINS

Back-adjustment works: the jump at each contract roll is removed from the return series, so later results are not driven by roll gaps.

Rows treated per data-quality flag (blue = changed, grey = kept as real)



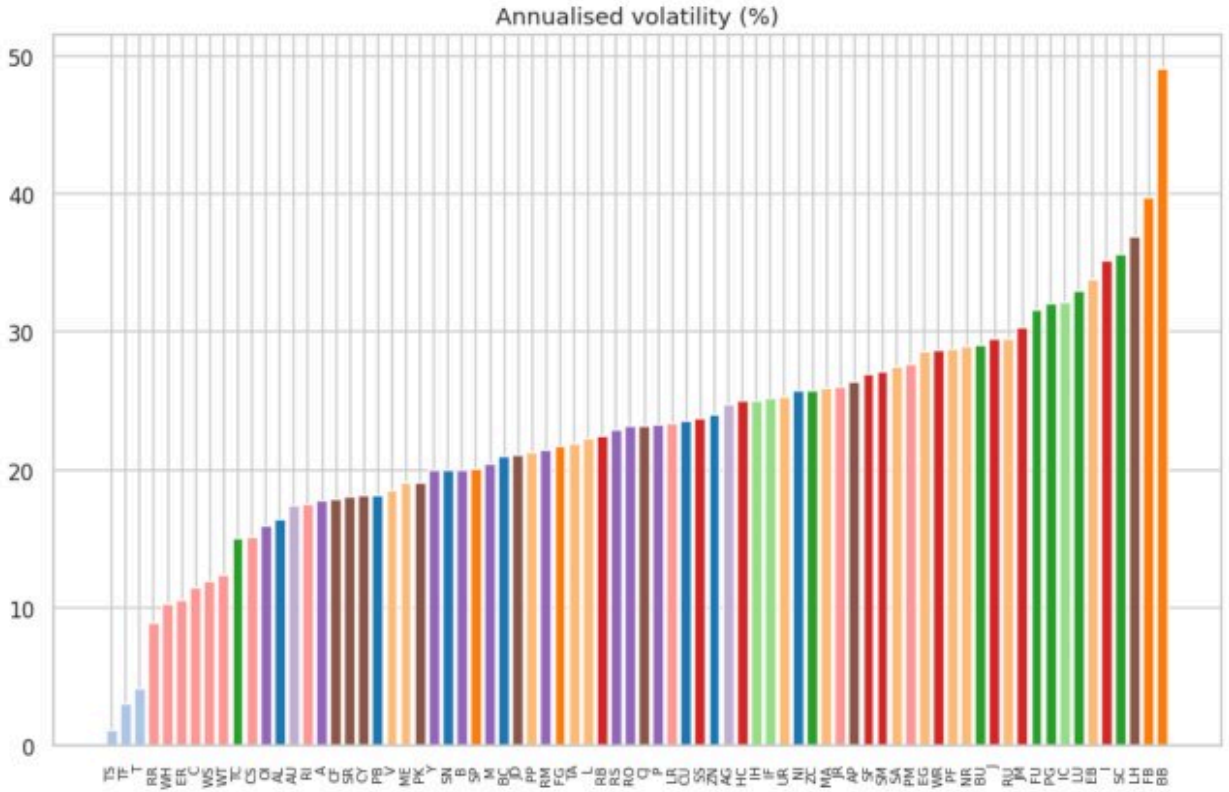
WHAT IT IS

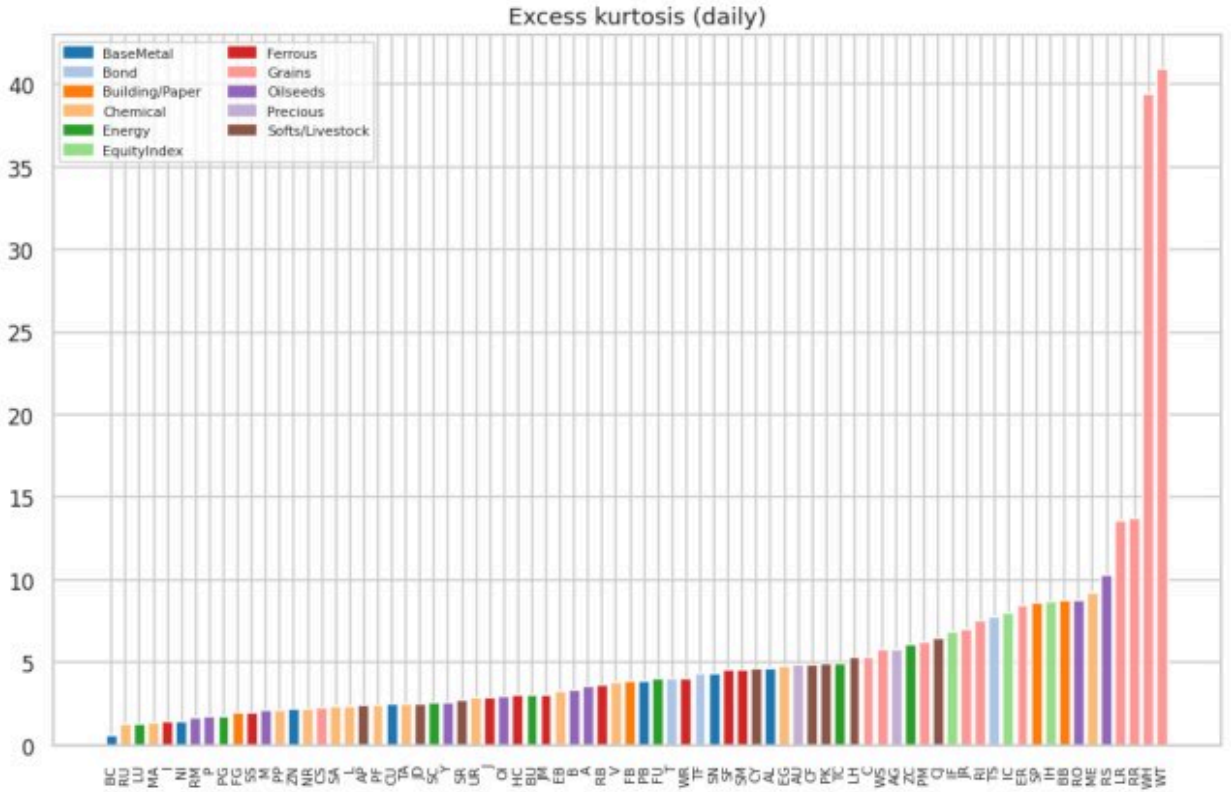
Number of rows treated per data-quality flag. Blue = changed, grey = kept because the event is real.

WHAT IT EXPLAINS

Every flag in Step 2f has an explicit rule. Errors (stale bars, calendar gaps, duplicate stamps) are removed; real extreme moves are kept and only clipped. No rule uses future data.

Annualised volatility (%)





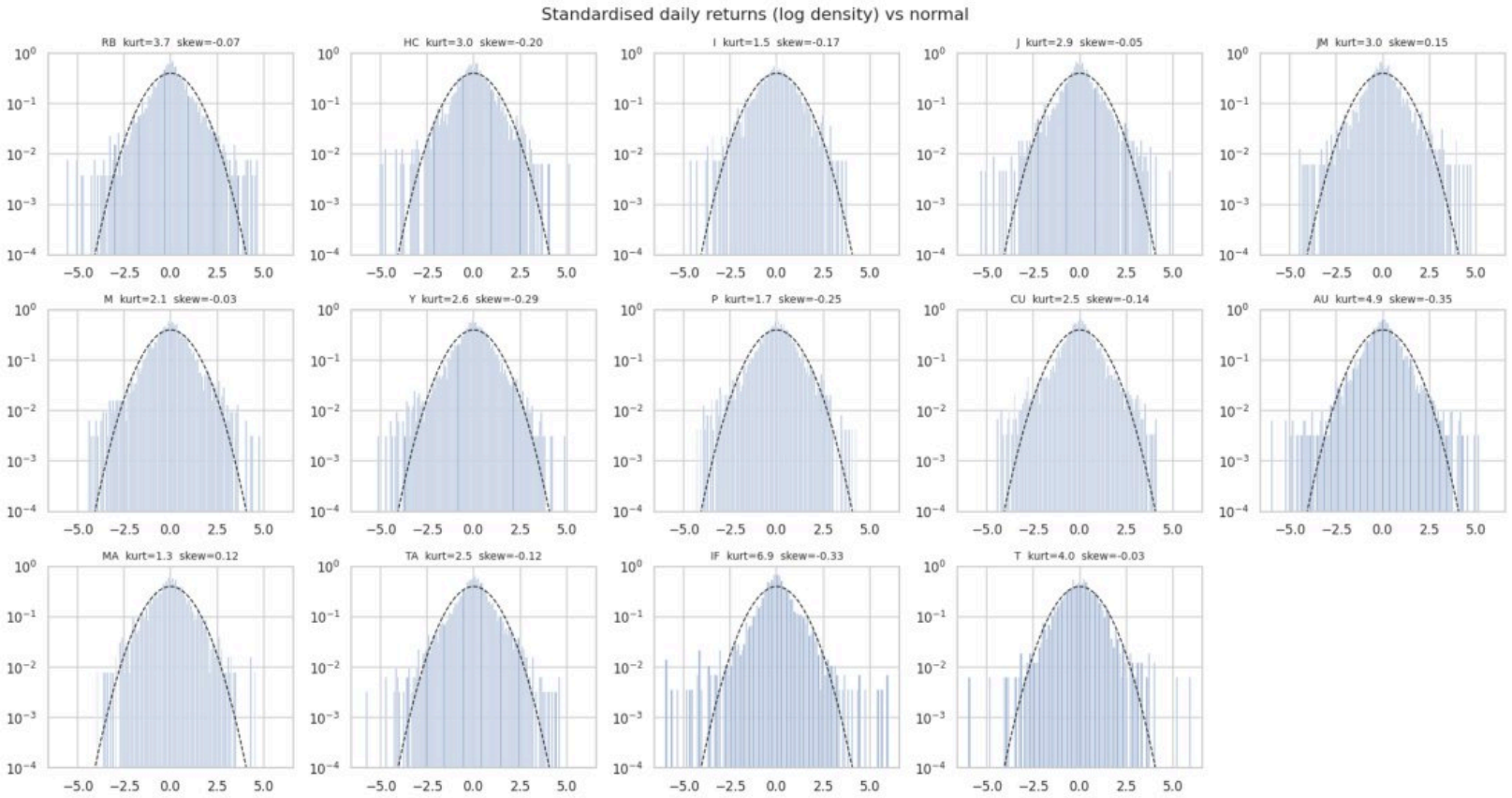
WHAT IT IS

Annualised volatility (%) of each product.

WHAT IT EXPLAINS

Products differ a lot in risk, so every signal is scaled by the product's own volatility before comparison.

Standardised daily returns (log density) vs normal



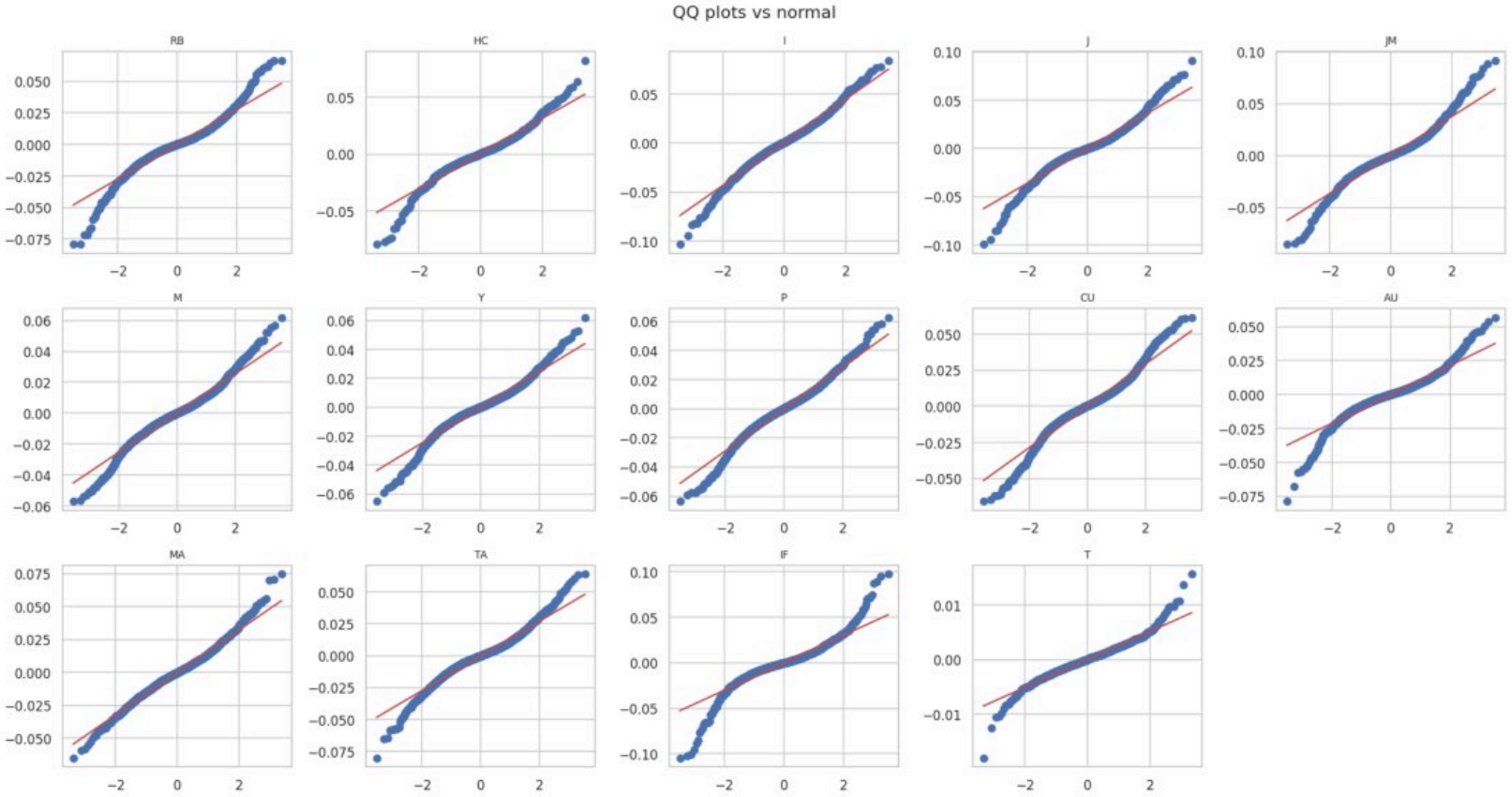
WHAT IT IS

Log-density of standardised daily returns against a normal curve.

WHAT IT EXPLAINS

Returns are fat-tailed: extreme days are far more frequent than a normal curve allows. Normal-based risk numbers understate tail losses.

QQ plots vs normal



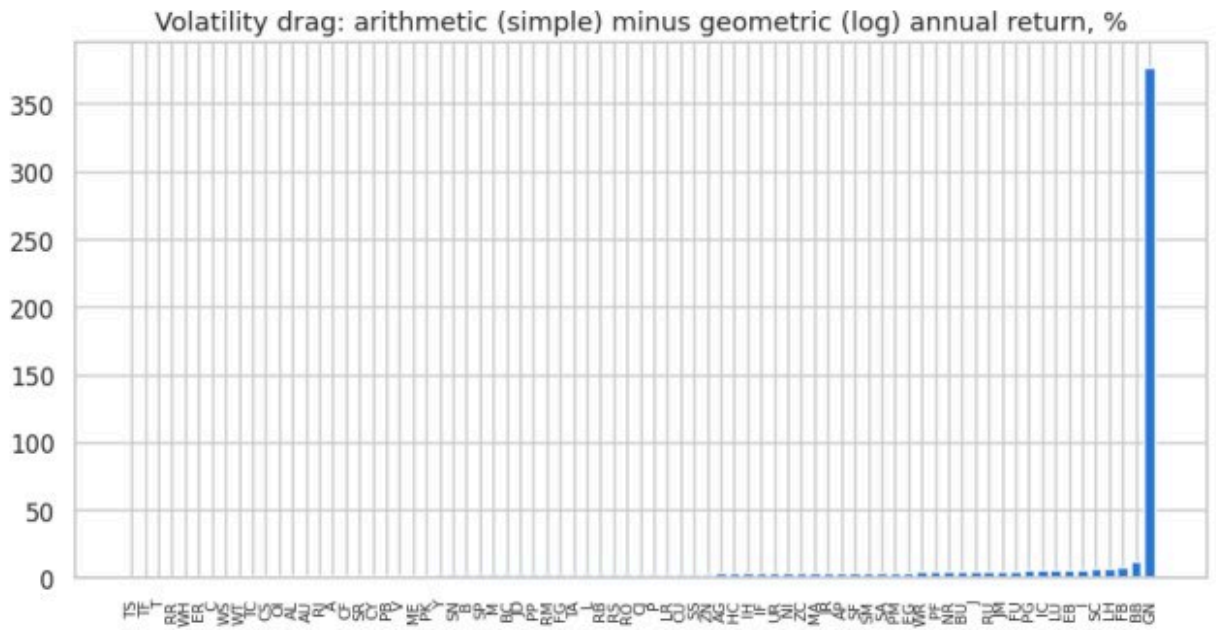
WHAT IT IS

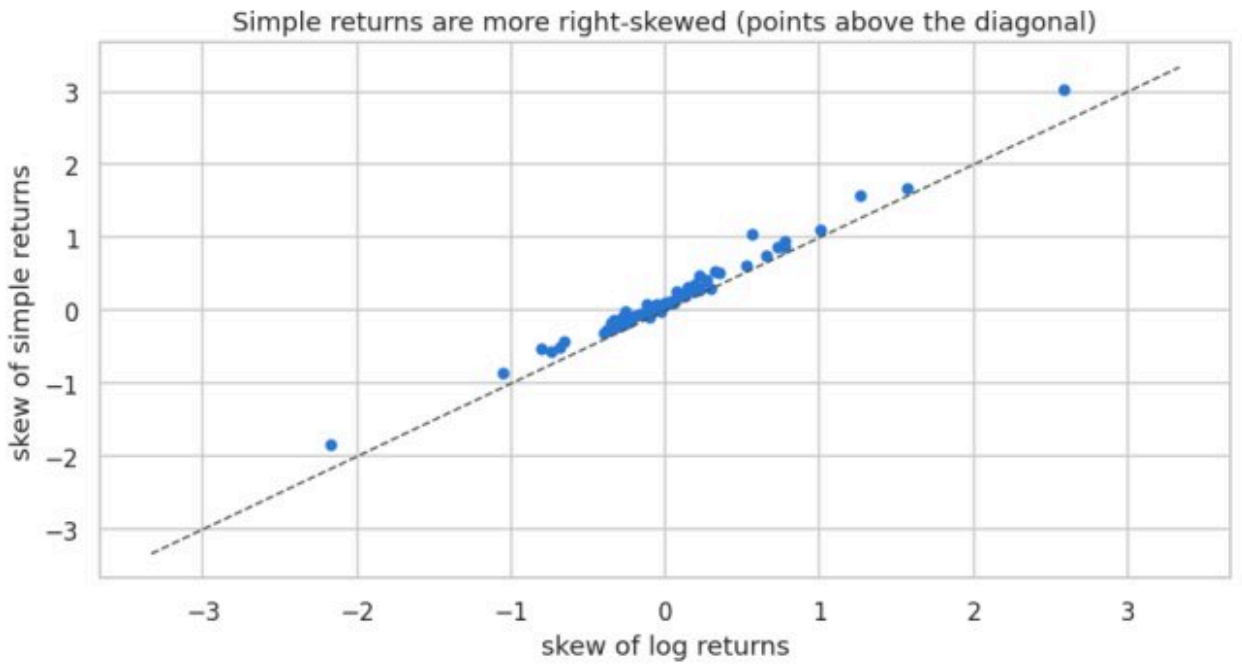
QQ plots of returns against the normal distribution.

WHAT IT EXPLAINS

Points leaving the line in the tails confirm the heavy tails seen in the density plot.

Volatility drag: arithmetic (simple) minus geometric (log) annual return, %





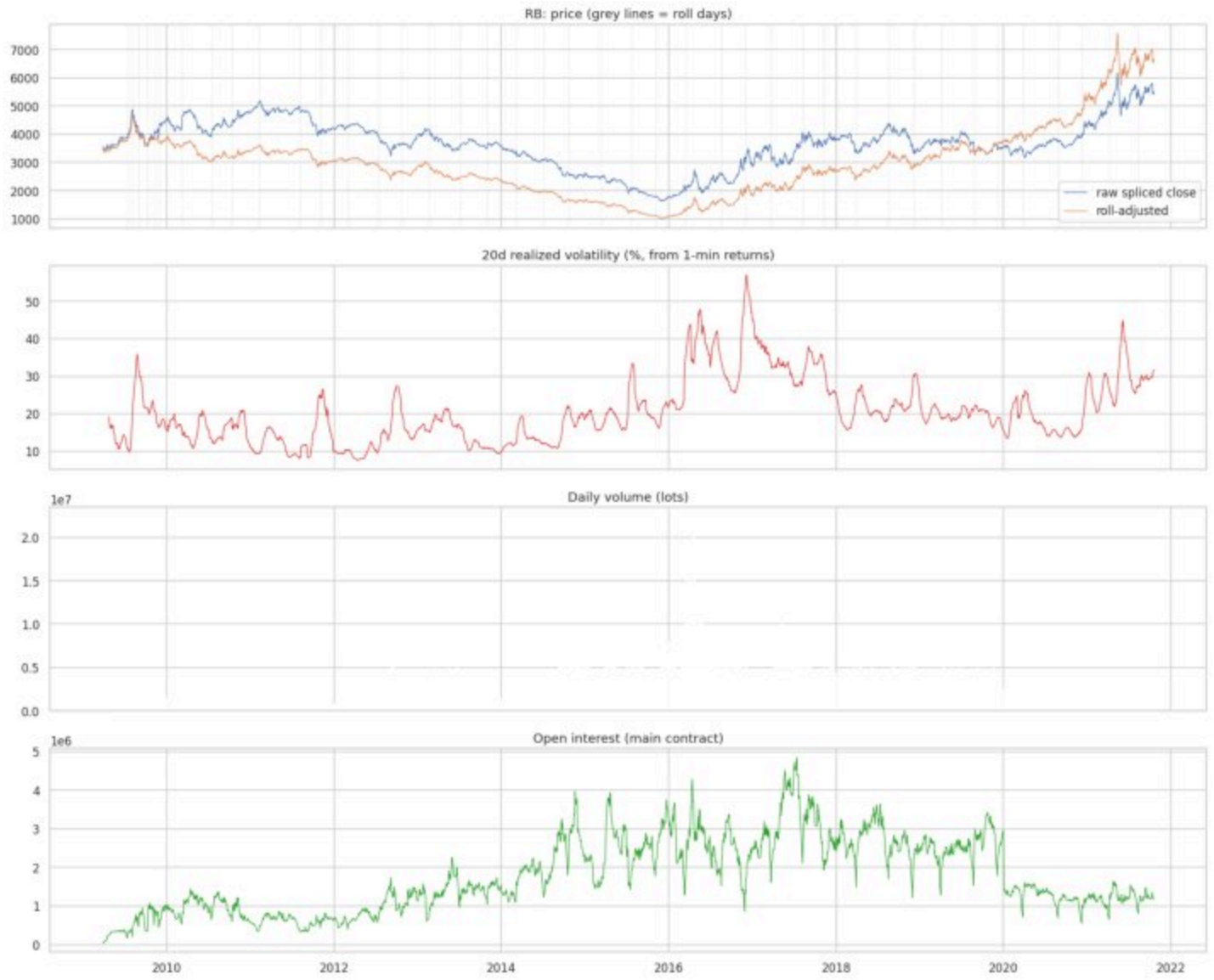
WHAT IT IS

Volatility drag: arithmetic minus geometric annual return (%).

WHAT IT EXPLAINS

How much compounding loses to volatility in each product. High-volatility products lose more of their simple average return.

RB: price (grey lines = roll days)



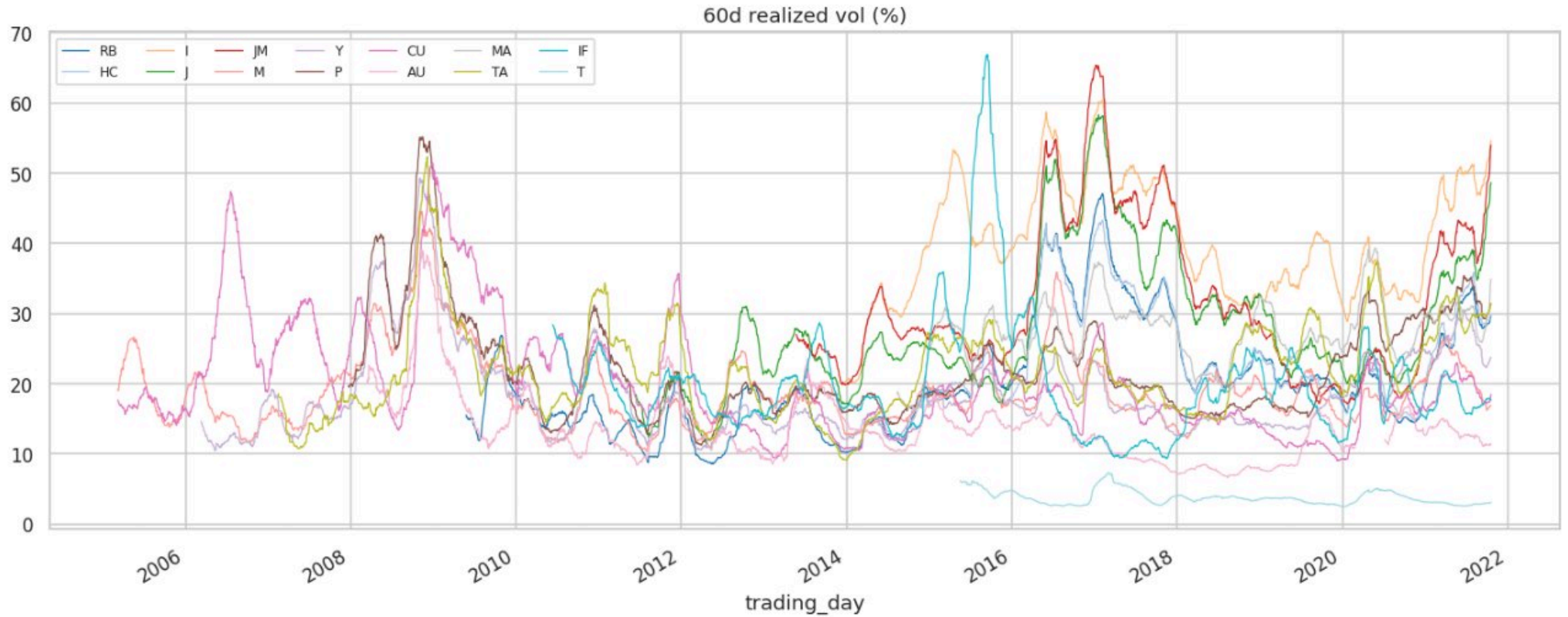
WHAT IT IS

Price of RB (rebar) with grey lines on roll days.

WHAT IT EXPLAINS

Example of the continuous series used throughout: the price path and where contracts were rolled.

60d realized vol (%)



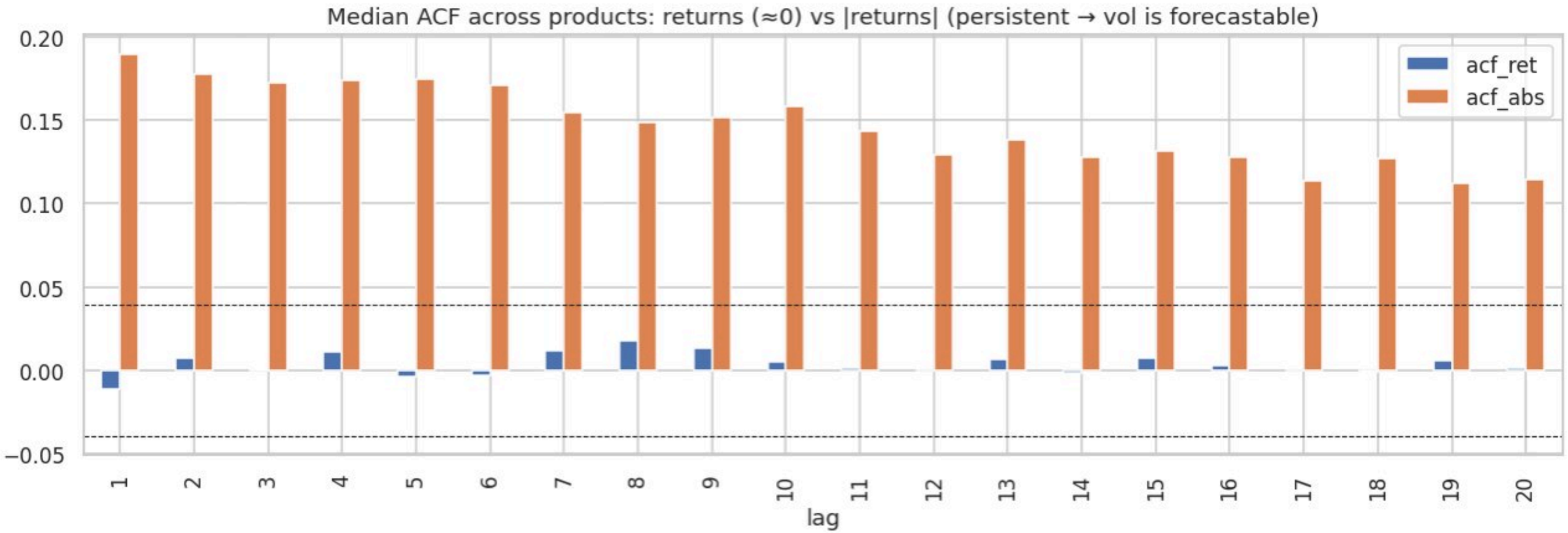
WHAT IT IS

60-day realised volatility (%) through time.

WHAT IT EXPLAINS

Volatility clusters: calm and turbulent spells last, which is why volatility is forecastable while returns are not.

Median ACF across products: returns (0) vs |returns| (persistent vol is forecastable)



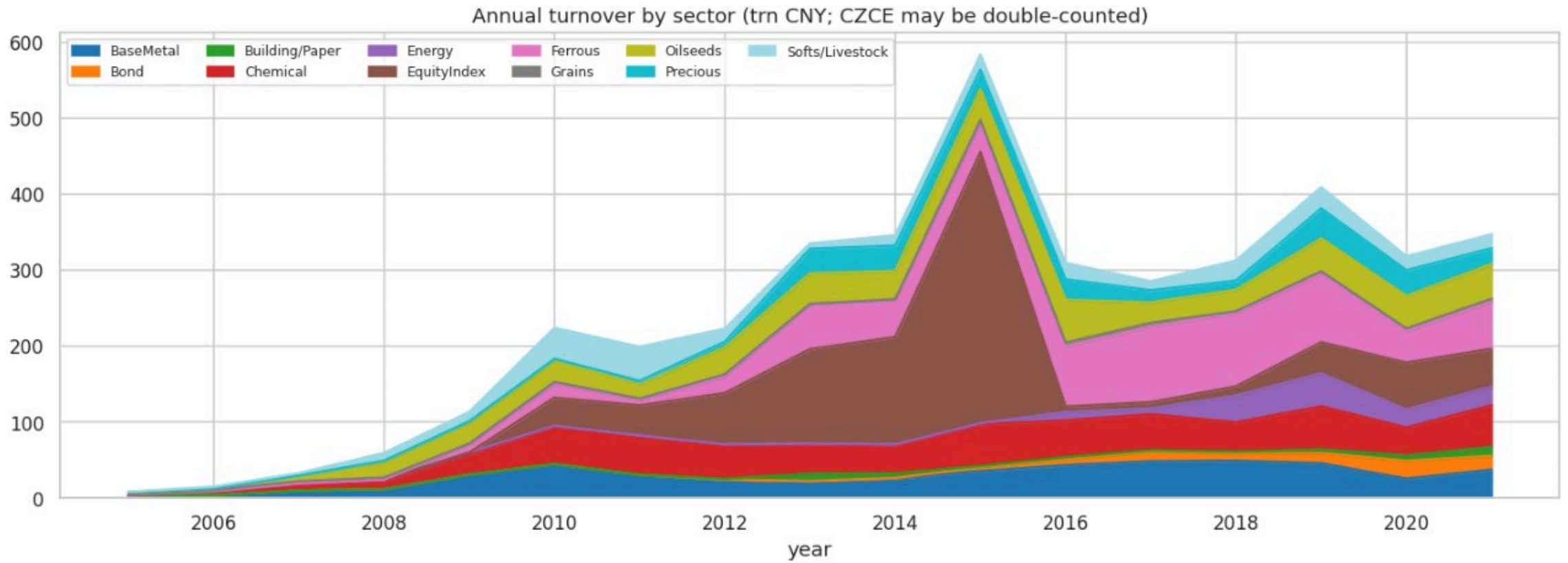
WHAT IT IS

Median autocorrelation across products for returns versus absolute returns.

WHAT IT EXPLAINS

Returns have almost no memory (about zero), absolute returns stay correlated for many lags. Volatility can be modelled; direction cannot be read from past returns alone.

Annual turnover by sector (trn CNY; CZCE may be double-counted)



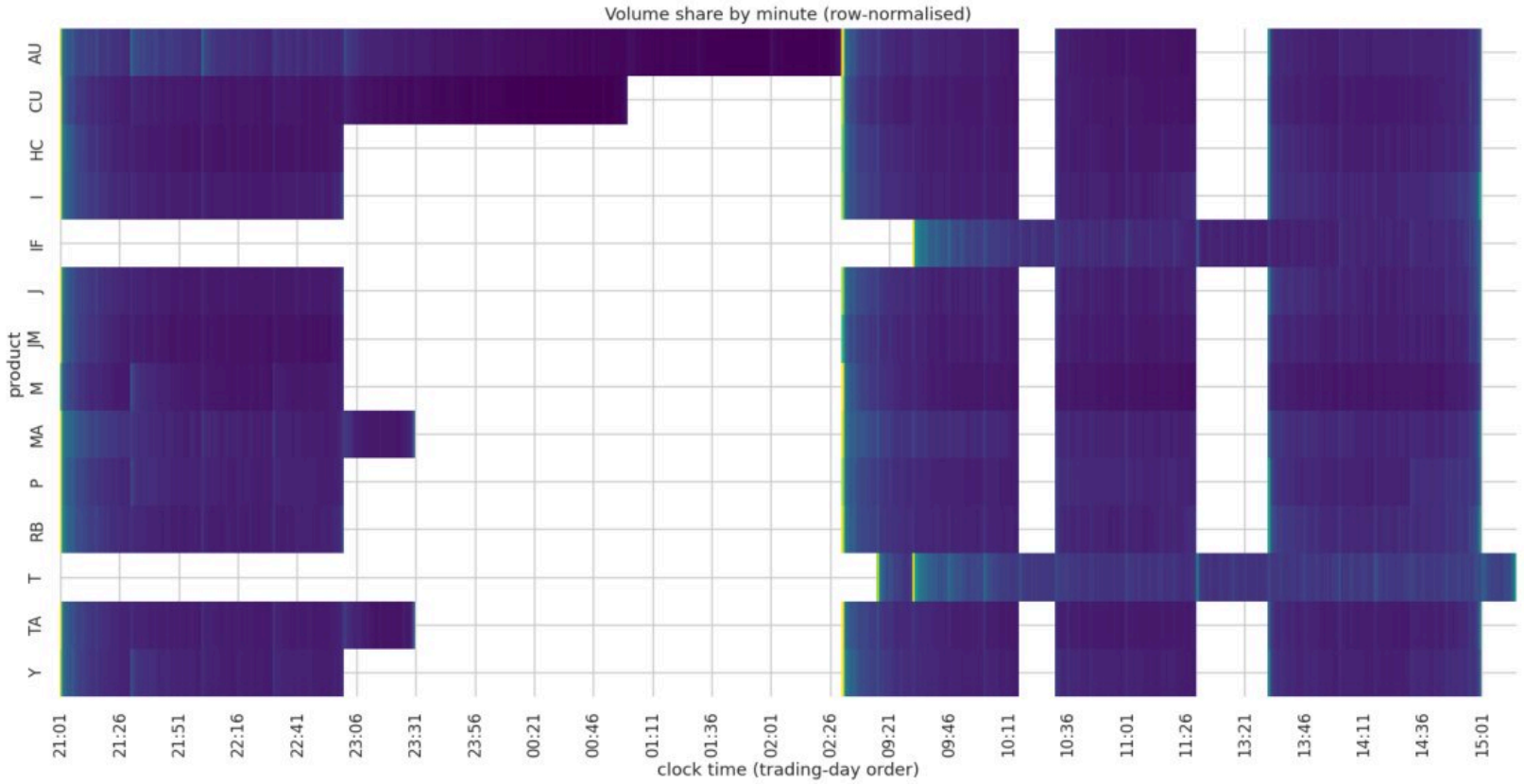
WHAT IT IS

Annual turnover by sector (trillion CNY).

WHAT IT EXPLAINS

Where the money trades. CZCE turnover may be double-counted, flagged in the data checks.

Volume share by minute (row-normalised)



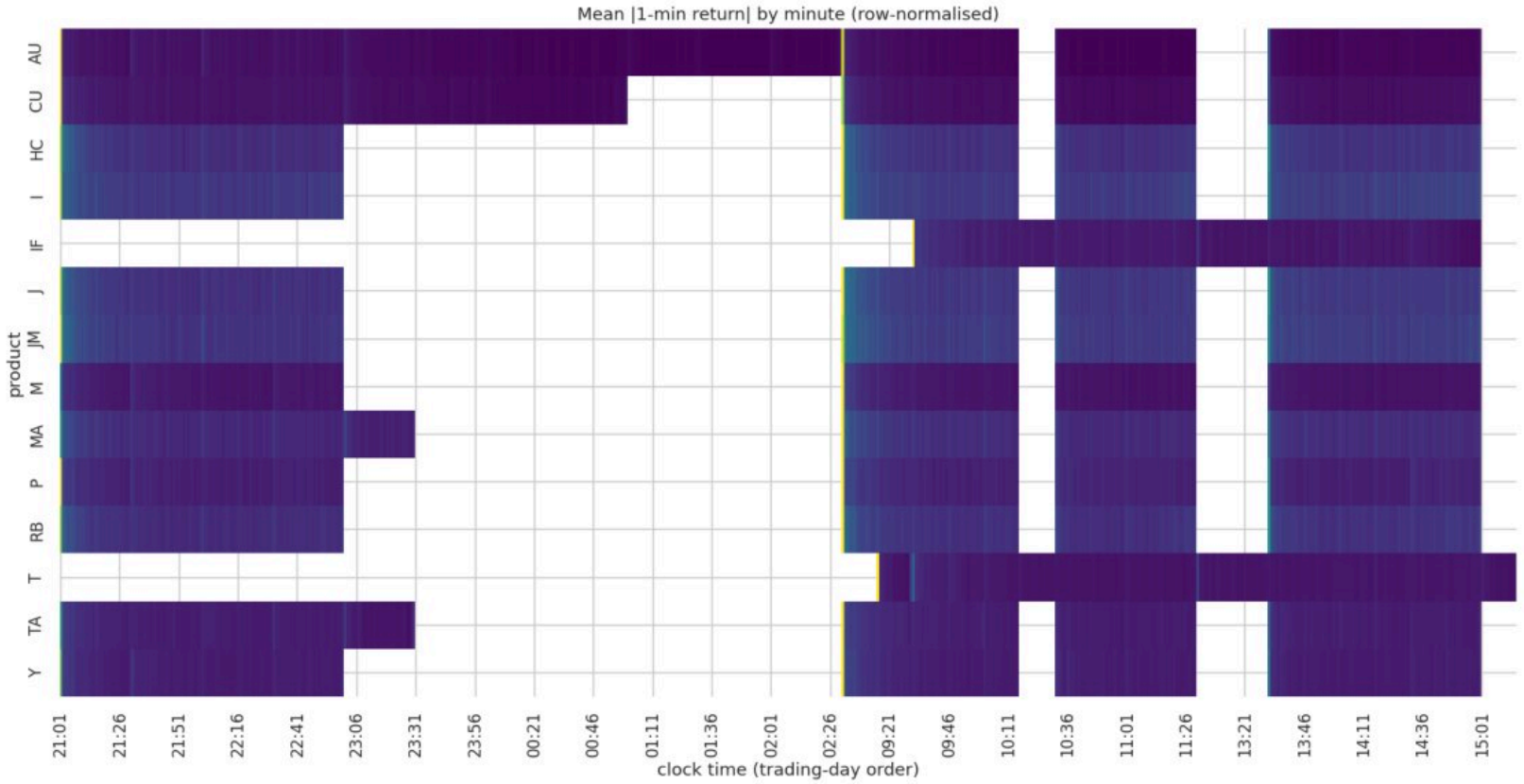
WHAT IT IS

Share of volume by minute of the day, each row normalised.

WHAT IT EXPLAINS

Trading activity is concentrated at the open and close. Volume features must be adjusted for time of day or they only measure the clock.

Mean |1-min return| by minute (row-normalised)



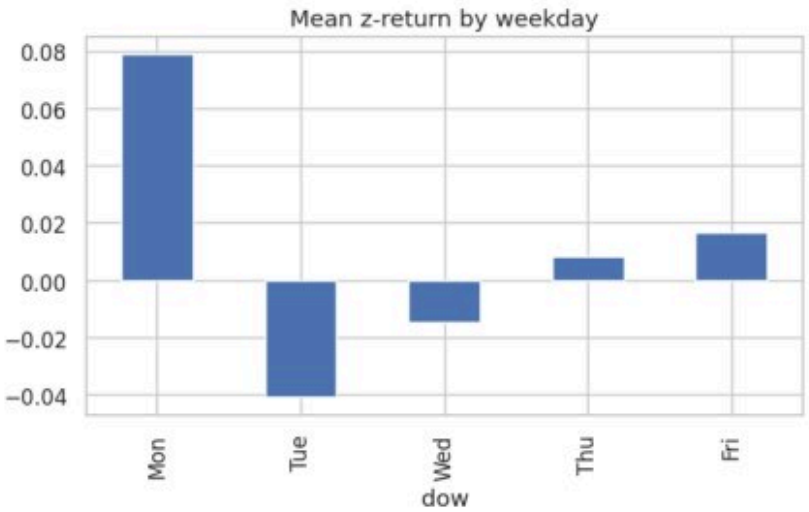
WHAT IT IS

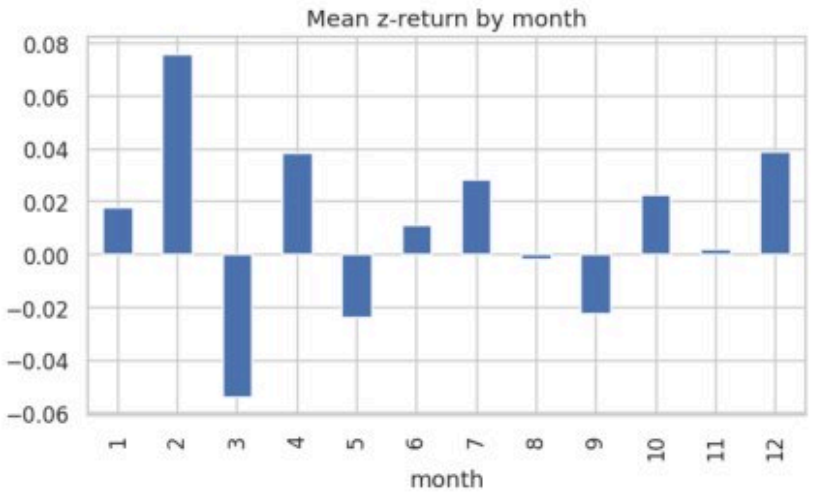
Mean absolute 1-minute return by minute, each row normalised.

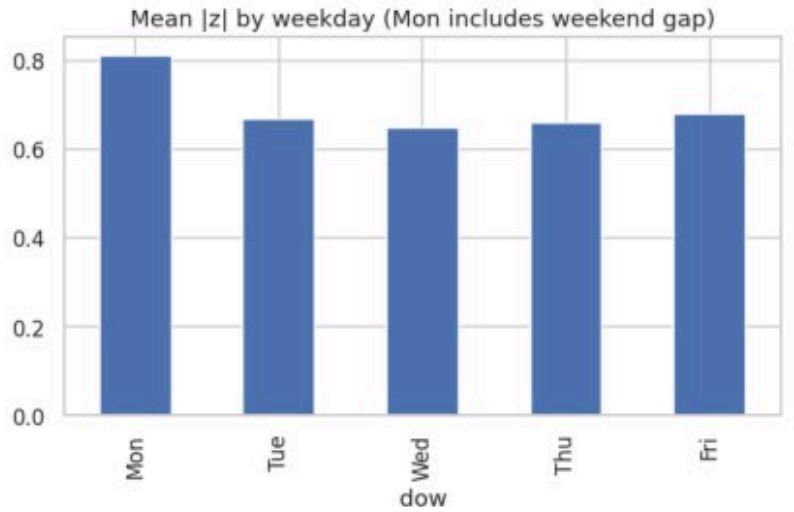
WHAT IT EXPLAINS

Volatility has the same intraday shape as volume: highest in the opening minutes. First-30-minute variance per bar is 2.2x the rest of the day.

Mean z-return by weekday







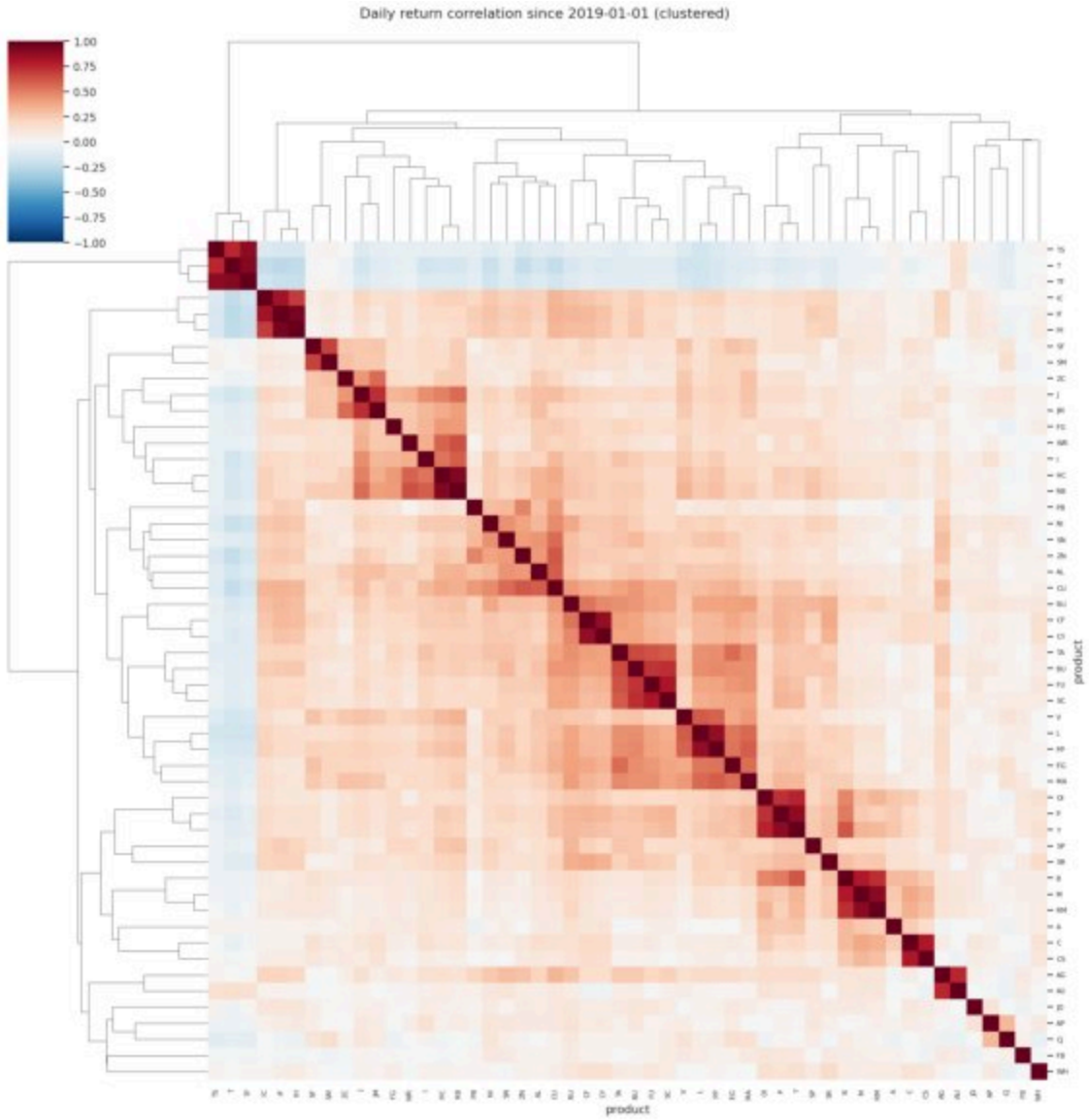
WHAT IT IS

Mean standardised return by weekday.

WHAT IT EXPLAINS

Checks for a day-of-week effect; any weekday pattern is small relative to its noise.

Daily return correlation since 2019-01-01 (clustered)



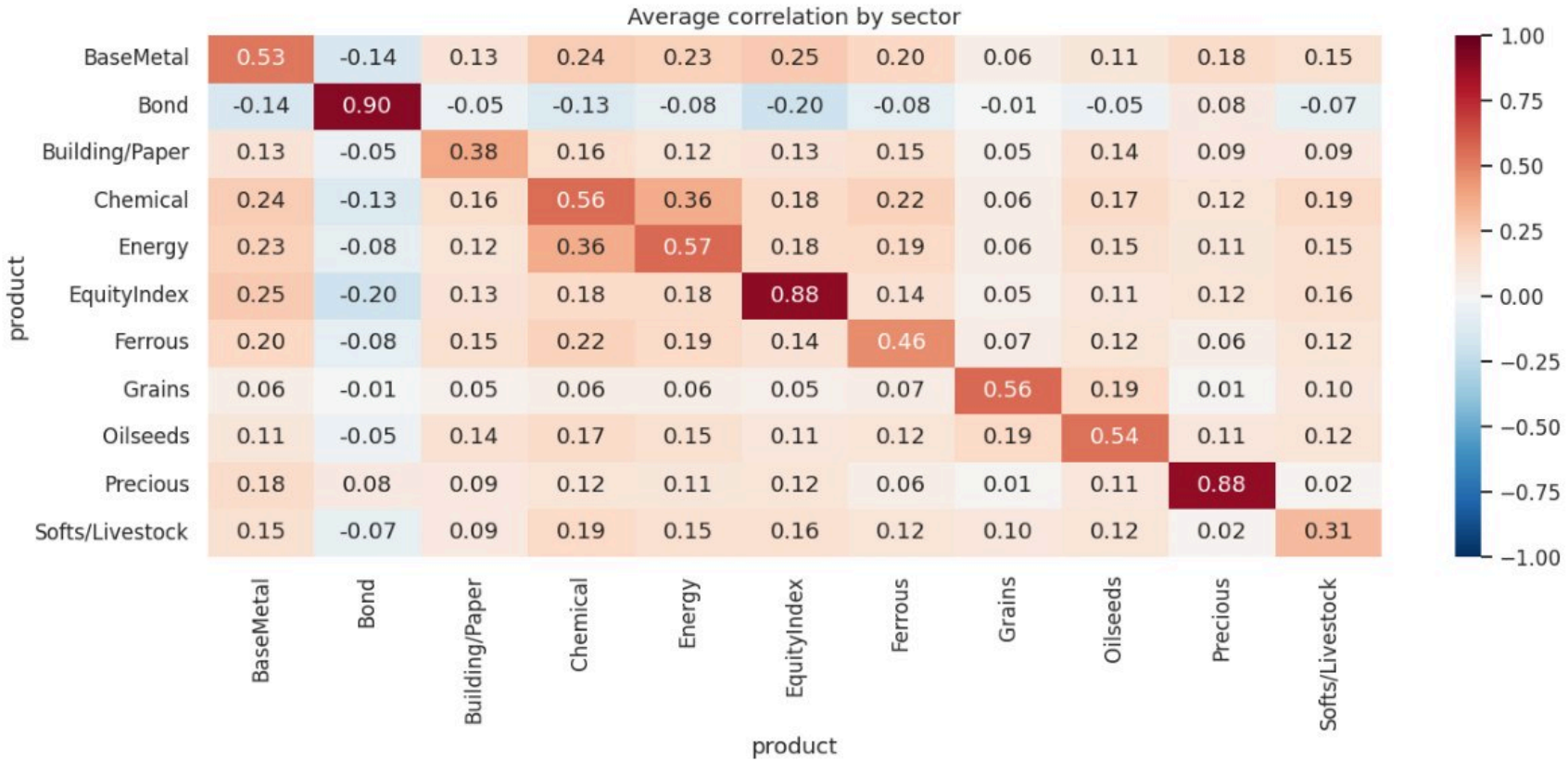
WHAT IT IS

Daily return correlation matrix since 2019, products clustered.

WHAT IT EXPLAINS

Products group into sector blocks; products in the same block move together and add little diversification.

Average correlation by sector



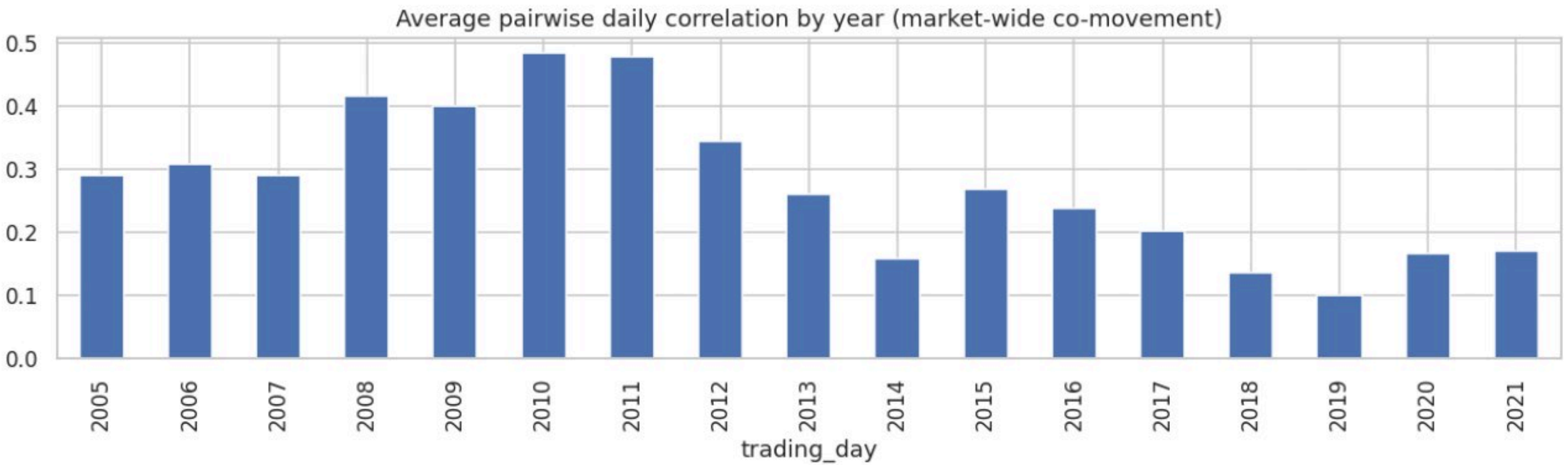
WHAT IT IS

Average correlation by sector.

WHAT IT EXPLAINS

Within-sector links are strongest, between-sector links weaker. Diversification has to come across sectors.

Average pairwise daily correlation by year (market-wide co-movement)



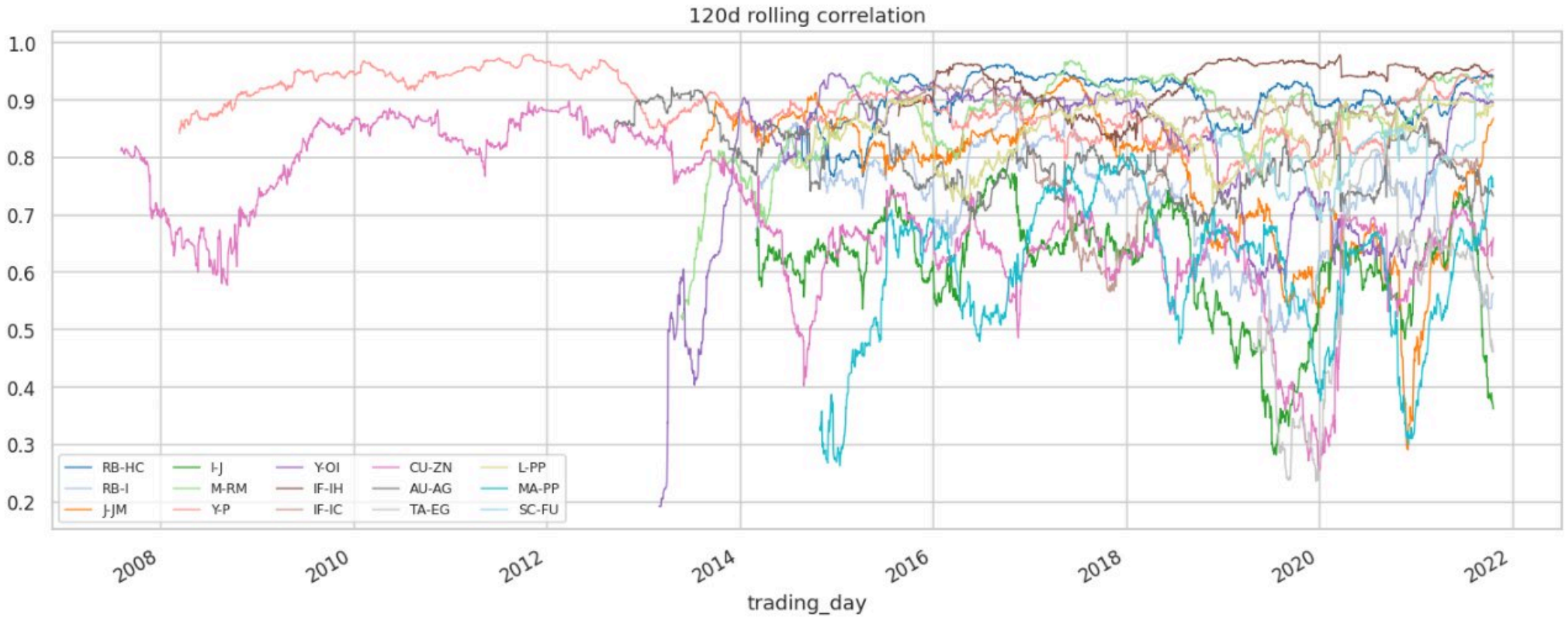
WHAT IT IS

Average pairwise daily correlation by year.

WHAT IT EXPLAINS

Market-wide co-movement rises and falls over time, so diversification is weaker in some years.

120d rolling correlation



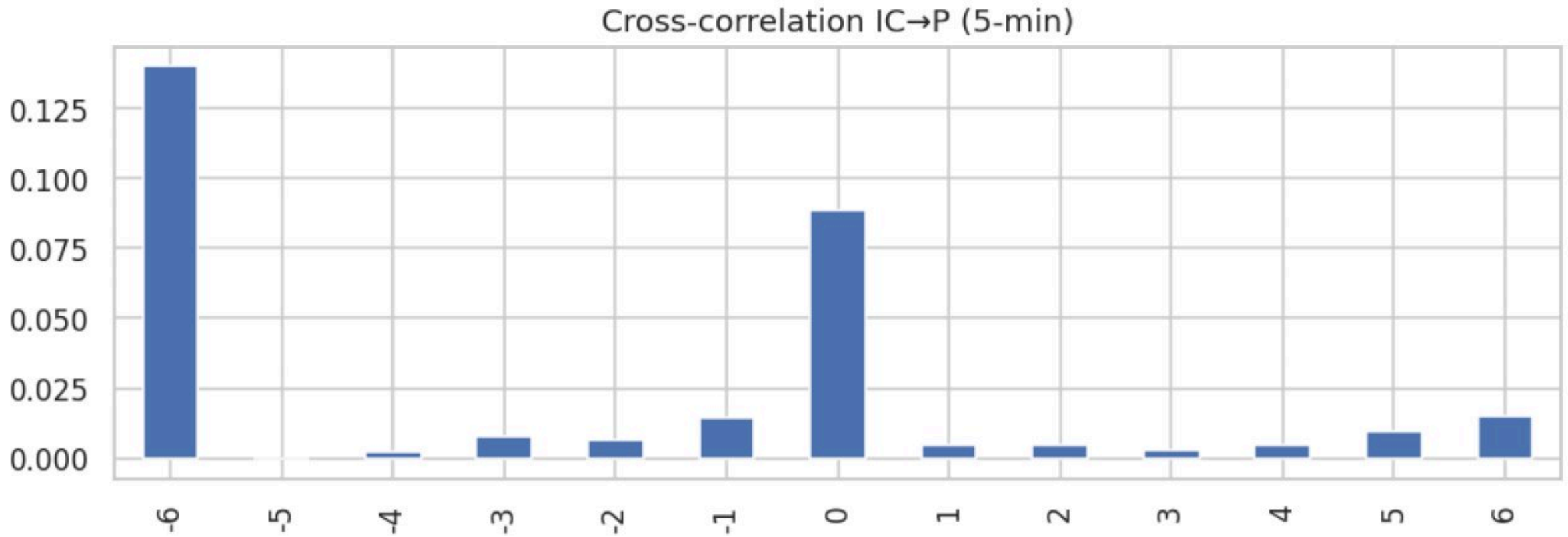
WHAT IT IS

120-day rolling correlation for chosen pairs.

WHAT IT EXPLAINS

Pair correlations drift; a relationship measured in one window is not guaranteed in the next.

Cross-correlation IC P (5-min)



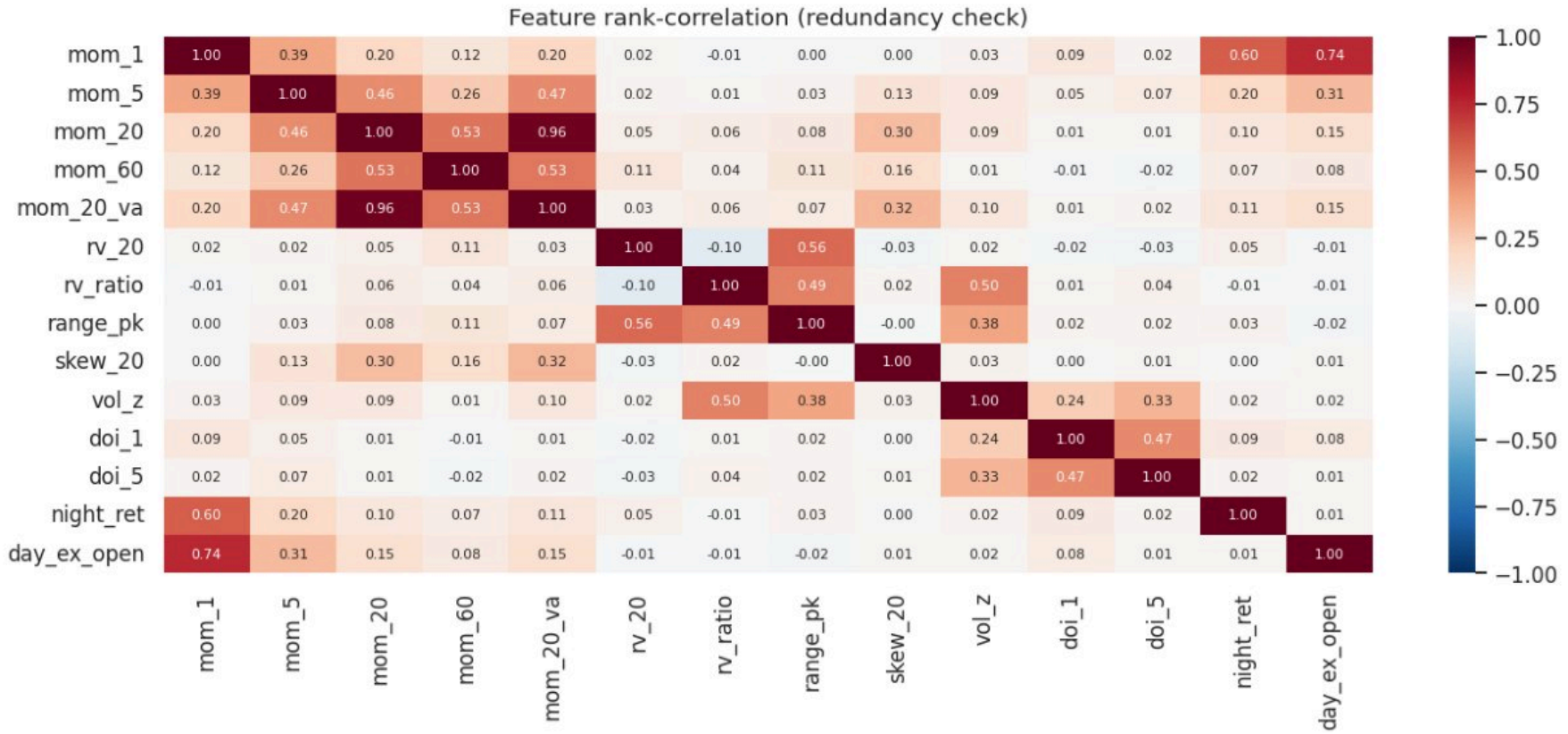
WHAT IT IS

Cross-correlation of 5-minute returns (IC) between leaders and followers.

WHAT IT EXPLAINS

Tests whether one product's move leads another's within minutes. Used as the starting point for the lead-lag work in Step 14.

Feature rank-correlation (redundancy check)



WHAT IT IS

Rank-correlation matrix between candidate features.

WHAT IT EXPLAINS

Redundancy check: features that are almost the same information are not counted as separate evidence.

Cumulative cross-sectional IC vs next-day return (straight line = stable signal)



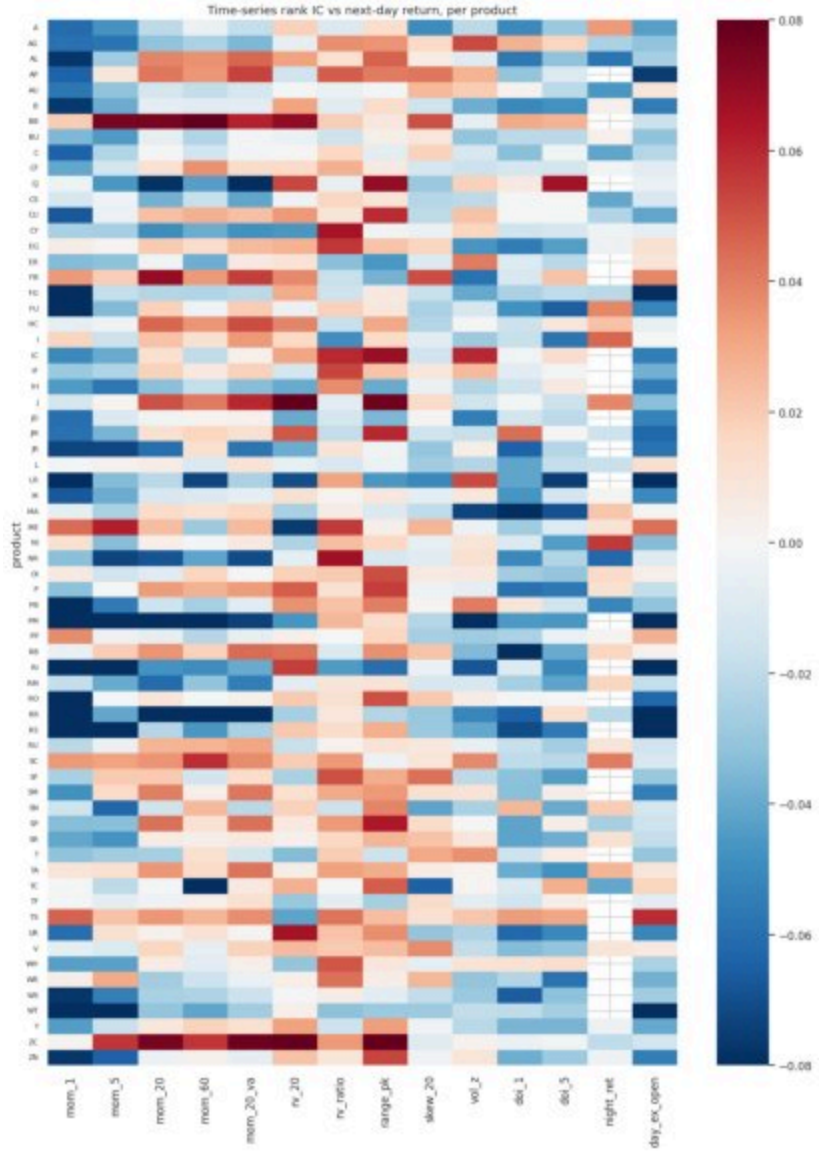
WHAT IT IS

Cumulative cross-sectional information coefficient (IC) against next-day return. A straight rising line would mean a stable signal.

WHAT IT EXPLAINS

Most features give a flat or wandering line: little stable cross-sectional daily predictability.

Time-series rank IC vs next-day return, per product



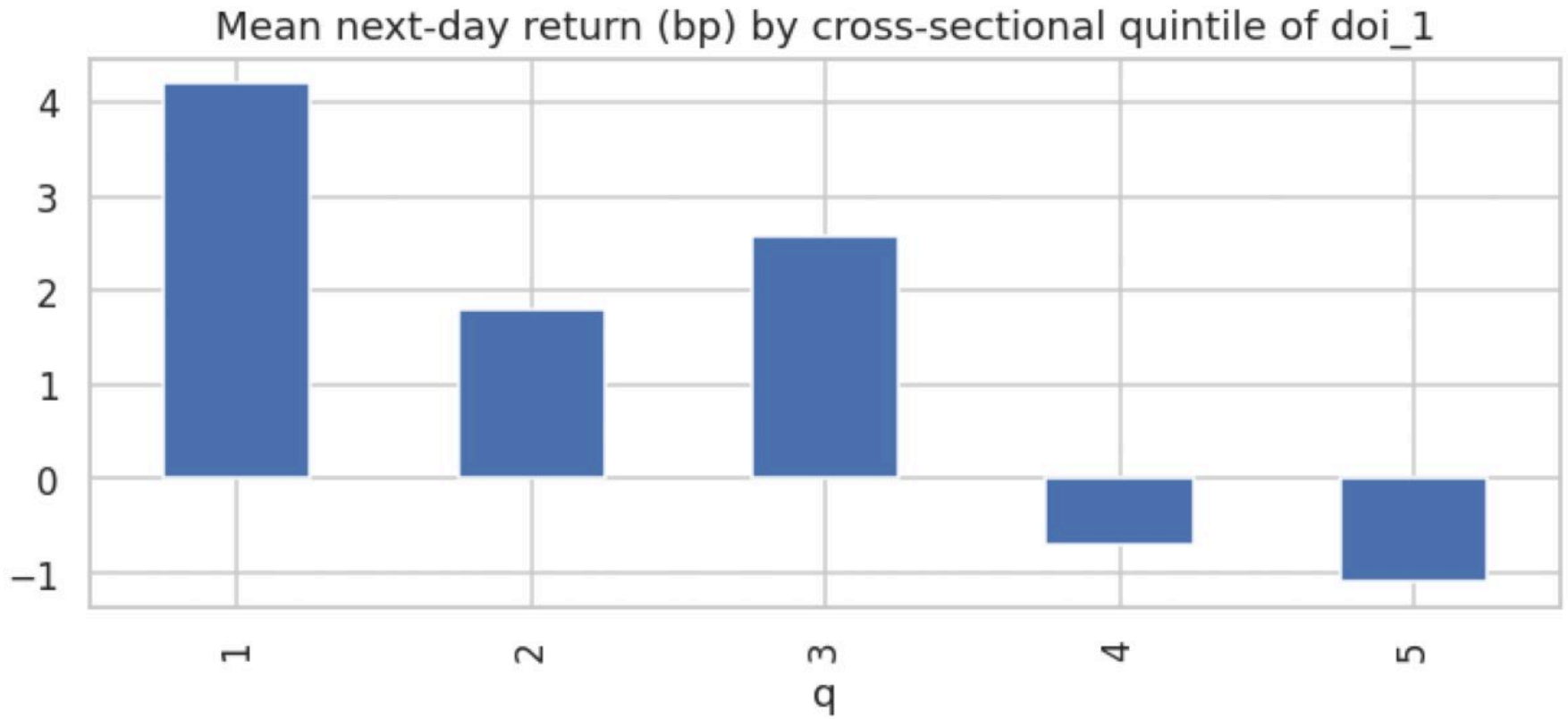
WHAT IT IS

Time-series rank IC against next-day return, per product.

WHAT IT EXPLAINS

Predictive power of a product's own signal. Values cluster near zero.

Mean next-day return (bp) by cross-sectional quintile of doi_1



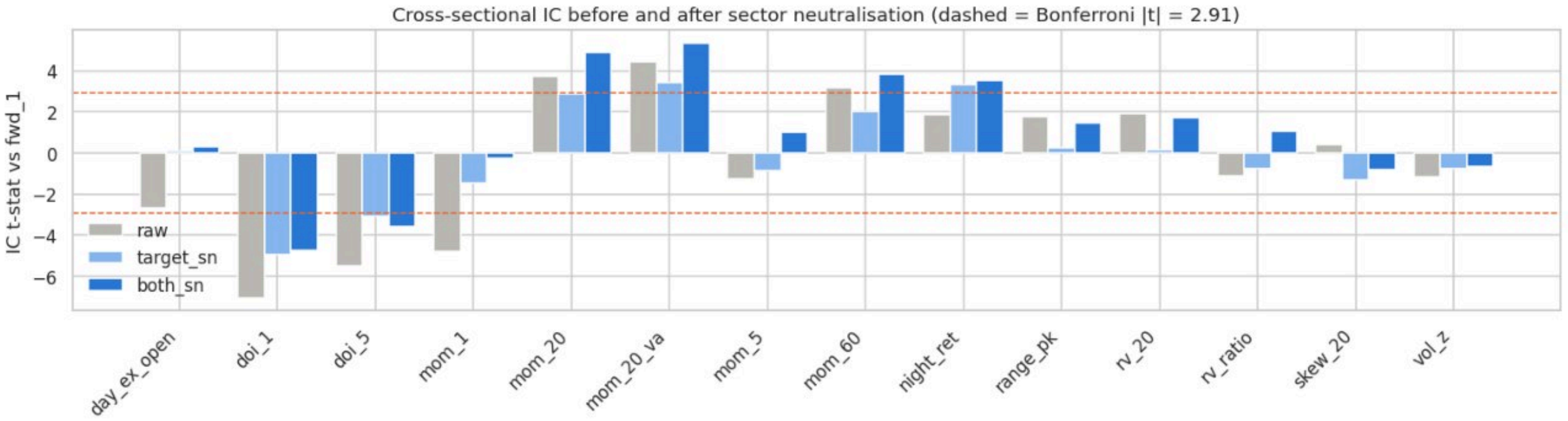
WHAT IT IS

Mean next-day return (bp) by cross-sectional quintile of the day's open-interest change.

WHAT IT EXPLAINS

If the signal worked, returns would step up from quintile 1 to 5. The pattern is weak.

Cross-sectional IC before and after sector neutralisation (dashed = Bonferroni |t| = 2.91)



WHAT IT IS

Cross-sectional IC before and after removing sector effects; the dashed line is the Bonferroni |t| = 2.91.

WHAT IT EXPLAINS

Whether a signal survives once sector moves are removed. Effects that vanish after neutralisation were sector bets, not product-level information.

PART B · FINDINGS

Part B in one page: what the series look like

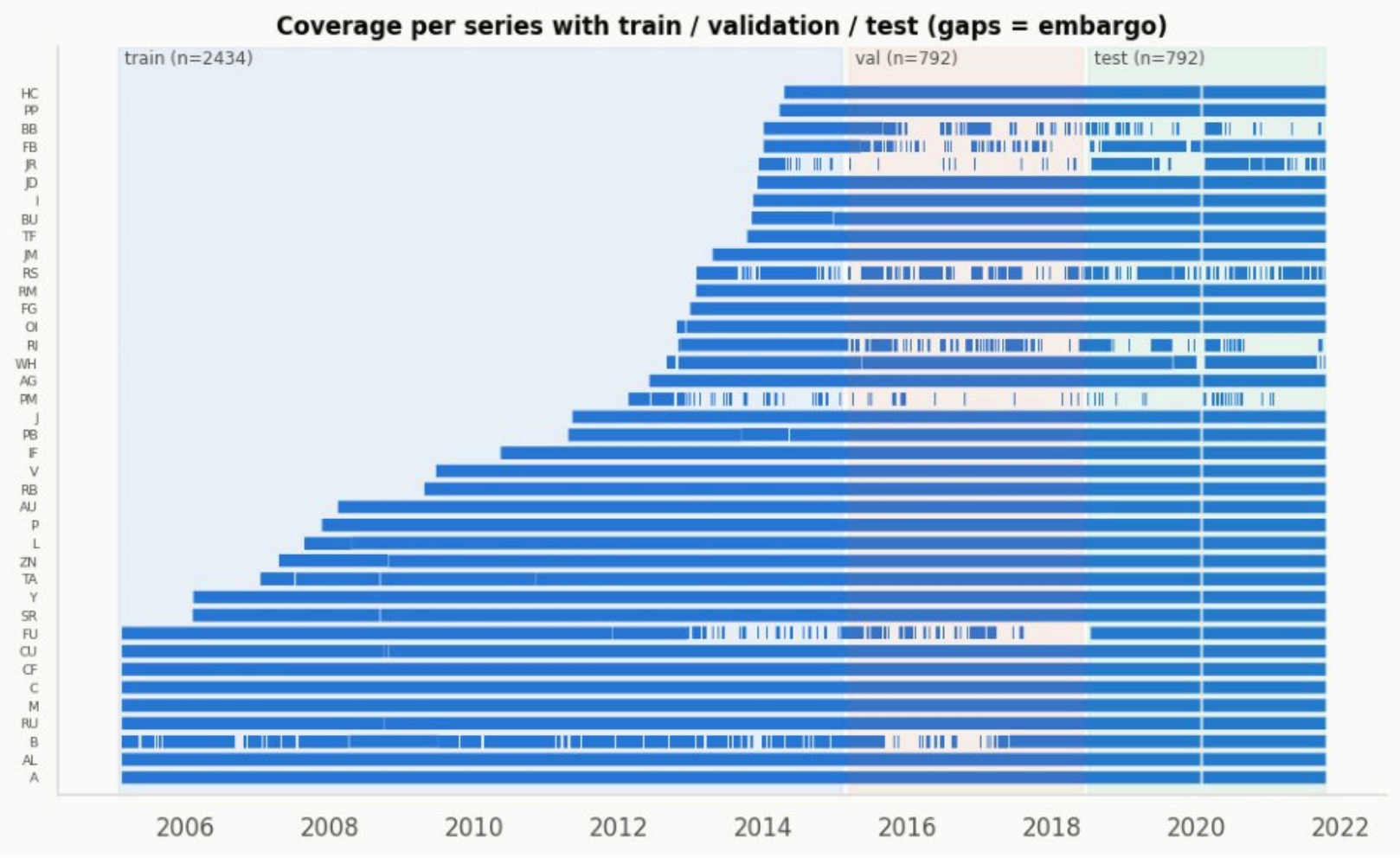
- **Prices non-stationary, returns stationary** in 96% of series. Median fractional-differencing order d* = 0.3.
- **Mean models give nothing:** best median test R-squared 0.0000.
- **Volatility:** HAR-RV wins in every time frame. 20 of 34 series have adequate residuals out of sample; the other 14 (AG, AL, BU, CF, CU, FU, I, JD, JM, PM, RI, RS, V, ZN) get less reliable tail and VaR numbers.
- **Tails and dependence:** heavy-tailed returns, one market factor = 41% of co-movement, Student-t copula wins.
- **Overnight puzzle:** overnight +5.3 bp/day vs intraday -0.8 bp on test, but no stable T+1 or regime pattern; all four regime hypotheses unsupported.
- **Night-session variance** improves HAR in HIGH volatility (t 4.9 on test): supported.

Part B

Univariate and multivariate properties

Stationarity, mean and volatility models, return distributions, tails, copulas and the overnight effect.

Coverage and splits



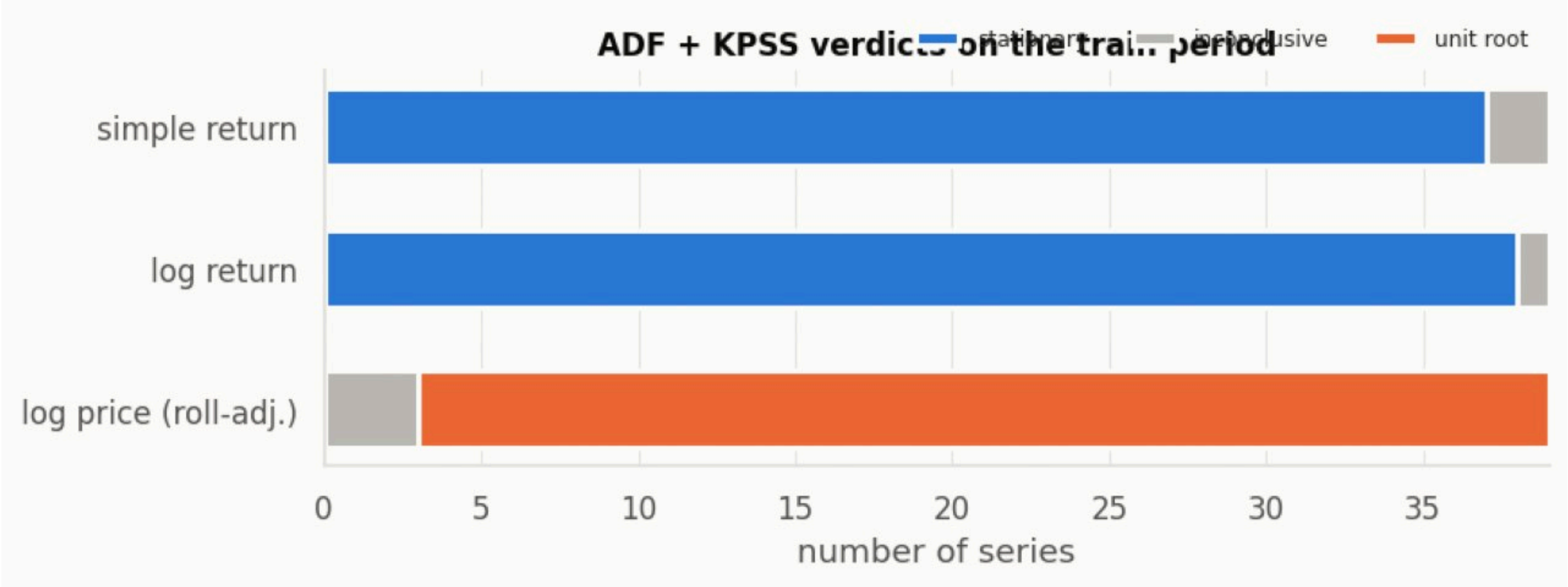
WHAT IT IS

Data coverage and the train / validation / test split.

WHAT IT EXPLAINS

One calendar split for all frequencies: train to 2015-02-09, validation 2015-03-17 to 2018-06-11, test from 2018-07-11 with a 20-day embargo. The test is read once.

Stationarity verdicts



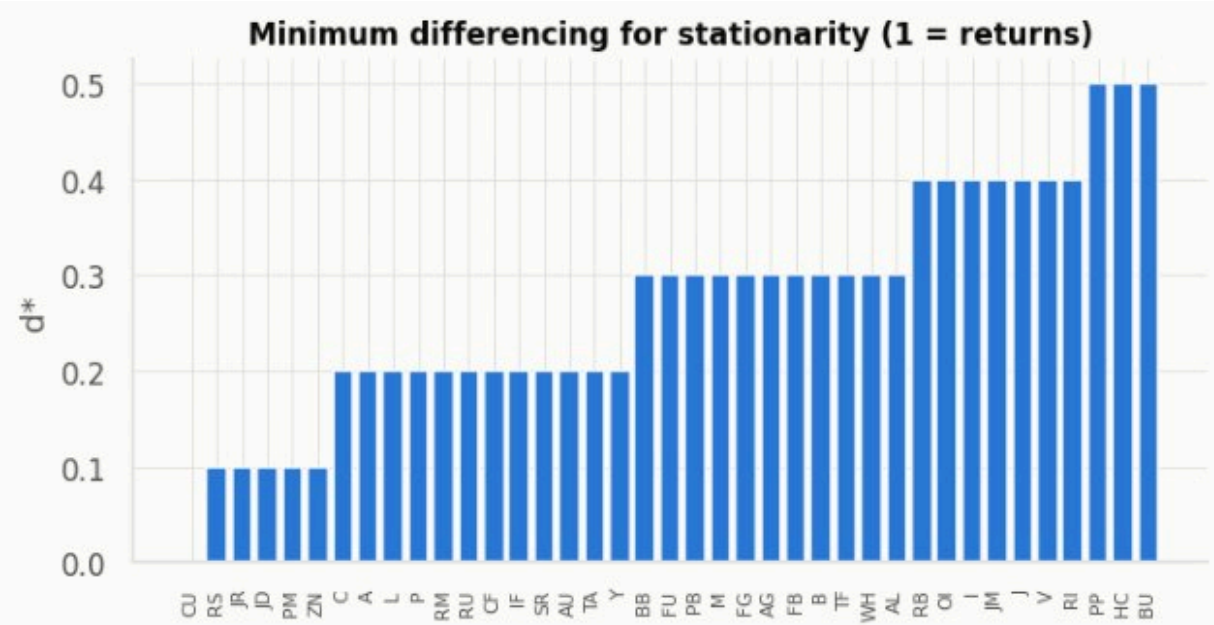
WHAT IT IS

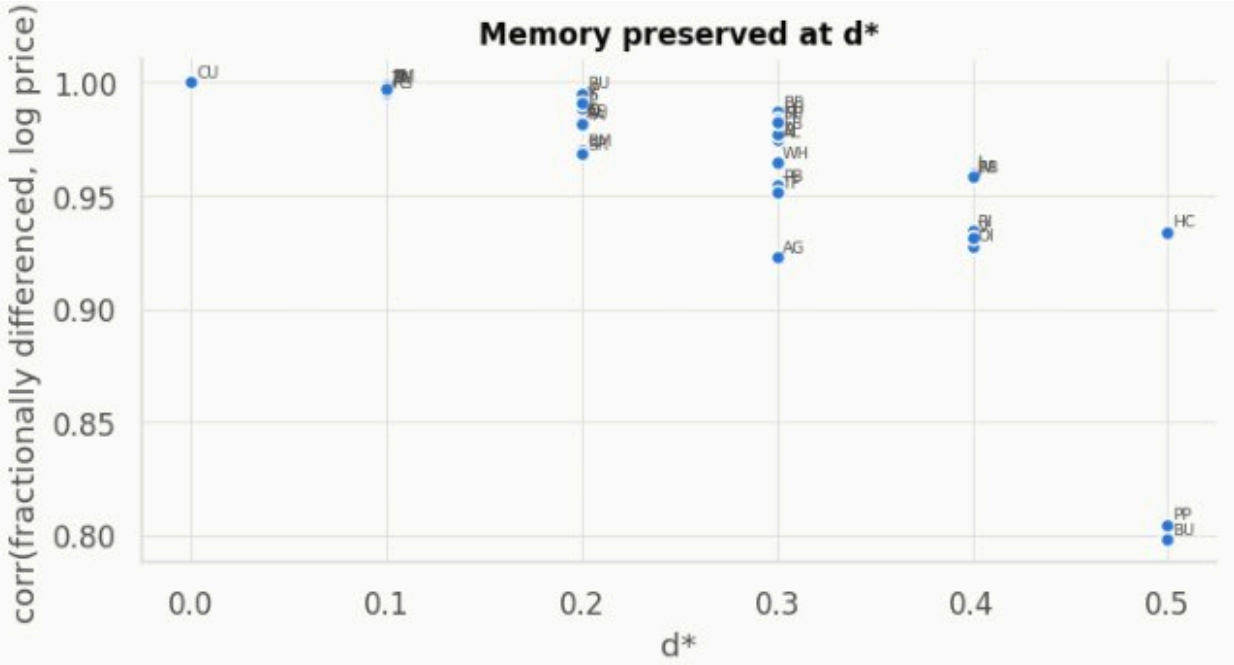
Stationarity verdicts from ADF and KPSS per series.

WHAT IT EXPLAINS

Prices are non-stationary, returns are stationary in 96% of series tested. Models are fitted on returns, not price levels.

Fractional differentiation





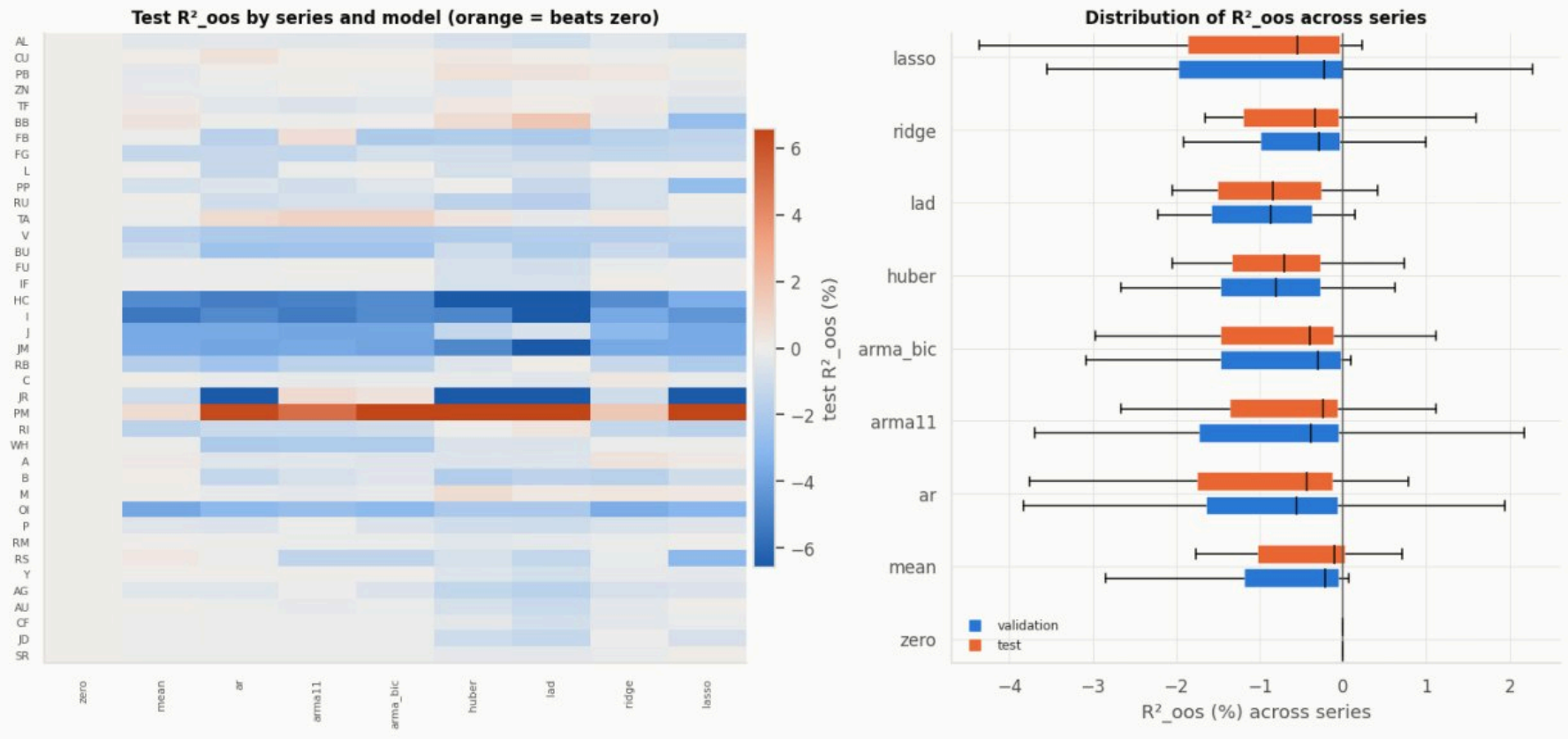
WHAT IT IS

Fractional differentiation: the order d* that makes a series stationary while keeping memory.

WHAT IT EXPLAINS

Median d* is 0.3, so a mild differencing keeps more information than taking plain returns.

Mean models: R² by series



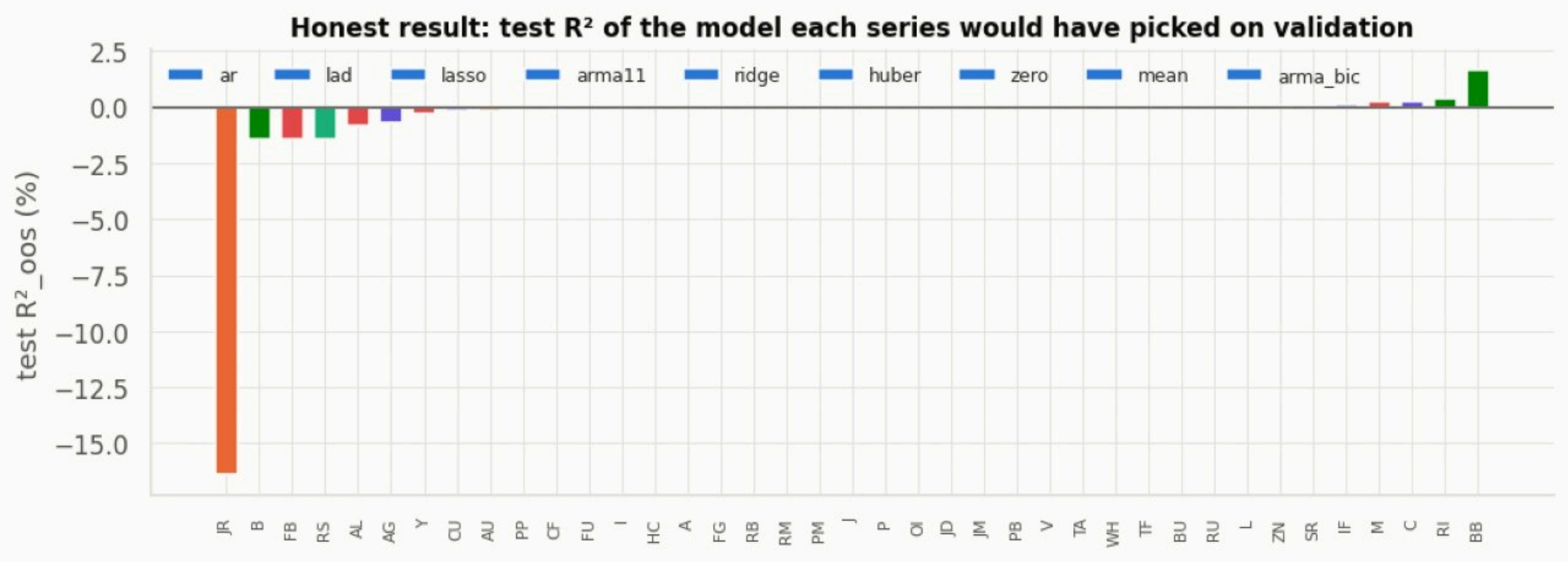
WHAT IT IS

R-squared of mean models by series.

WHAT IT EXPLAINS

Linear mean models (AR/ARMA type) explain essentially none of the next return.

Best-by-validation model, test R²



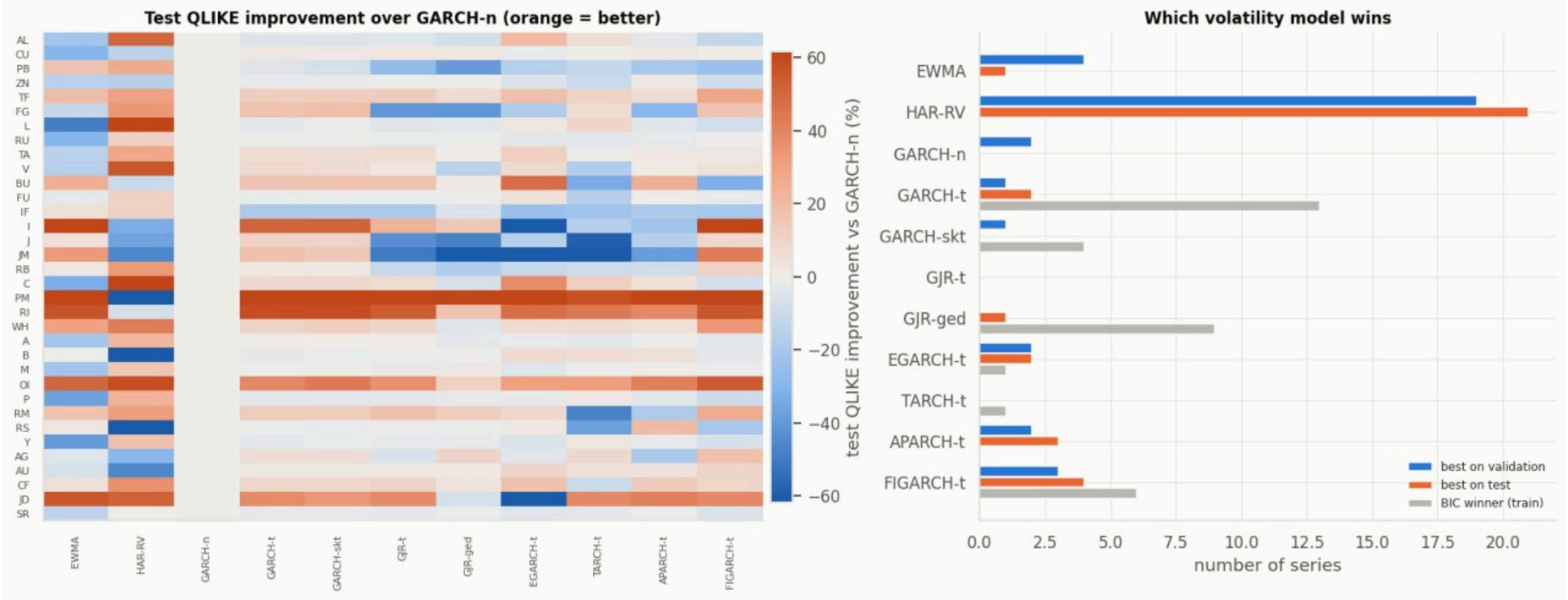
WHAT IT IS

Test R-squared of the model chosen on validation.

WHAT IT EXPLAINS

Best median out-of-sample R-squared is 0.0000. No linear own-history model beats zero, so edge must come from relationships, factors and conditions.

Volatility model comparison



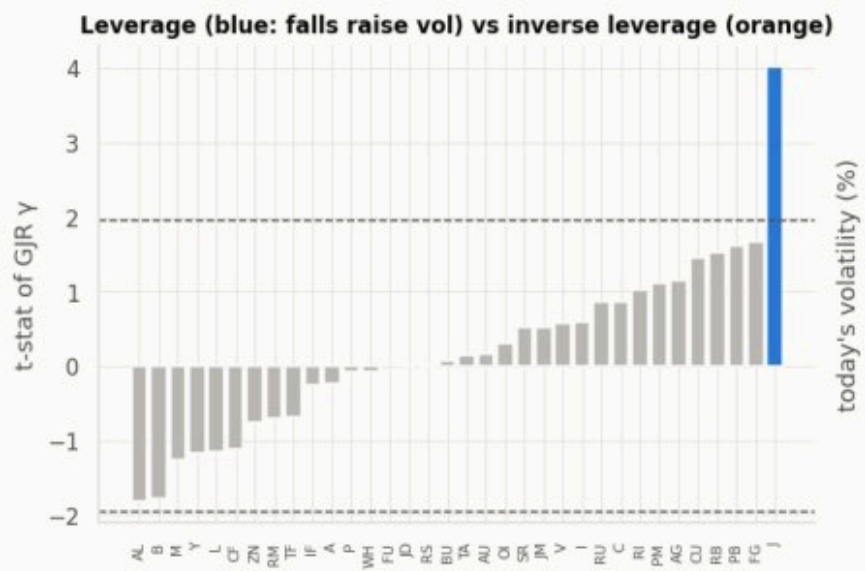
WHAT IT IS

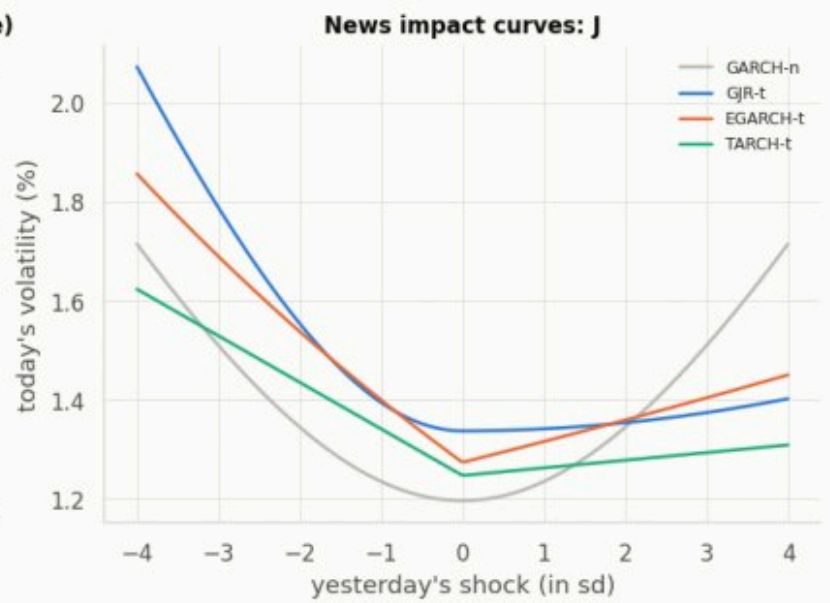
Comparison of volatility models (GARCH family, HAR-RV and others) by out-of-sample loss.

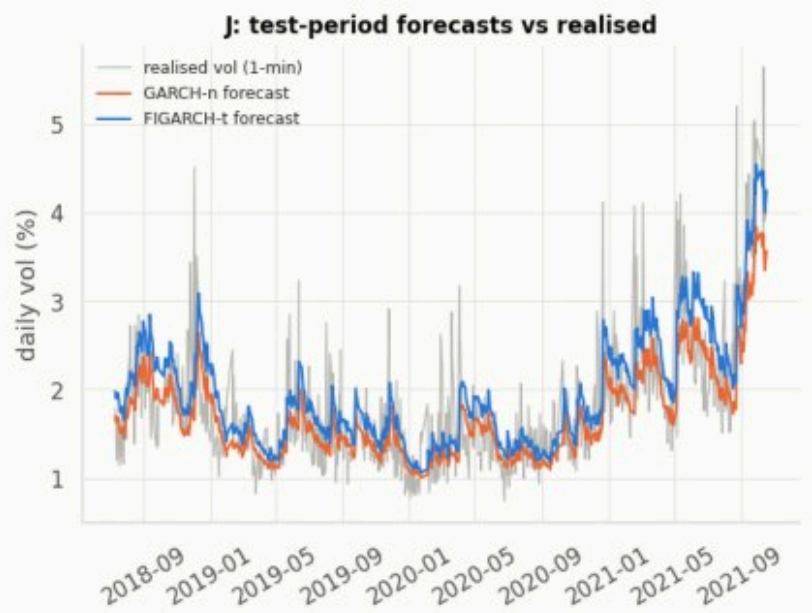
WHAT IT EXPLAINS

Volatility is predictable. HAR-RV is the model that holds up in every regime and time frame.

Leverage, news impact, forecasts







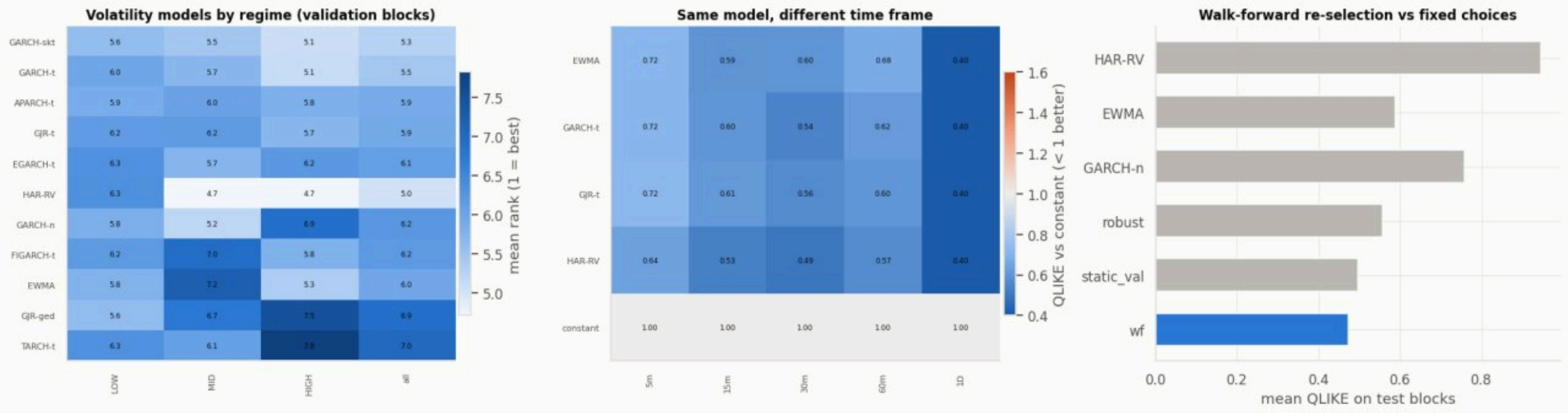
WHAT IT IS

Leverage effect, news-impact curves and volatility forecasts.

WHAT IT EXPLAINS

Shows how bad and good news change next-day volatility and what the forecasts look like.

Volatility models: regimes, time frames, walk-forward selection



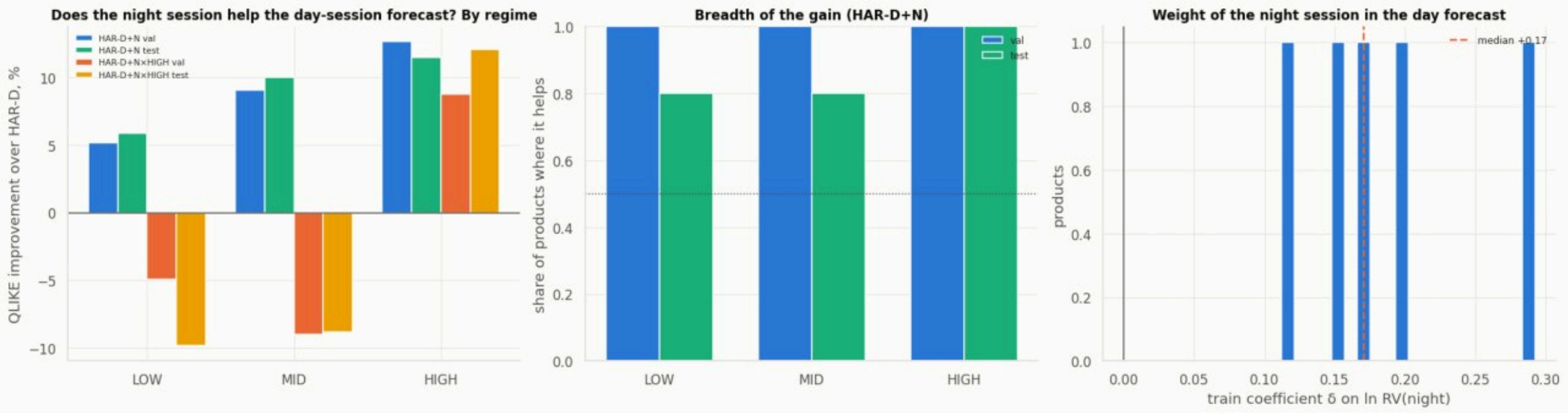
WHAT IT IS

Volatility models by regime, time frame and walk-forward selection.

WHAT IT EXPLAINS

HAR-RV wins in 5m, 15m, 30m, 60m and 1D. Re-selecting the model each block (0.473 QLIKE) does slightly better than the single validation winner (0.496).

HAR-RV with a night-session term, by volatility regime



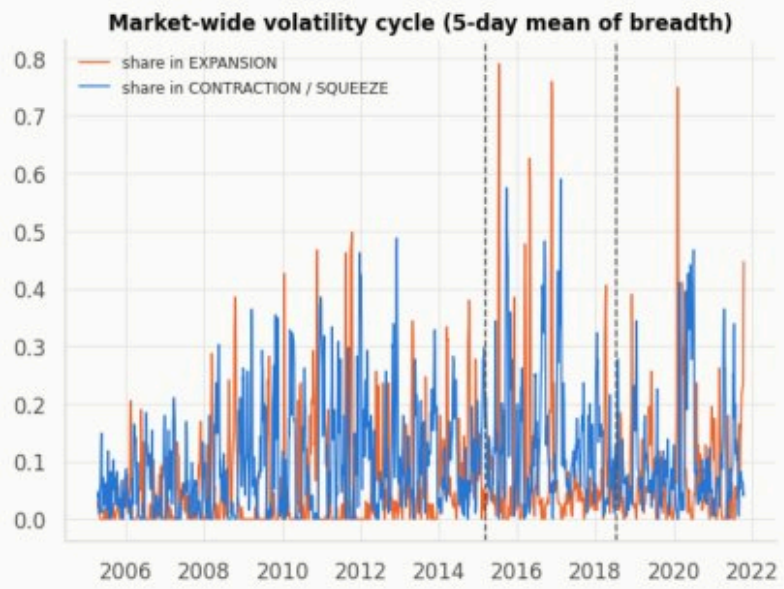
WHAT IT IS

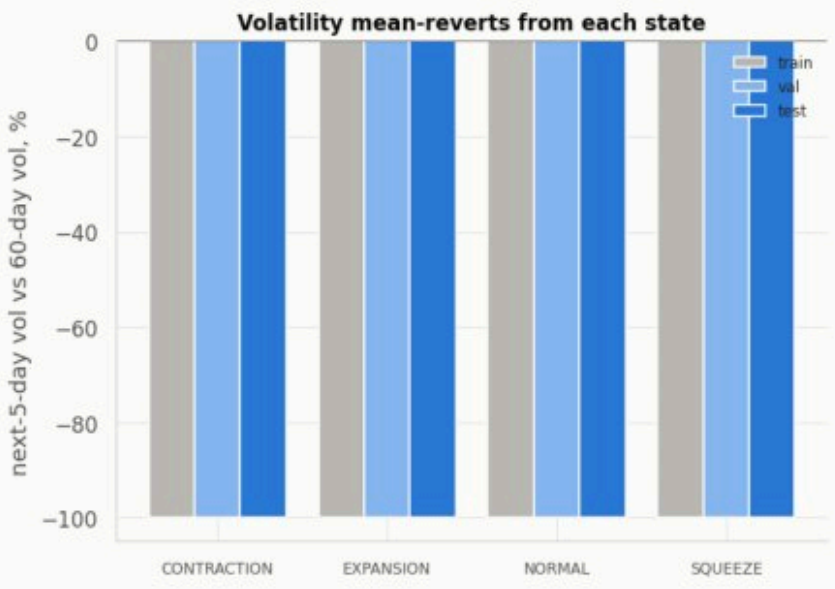
HAR-RV with an added night-session variance term, by volatility regime.

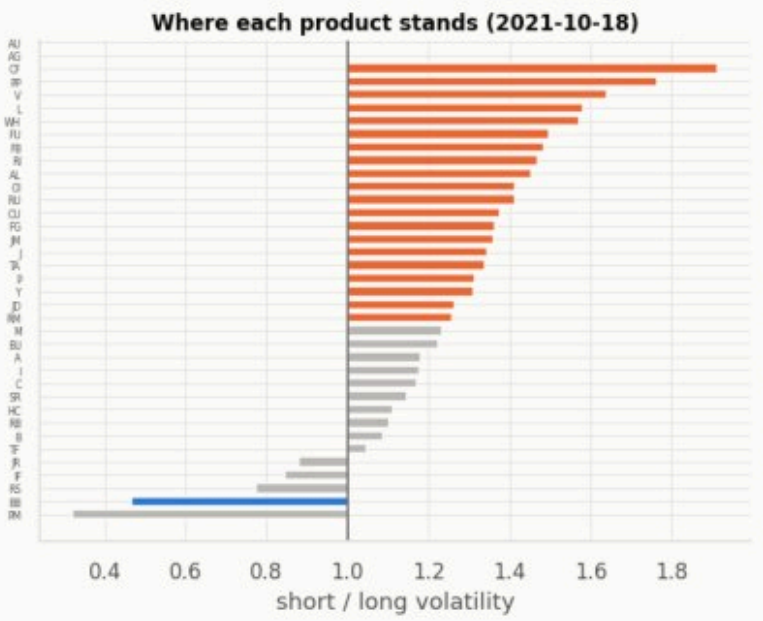
WHAT IT EXPLAINS

The night-session term improves day-session QLIKE by about 11.5% on test in HIGH volatility (t 4.9) against 5.9% in LOW: claim supported.

Volatility expansion / contraction monitor







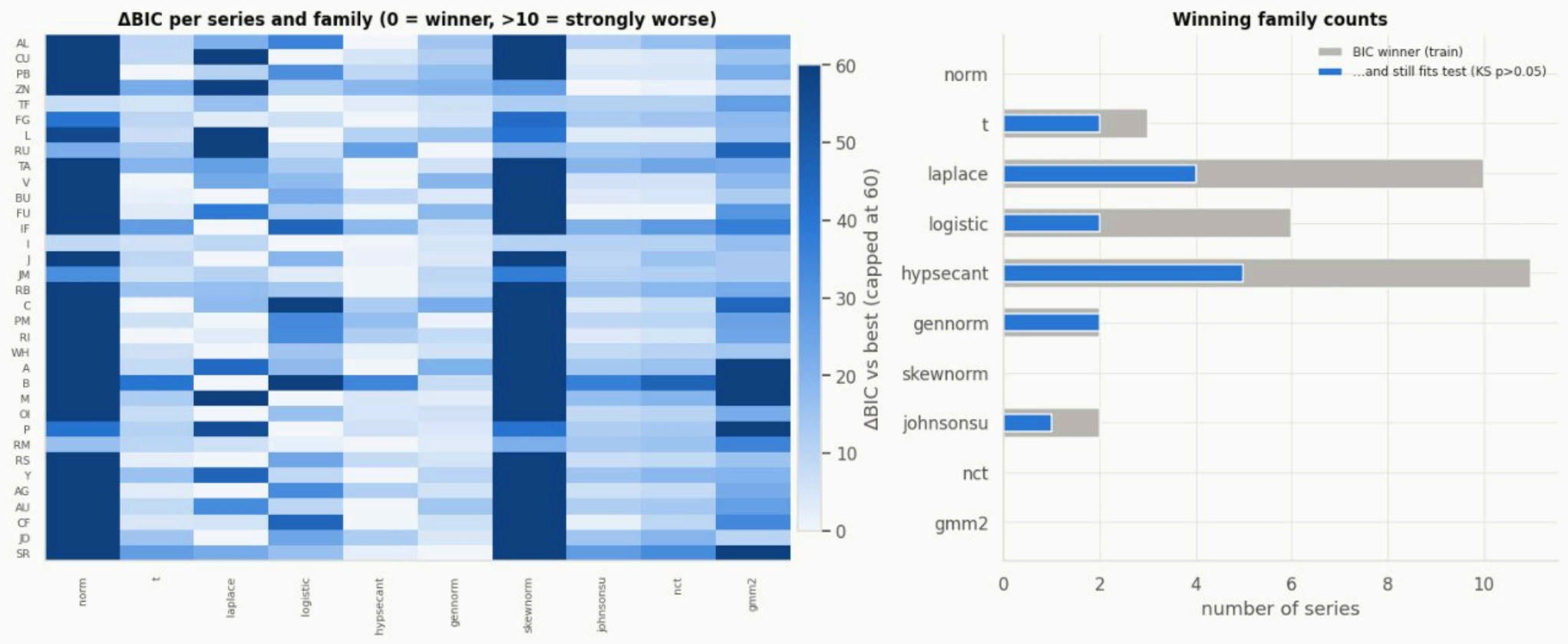
WHAT IT IS

Volatility expansion / contraction monitor.

WHAT IT EXPLAINS

Marks which products are expanding or squeezed now relative to their own history; feeds the board in Part H.

Distribution selection



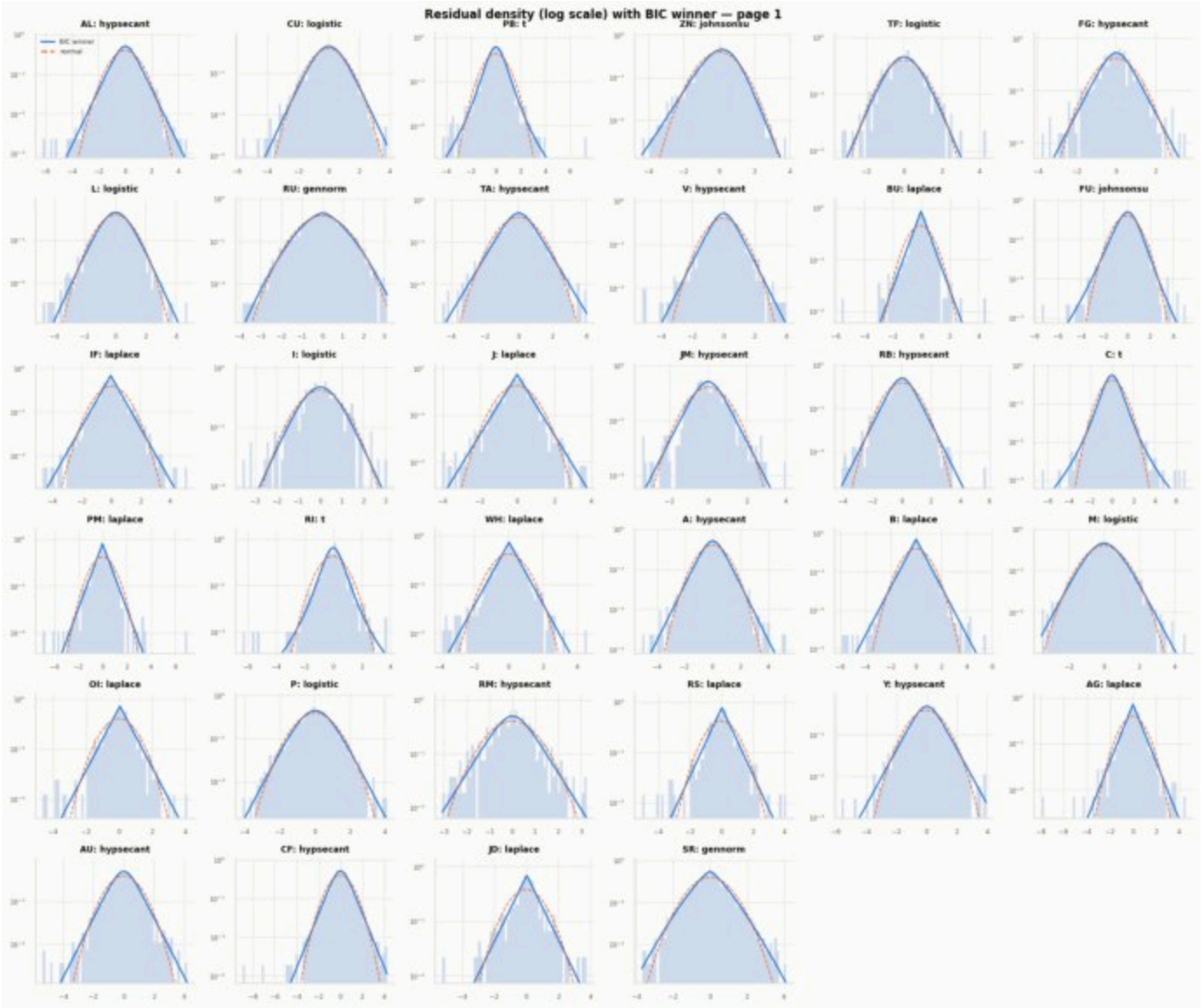
WHAT IT IS

Distribution selection by BIC across candidate families.

WHAT IT EXPLAINS

Hyperbolic-secant, Laplace, logistic and t beat the normal; mean out-of-sample gain is 0.050 log-likelihood per observation.

Residual density (log scale) with BIC winner | page 1



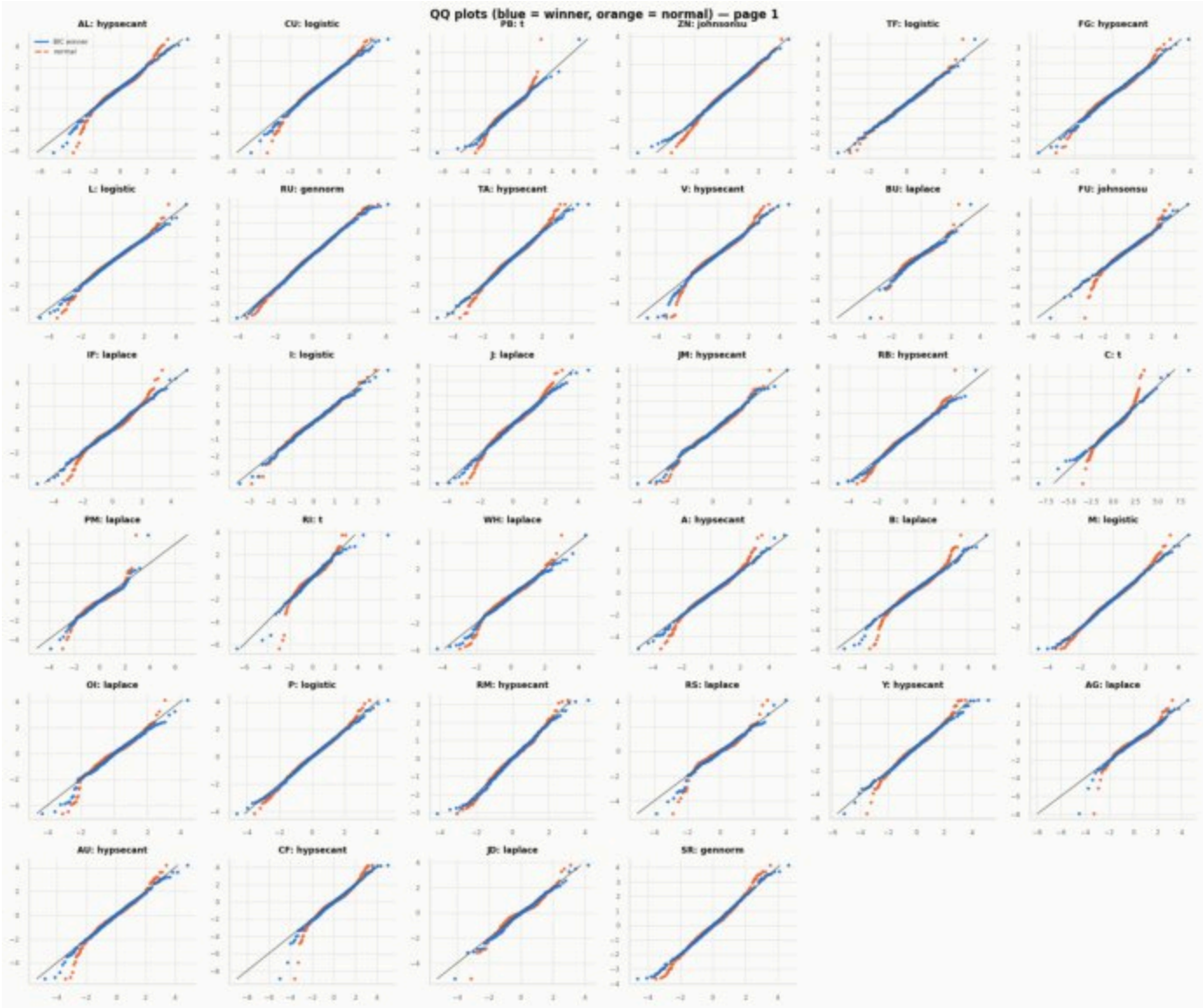
WHAT IT IS

Residual density (log scale) with the BIC-winning distribution.

WHAT IT EXPLAINS

The winning family tracks the tails far better than a normal curve.

QQ plots (blue = winner, orange = normal) | page 1



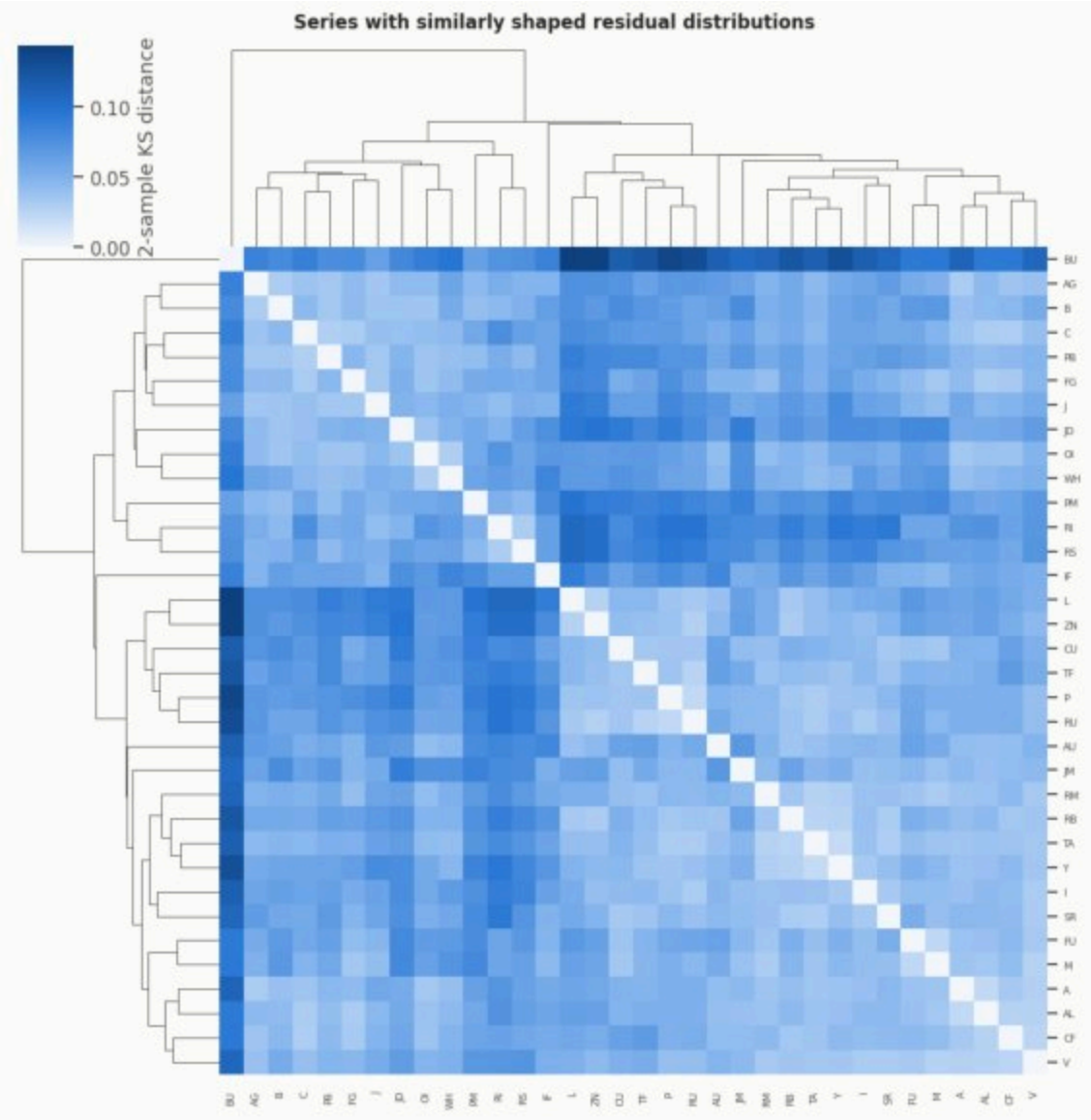
WHAT IT IS

QQ plots of residuals: blue = winning family, orange = normal.

WHAT IT EXPLAINS

Blue hugs the line, orange does not: the chosen family describes the tails.

Shape-distance clustermap



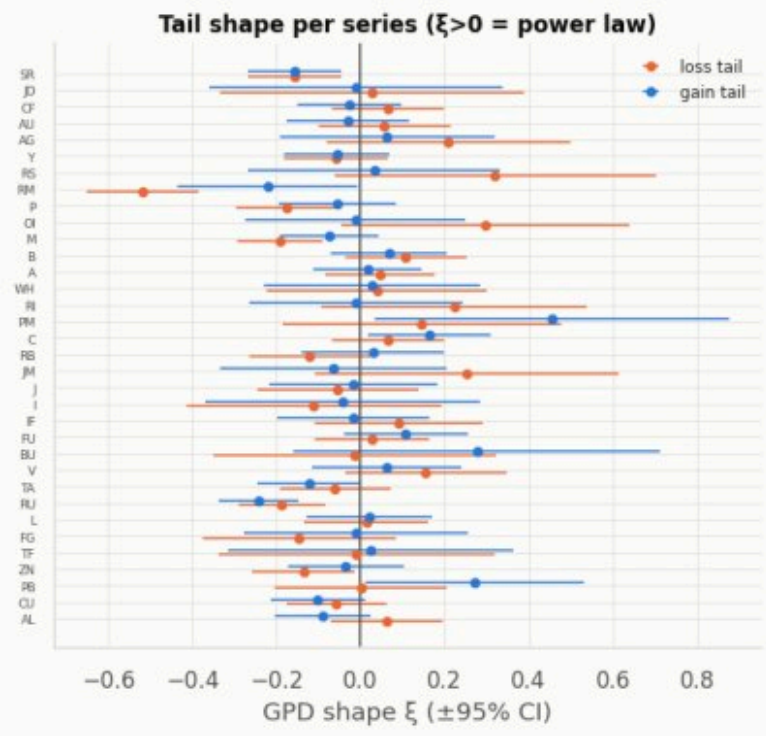
WHAT IT IS

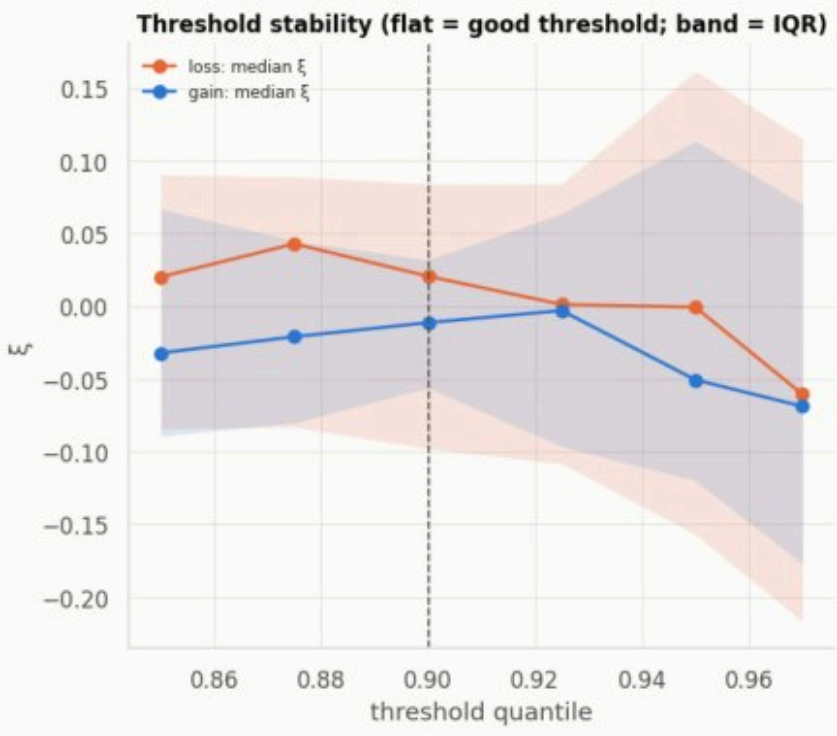
Clustermap of distribution shape distances between products.

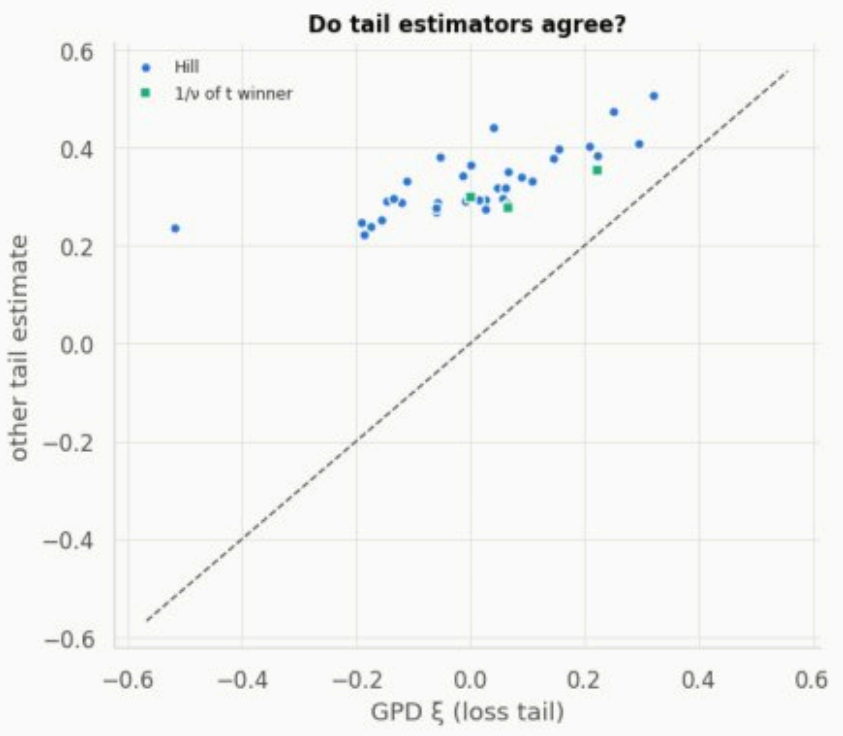
WHAT IT EXPLAINS

Products with similar return shape can share risk assumptions.

GPD tail shape, threshold stability, estimator agreement







WHAT IT IS

Generalised Pareto tail shape, threshold stability and estimator agreement.

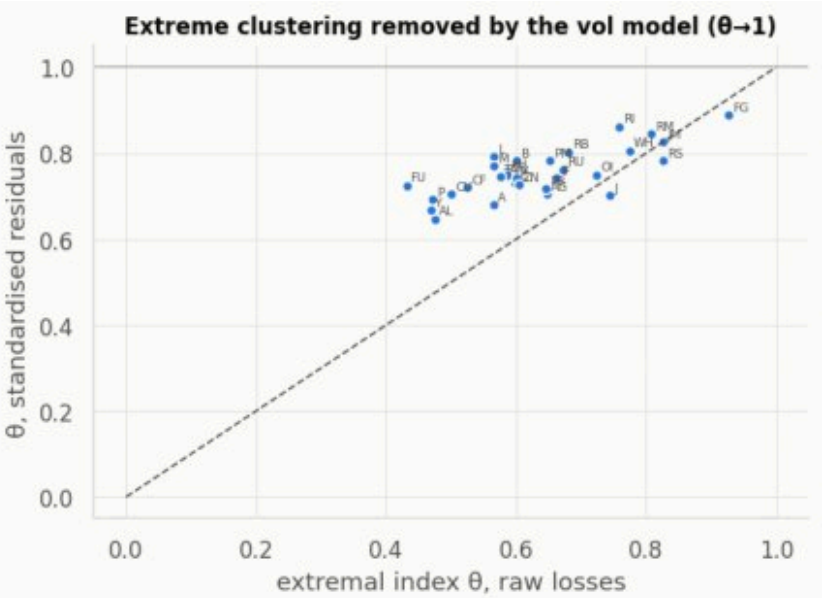
WHAT IT EXPLAINS

Median tail shape is about -0.01, i.e. not heavier than exponential. Tail estimates stay stable when the threshold moves.

Mean excess, return levels, extremal index







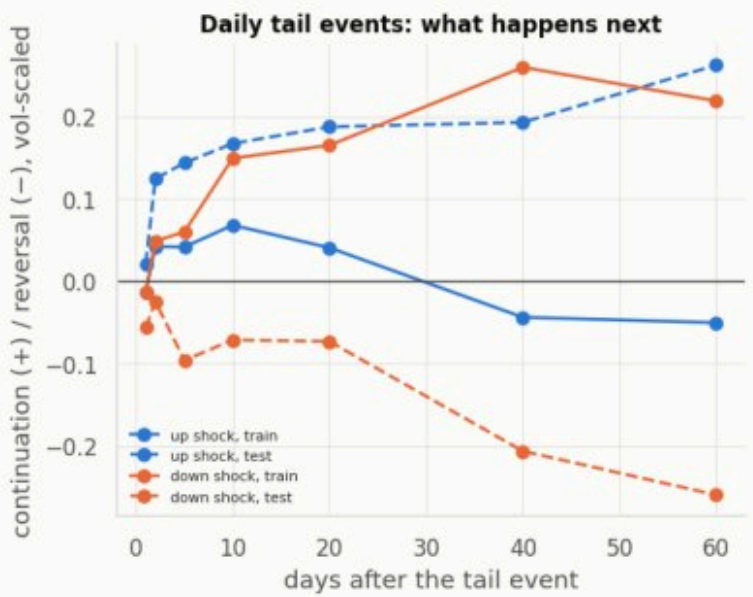
WHAT IT IS

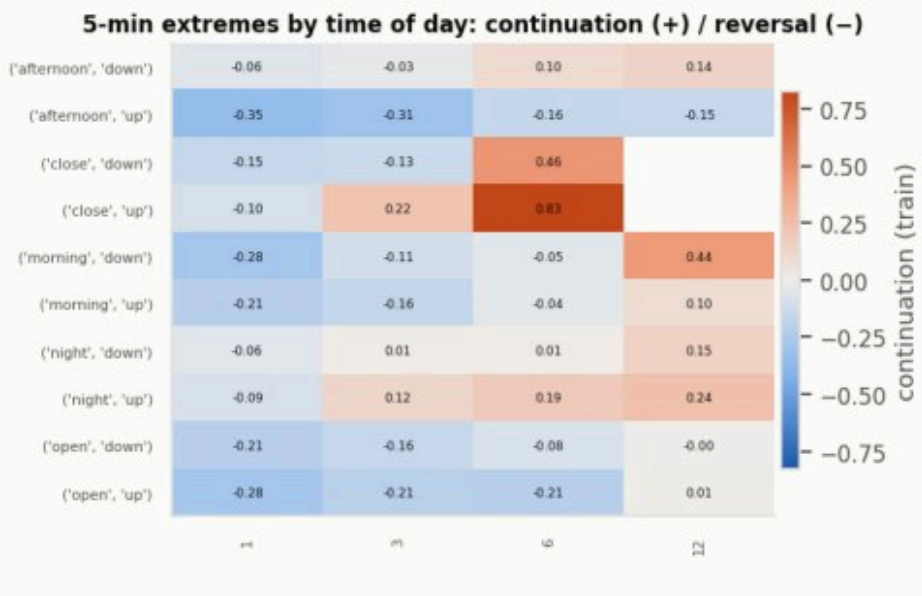
Mean excess, return levels and extremal index.

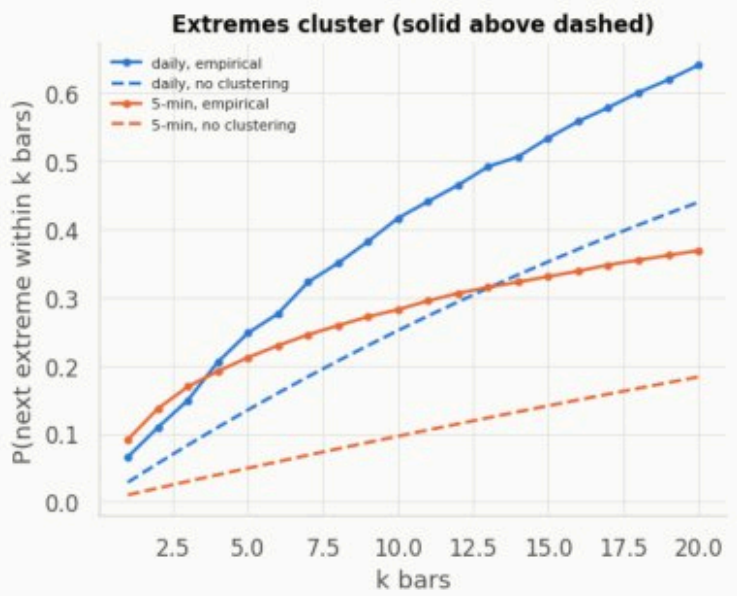
WHAT IT EXPLAINS

How far losses go beyond a threshold, how large a 1-in-N day is, and how much extremes cluster.

EVT opportunities: event study, time buckets, clustering







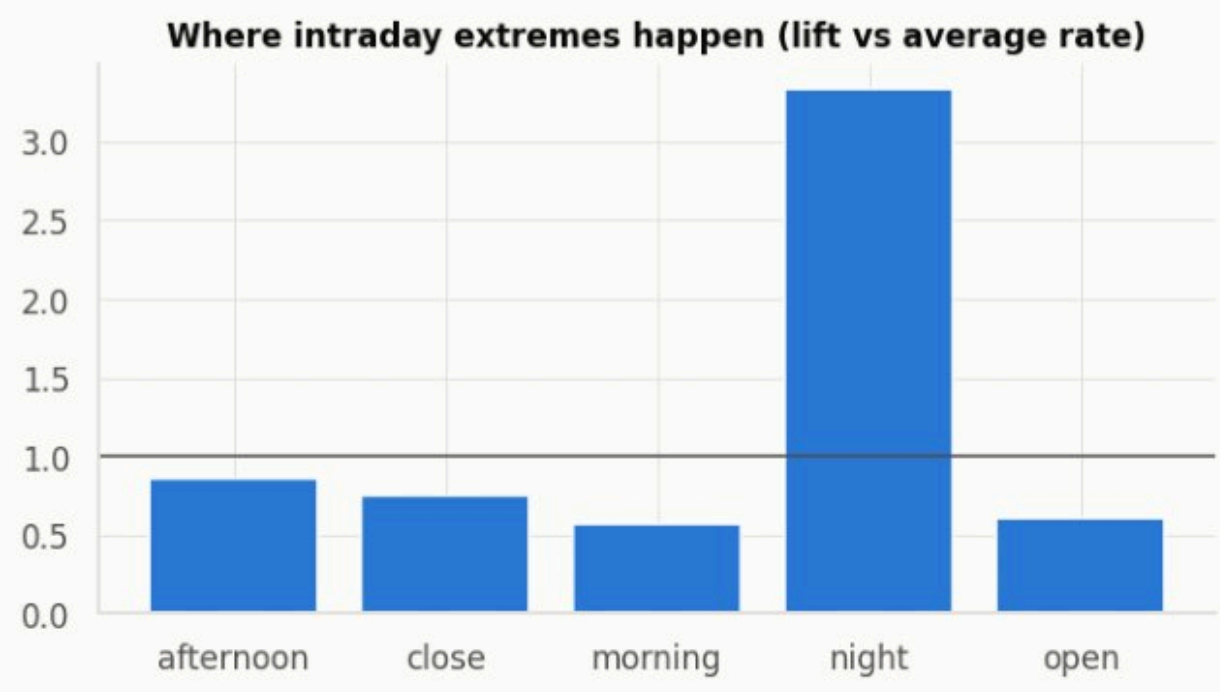
WHAT IT IS

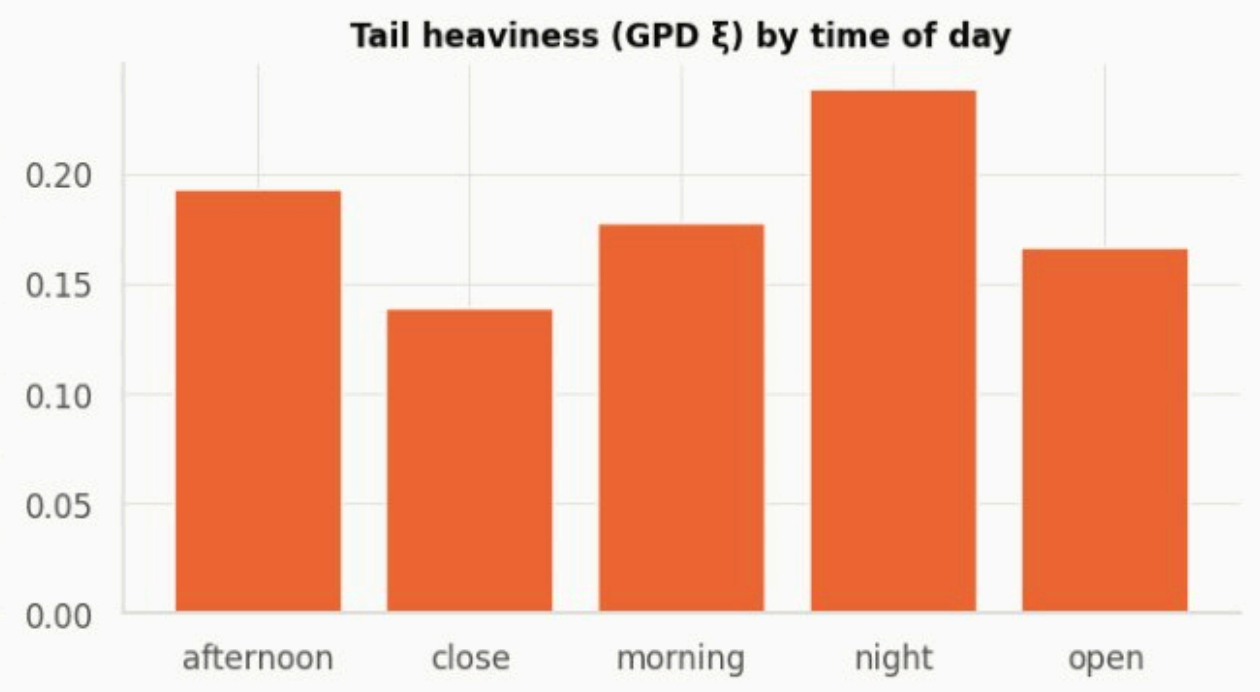
Extreme-value event study, time-of-day buckets and clustering.

WHAT IT EXPLAINS

After a daily shock the continuation is small: up shocks continue slightly (+0.04 to +0.17 by horizon), down shocks do not.

EVT by time-of-day bucket





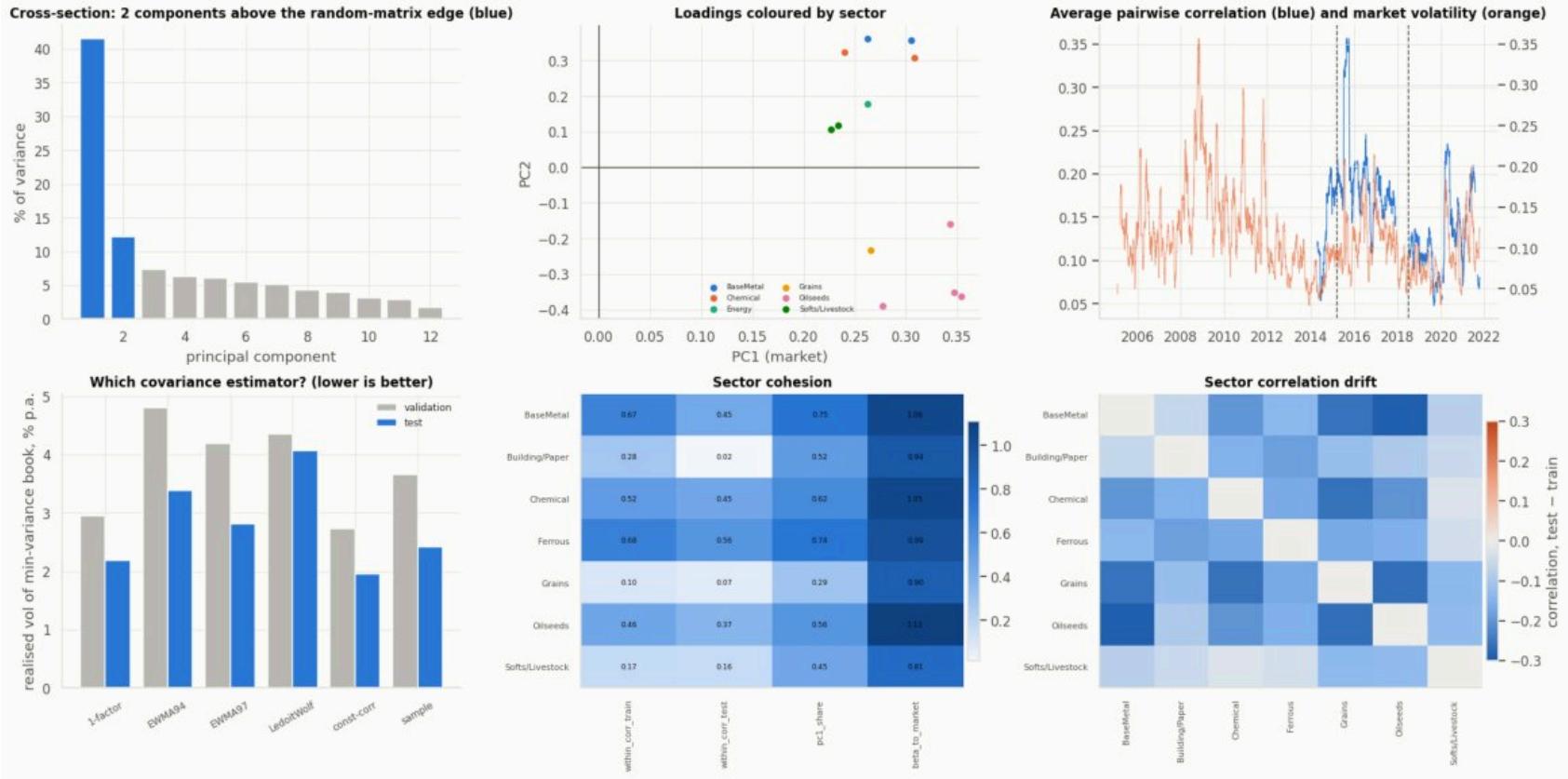
WHAT IT IS

EVT events by time-of-day bucket.

WHAT IT EXPLAINS

Tail events are not spread evenly across the day; buckets differ in frequency.

Multivariate structure: cross-section, covariance, sectors



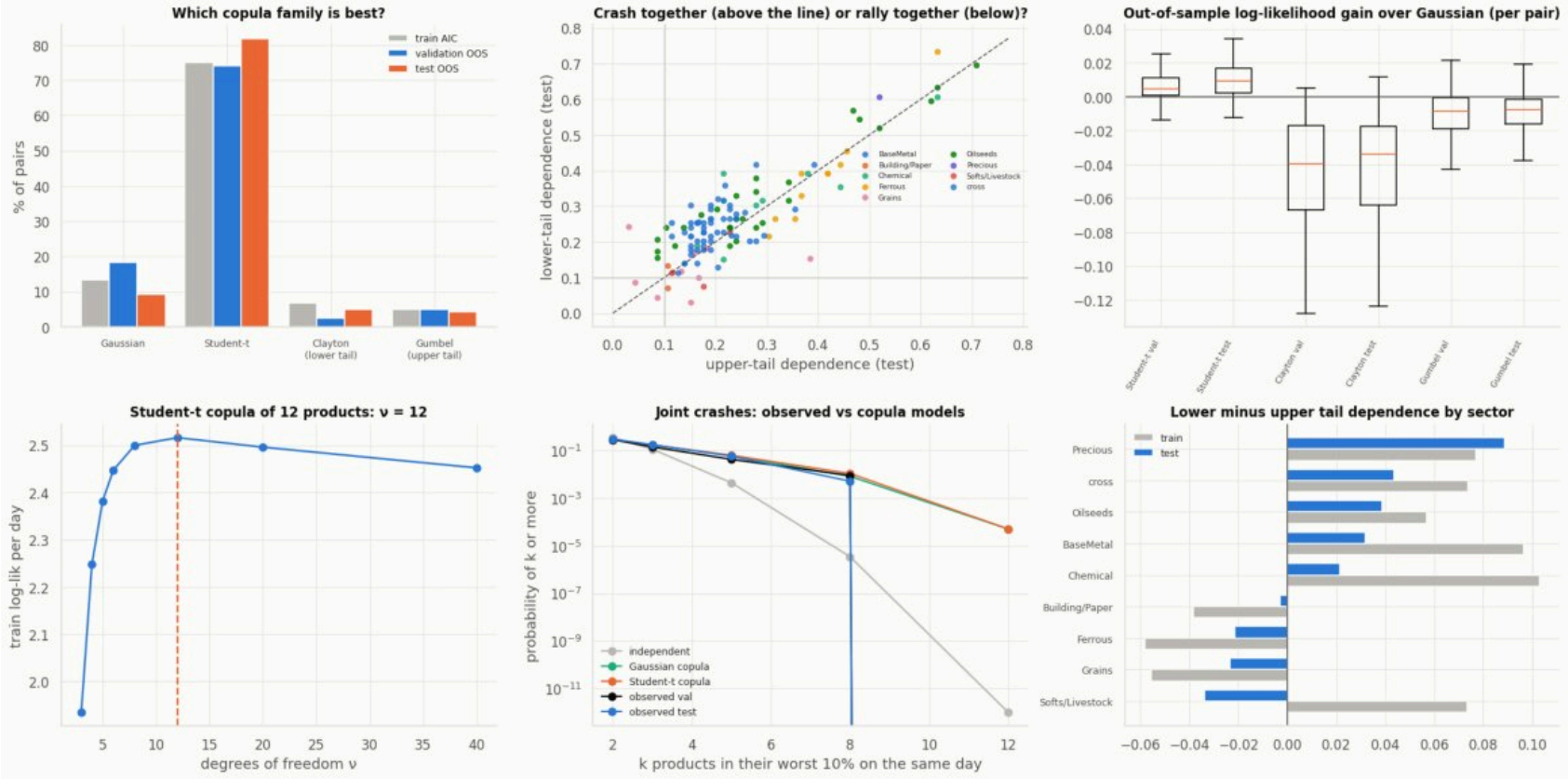
WHAT IT IS

Cross-section structure, covariance and sectors.

WHAT IT EXPLAINS

One market factor explains 41% of daily co-movement. The 1-factor covariance estimator wins on validation.

Copulas: families, tail asymmetry, joint extremes



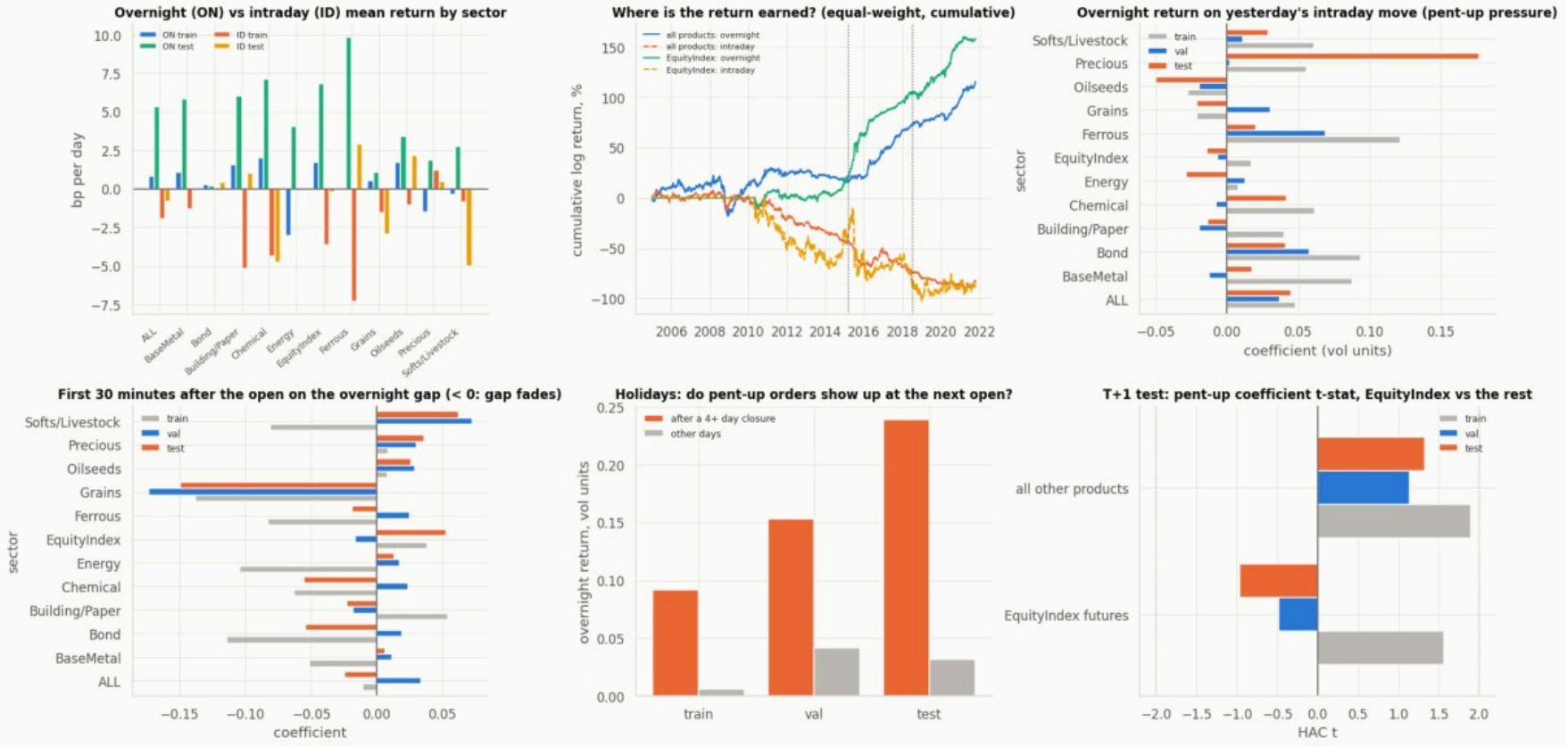
WHAT IT IS

Copula families, tail asymmetry and joint extremes.

WHAT IT EXPLAINS

Student-t copula beats Gaussian out of sample. Products crash together more than a Gaussian model says.

Overnight returns and the T+1 effect



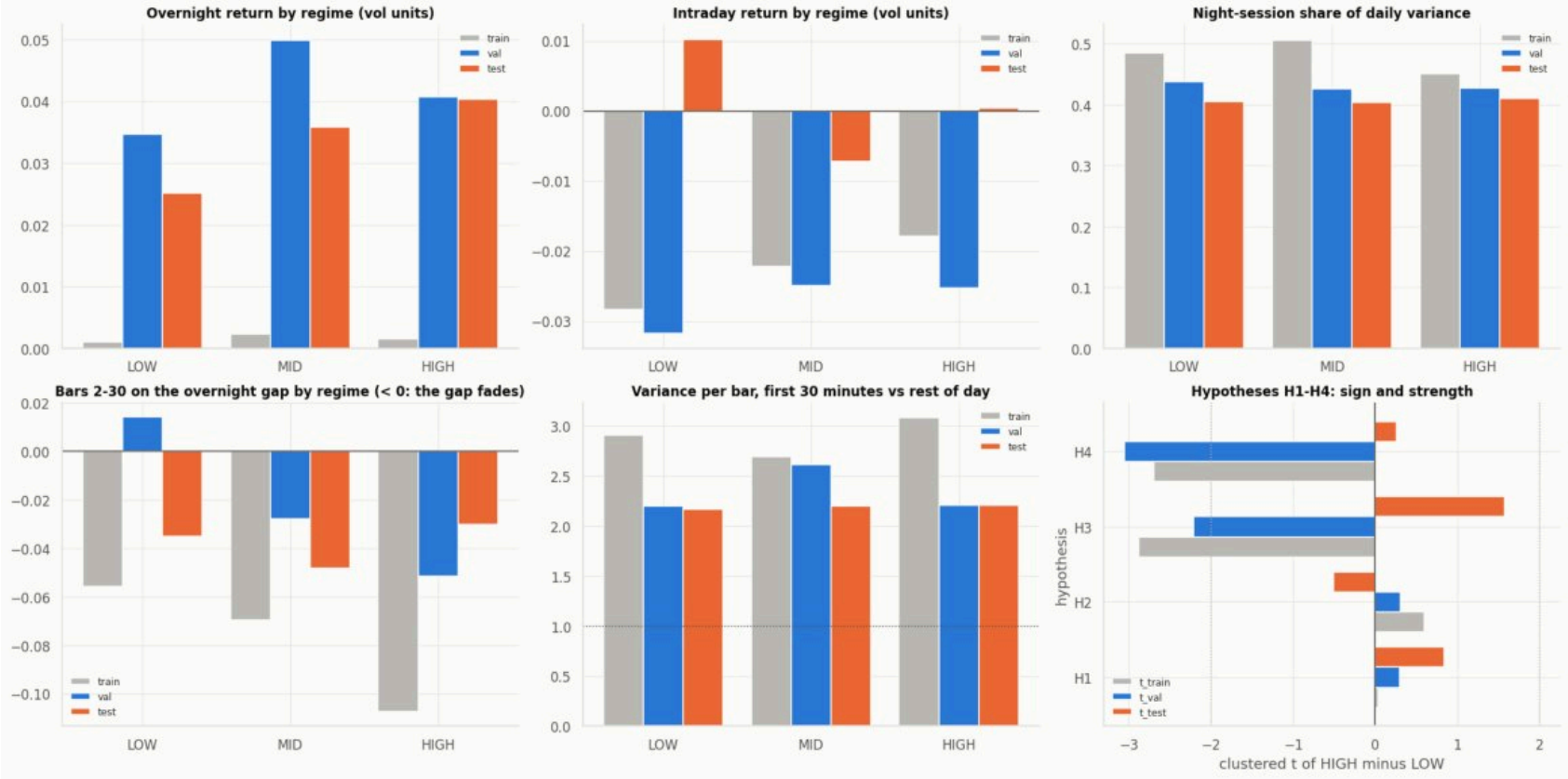
WHAT IT IS

Overnight returns and the T+1 effect.

WHAT IT EXPLAINS

Overnight returns are positive and intraday returns negative on test (ON +5.3 bp, ID -0.8 bp per day), but no stable pent-up-pressure or T+1 pattern exists.

Overnight versus day returns by volatility regime and the opening reversal



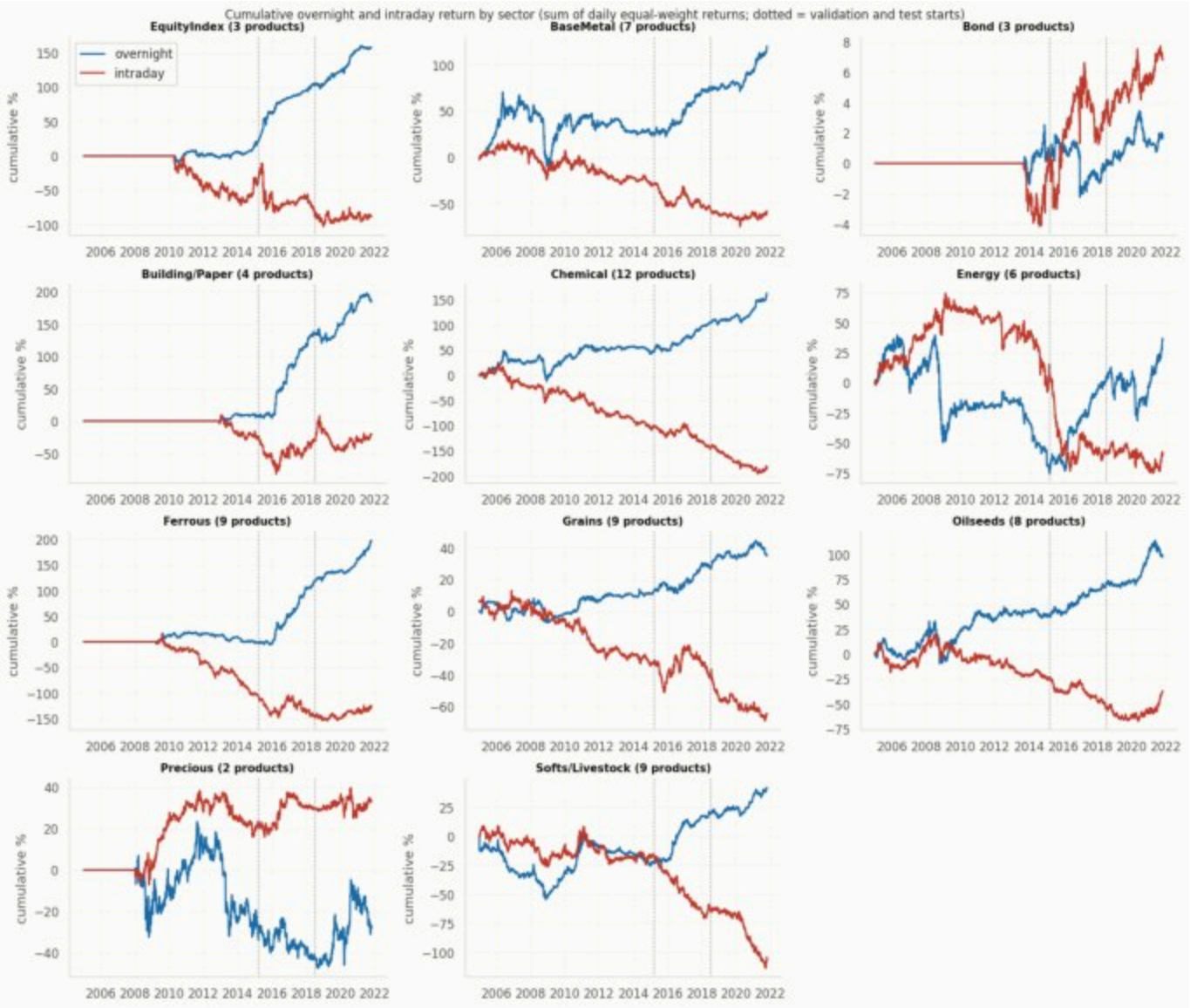
WHAT IT IS

Overnight versus day returns by volatility regime and the opening reversal.

WHAT IT EXPLAINS

The four regime hypotheses (overnight pays more in HIGH, etc.) are all unsupported. The opening reversal trade has Sharpe +1.80 train, +1.28 validation, +0.21 test.

Cumulative overnight and intraday return by sector (sum of daily equal-weight returns; dotted =



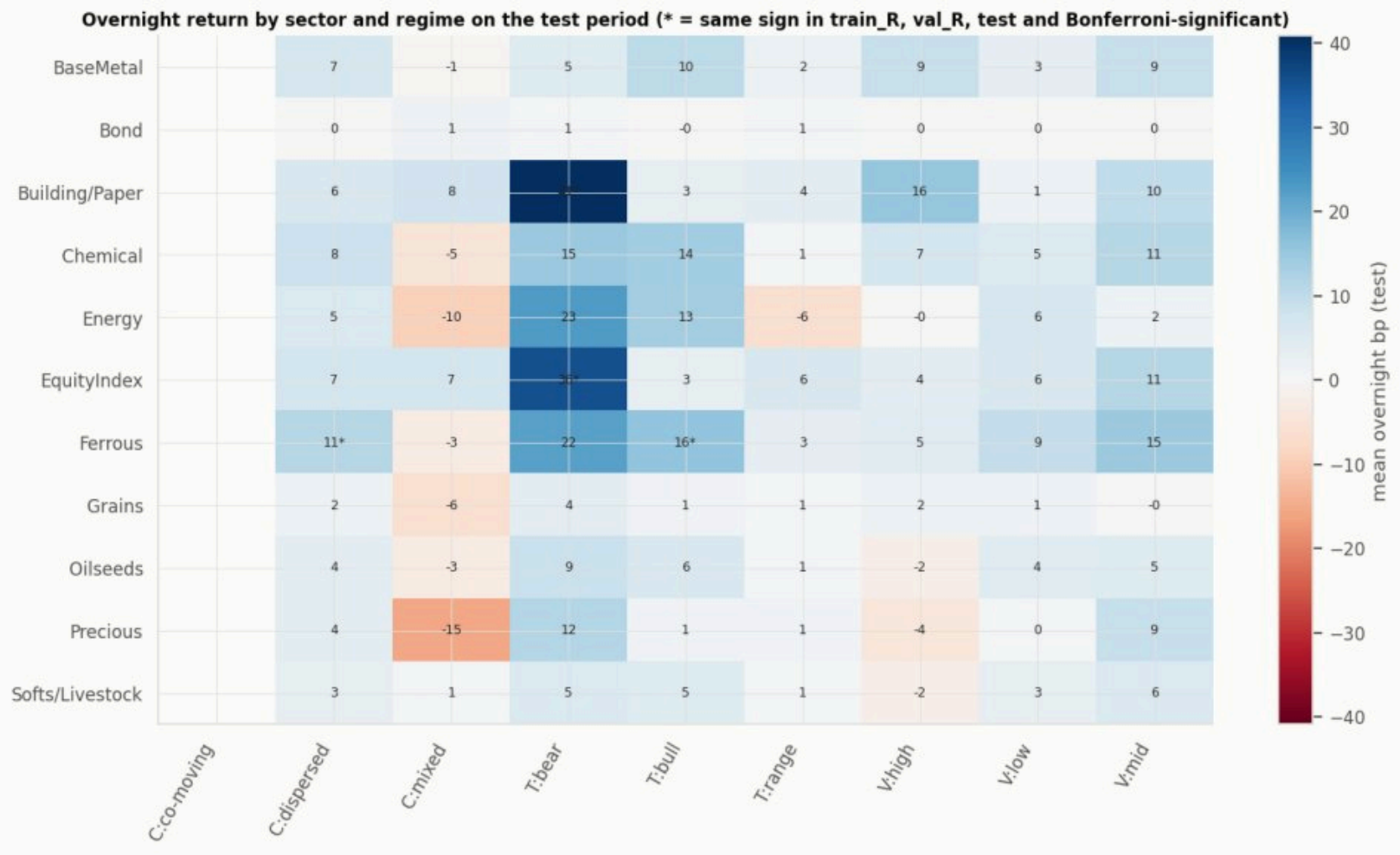
WHAT IT IS

Cumulative overnight and intraday return by sector. Dotted line = split dates.

WHAT IT EXPLAINS

Most of the equal-weight gain sits in the overnight leg, by sector, over time.

Overnight return by sector and regime on the test period (* = same sign in train_R, val_R, test



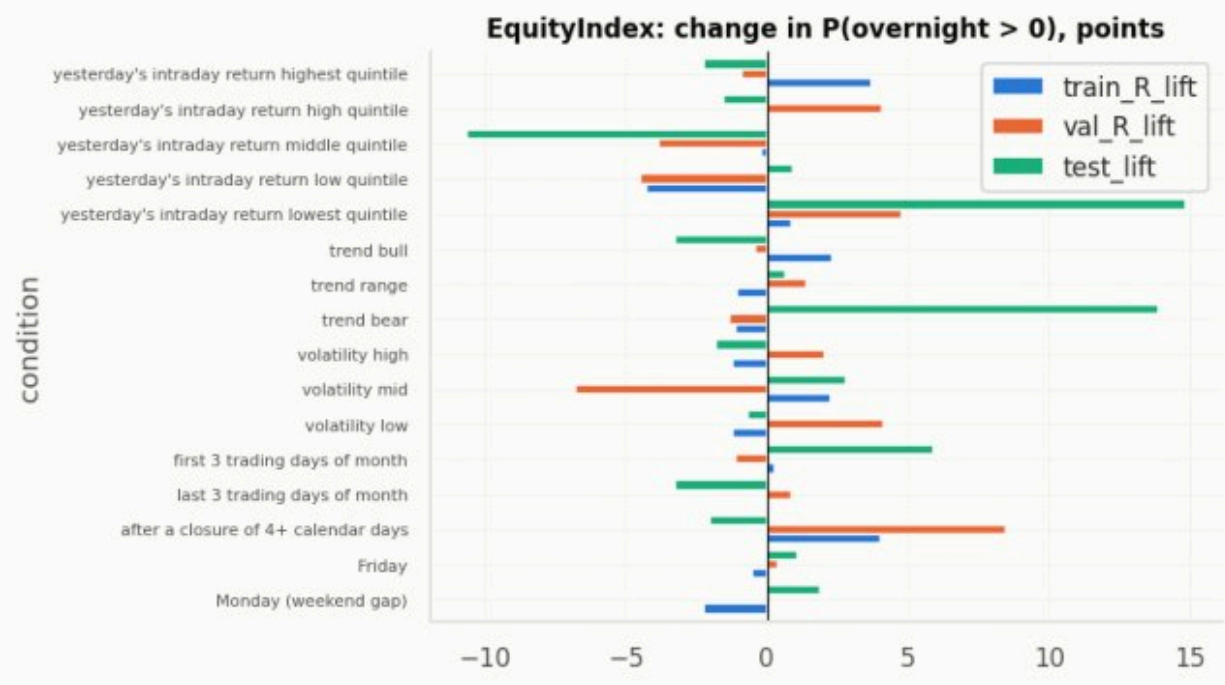
WHAT IT IS

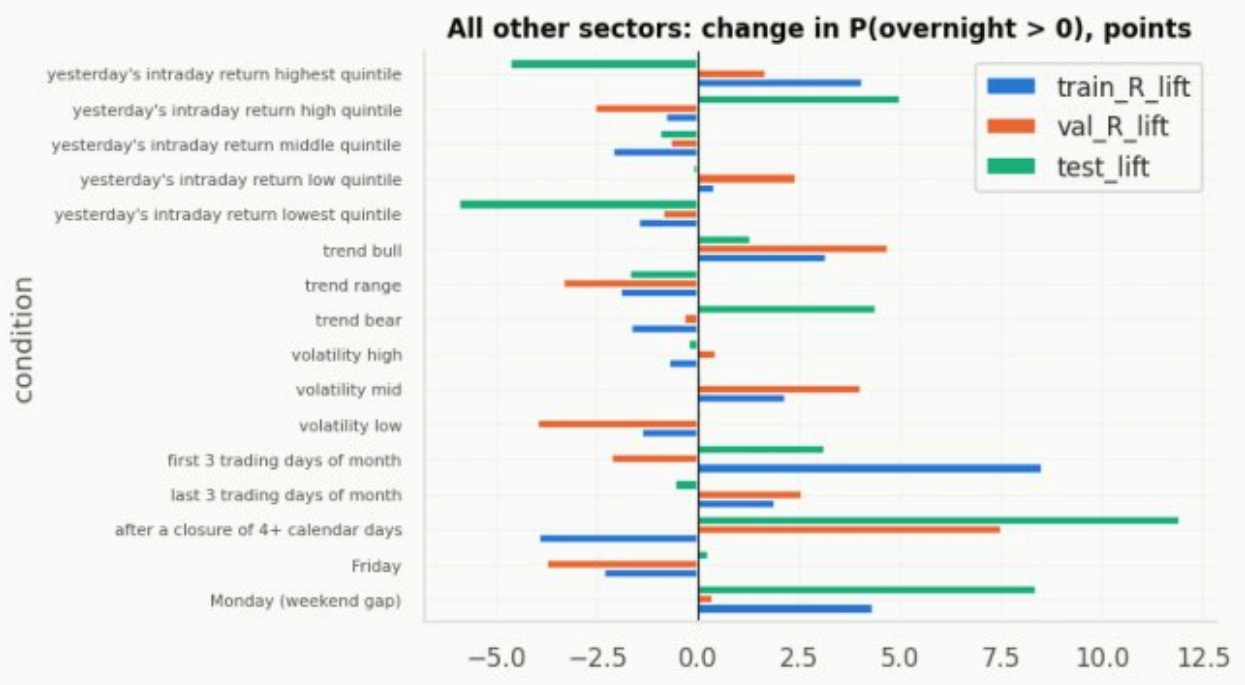
Overnight return by sector and regime on test. A star marks the same sign in train, validation and test.

WHAT IT EXPLAINS

Shows which sector-regime cells are consistent. Few survive the Bonferroni bar.

EquityIndex: change in P(overnight > 0), points





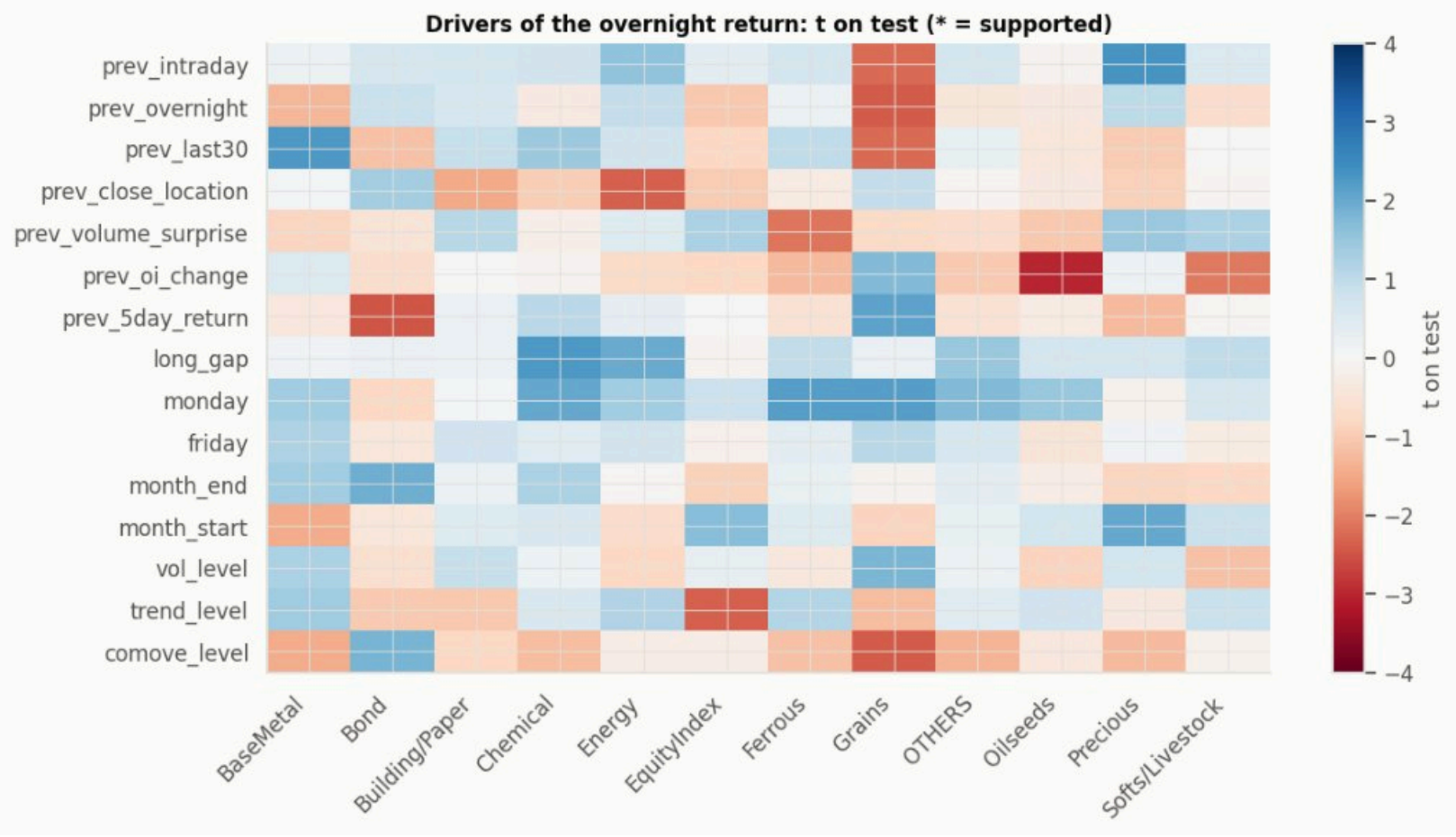
WHAT IT IS

EquityIndex: change in the probability that the overnight return is positive, in points.

WHAT IT EXPLAINS

Tests whether equity-index futures behave differently overnight because of the T+1 stock link. No stable difference is found.

Drivers of the overnight return: t on test (* = supported)



WHAT IT IS

Drivers of the overnight return: t on test, star = supported.

WHAT IT EXPLAINS

Candidate drivers (gap, prior-day move, volume, etc.). None is supported in all three periods.

PART C · FINDINGS

Part C in one page: volume, open interest and relationships

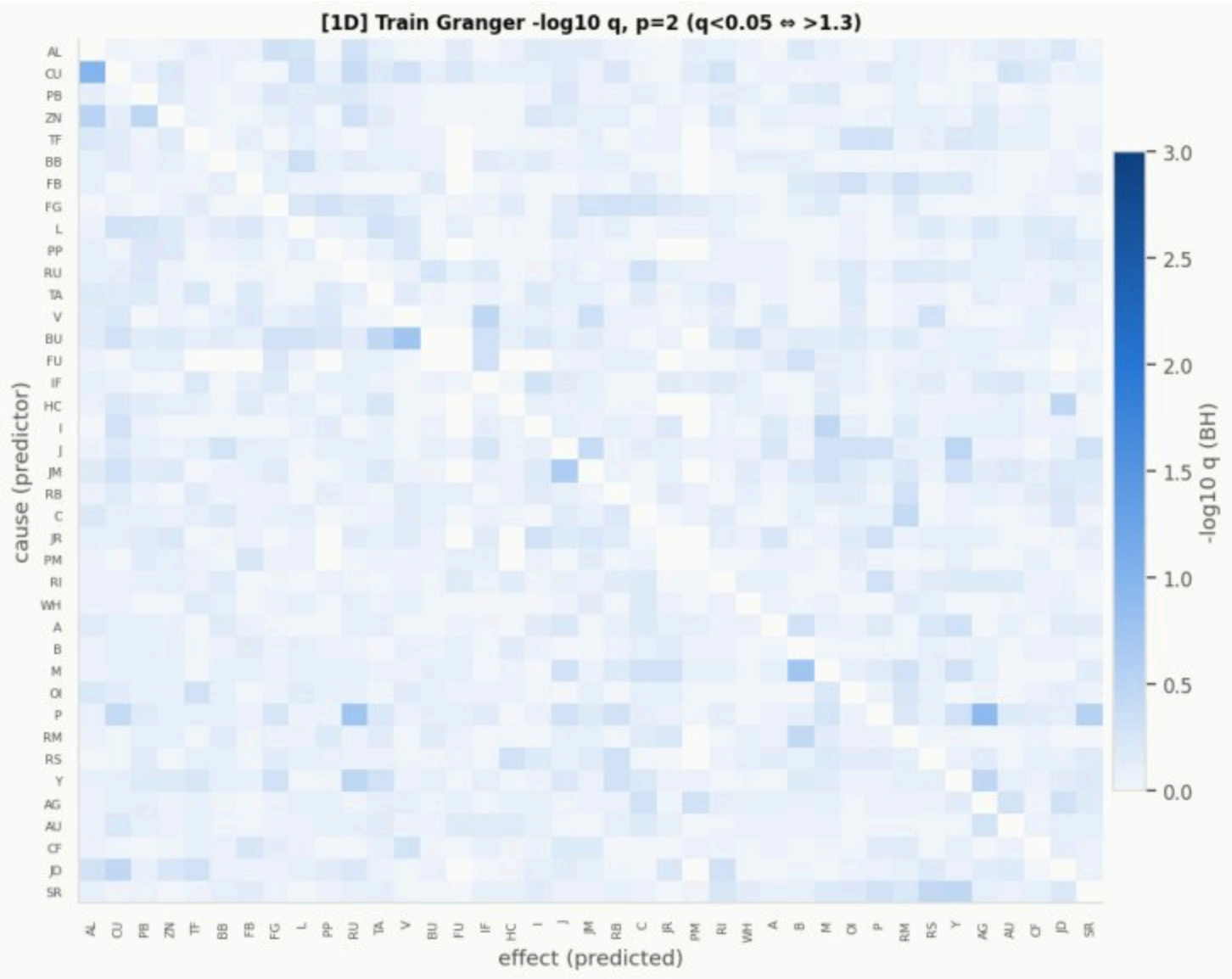
- **Price-volume geometry:** 5 of 33 pairs confirmed, all about size of the next move. Map not stable.
- **Topology:** loops are close to chance (9% vs 5%); no topology feature adds to trailing volatility.
- **Volume + OI:** 2 of 9 state calls confirmed on test (short covering and new longs, low volume). Volume and OI together predict size of move (t 5.2 on test).
- **Trend phases:** 2273 episodes; 1 causal rule supported (rise coil, t 2.99).
- **Flow indicators (PMF, QFF, Wasserstein) and ECHO:** none confirmed; the three flow signals are effectively one family.
- **Granger links:** 27 replicate out of sample; fundamental chains: 0 of 10.

Part C

Relationships between products

Relationships between products: Granger lead-lag, cointegration, tail dependence and fundamental chains. The price-volume studies (18c to 18v) sit in Part D below.

[1D] Granger train heatmap



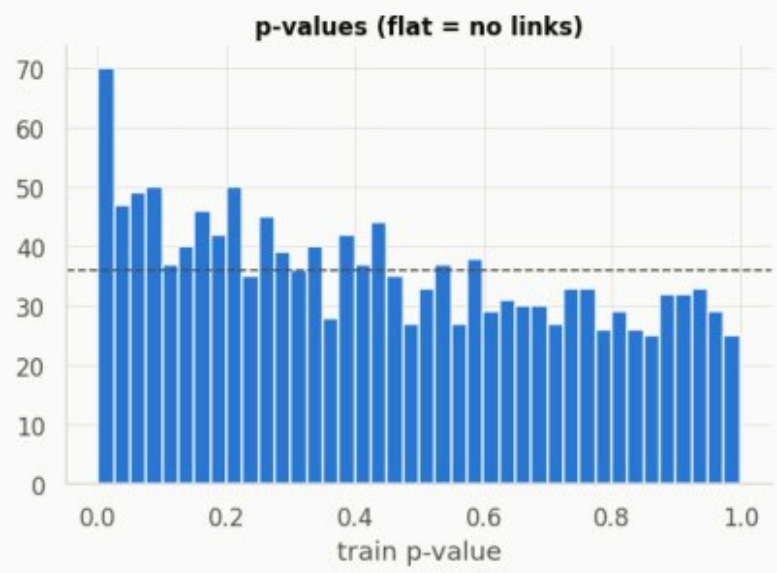
WHAT IT IS

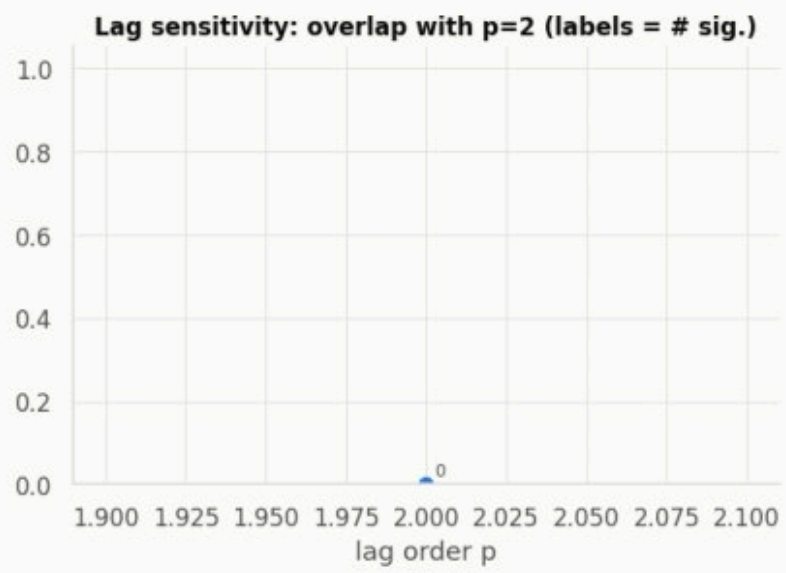
Granger-causality heatmap on daily data, train period.

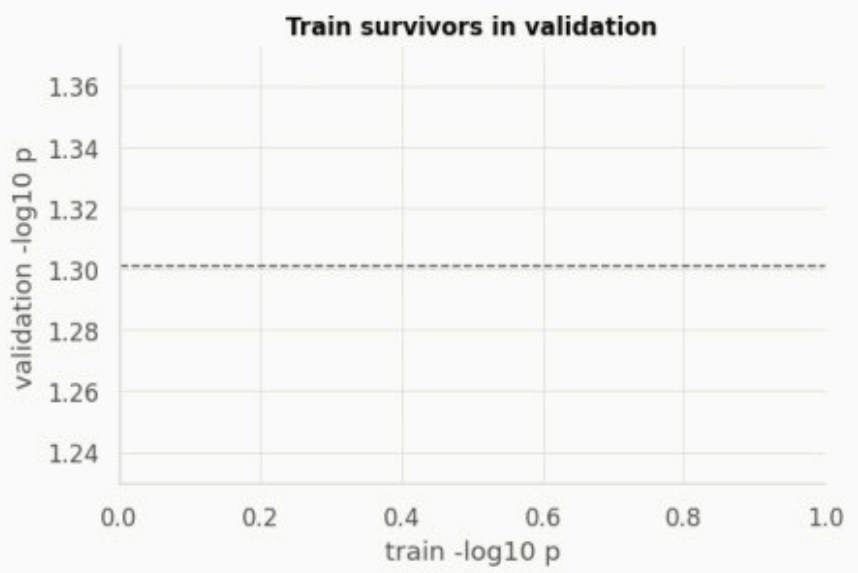
WHAT IT EXPLAINS

Which products lead which others. Starting list before out-of-sample checks.

[1D] p-values, lag sensitivity, validation







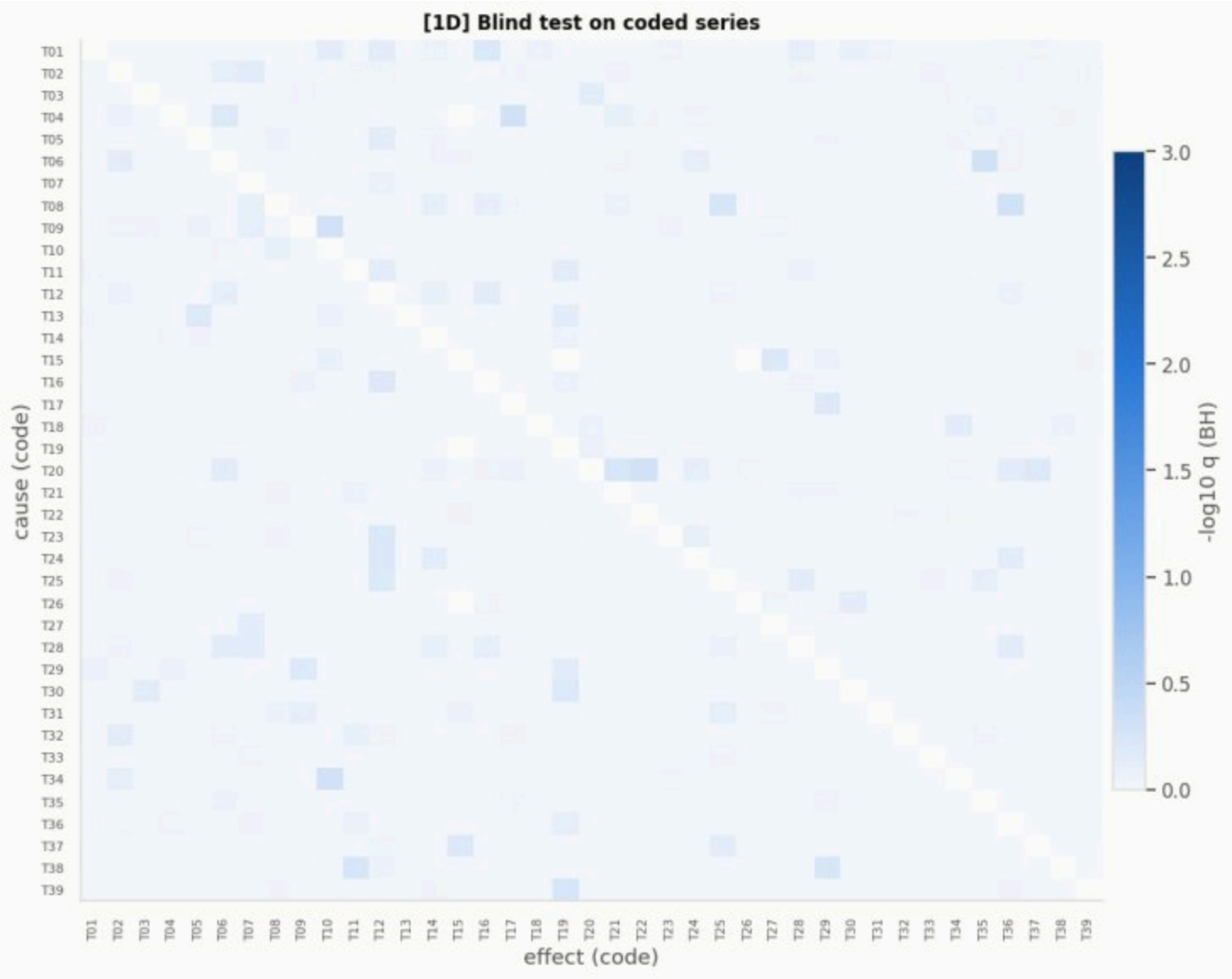
WHAT IT IS

p-values, lag sensitivity and validation for daily links.

WHAT IT EXPLAINS

Links are kept only if they stay significant when the lag changes and in validation.

[1D] Blind test heatmap (coded)



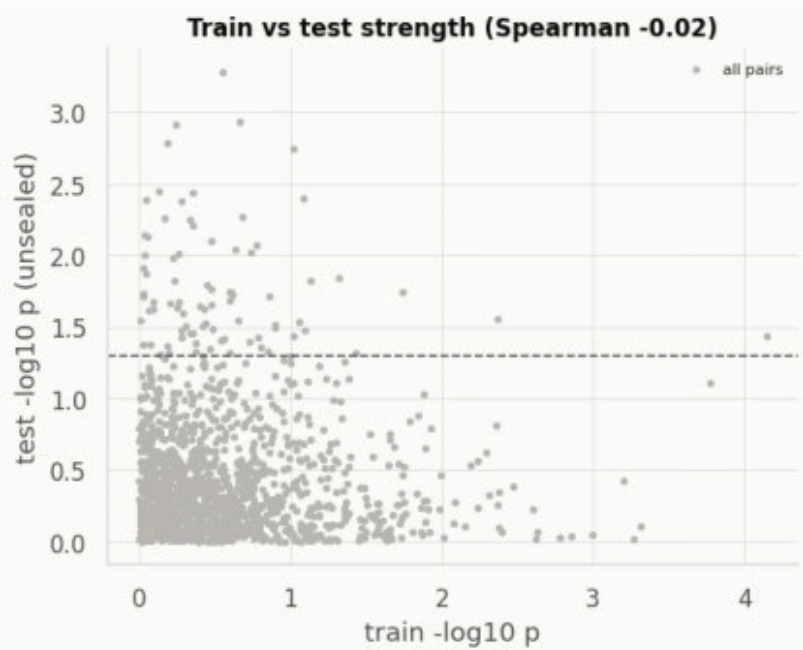
WHAT IT IS

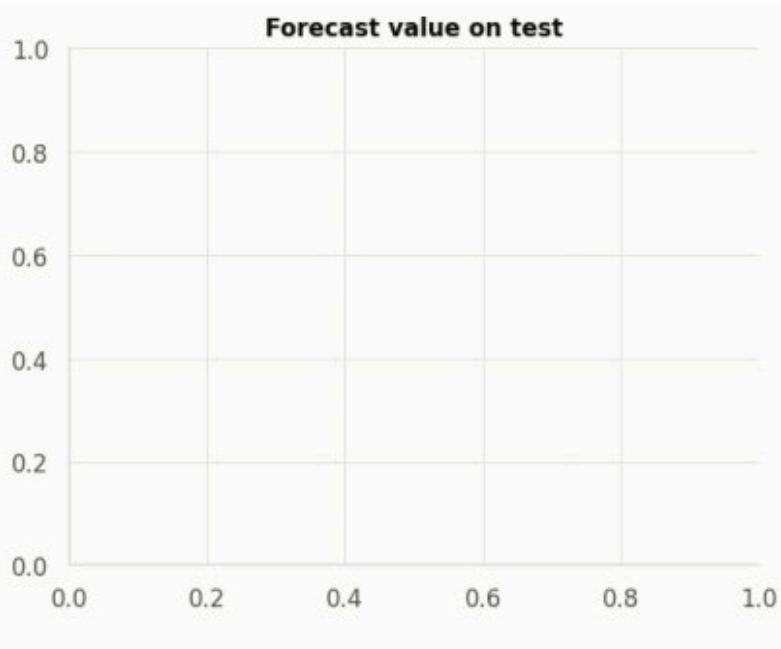
Blind test heatmap with coded products.

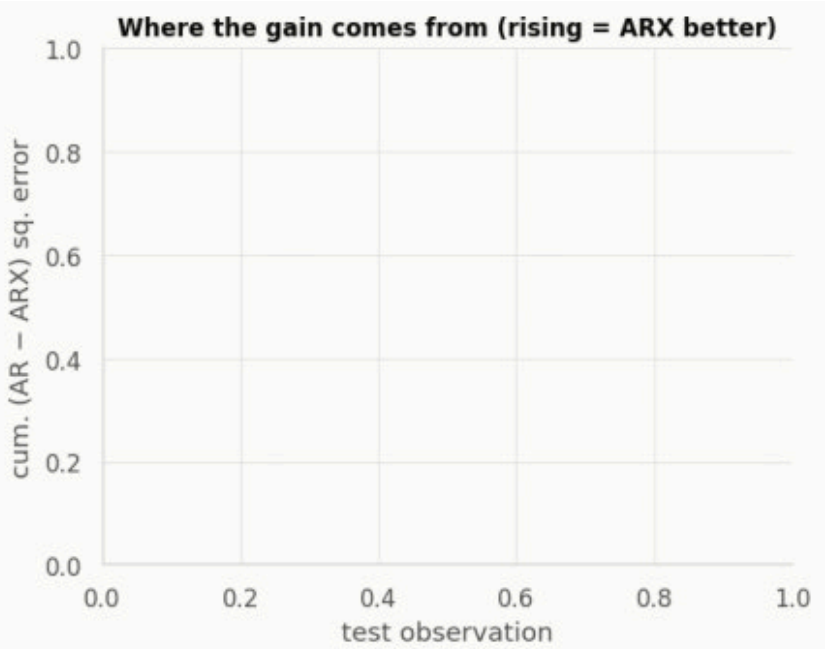
WHAT IT EXPLAINS

The test-period map of the same links, with product names hidden as codes so selection cannot be steered.

[1D] Replication and forecast value







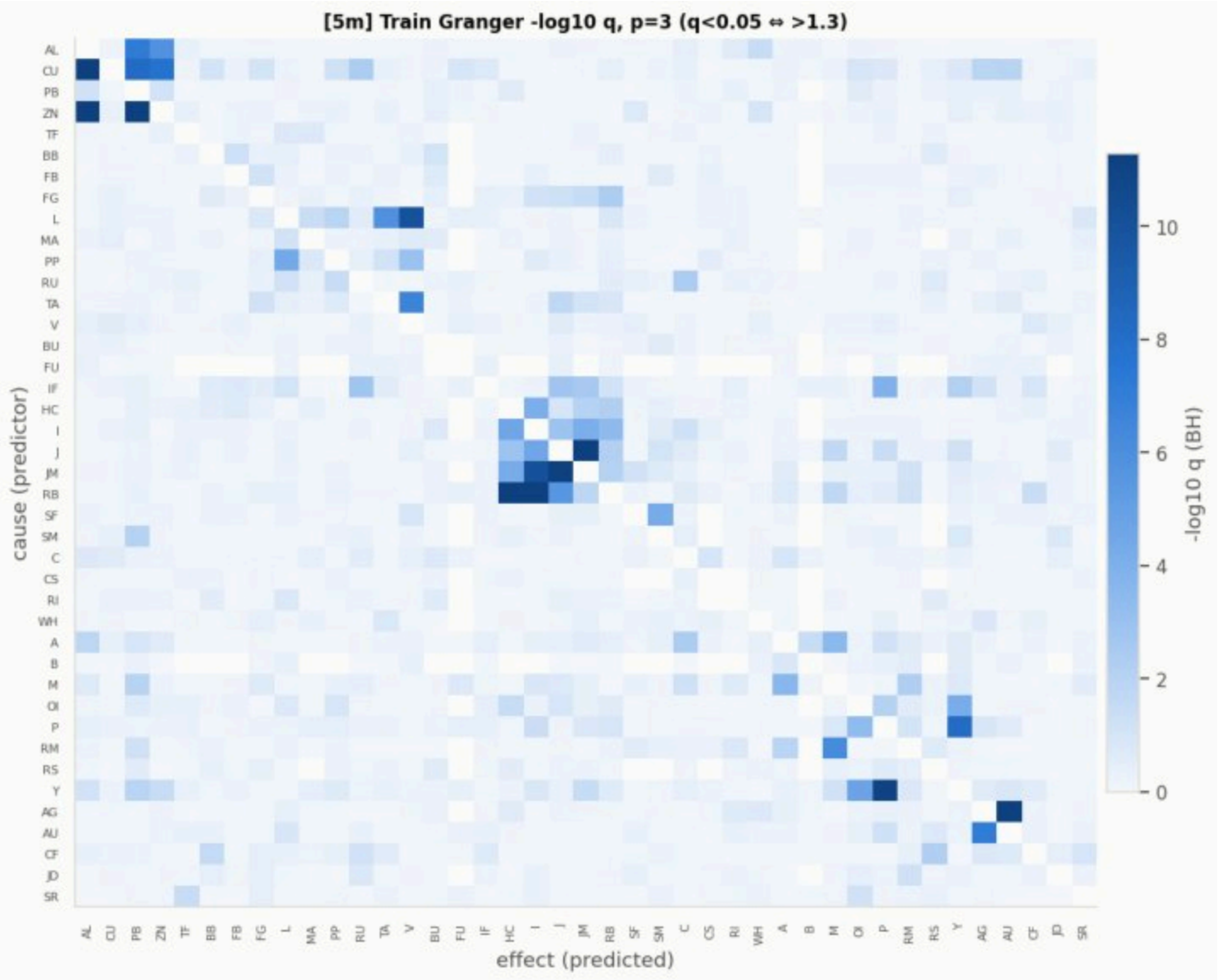
WHAT IT IS

Replication and forecast value of daily links.

WHAT IT EXPLAINS

How many links replicate out of sample and whether they improve a forecast.

[5m] Granger train heatmap



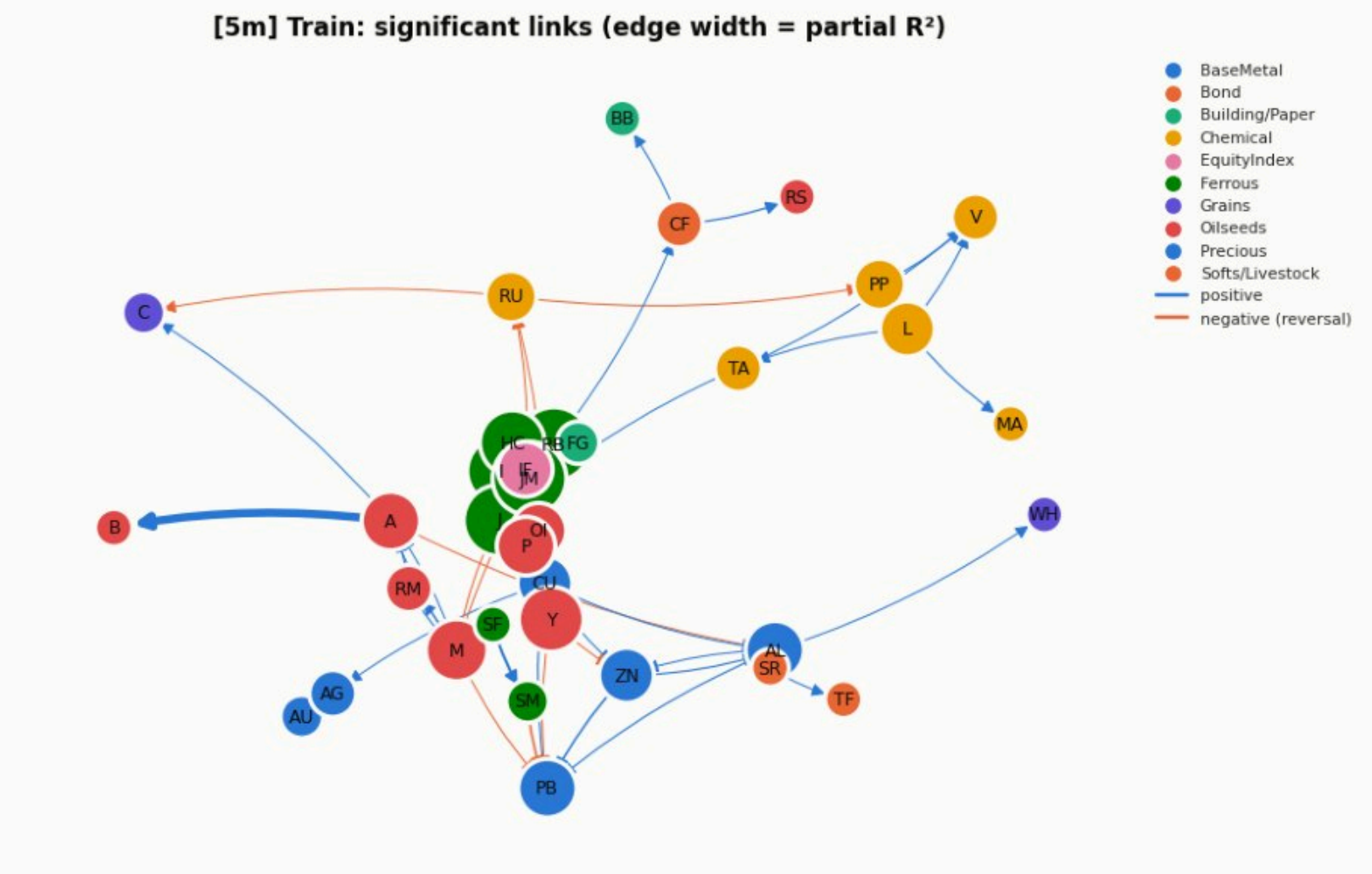
WHAT IT IS

5-minute Granger heatmap, train period.

WHAT IT EXPLAINS

Intraday leader-follower links, train period.

[5m] Train: significant links (edge width = partial R²)



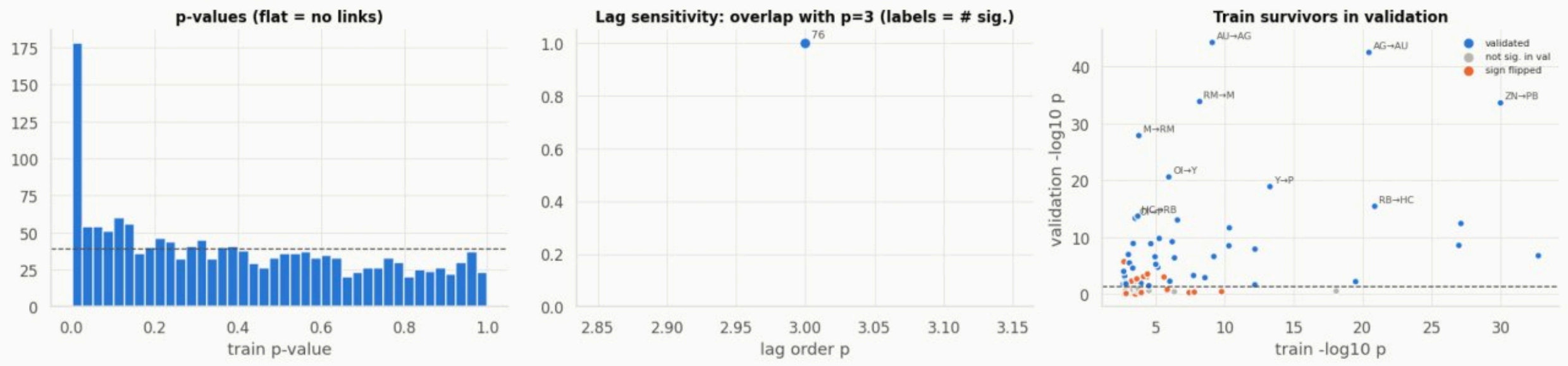
WHAT IT IS

Significant 5-minute links in train; edge width = partial R-squared.

WHAT IT EXPLAINS

The network of leading products; thicker edges explain more.

[5m] p-values, lag sensitivity, validation



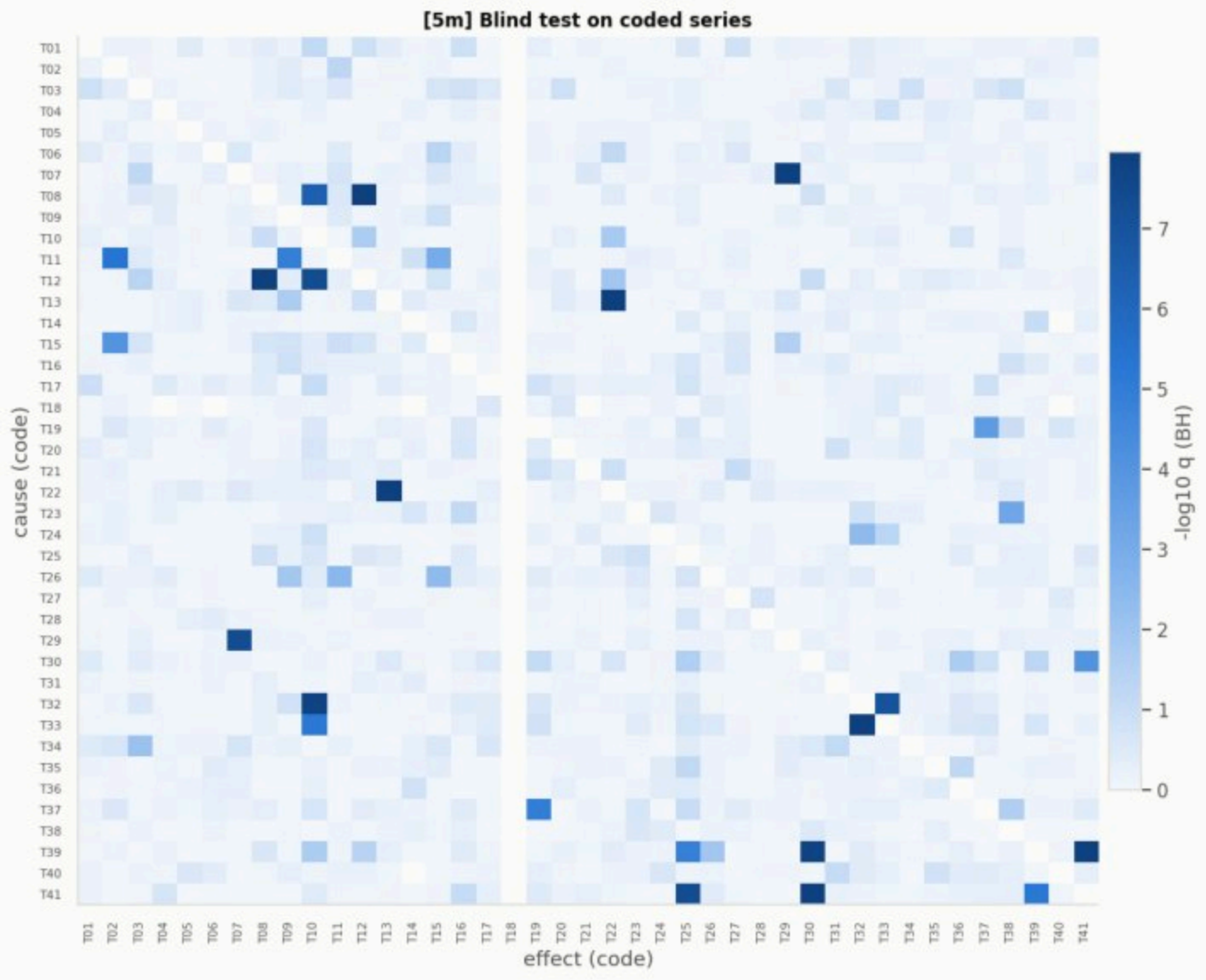
WHAT IT IS

p-values, lag sensitivity and validation for 5-minute links.

WHAT IT EXPLAINS

Same robustness checks at 5 minutes.

[5m] Blind test heatmap (coded)



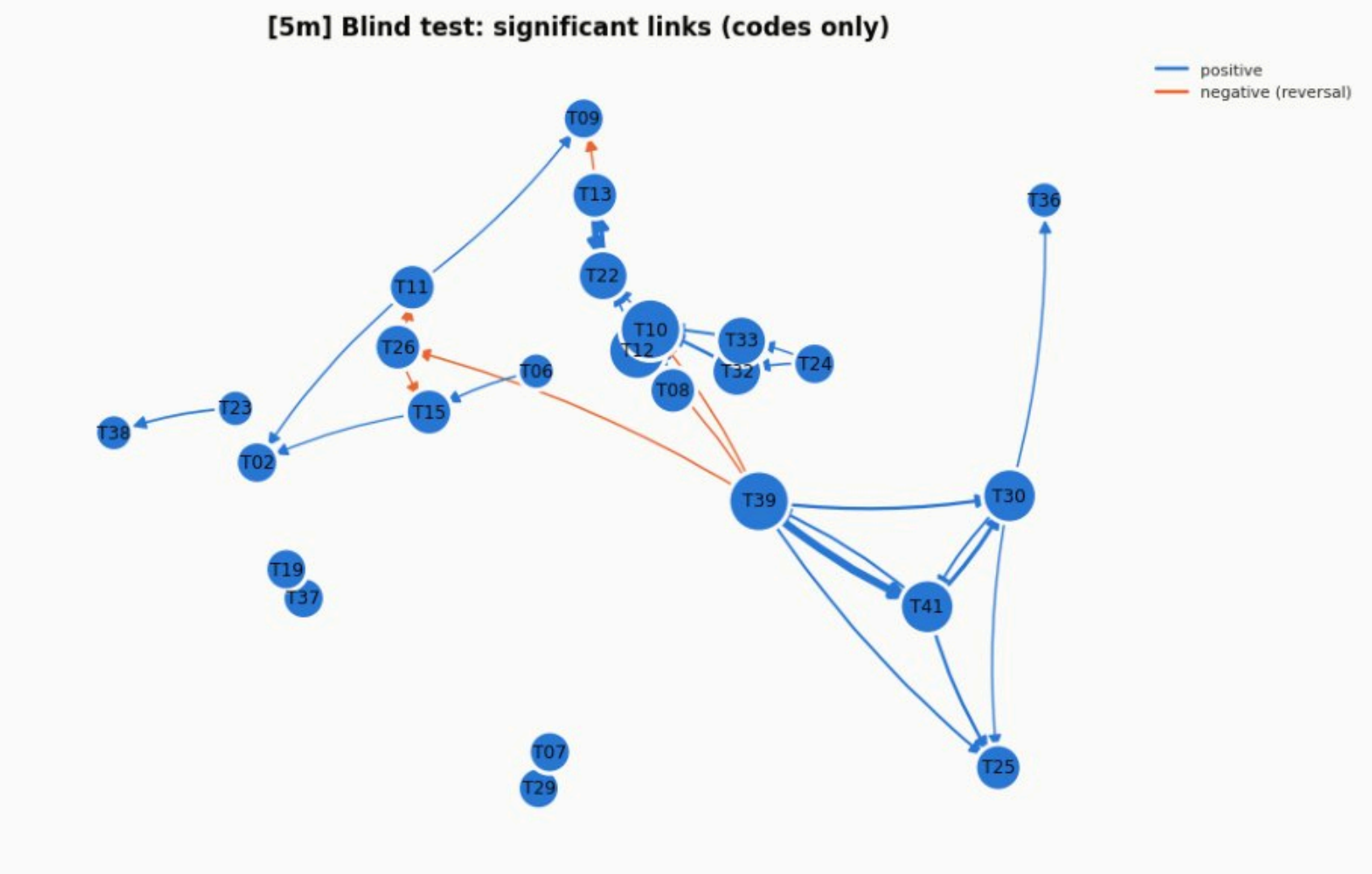
WHAT IT IS

Blind test heatmap for 5-minute links (coded).

WHAT IT EXPLAINS

Out-of-sample map, names hidden.

[5m] Blind test: significant links (codes only)



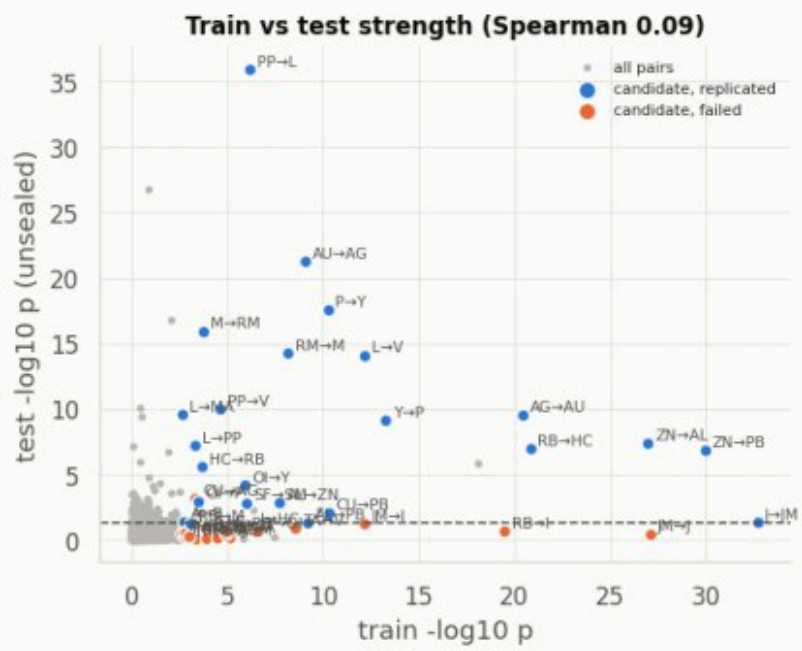
WHAT IT IS

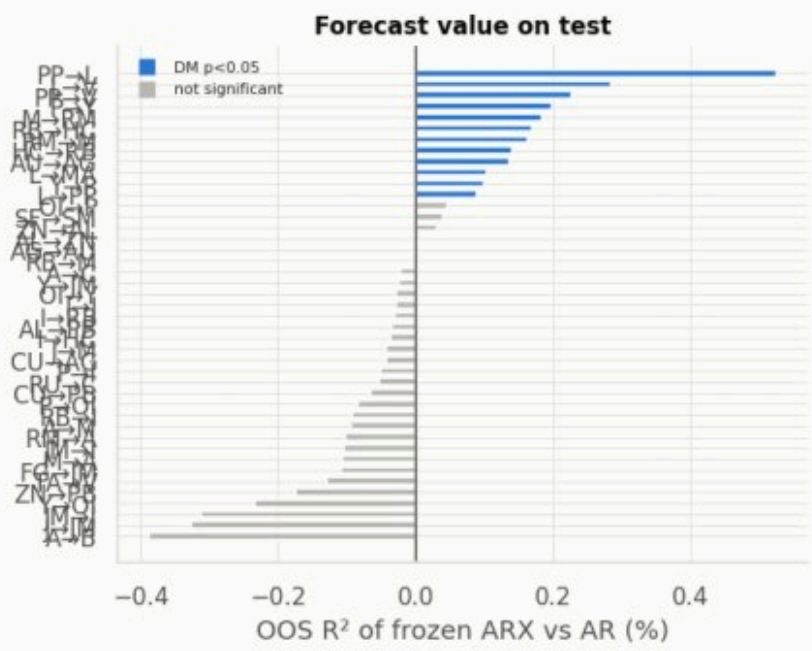
Blind test: significant links, codes only.

WHAT IT EXPLAINS

The network that actually survives out of sample.

[5m] Replication and forecast value







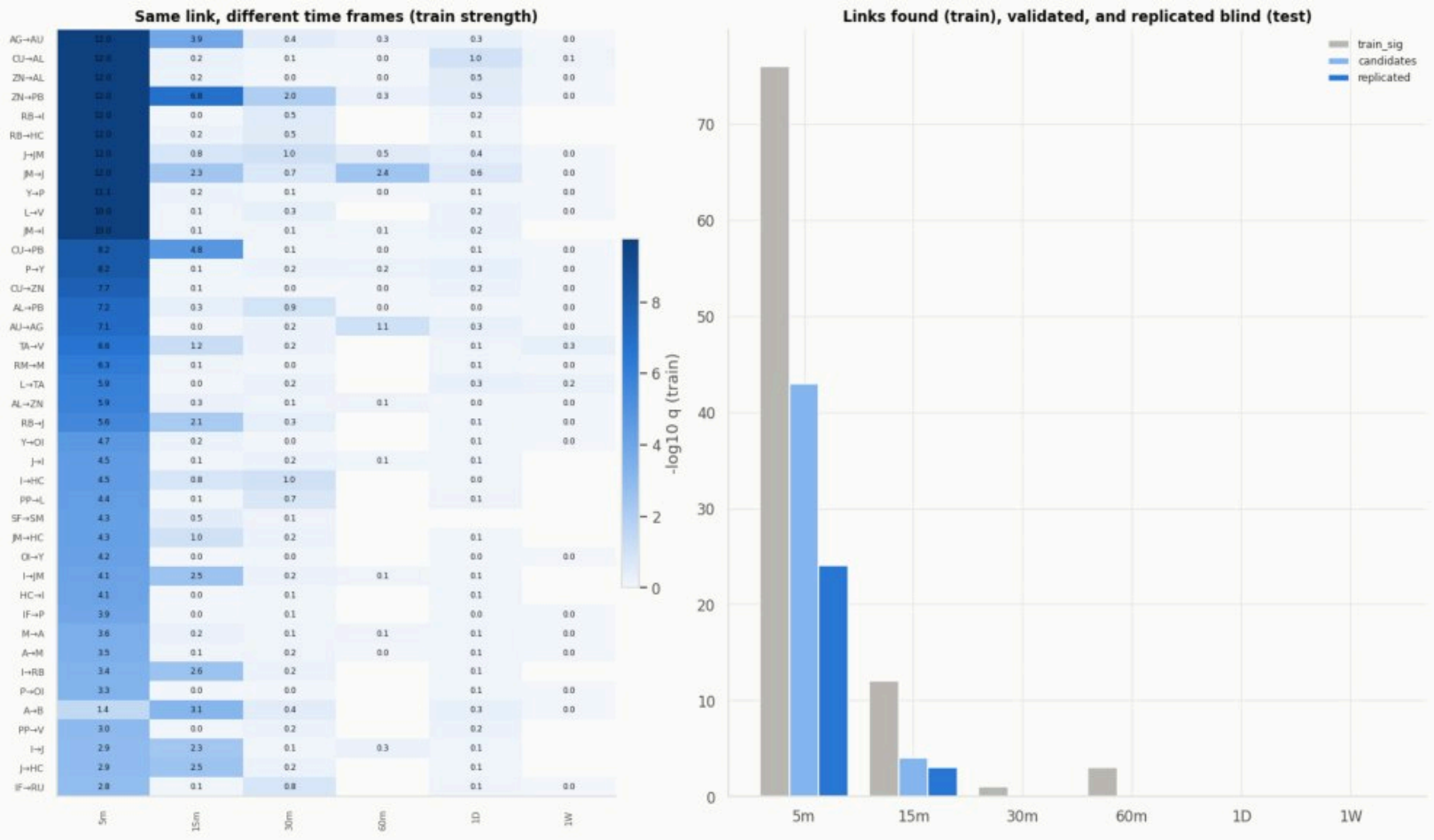
WHAT IT IS

Replication and forecast value of 5-minute links.

WHAT IT EXPLAINS

Granger links that replicate: 27 in total; they are real but small.

Granger across time frames



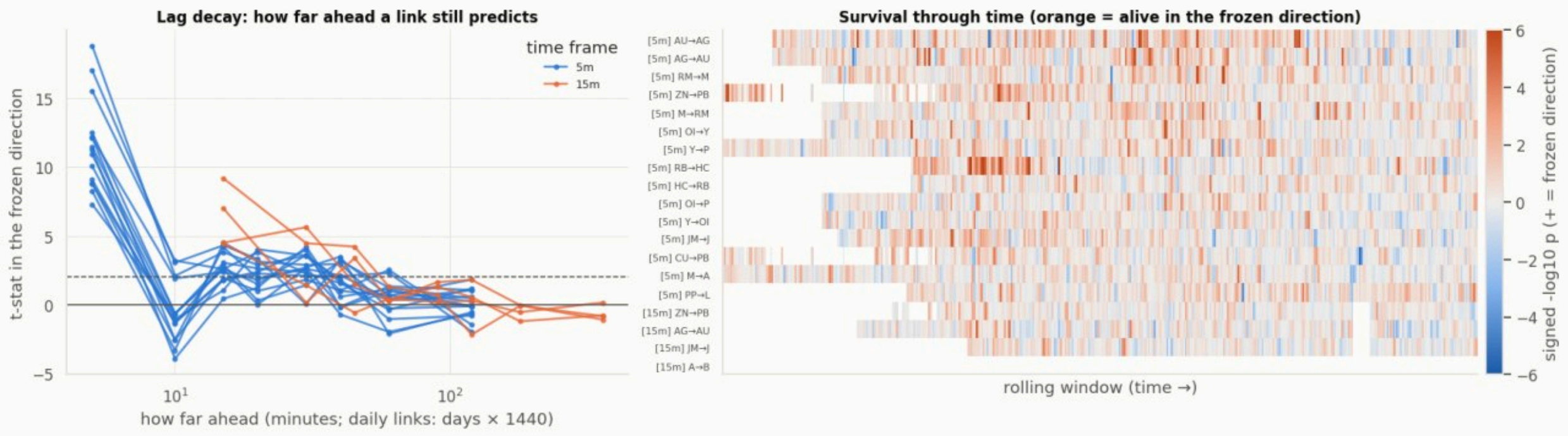
WHAT IT IS

Granger links across time frames.

WHAT IT EXPLAINS

Compares how many links exist at 5m, 15m, 30m, 60m and daily.

Relationship survival: lag decay and through time



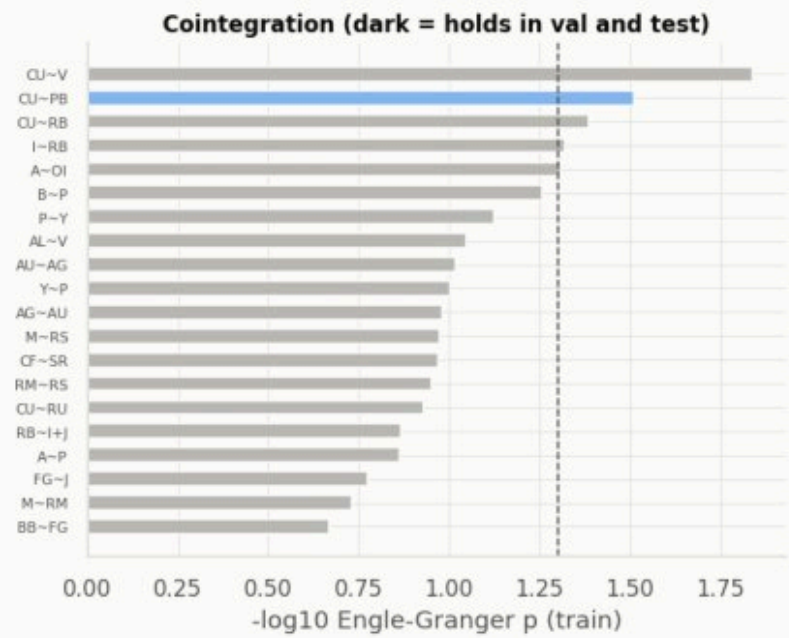
WHAT IT IS

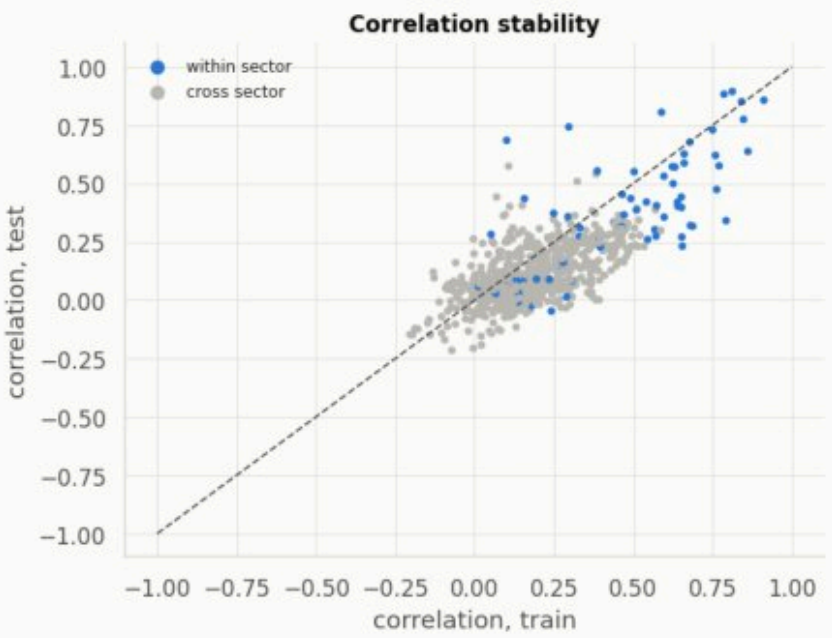
Relationship survival: lag decay and persistence through time.

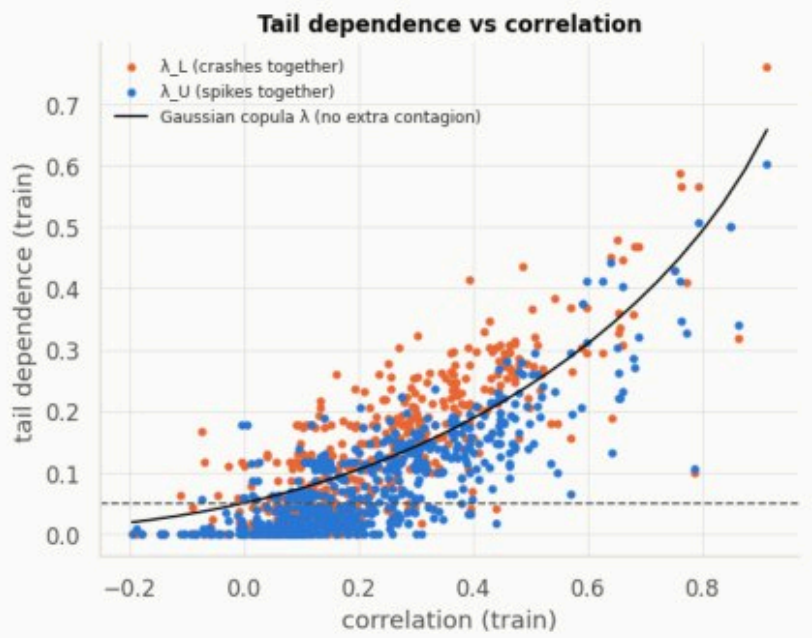
WHAT IT EXPLAINS

Links fade quickly with lag and are not constant through time.

Cointegration, correlation stability, tail dependence







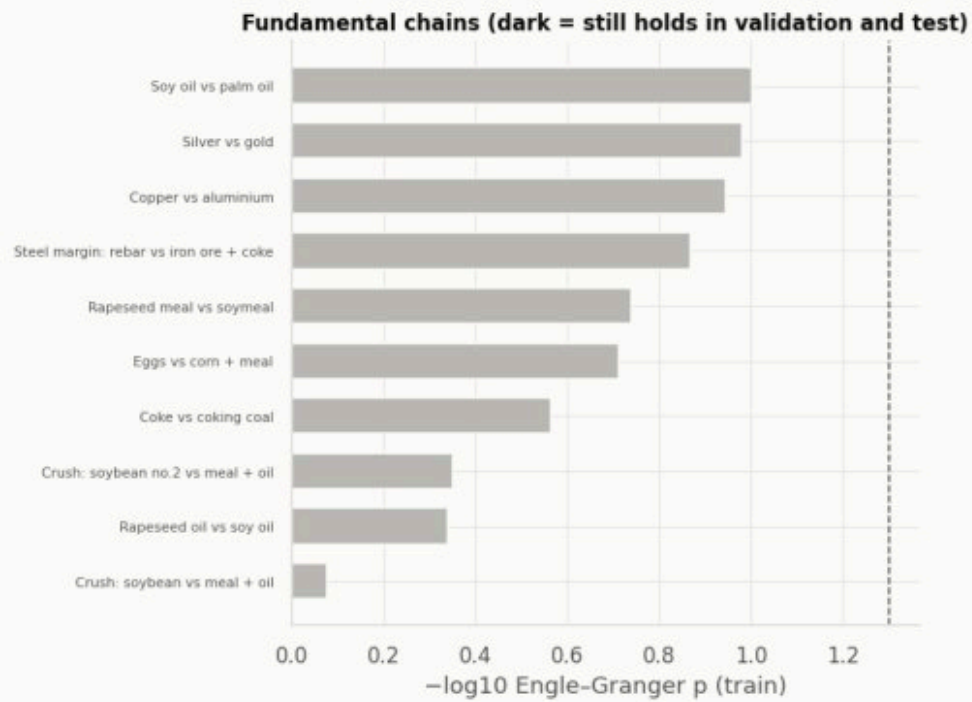
WHAT IT IS

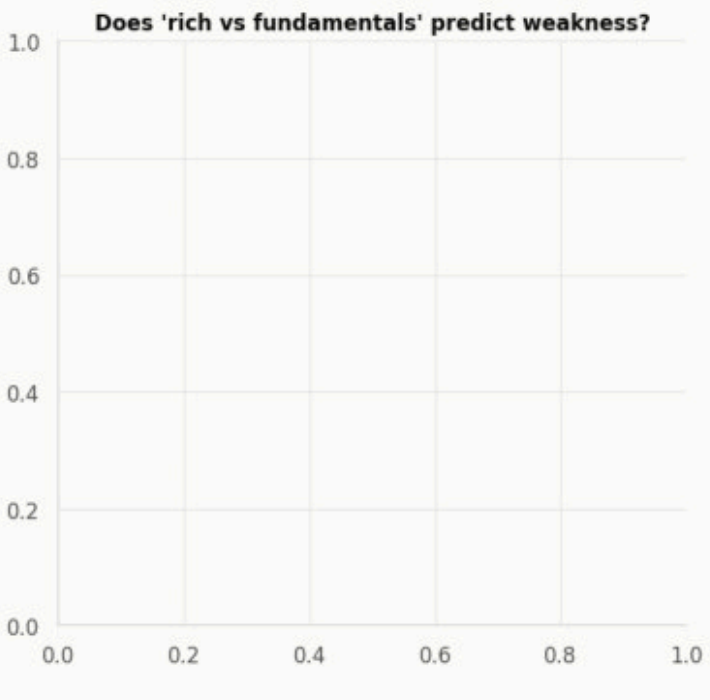
Cointegration, correlation stability and tail dependence for pairs.

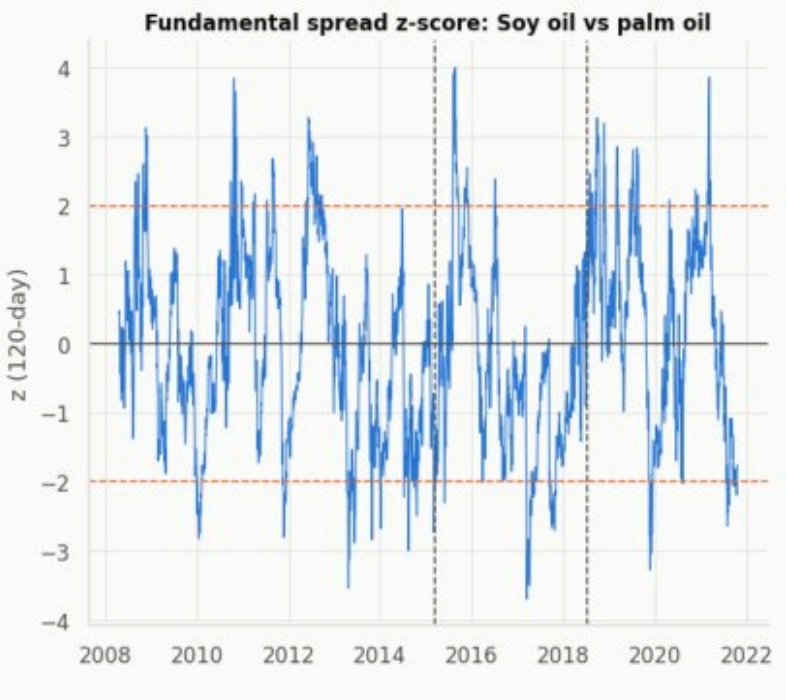
WHAT IT EXPLAINS

Pairs that move together in the long run, how stable their correlation is, and how much they crash together.

Fundamental-linked relationships







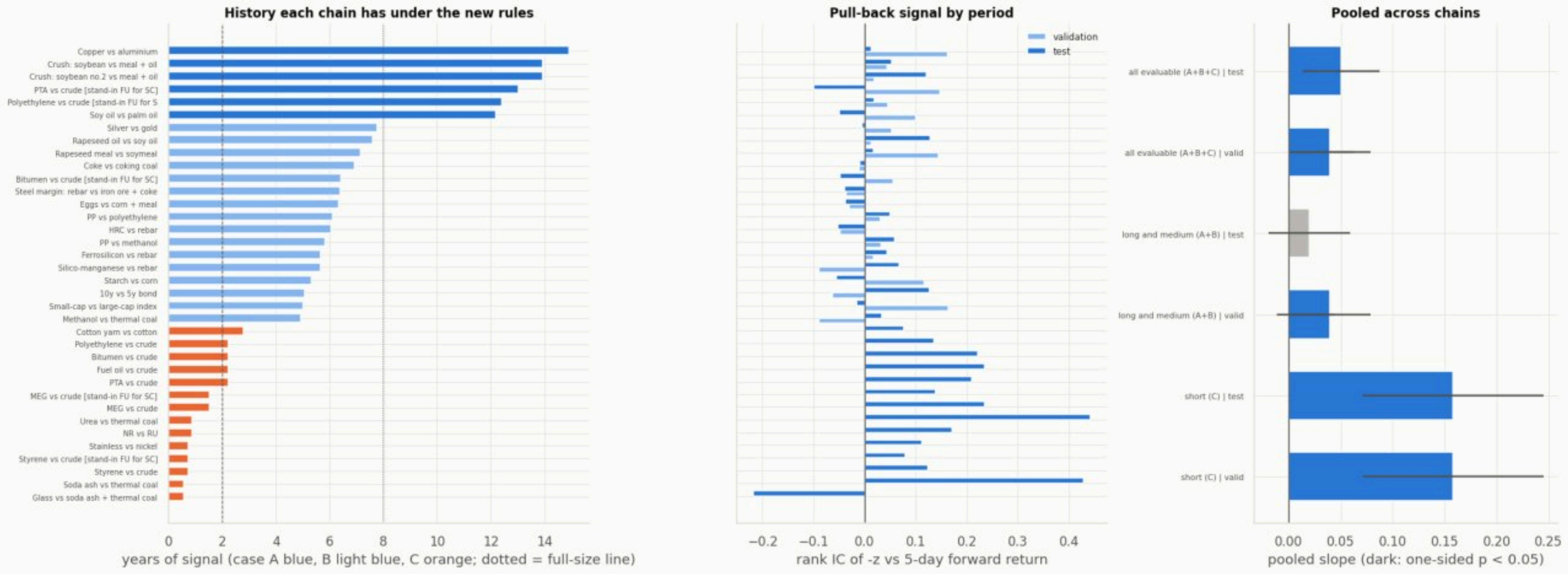
WHAT IT IS

Fundamental-linked relationships (e.g. input-output chains).

WHAT IT EXPLAINS

Tests whether economically linked products lead each other as theory says.

Fundamental chains 16b: history, pull-back signal, pooled test



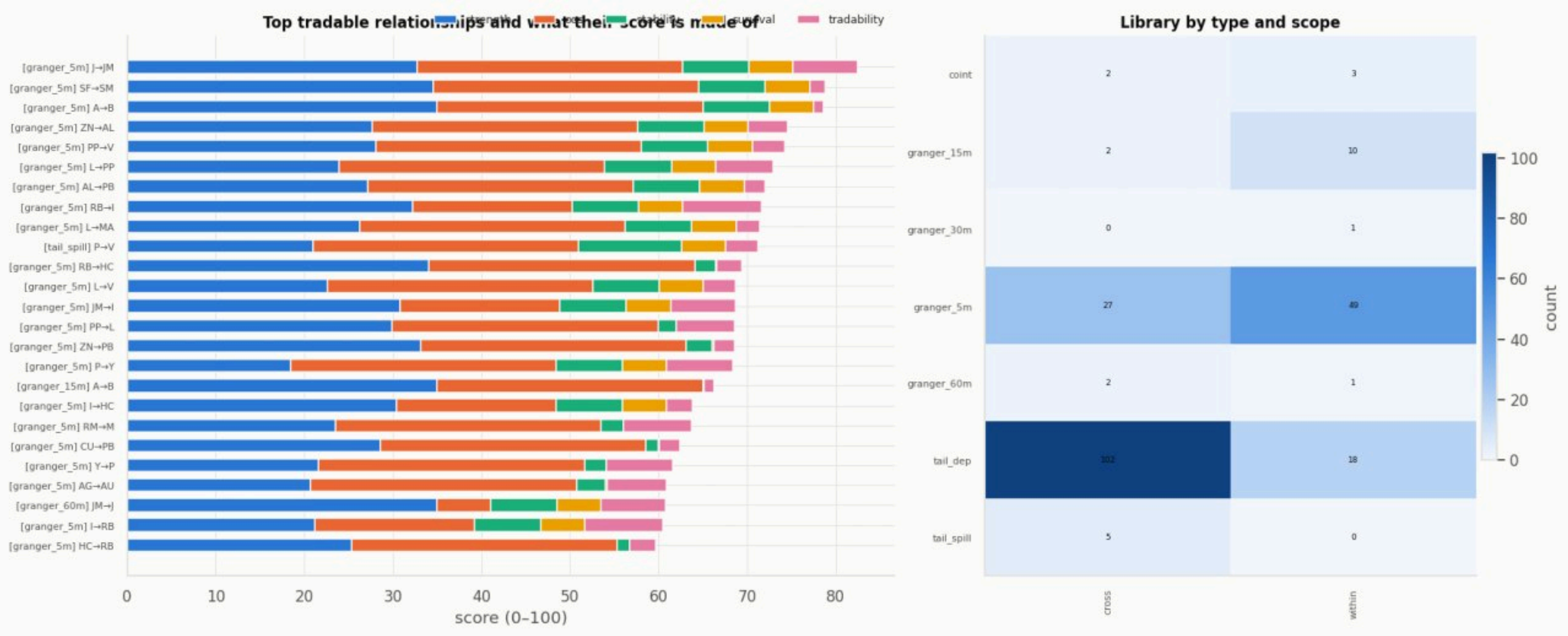
WHAT IT IS

Fundamental chains 16b: history, pull-back signal and pooled test.

WHAT IT EXPLAINS

0 of 10 fundamental chains replicate out of sample.

Relationship library scorecard



WHAT IT IS

Relationship library scorecard.

WHAT IT EXPLAINS

One scorecard of every relationship tested and its out-of-sample status.

PART D · MAP OF STEP 18

The Step 18 series: every idea tested, in order

Step	Idea	Result
18c-d	Price-volume geometry and topology	Size of move only; unstable map
18e-f	Volume x OI states, trend phases	2 of 9 states; 1 rule
18g-i	Quantile flow, ECHO	None confirmed
18k-m	Candle anatomy, learned thresholds, close location	Range yes, direction no; 5m reversal ~0.7 bp
18n-p	Pasted framework, Stage 2 vs VCP, incremental volume	Pasted signs wrong; VCP no better than Stage 2
18q-r	Per-instrument rules, direction by horizon	0 rules; no horizon pays cost
18s	Sectors and regimes, variance ratio	11 stable rules, all reversion
18t	Daily Stage 2 as a portfolio	Best candidate, not yet robust
18u-v	RSI entries, RSI inside Stage 2	0 pass

PART D · FINDINGS

Part D in one page: where direction pays

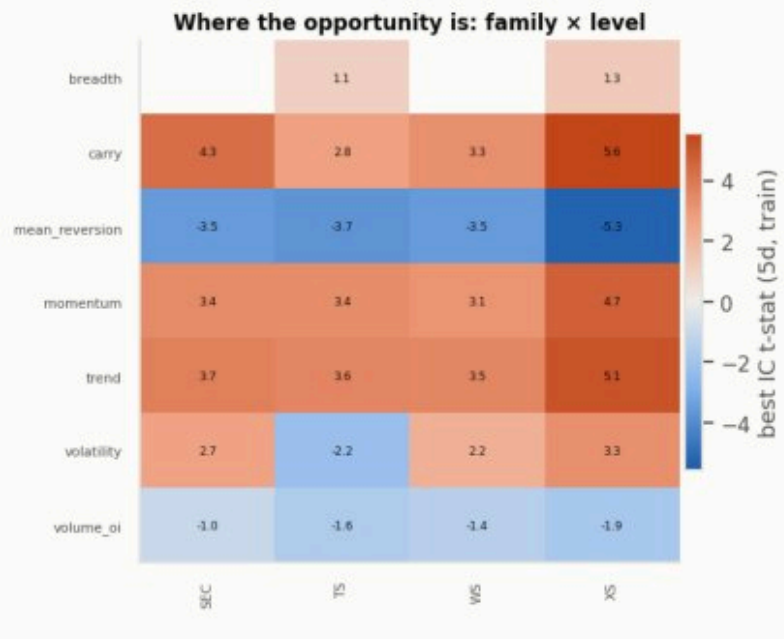
- **Thresholds are market-specific:** learned per regime on train_R, 204 findings pass (Bonferroni |t| > 4.15), nearly all about the next range (range 0-30m: 117; 30-60m: 72), only 15 about signed return.
- **Close location:** 0 of 167 cells beat their parent in all three periods; 5m break-even cost 0.66 bp.
- **Direction by horizon:** 380 cells, 2 stable, 0 beat the 6 bp cost. No horizon between 5 minutes and 20 days is found where direction pays.
- **Sectors:** 245 rules, 11 stable, all reversion; 0 of 66 sector x regime direction cells hold in all periods.
- **VCP vs plain Stage 2:** the VCP beats its parent in 0 of 6 cells.
- **Stage 2 + breakout, daily:** the single survivor (next slides).

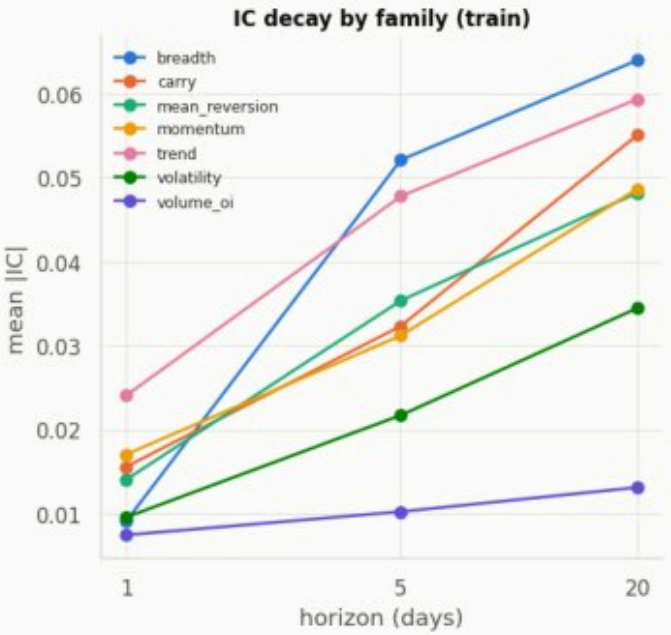
Part D

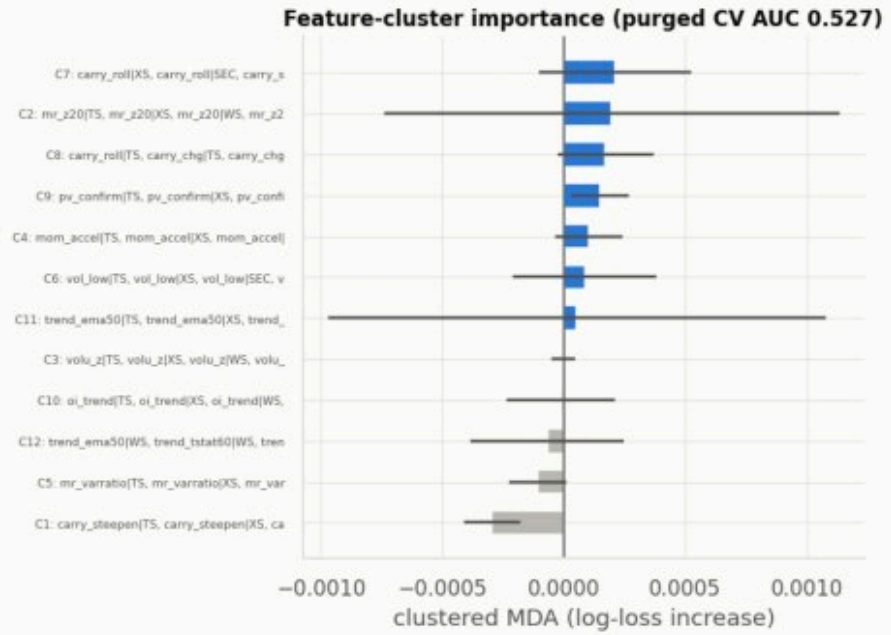
Factors and opportunity ranks

Factor library, price-volume studies, candle anatomy, thresholds, Stage 2 and RSI, then the opportunity ranks.

Factor library: family × level, IC decay, clustered MDA







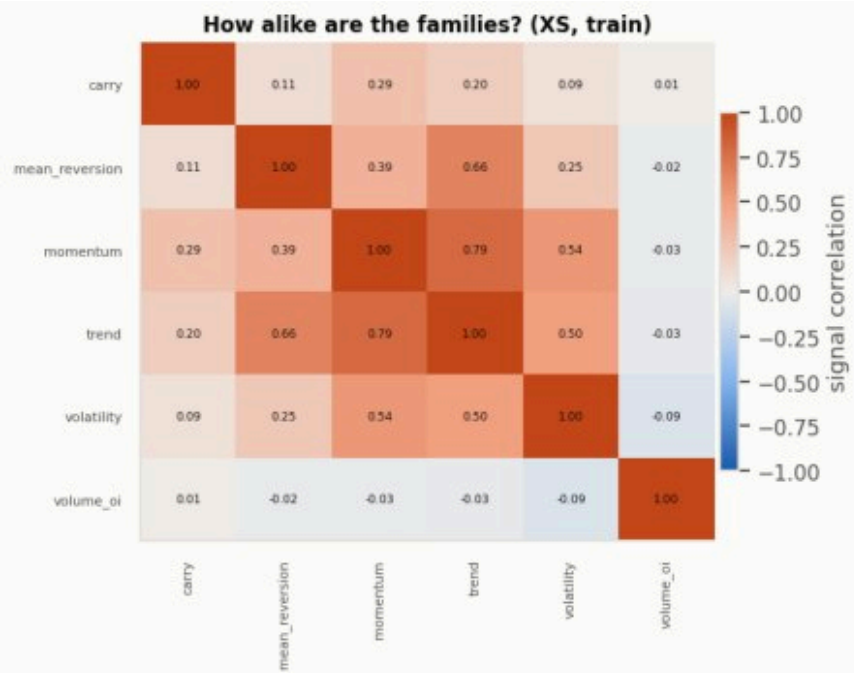
WHAT IT IS

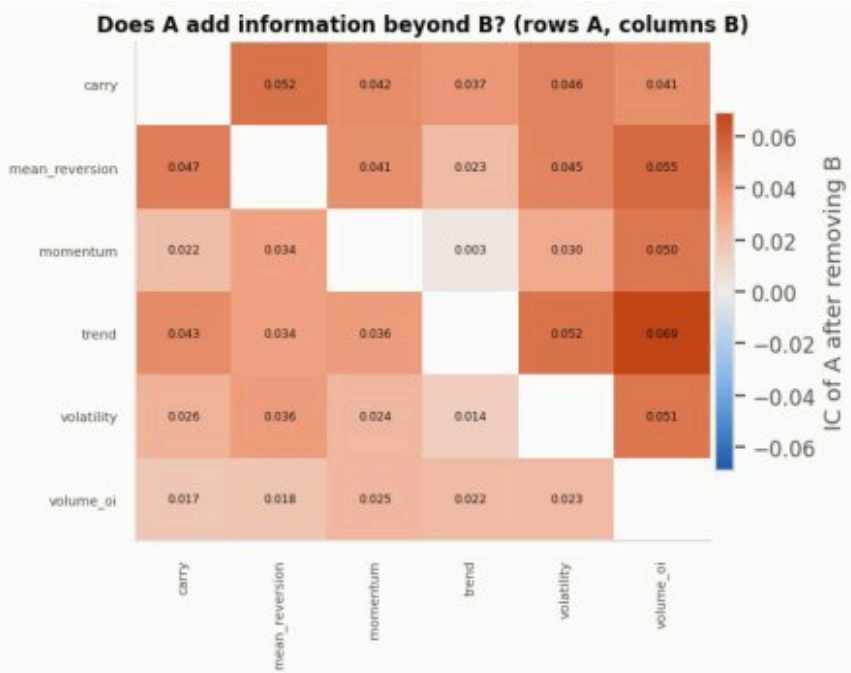
Factor library: family x level, IC decay and clustered MDA importance.

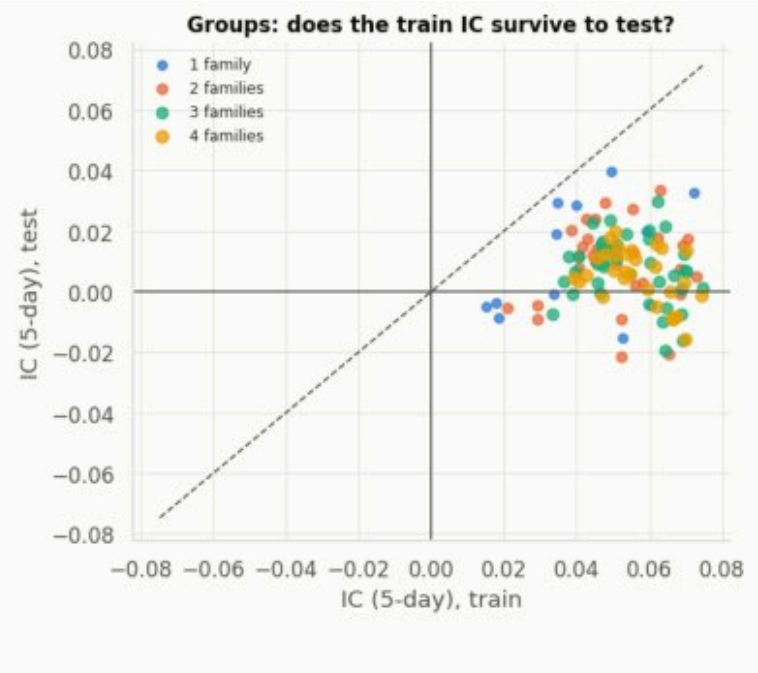
WHAT IT EXPLAINS

Which factor families carry information and how fast their IC decays with horizon.

Factor families in pairs, triplets and quadruplets







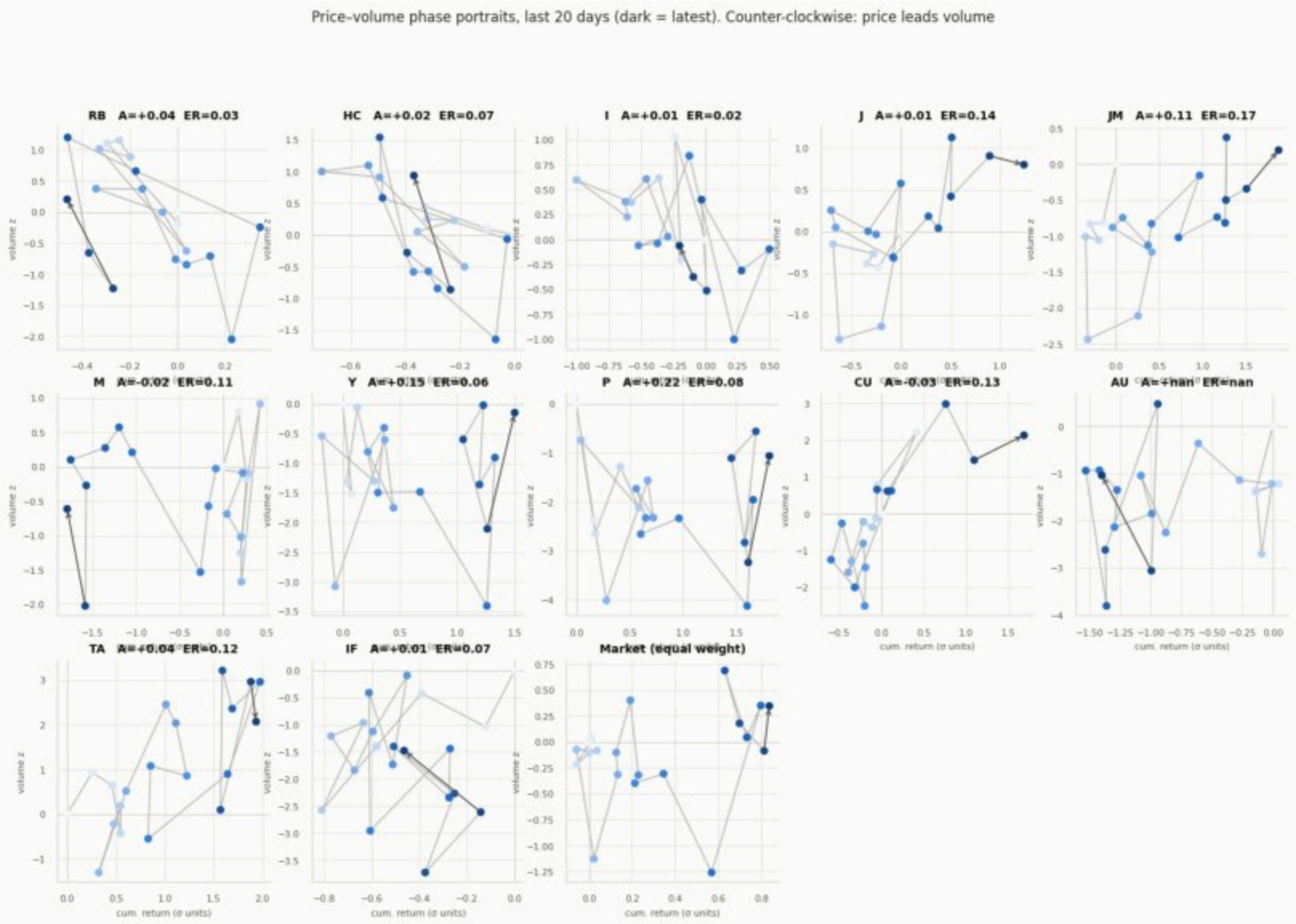
WHAT IT IS

Factor families in pairs, triplets and quadruplets.

WHAT IT EXPLAINS

Whether combining families adds information beyond single families.

Price volume phase portraits of the deep-dive products



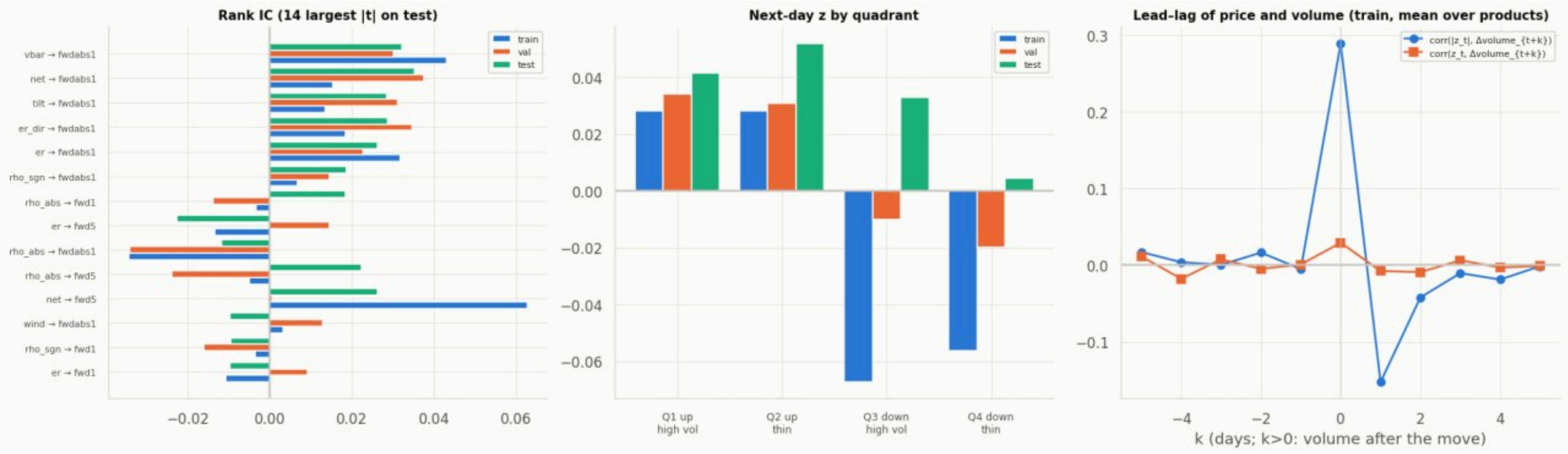
WHAT IT IS

Price-volume phase portraits of the deep-dive products.

WHAT IT EXPLAINS

Each product traced in a price-change vs volume-change plane; loops and spirals describe how price and volume interact.

Price volume geometry: predictive power, quadrants, lead lag



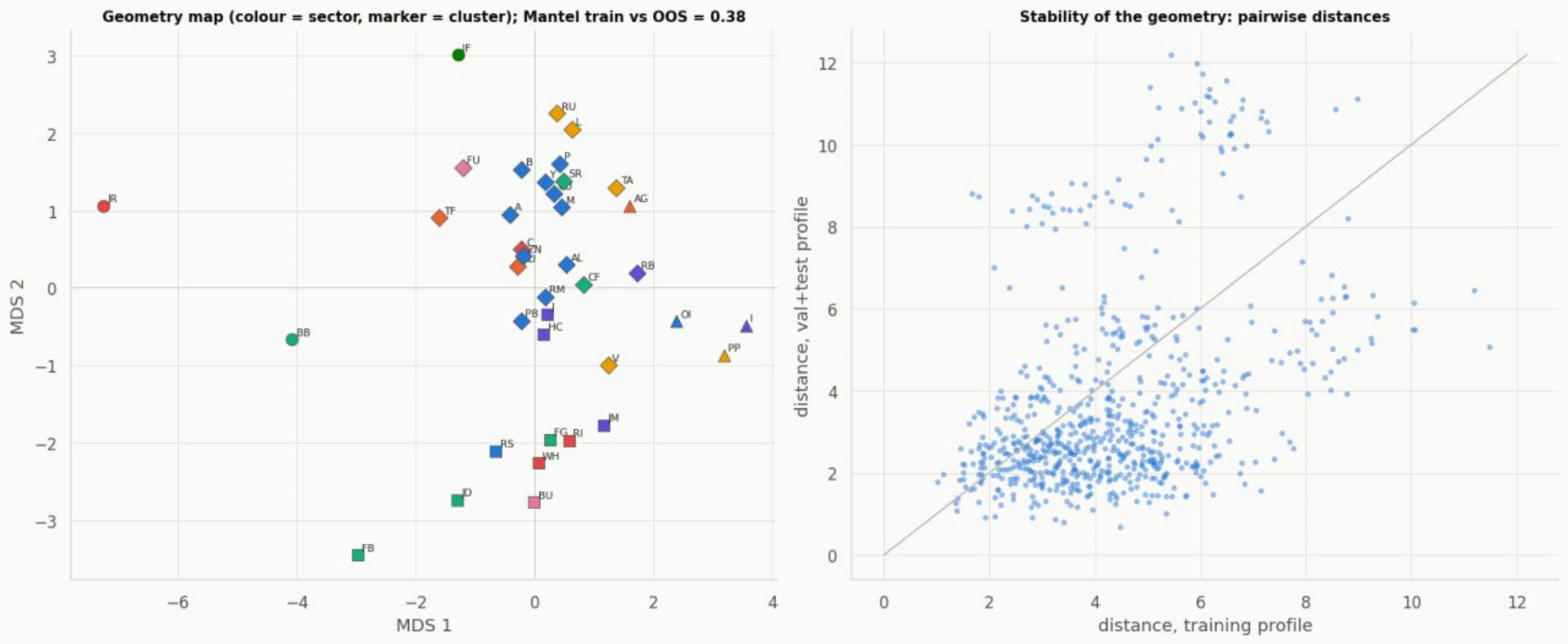
WHAT IT IS

Price-volume geometry: predictive power, quadrants and lead-lag.

WHAT IT EXPLAINS

5 of 33 feature-label pairs are confirmed out of sample, all about the size of the next move (efficiency and net area), not direction.

Geometry map of the products and its stability



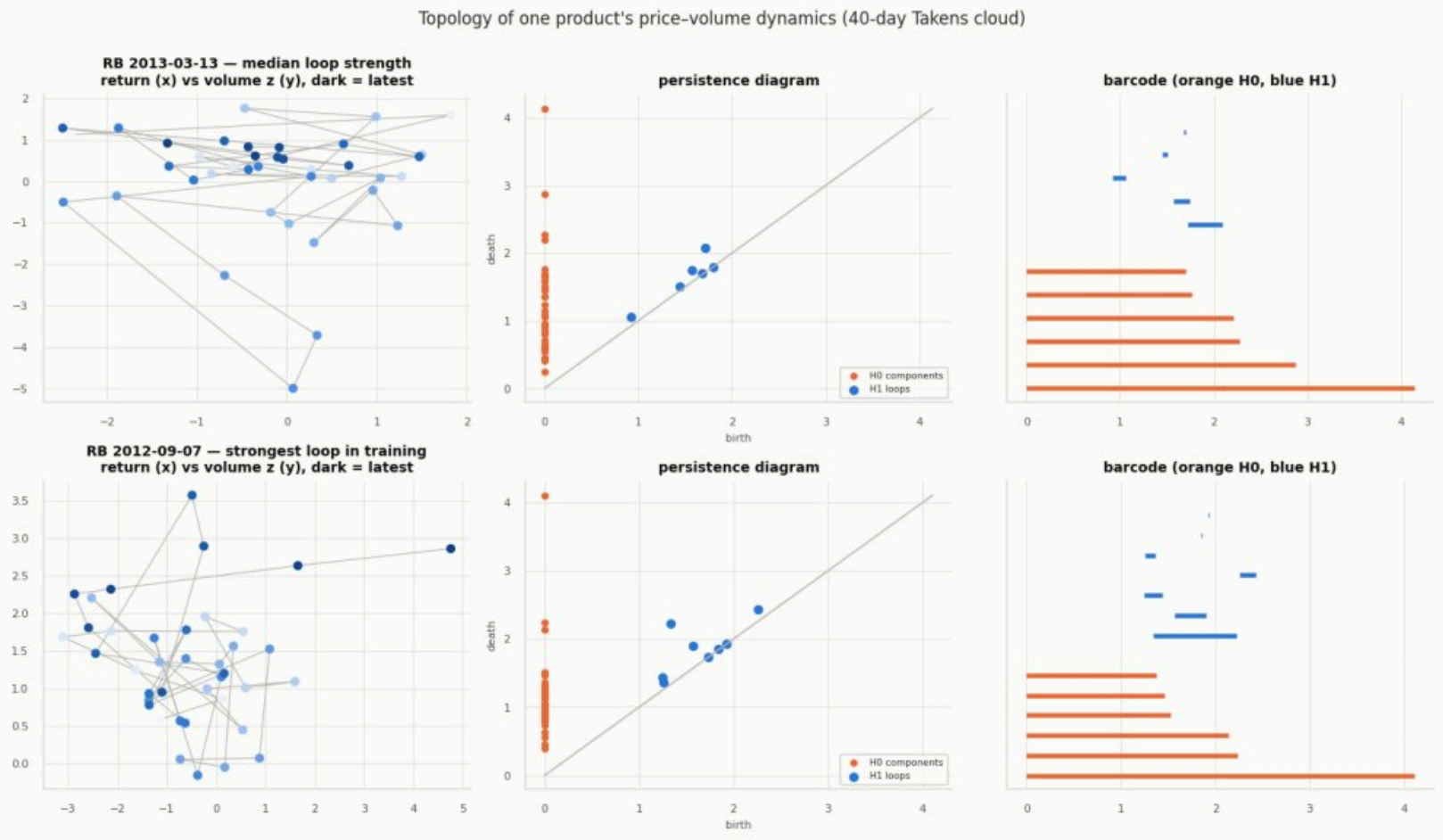
WHAT IT IS

Geometry map of the products and its stability.

WHAT IT EXPLAINS

Products cluster into four groups, but the map is not clearly stable over time (Mantel 0.38).

Persistence of price volume dynamics: typical vs loop-rich window



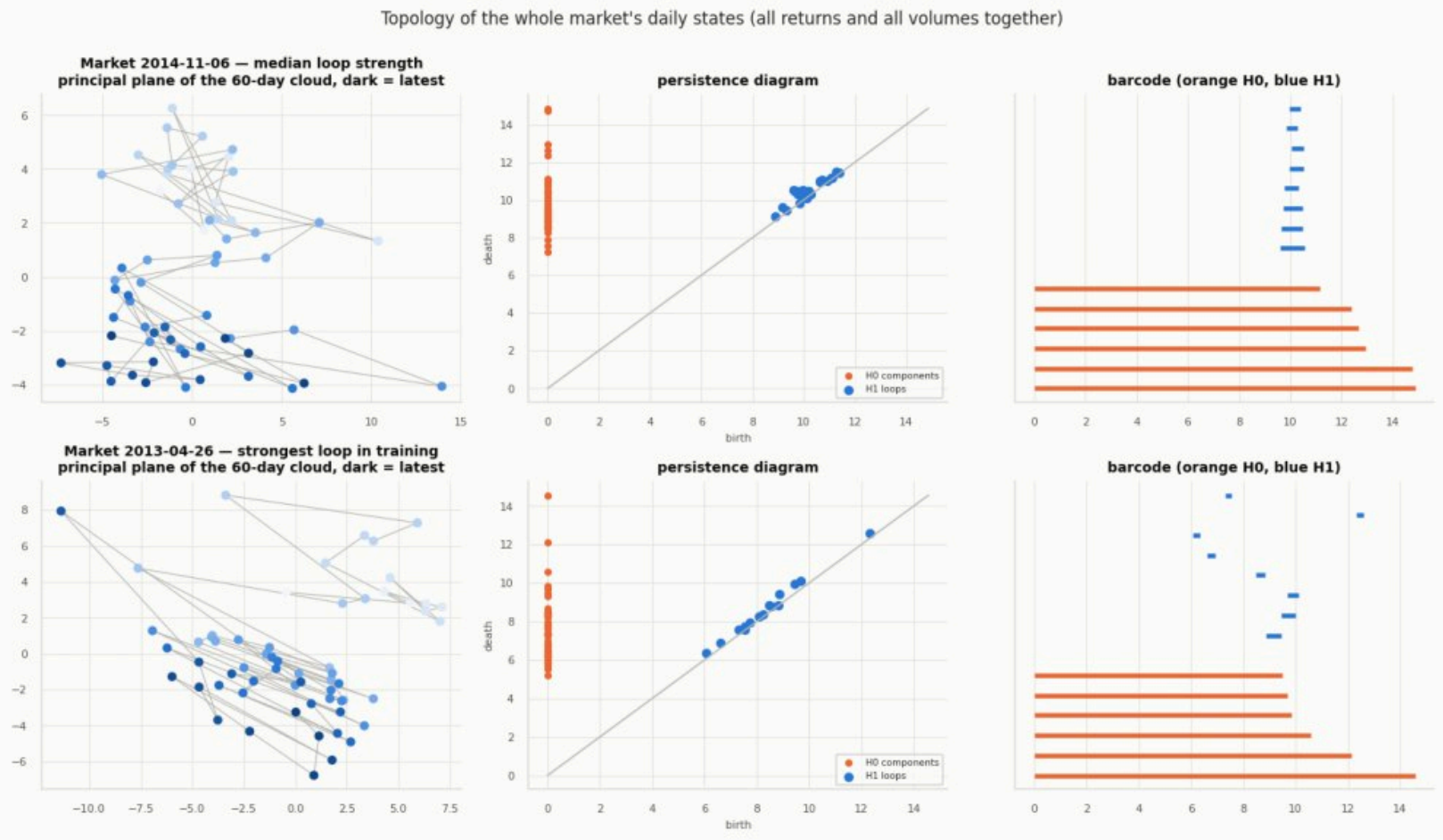
WHAT IT IS

Persistence of price-volume dynamics: typical window versus loop-rich window.

WHAT IT EXPLAINS

Do the loop shapes found in one period repeat in the next? Largely no.

Persistence of the market state cloud



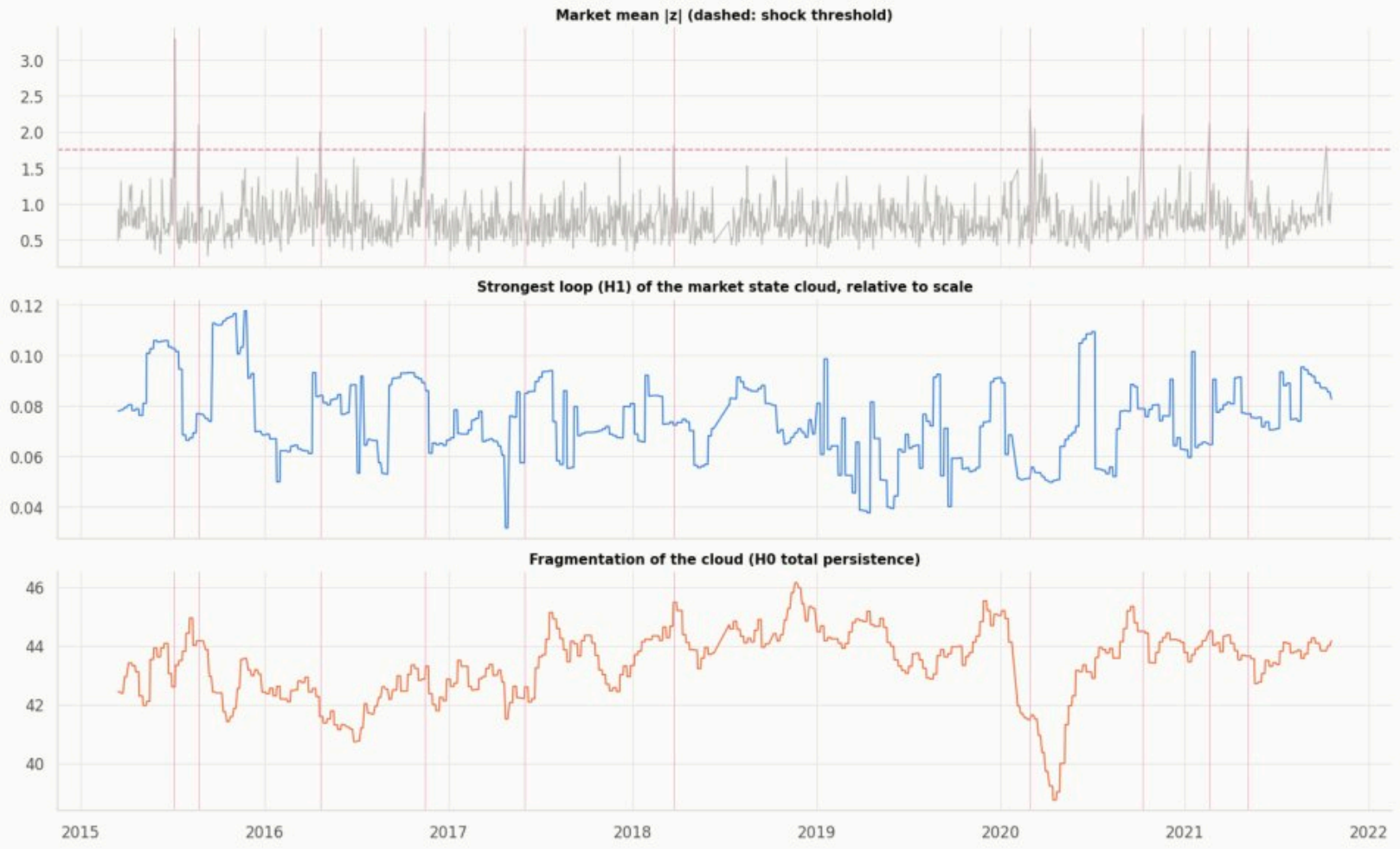
WHAT IT IS

Persistence of the market state cloud.

WHAT IT EXPLAINS

The market state cloud overlaps strongly across periods.

Market topology over time (validation + test) with shock dates



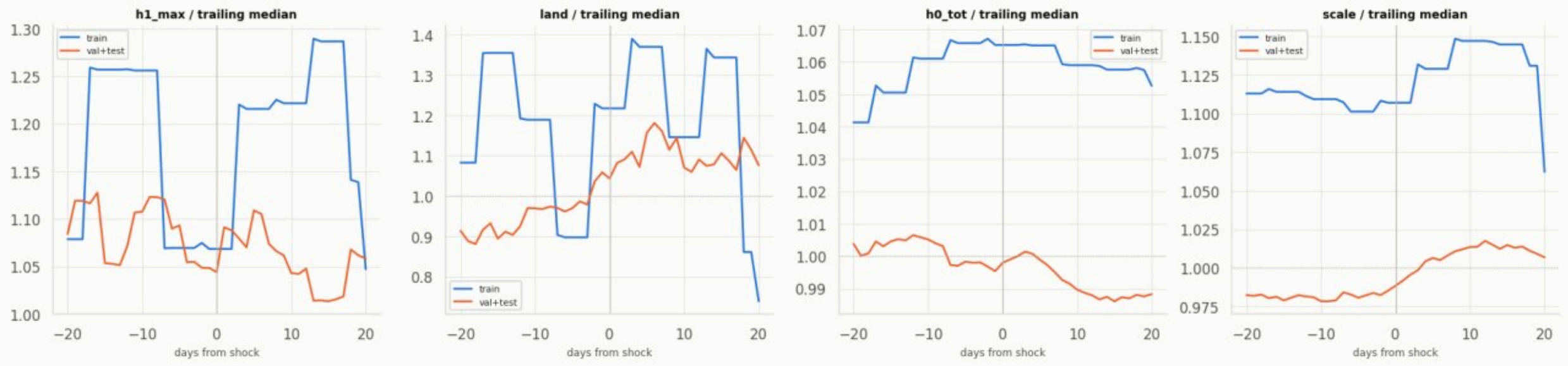
WHAT IT IS

Market topology over time (validation + test) with shock dates.

WHAT IT EXPLAINS

Topology measures through time with volatility shocks marked.

How topology moves around volatility shocks



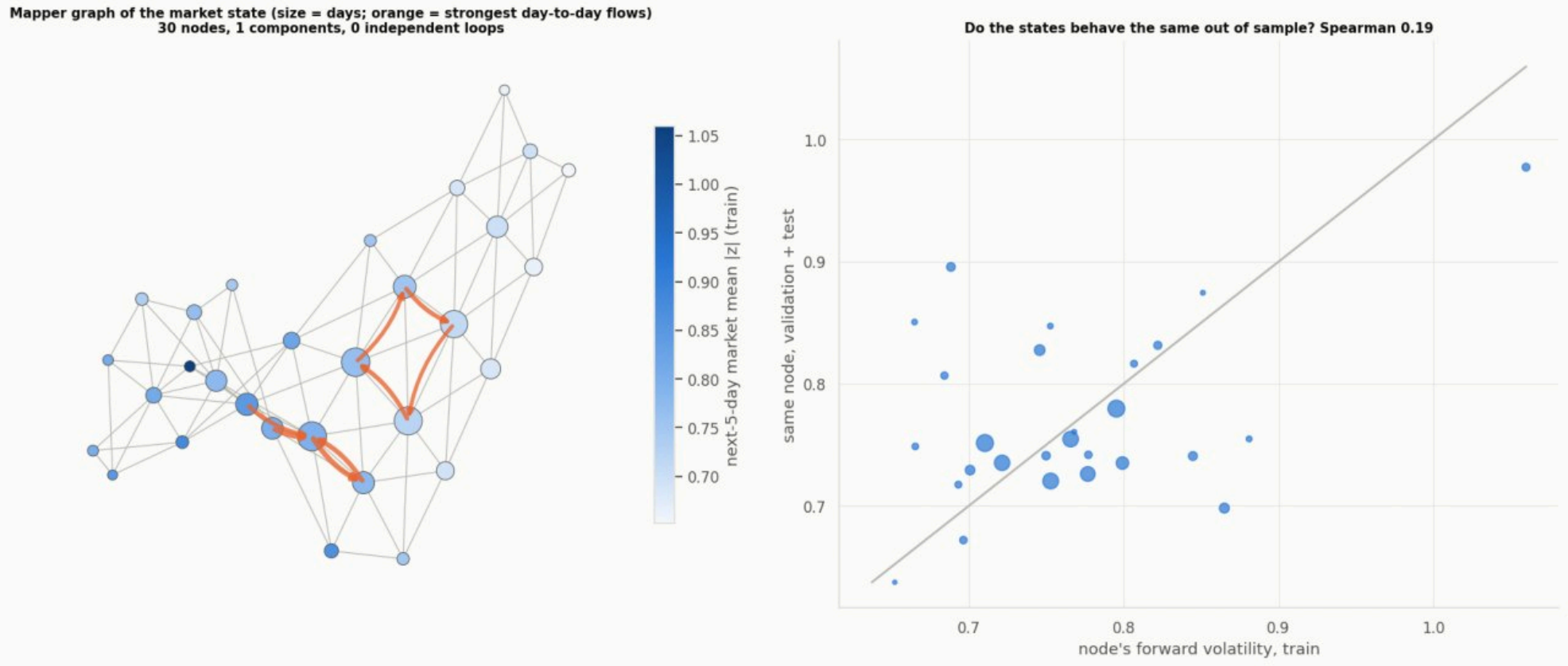
WHAT IT IS

How topology moves around volatility shocks.

WHAT IT EXPLAINS

No topology feature leads volatility shocks beyond chance.

Mapper graph of the market state space and its out-of-sample check



WHAT IT IS

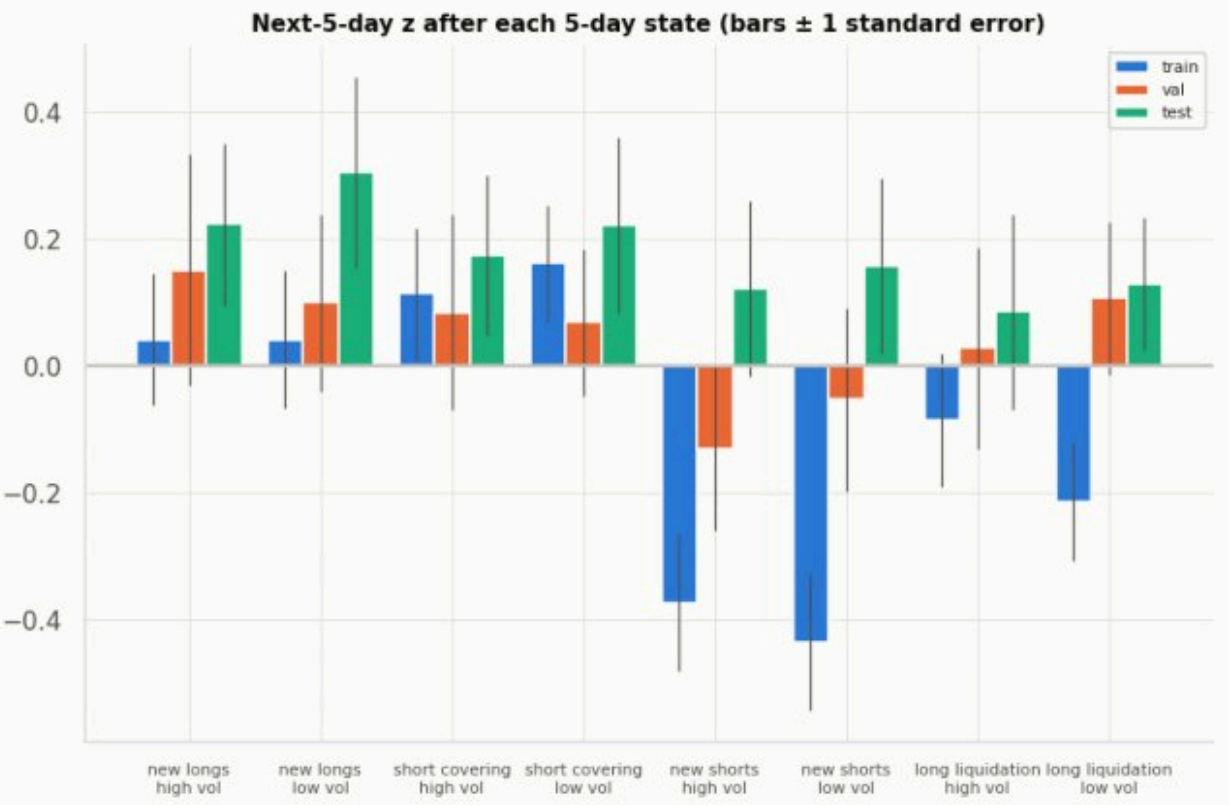
Mapper graph of the market state space and its out-of-sample check.

WHAT IT EXPLAINS

30 nodes, 0 independent loops; the market stays in the same node 75% of days. Node means explain none of out-of-sample volatility beyond trailing volatility.

What each volume x open-interest state does to price





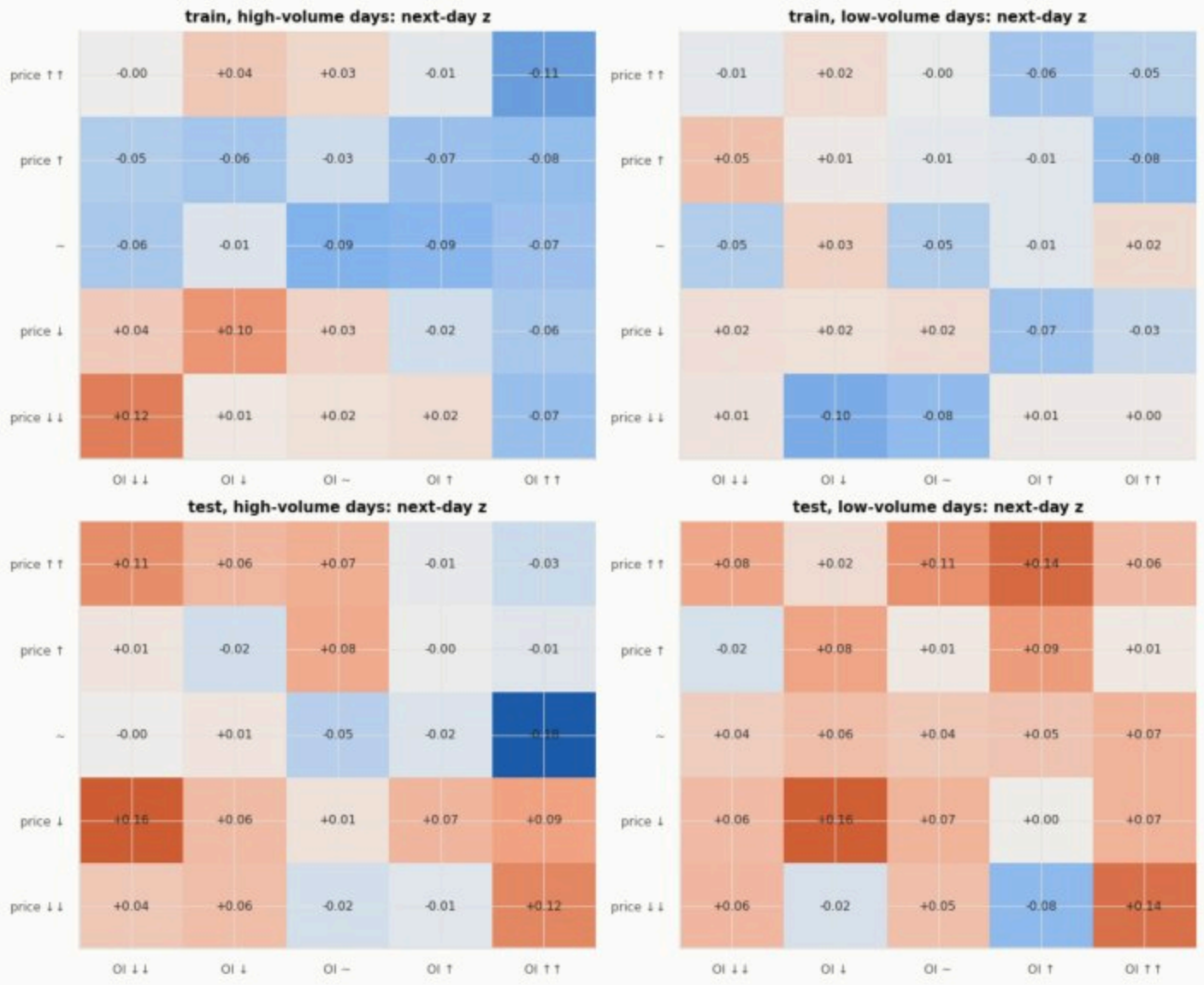
WHAT IT IS

What each volume x open-interest state does to price.

WHAT IT EXPLAINS

New longs, new shorts, short covering and long liquidation states, with next-day move.

Next-day return by price move and OI change, high vs low volume



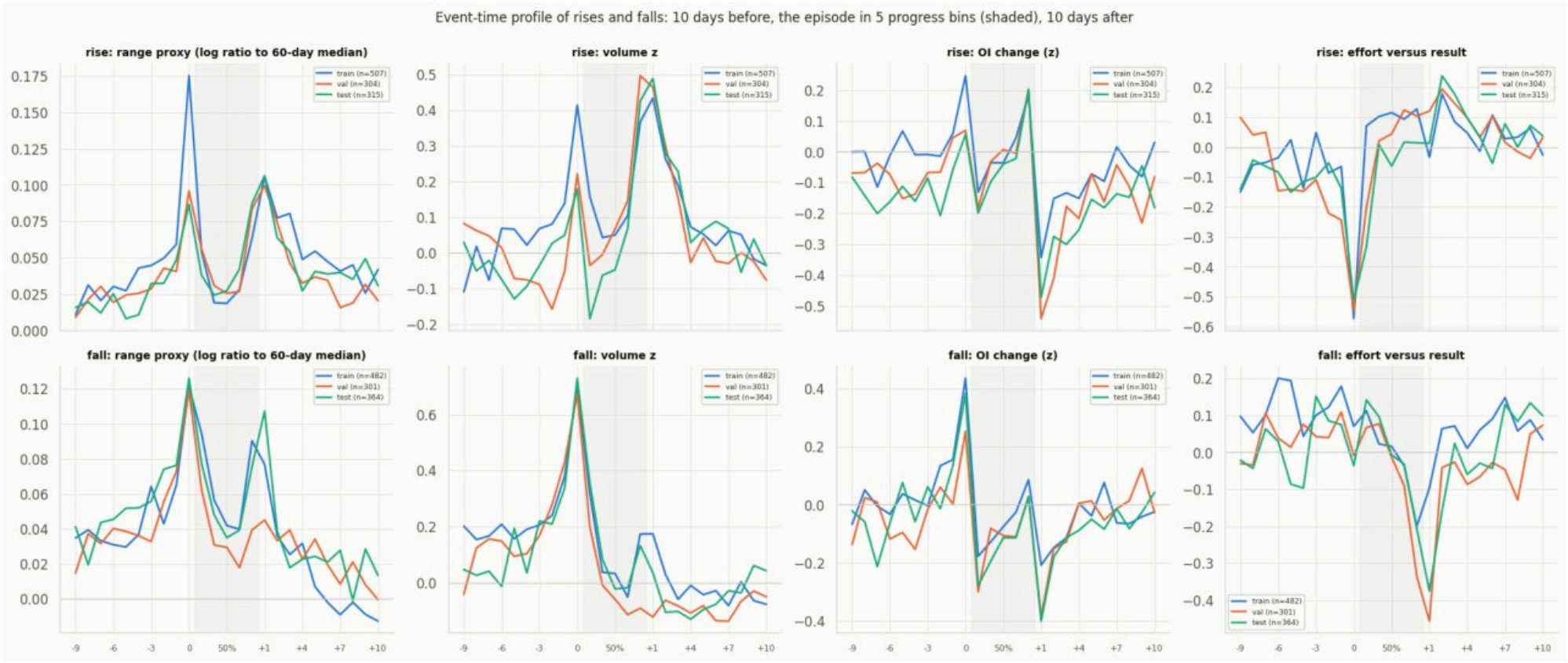
WHAT IT IS

Next-day return by price move and OI change, high versus low volume.

WHAT IT EXPLAINS

Of 9 state calls, 2 are confirmed on test: short covering on low volume and new longs on low volume, both continuation.

Volume, OI and range before, during and after trend episodes



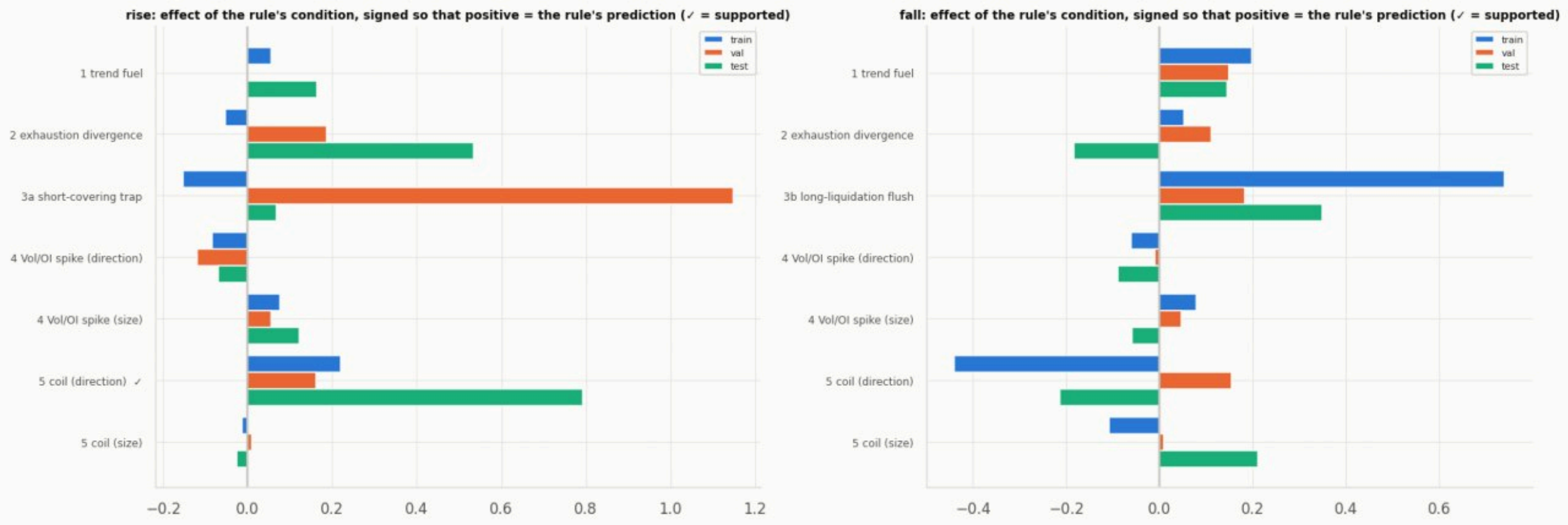
WHAT IT IS

Volume, OI and range before, during and after trend episodes.

WHAT IT EXPLAINS

The anatomy of 2273 trend episodes (1126 rises, 1147 falls). Four phase signatures hold in train, validation and test, about range during the trend and the state after it.

Do the volume/OI rules hold? Effect in the direction each rule predicts



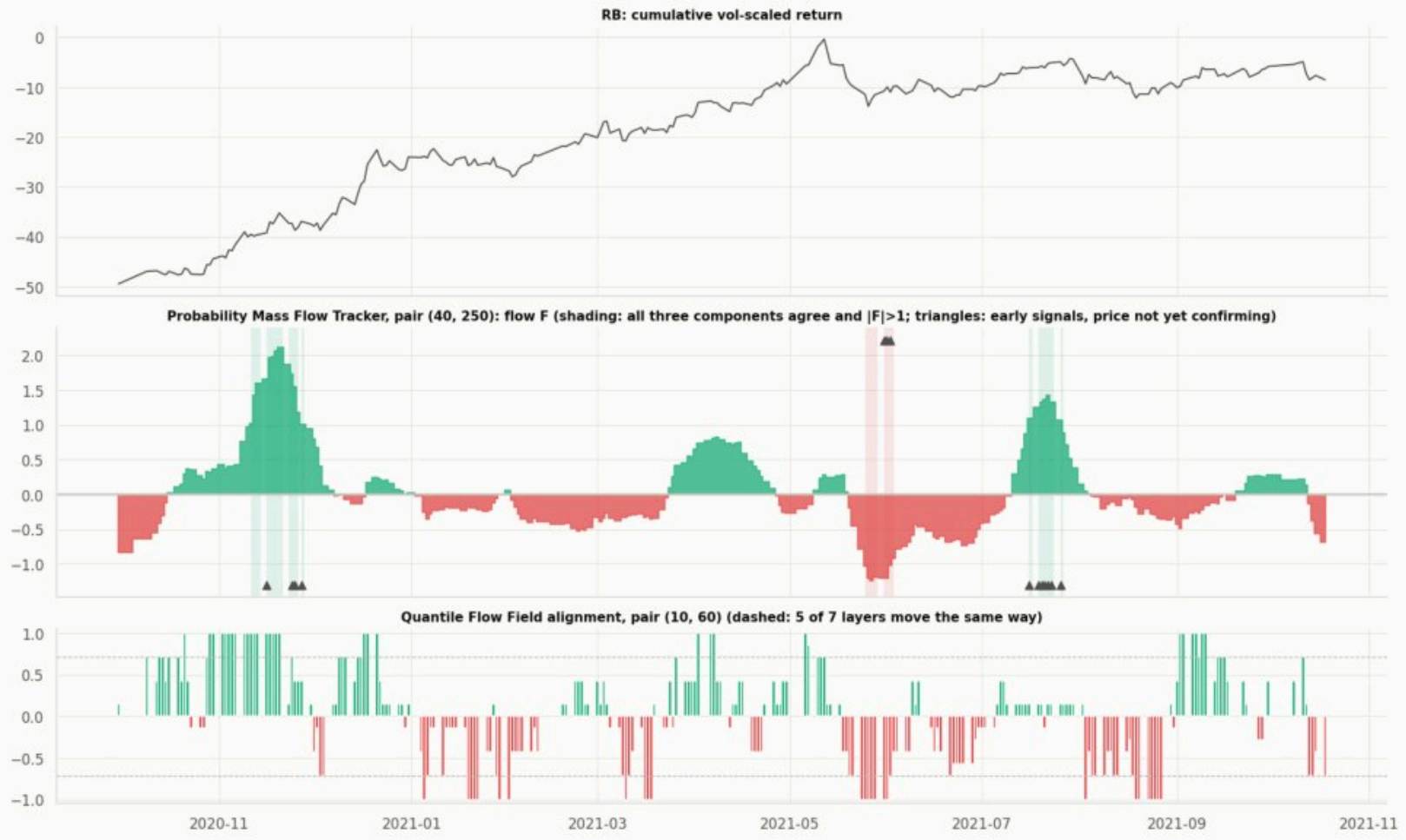
WHAT IT IS

Do the volume/OI rules hold? Effect in the direction each rule predicts.

WHAT IT EXPLAINS

Of the causal rules, one is supported: the rise coil (direction) rule, t 2.99 on test.

Probability Mass Flow Tracker and Quantile Flow Field for one product



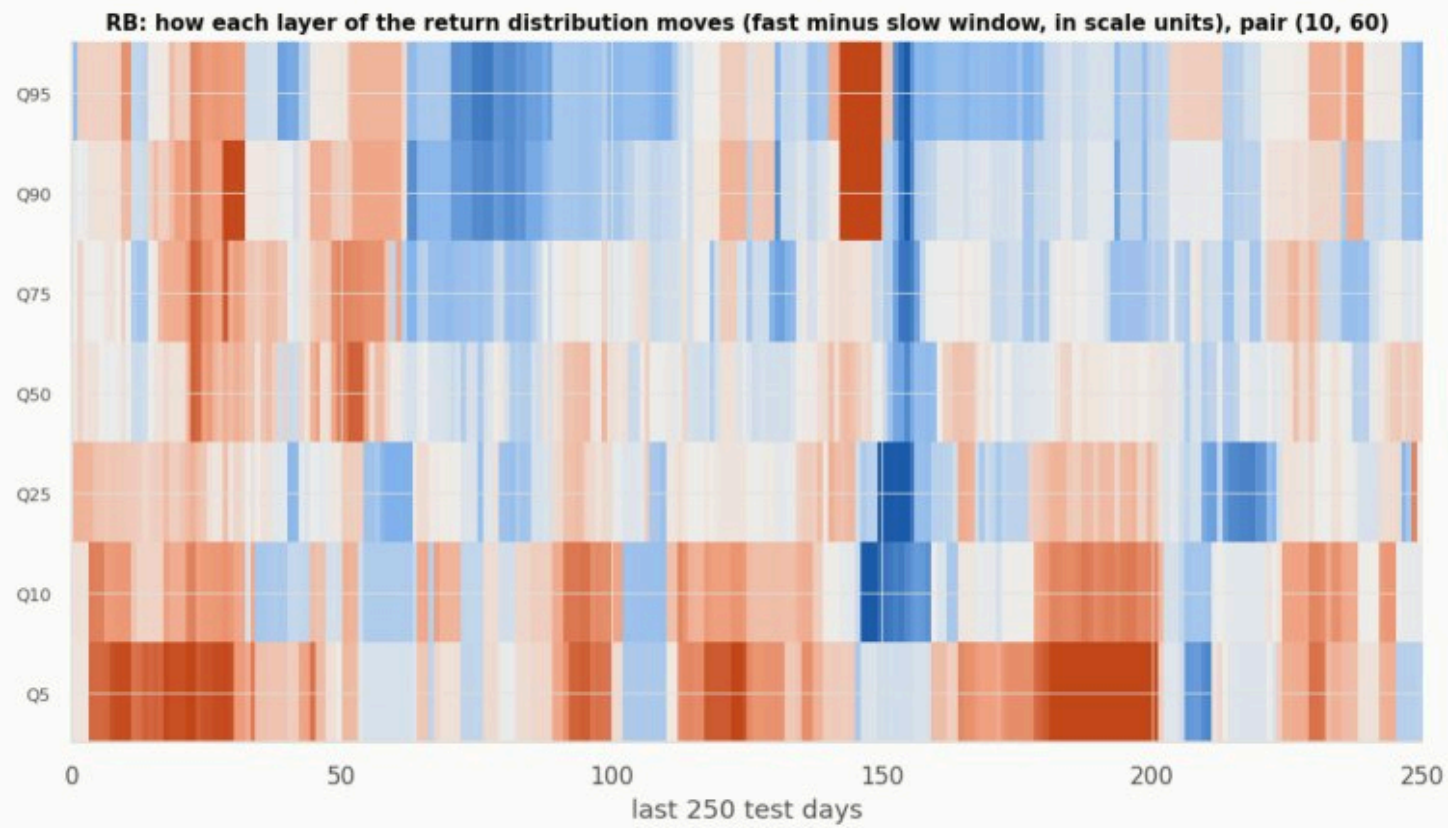
WHAT IT IS

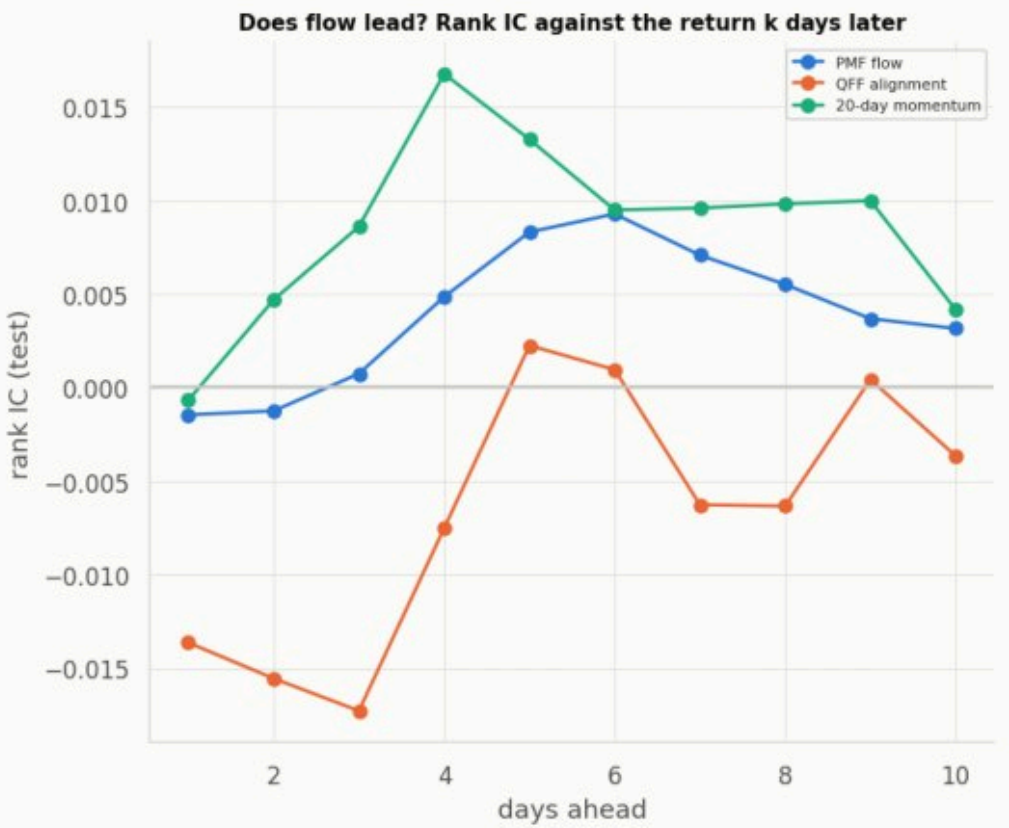
Probability Mass Flow Tracker and Quantile Flow Field for one product.

WHAT IT EXPLAINS

Two distribution-flow indicators drawn on one product's history.

Layers of the return distribution and lead-lag of the flow signals





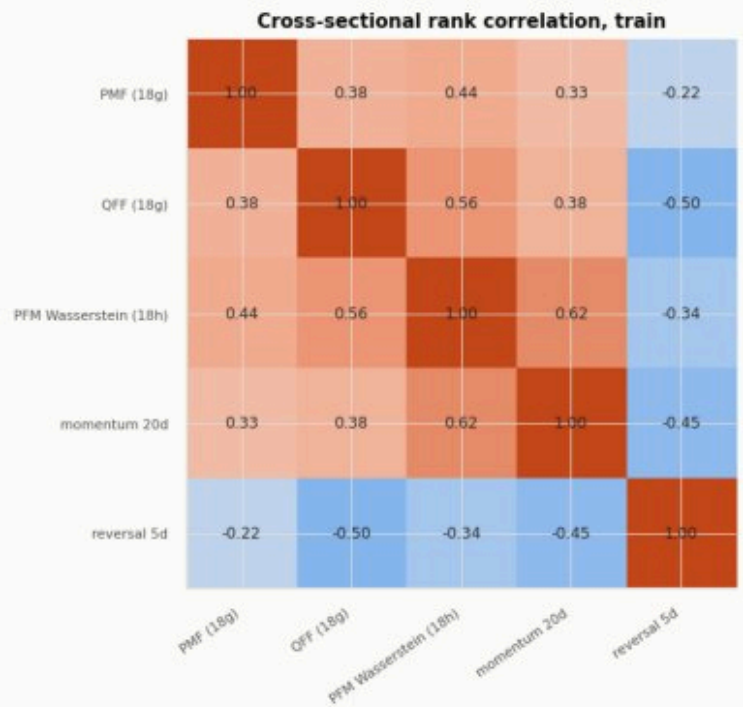
WHAT IT IS

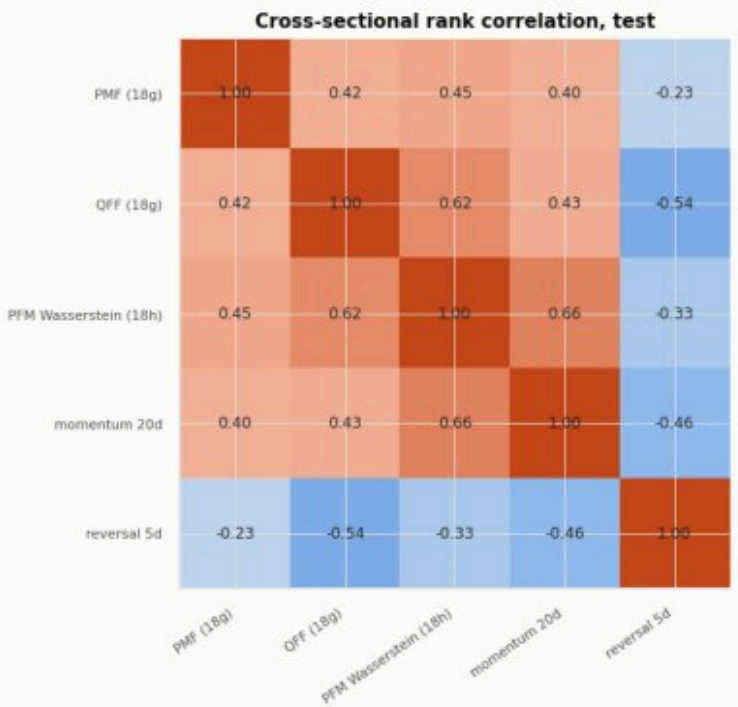
Layers of the return distribution and lead-lag of the flow signals.

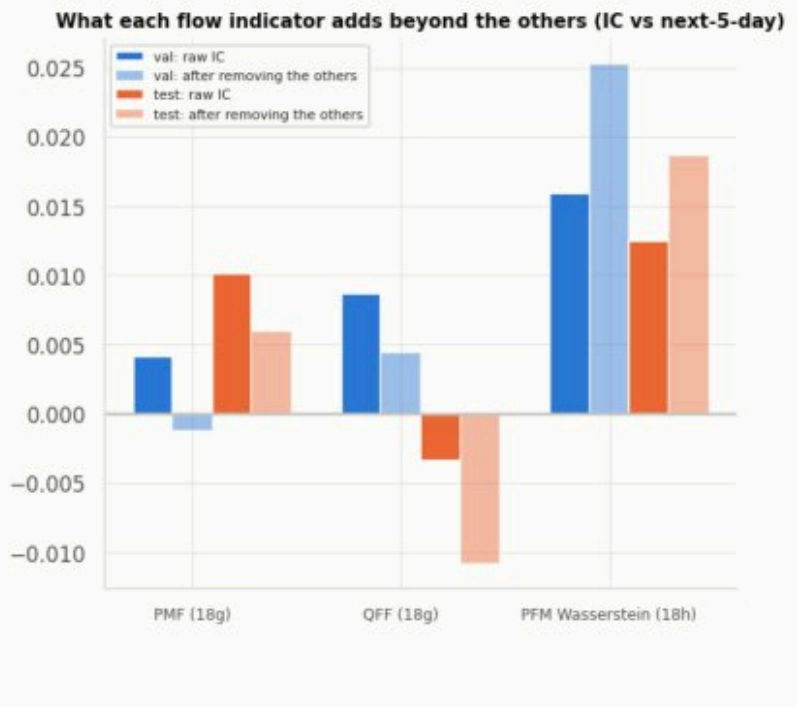
WHAT IT EXPLAINS

None of the flow signals is confirmed on test.

How much do the three flow indicators overlap?







WHAT IT IS

Overlap of the three flow indicators.

WHAT IT EXPLAINS

They are one family (effective 1.98 independent signals of 3): do not count them as three confirmations.

ECHO: balance, injection and dissipation for one product



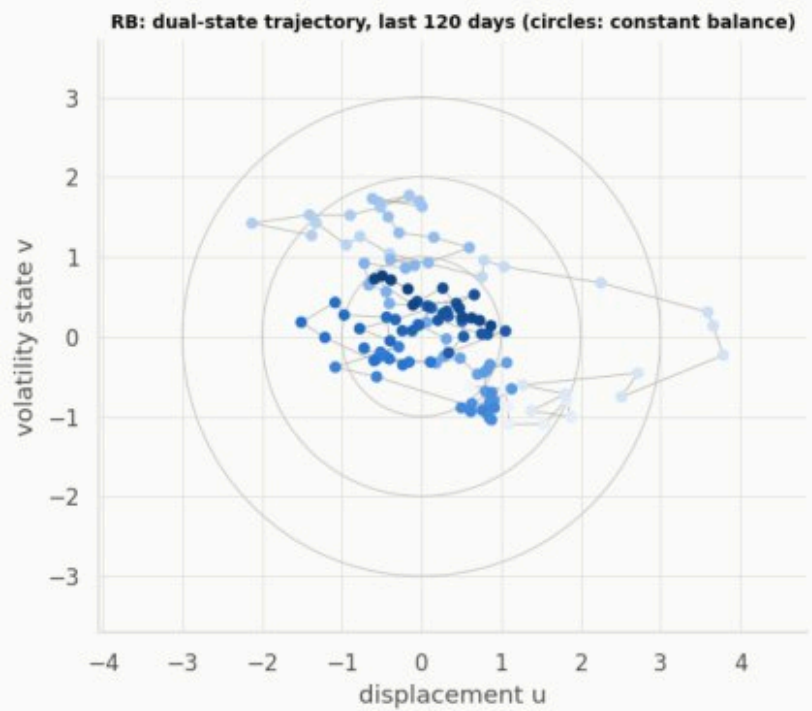
WHAT IT IS

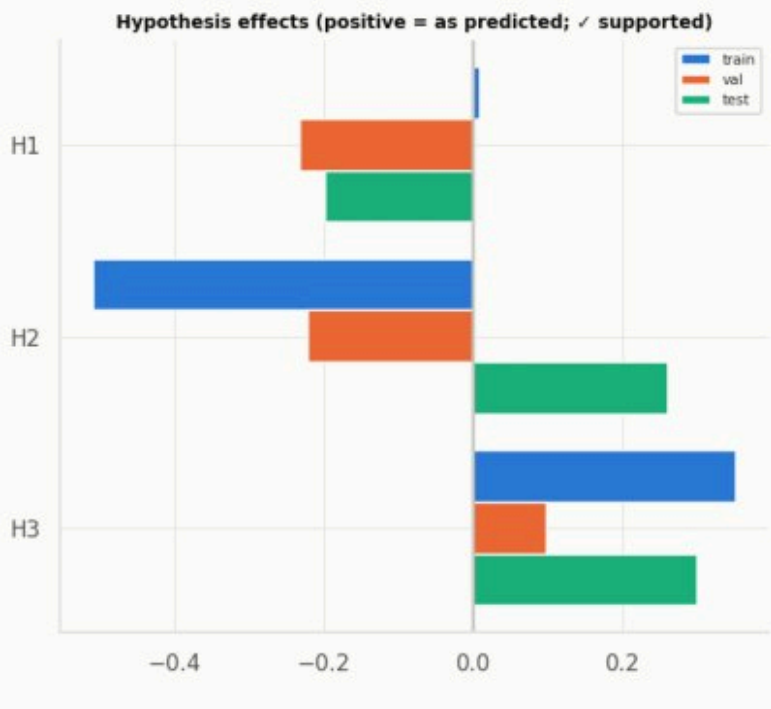
ECHO field momentum: balance, injection and dissipation for one product.

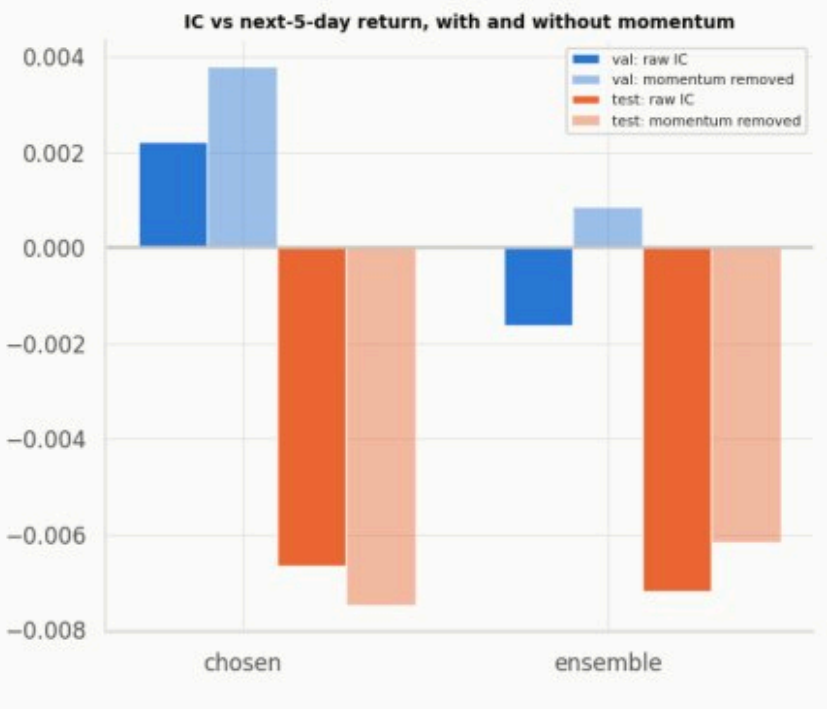
WHAT IT EXPLAINS

The ECHO indicator on one product, with its three components.

ECHO: phase portrait, hypotheses, and value beyond momentum







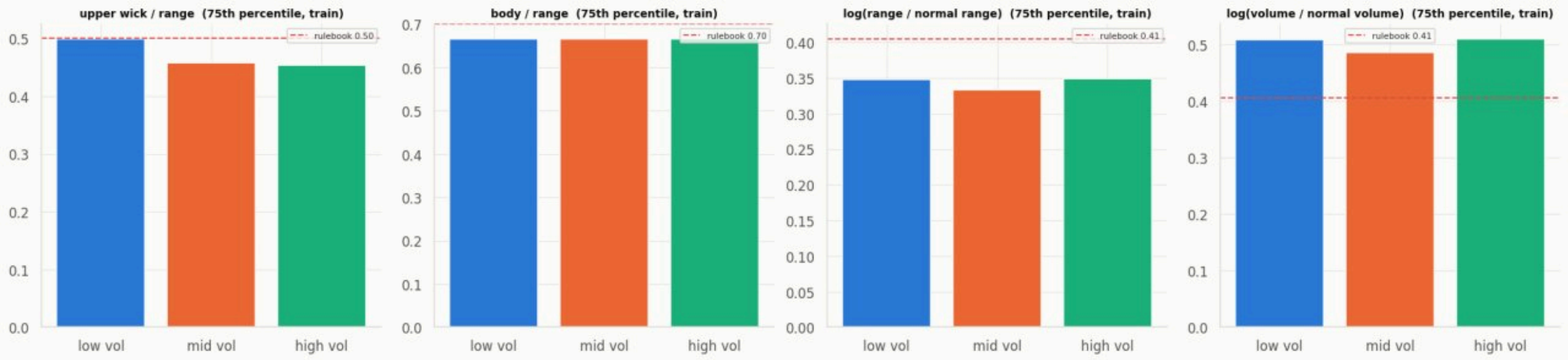
WHAT IT IS

ECHO: phase portrait, hypotheses and value beyond momentum.

WHAT IT EXPLAINS

ECHO adds nothing beyond 20-day momentum (test IC -0.007, t -0.8).

Learned candle thresholds by volatility regime against the rulebook values



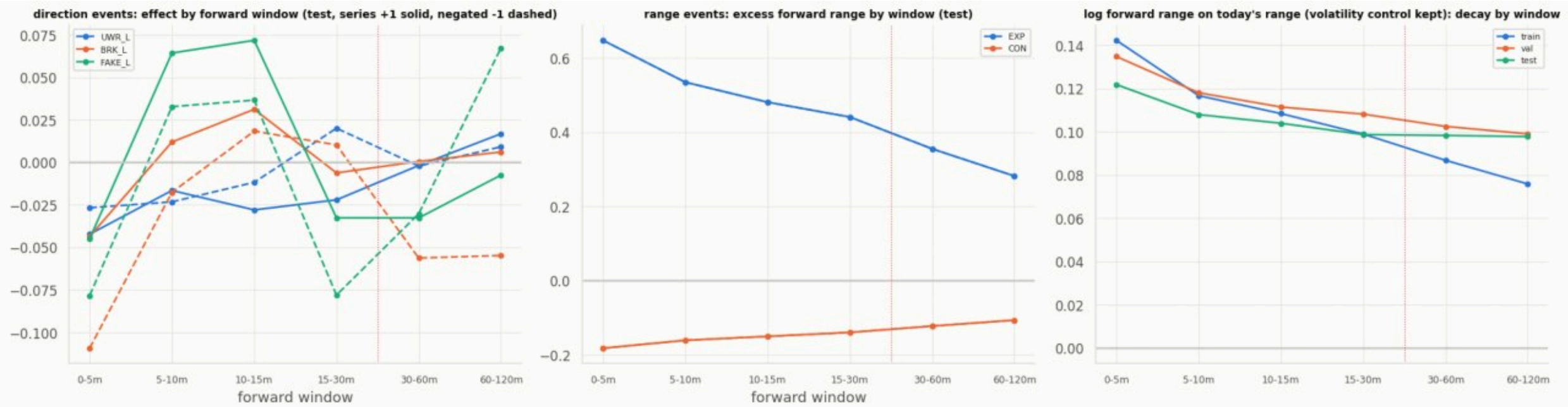
WHAT IT IS

Candle thresholds learned per volatility regime against the rulebook values.

WHAT IT EXPLAINS

Thresholds are learned from train quantiles, not assumed, and differ from the pasted rule values.

How long a candle's information lasts: forward windows up to 120 minutes (dotted line = 30 minu



WHAT IT IS

How long a candle's information lasts, with forward windows up to 120 minutes. Dotted = 30 minutes.

WHAT IT EXPLAINS

The effect is mostly gone beyond the 30-minute mark.

Bullish structures versus their mirror image on the negated series



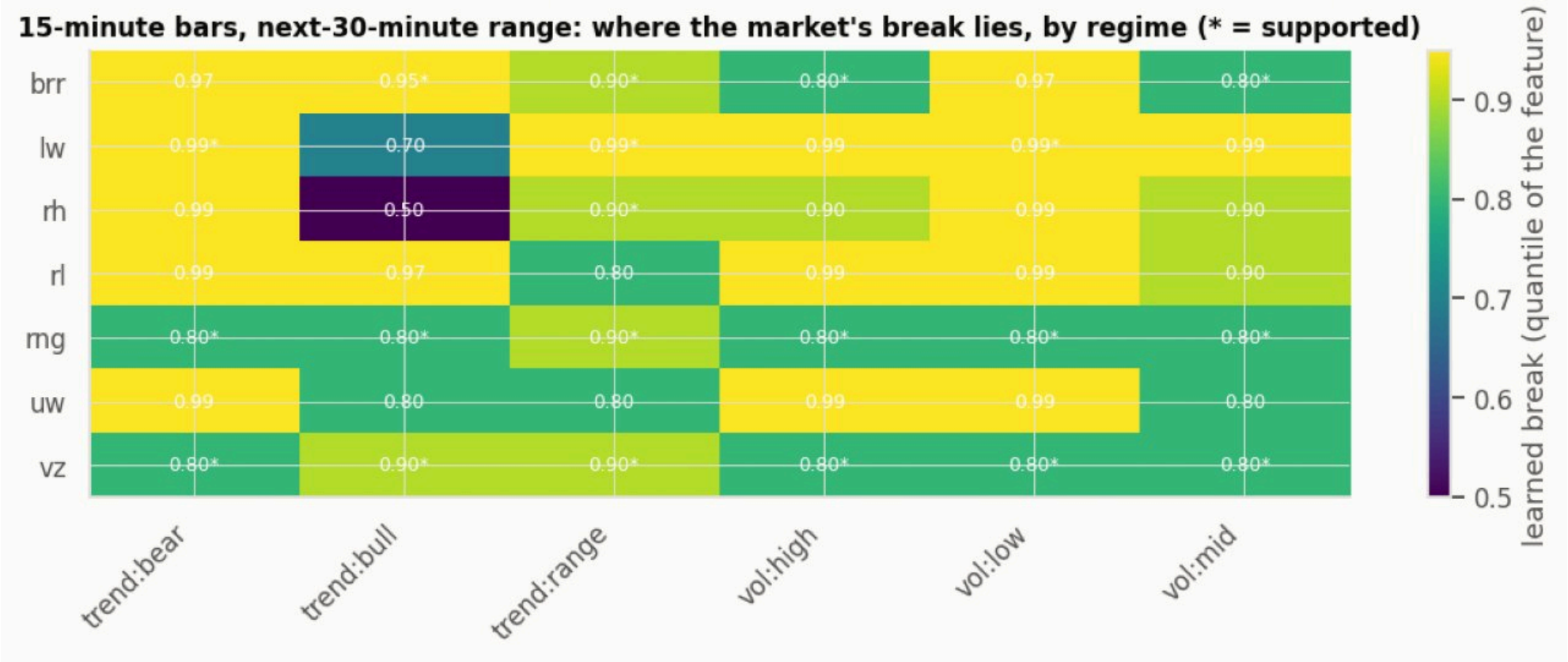
WHAT IT IS

Bullish candle structures versus their mirror image on the negated series.

WHAT IT EXPLAINS

Expansion candles show up in the series and in its negation alike, so they measure size of move, not direction.

15-minute bars, next-30-minute range: where the market's break lies, by regime (* = supported)



WHAT IT IS

15-minute bars, next-30-minute range: where the market's own break lies by regime. Star = supported.

WHAT IT EXPLAINS

Learned break points of the feature by regime; supported cells are about range.

Response curves, 15-minute bars, range: the market's own shape, solid = train_R, dashed = test



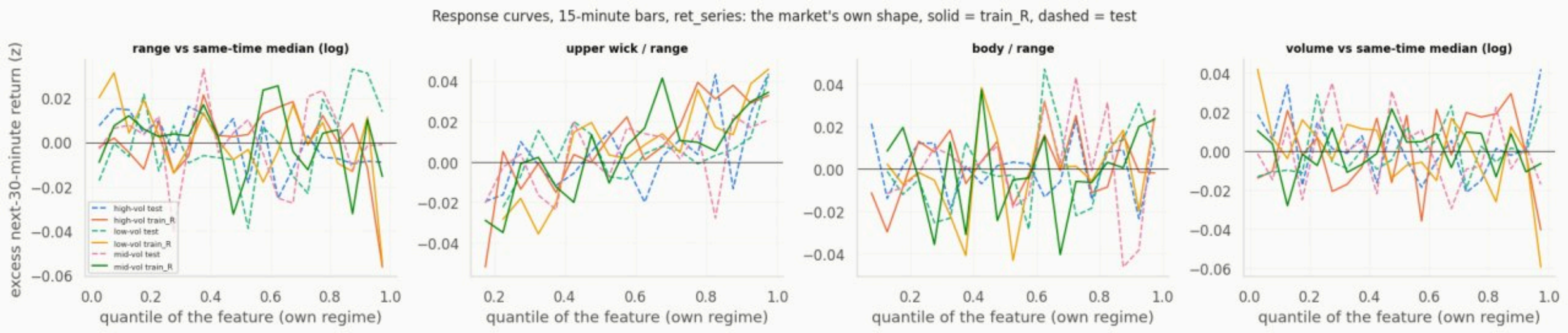
WHAT IT IS

Response curves, 15-minute bars, range. Solid = train_R, dashed = test.

WHAT IT EXPLAINS

The shape of the response is stable between train and test for range.

Response curves, 15-minute bars, ret_series: the market's own shape, solid = train_R, dashed =



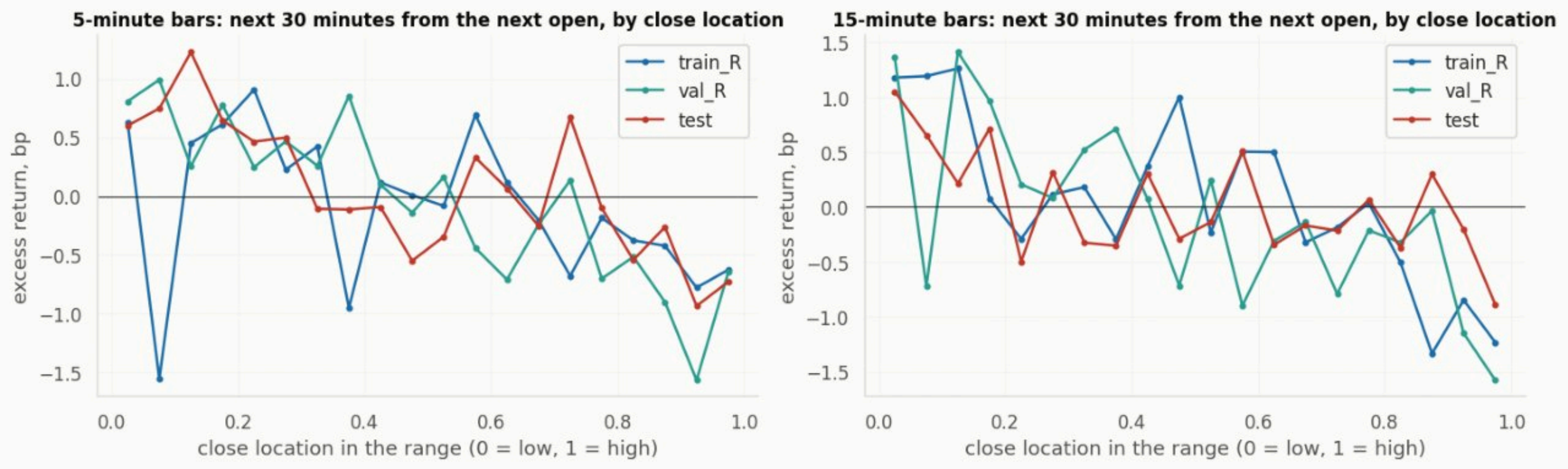
WHAT IT IS

Response curves, 15-minute bars, signed return. Solid = train_R, dashed = test.

WHAT IT EXPLAINS

For signed return the curves are flat or unstable: no direction information.

5-minute bars: next 30 minutes from the next open, by close location



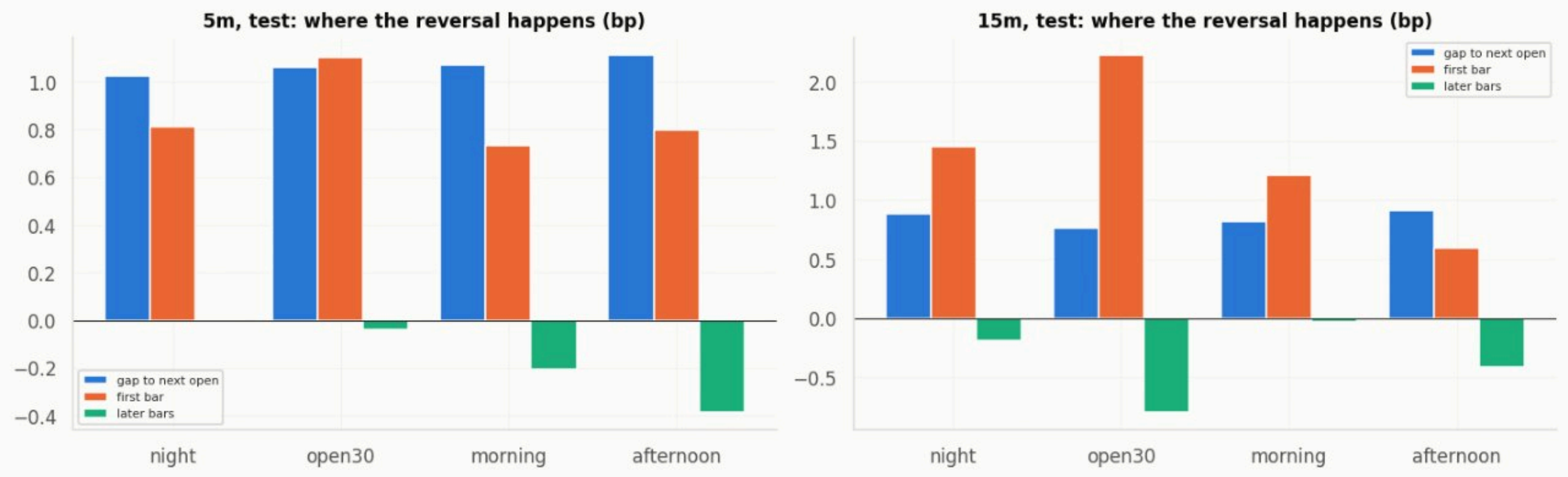
WHAT IT IS

5-minute bars: next 30 minutes from the next open, by close location in the bar.

WHAT IT EXPLAINS

Closing at the high or low of a bar tends to reverse at 5 minutes, which is largely the bid-ask bounce.

5m, test: where the reversal happens (bp)



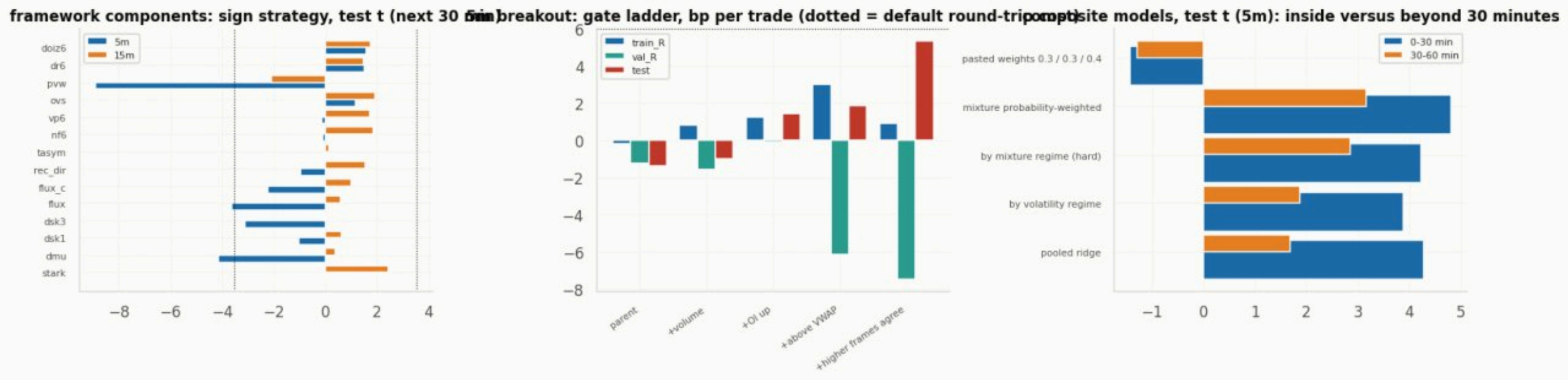
WHAT IT IS

5m, test: where the reversal happens (bp).

WHAT IT EXPLAINS

The reversal is small, about 0.66 bp break-even cost, far below the 6 bp round trip.

framework components: sign strategy, test t (next 30 min)



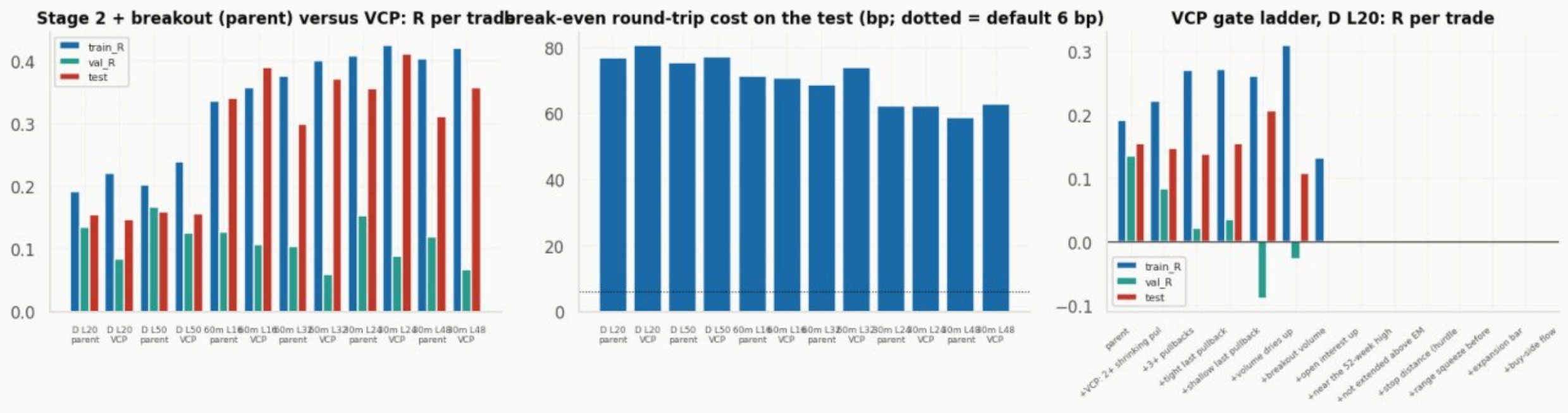
WHAT IT IS

Framework components: sign strategy, test t over the next 30 minutes.

WHAT IT EXPLAINS

Sign of each pasted component on test. Many of the pasted signs are wrong, so the pasted framework is not followed.

Stage 2 + breakout (parent) versus VCP: R per trade



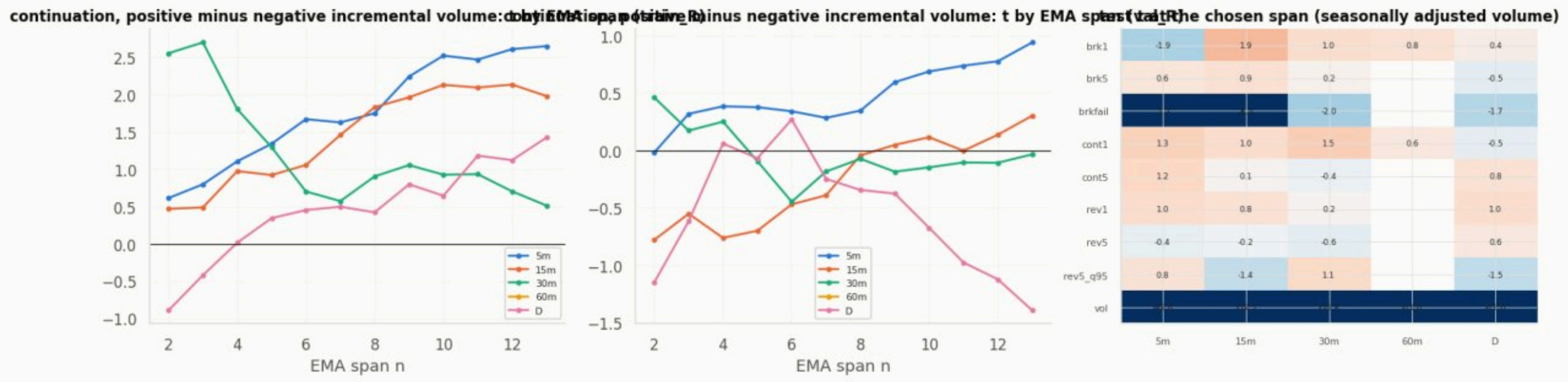
WHAT IT IS

Stage 2 + breakout (parent) versus VCP: R per trade.

WHAT IT EXPLAINS

The full volatility-contraction pattern does not beat plain Stage 2 + breakout in train, validation and test in any of 6 cells.

continuation, positive minus negative incremental volume: t by EMA span (train_R)



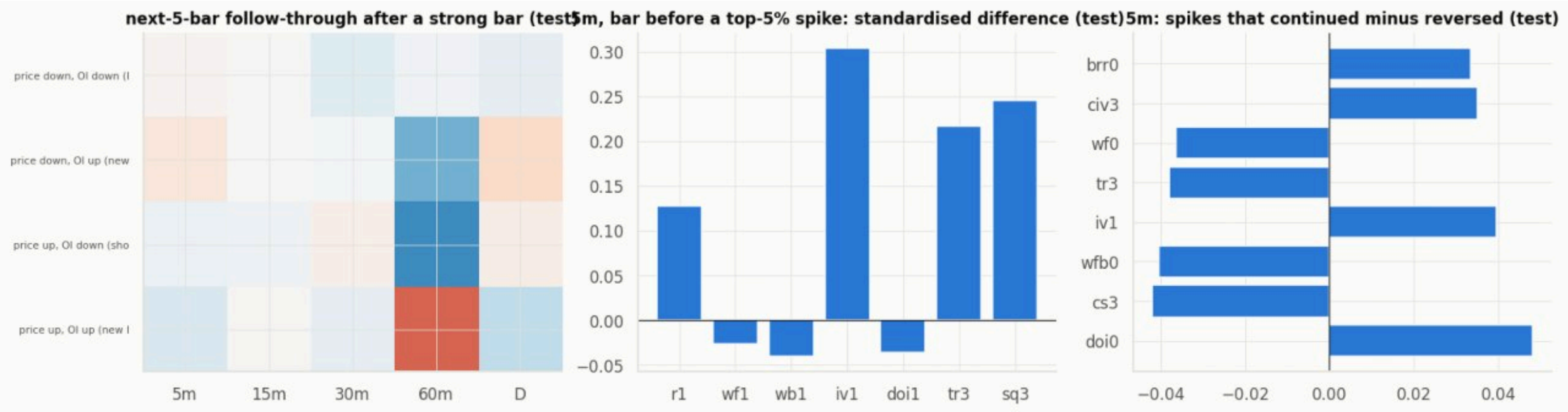
WHAT IT IS

Continuation, positive minus negative incremental volume: t by EMA span (train_R).

WHAT IT EXPLAINS

Incremental volume is tested across spans 2-13; most supported rules concern size of move, none clear cost.

next-5-bar follow-through after a strong bar (test)



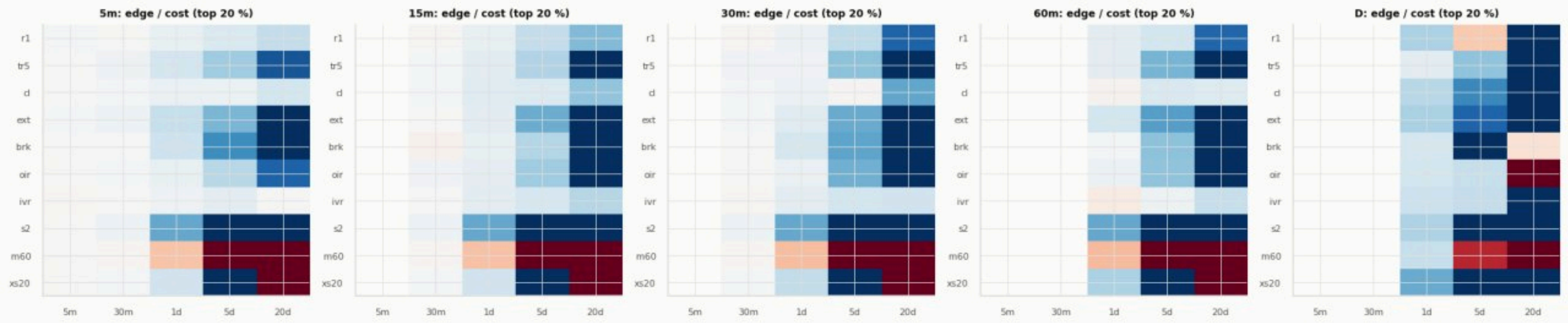
WHAT IT IS

Next-5-bar follow-through after a strong bar (test).

WHAT IT EXPLAINS

Per-instrument rules and the bars before a spike. 0 of 40 return x OI states have a stable follow-through.

5m: edge / cost (top 20 %)



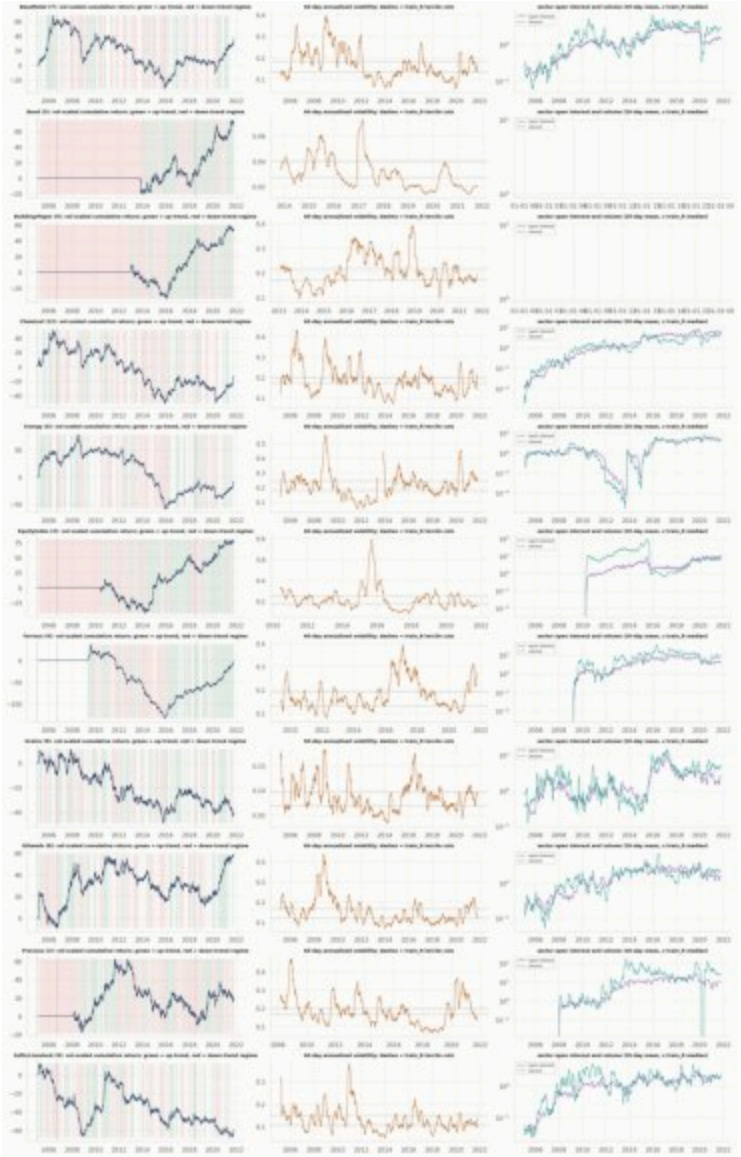
WHAT IT IS

5m: edge / cost for the top 20% of signals.

WHAT IT EXPLAINS

Gross edge of the strongest signals divided by trading cost. Direction signals under 30 minutes do not pay once cost is included.

BaseMetal (7): vol-scaled cumulative return; green = up-trend, red = down-trend regime



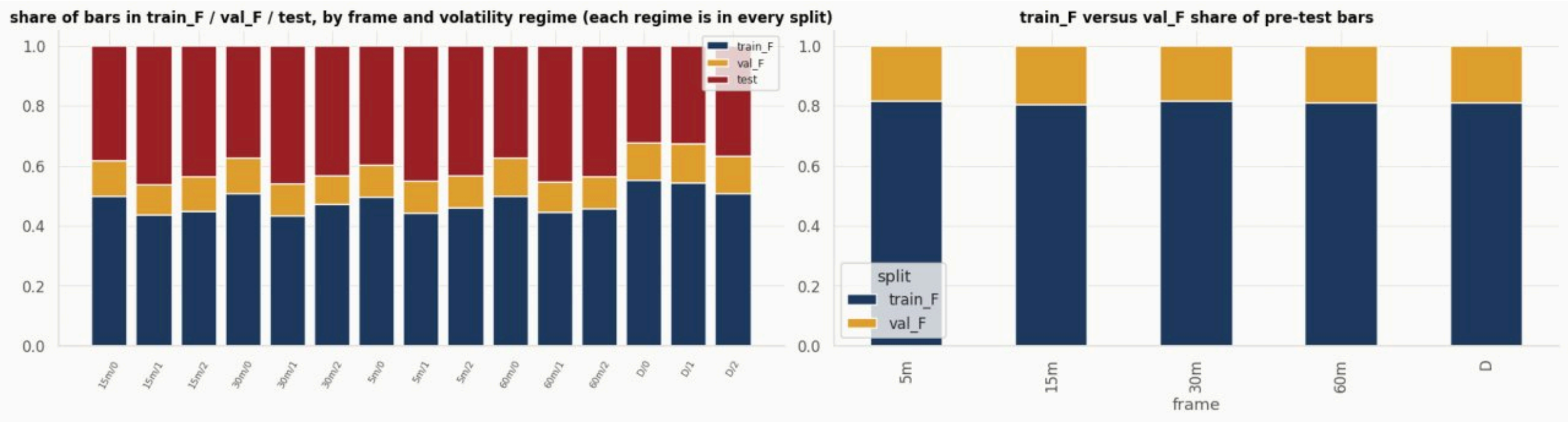
WHAT IT IS

BaseMetal (7 products): volatility-scaled cumulative return. Green = up-trend regime, red = down-trend regime.

WHAT IT EXPLAINS

Visual check of how a sector's price path splits into regimes.

share of bars in train_F / val_F / test, by frame and volatility regime (each regime is in ever



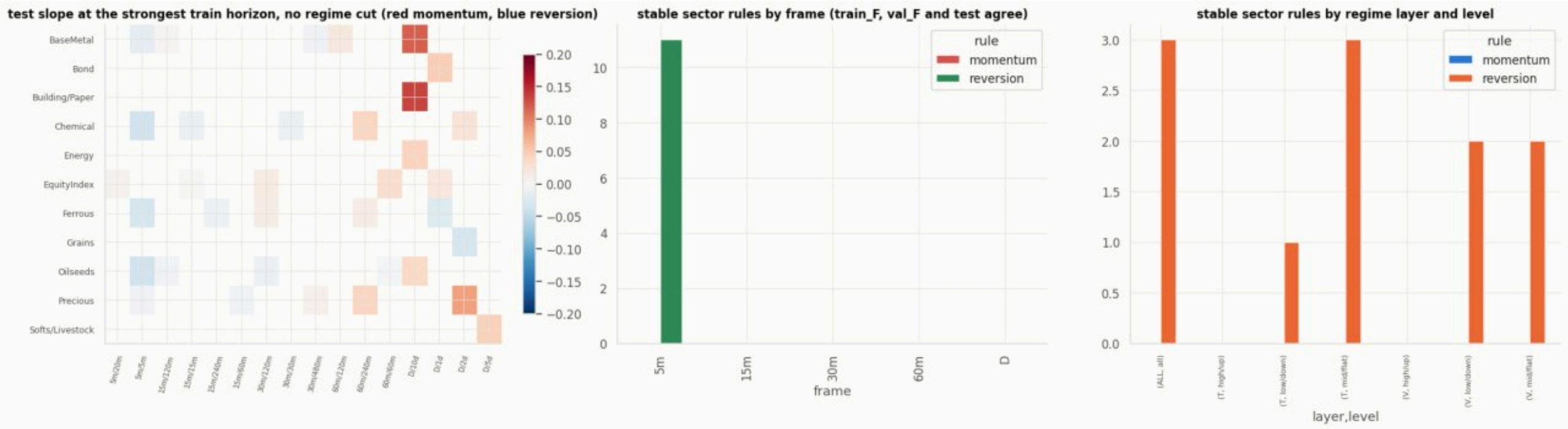
WHAT IT IS

Share of bars in train_F / val_F / test by time frame and volatility regime.

WHAT IT EXPLAINS

Each regime appears in train, validation and test, so a rule is tested in every regime.

test slope at the strongest train horizon, no regime cut (red momentum, blue reversion)



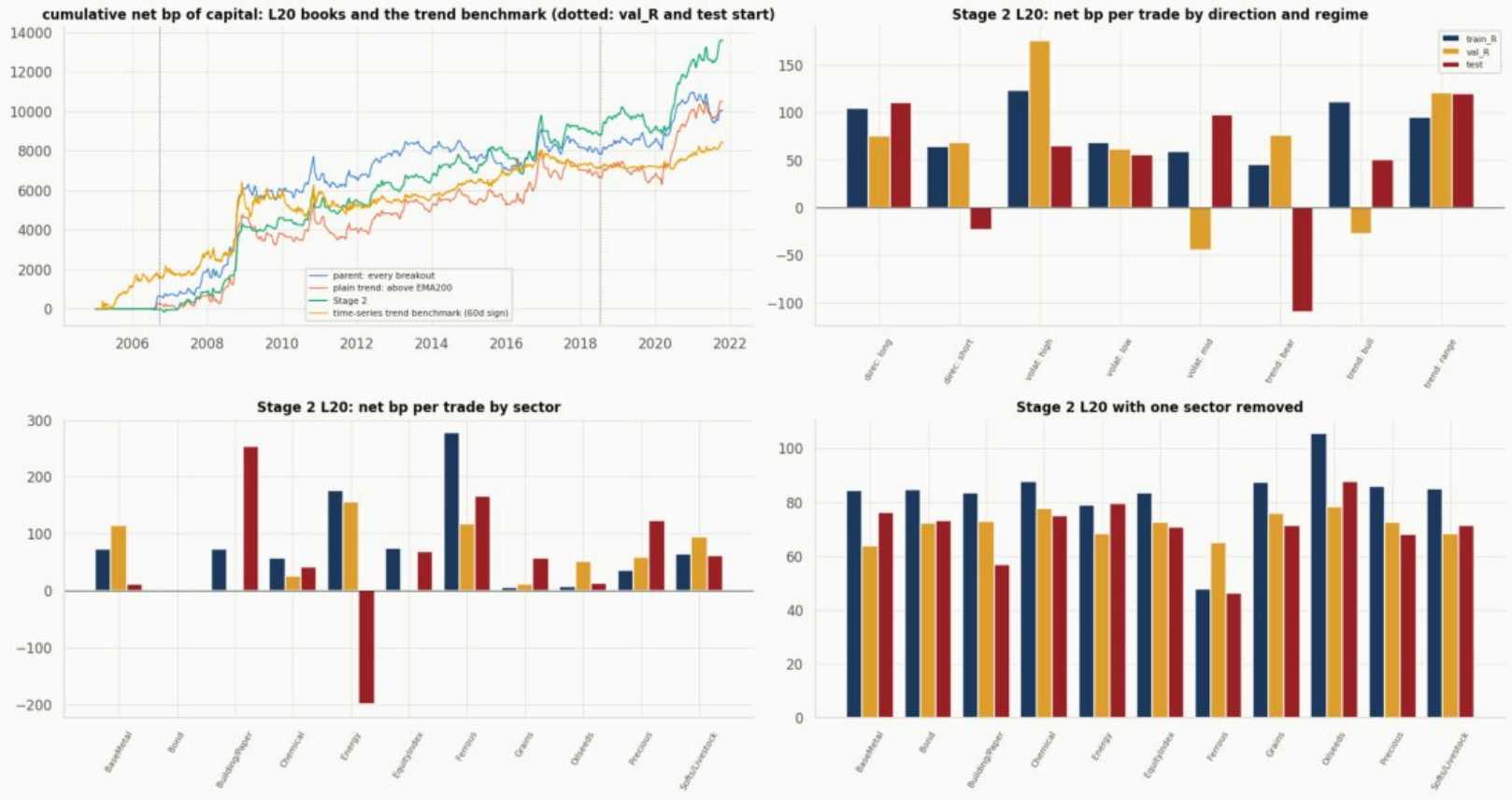
WHAT IT IS

Test slope at the strongest train horizon, no regime cut. Red = momentum, blue = reversion.

WHAT IT EXPLAINS

Where the next move continues or reverses the last. Of 11 stable sector rules, all 11 are reversion, none momentum.

cumulative net bp of capital: L20 books and the trend benchmark (dotted: val_R and test start)



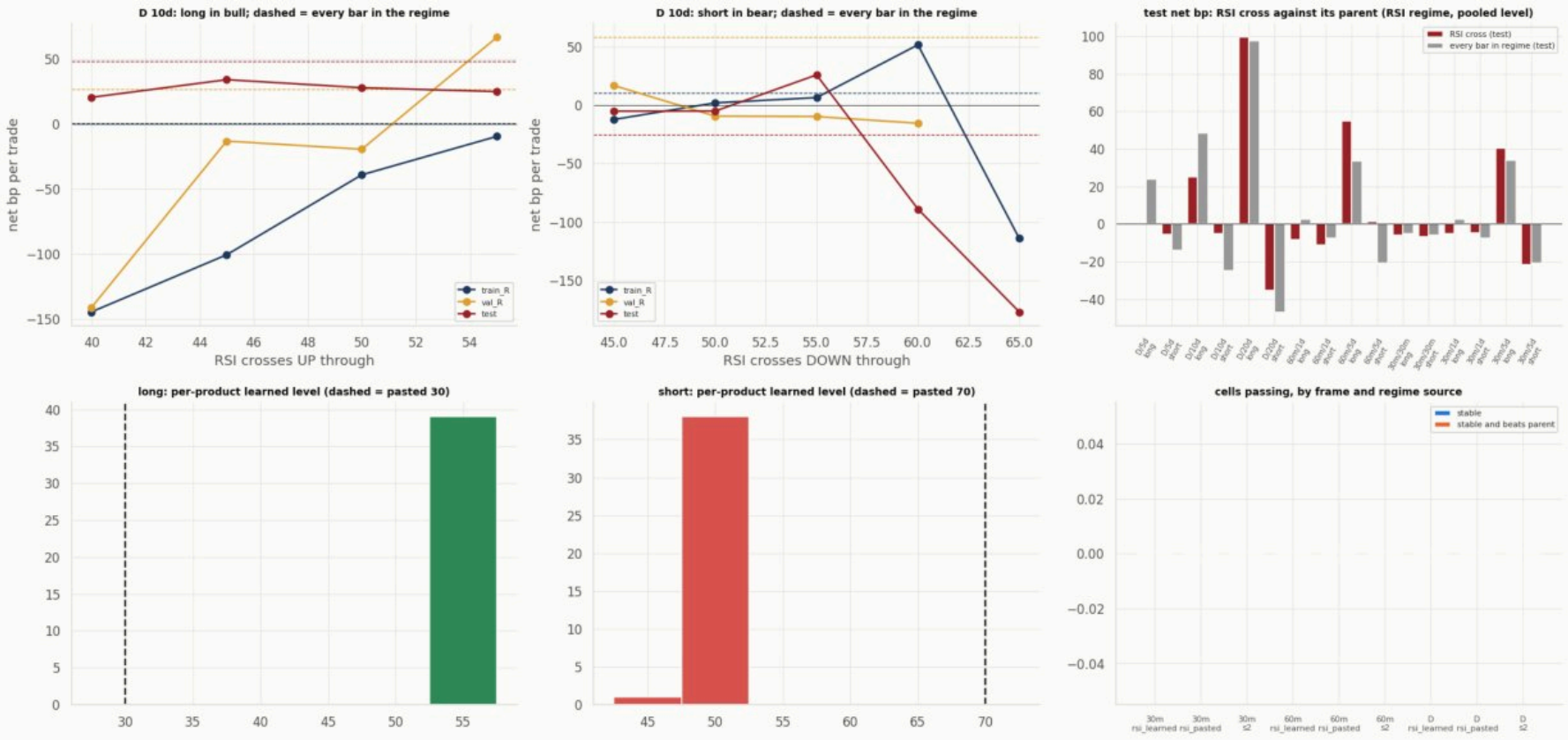
WHAT IT IS

Cumulative net bp of capital for the L20 Stage 2 books and the trend benchmark. Dotted = start of val_R and test.

WHAT IT EXPLAINS

The daily Stage 2 breakout as a portfolio: net of 6 bp cost, test Sharpe 4.10 against 1.16 for the 60-day trend benchmark.

D 10d: long in bull; dashed = every bar in the regime



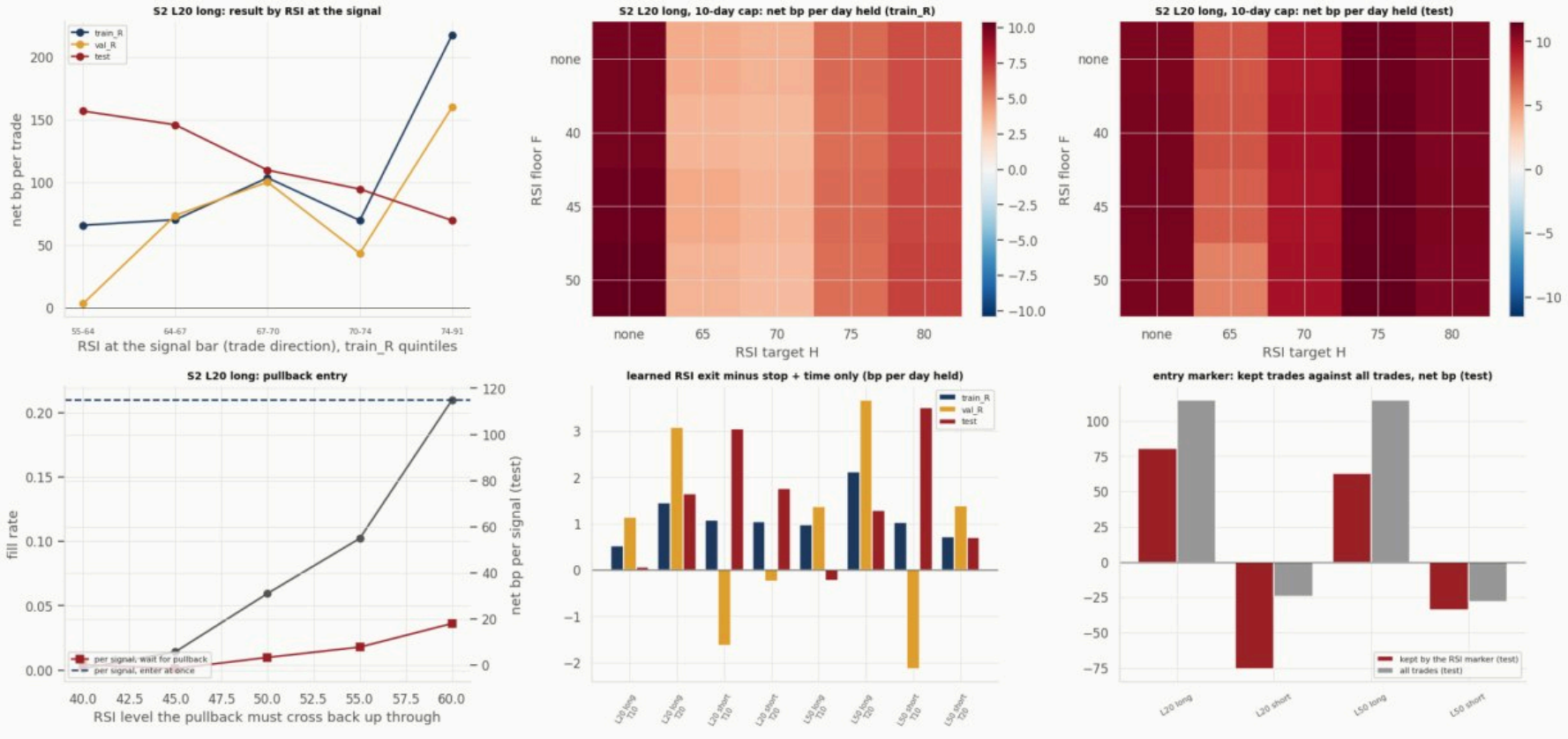
WHAT IT IS

D 10d: long in bull; dashed = every bar in the regime.

WHAT IT EXPLAINS

RSI-marked entries compared with simply entering on every bar in the same regime. RSI adds nothing.

S2 L20 long: result by RSI at the signal



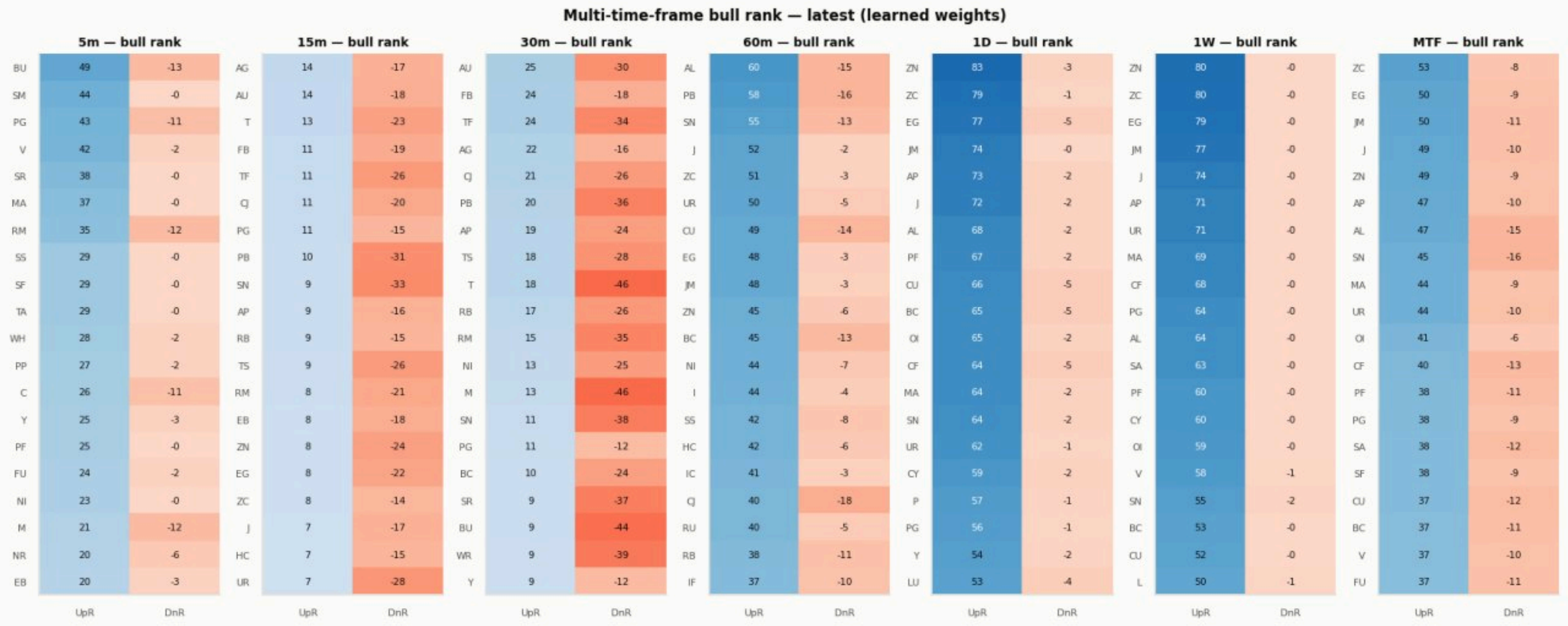
WHAT IT IS

S2 L20 long: result by RSI at the signal.

WHAT IT EXPLAINS

Whether the RSI at the breakout separates good from bad Stage 2 trades. It does not in all three periods.

Multi-time-frame bull rank | latest



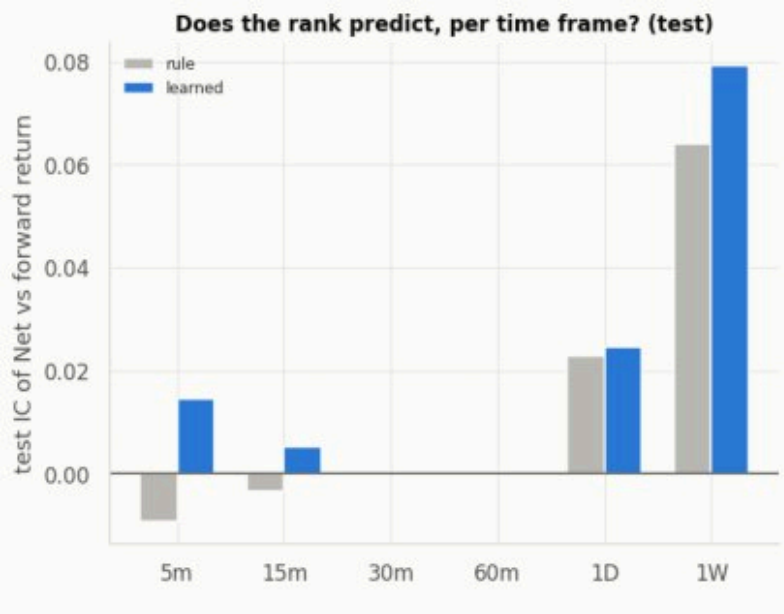
WHAT IT IS

Multi-time-frame bull rank, latest date.

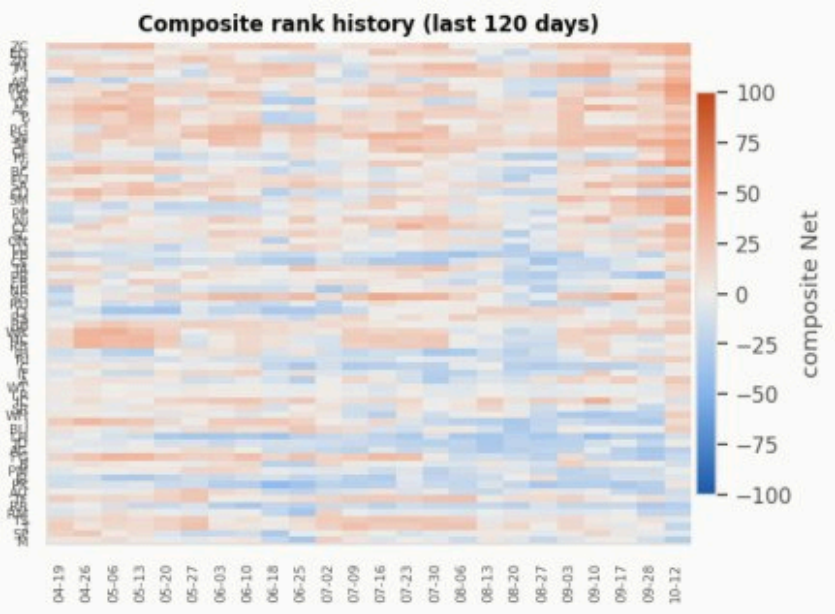
WHAT IT EXPLAINS

Ranking of products for bull candidates on the last date, from ex-ante scores only.

Rank validation and history







WHAT IT IS

Rank validation and history.

WHAT IT EXPLAINS

Whether the ranks predicted later returns: they are ranks to watch, not proof.

PART D · STEP 18T AND 18V VERDICT

The one live candidate: daily Stage 2 breakout

Cohort (net of 6 bp)	train_R	val_R	test
L20, per trade (18t)	83 bp	72 bp	71 bp (t 2.3)
L50, per trade (18t)	89 bp	89 bp	69 bp (t 2.0)
S2 L20 long (18v)	105 bp	77 bp	115 bp
S2 L20 short (18v)	65 bp	71 bp	-24 bp

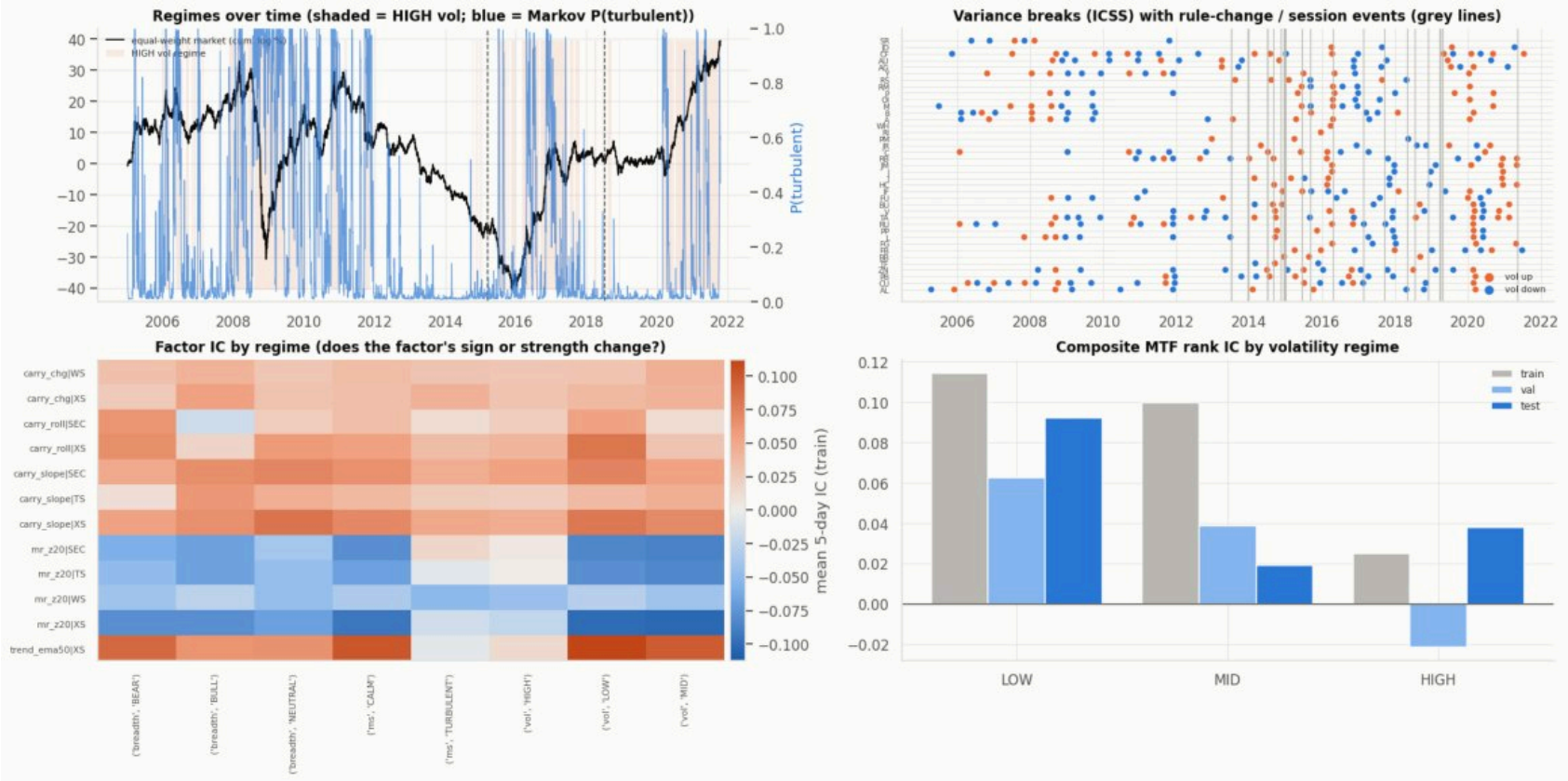
- Portfolio test Sharpe 4.10 against 1.16 for a 60-day trend benchmark; 7 of 8 sectors hold; dropping any one sector keeps it positive.
- **Not robust:** it does not beat the plain above-EMA200 trend filter in all three periods, test t is only 2.3 (bar 2.24), and shorts fail on test. Long only, range regime is the version to build on.
- **RSI (18v) adds nothing:** 0 of 8 exit rules, 0 of 4 entry markers, 0 of 2 pullback entries beat the plain trade.

Part E

Regimes

The regime today and regime-dependent factors.

Regimes, breaks, regime-dependent factors



WHAT IT IS

Regimes, structural breaks and regime-dependent factors.

WHAT IT EXPLAINS

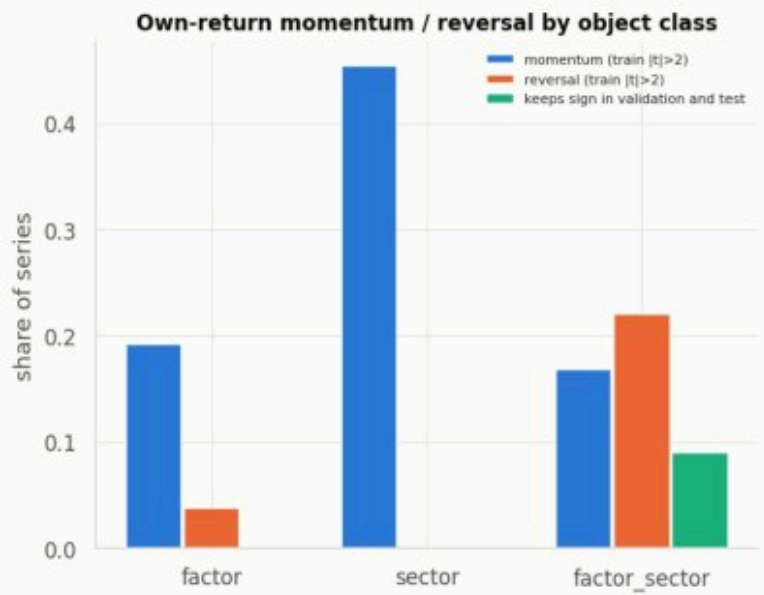
The market regime today is HIGH volatility / BULL breadth; some factors change sign between regimes.

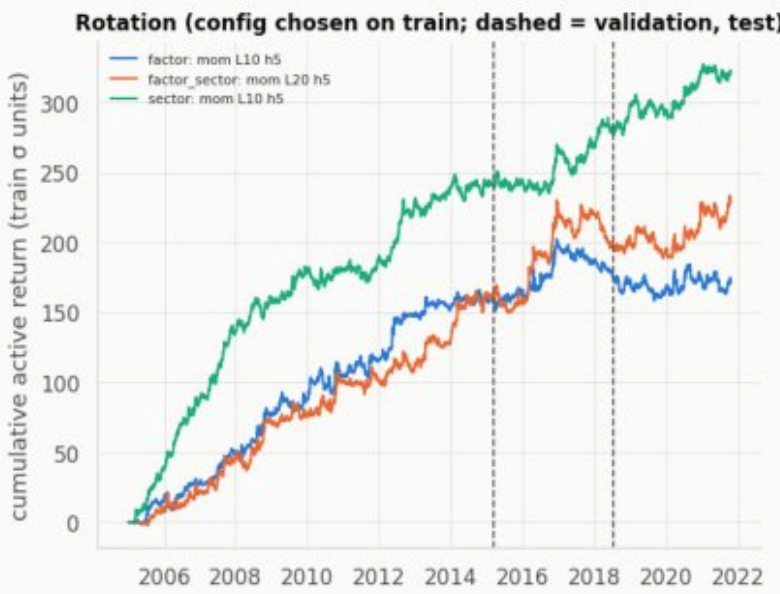
Part F

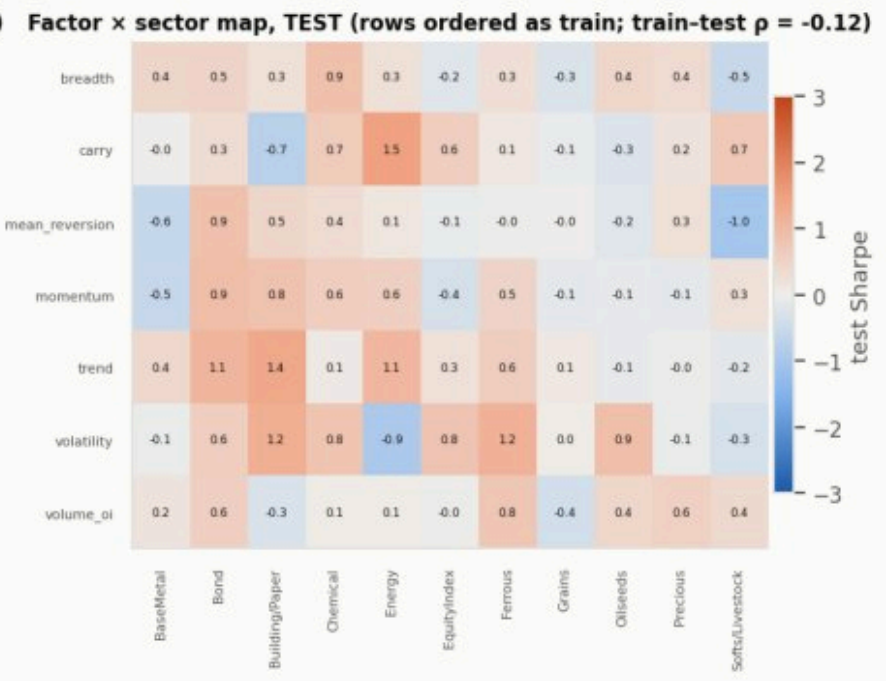
Dynamics and monitoring

Dynamics, drift and decay monitoring of the edges.

Factor, sector and factor × sector dynamics







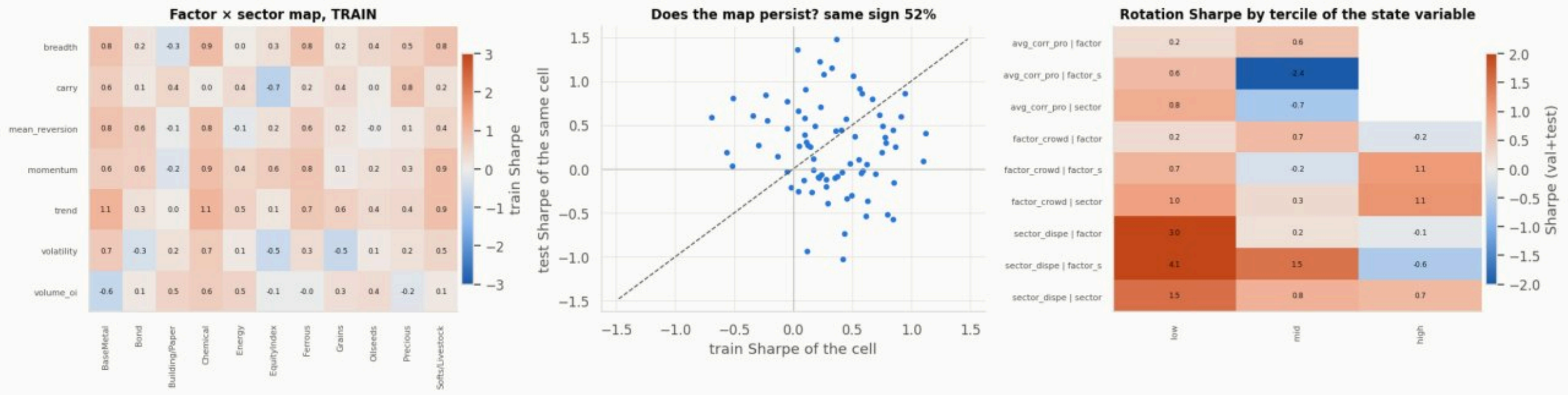
WHAT IT IS

Factor, sector and factor x sector dynamics.

WHAT IT EXPLAINS

How factor effectiveness moves through time and by sector.

Factor × sector map and state variables



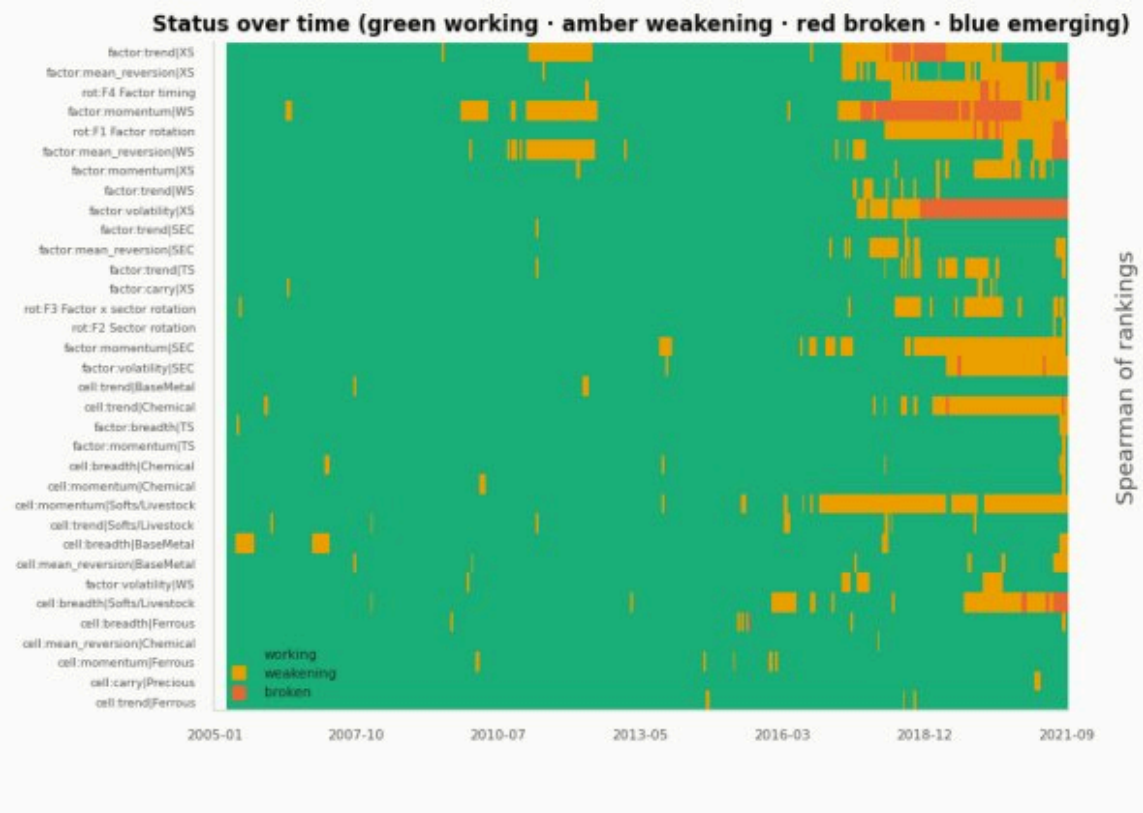
WHAT IT IS

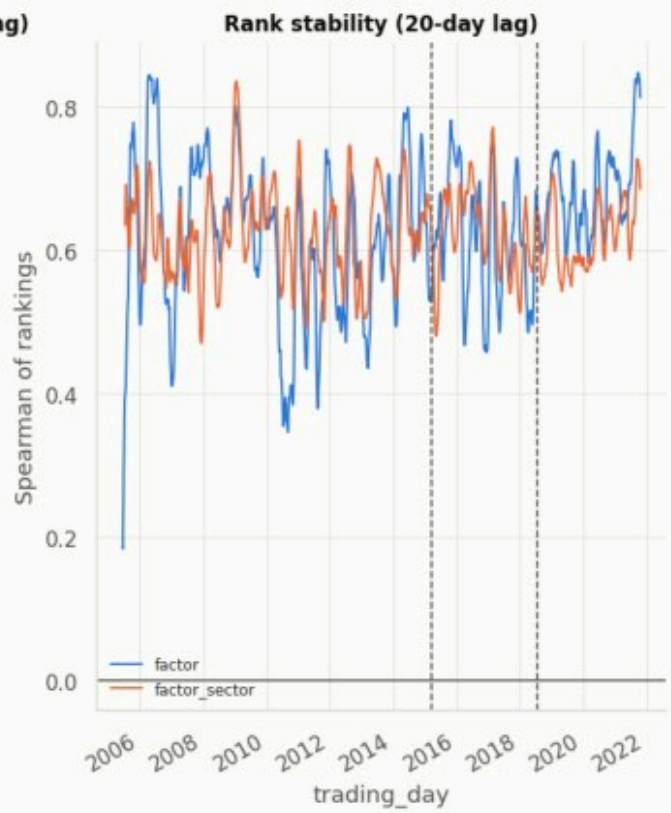
Factor x sector map and state variables.

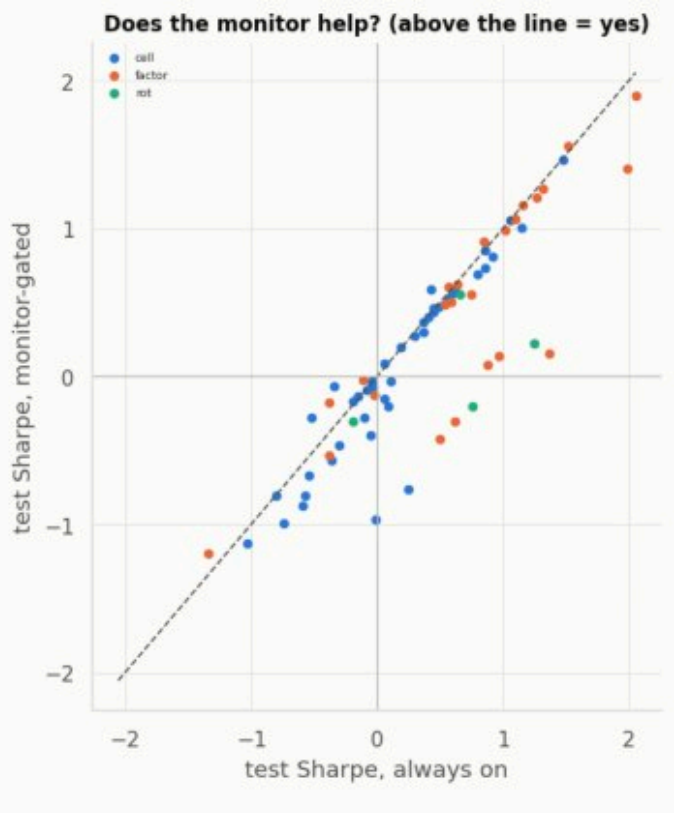
WHAT IT EXPLAINS

Which factor works in which sector now.

Drift and decay monitor







WHAT IT IS

Drift and decay monitor.

WHAT IT EXPLAINS

Edges that were found are tracked for decay and alarm.

Cumulative edge and CUSUM for edges that raised an alarm



WHAT IT IS

Cumulative edge and CUSUM for edges that raised an alarm.

WHAT IT EXPLAINS

Shows when an edge stopped working.

PART G · FINDINGS

Part G in one page: strategies and portfolios

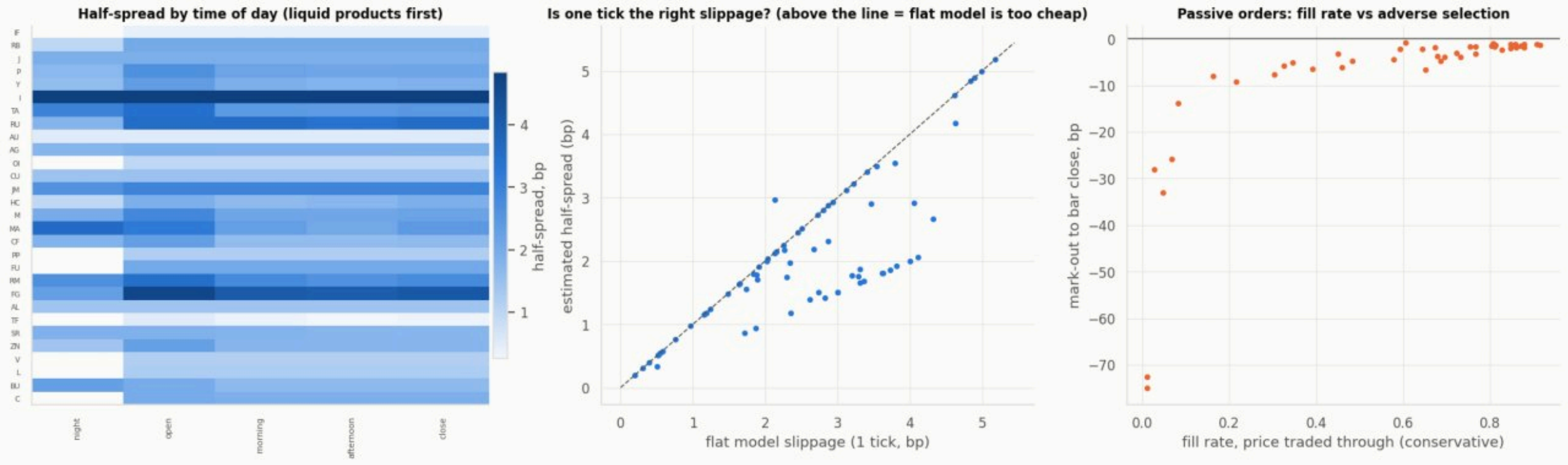
- **Strategies:** none clears all thresholds. Selection risk: PBO 0.16, deflated-Sharpe hurdle 2.03 for 45 trials.
- **Execution:** median half-spread 1.84 bp; B breadth TS, B trend TS and D regime-switching drop once realistic costs apply.
- **Portfolios:** test Sharpe 0.29-0.57 across schemes; the tangency portfolio is 2.59 on validation and 0.11 on test, a picture of selection risk. Capacity median CNY 12.55 bn at 5% participation.
- **Investor profiles:** 6 of 12 admitted, none profitable on test.
- **Best recipe chosen without the test:** band 4.0, HRP, test Sharpe +0.46, drawdown -21.5%.
- **Risk:** BestVol-winner VaR has exceedance 1.0% against 1.0% nominal.

Part G

Strategies, portfolio and risk

Execution, strategy library, portfolio construction, investor profiles and VaR / ES backtests.

Execution model: spread, passive fills, impact



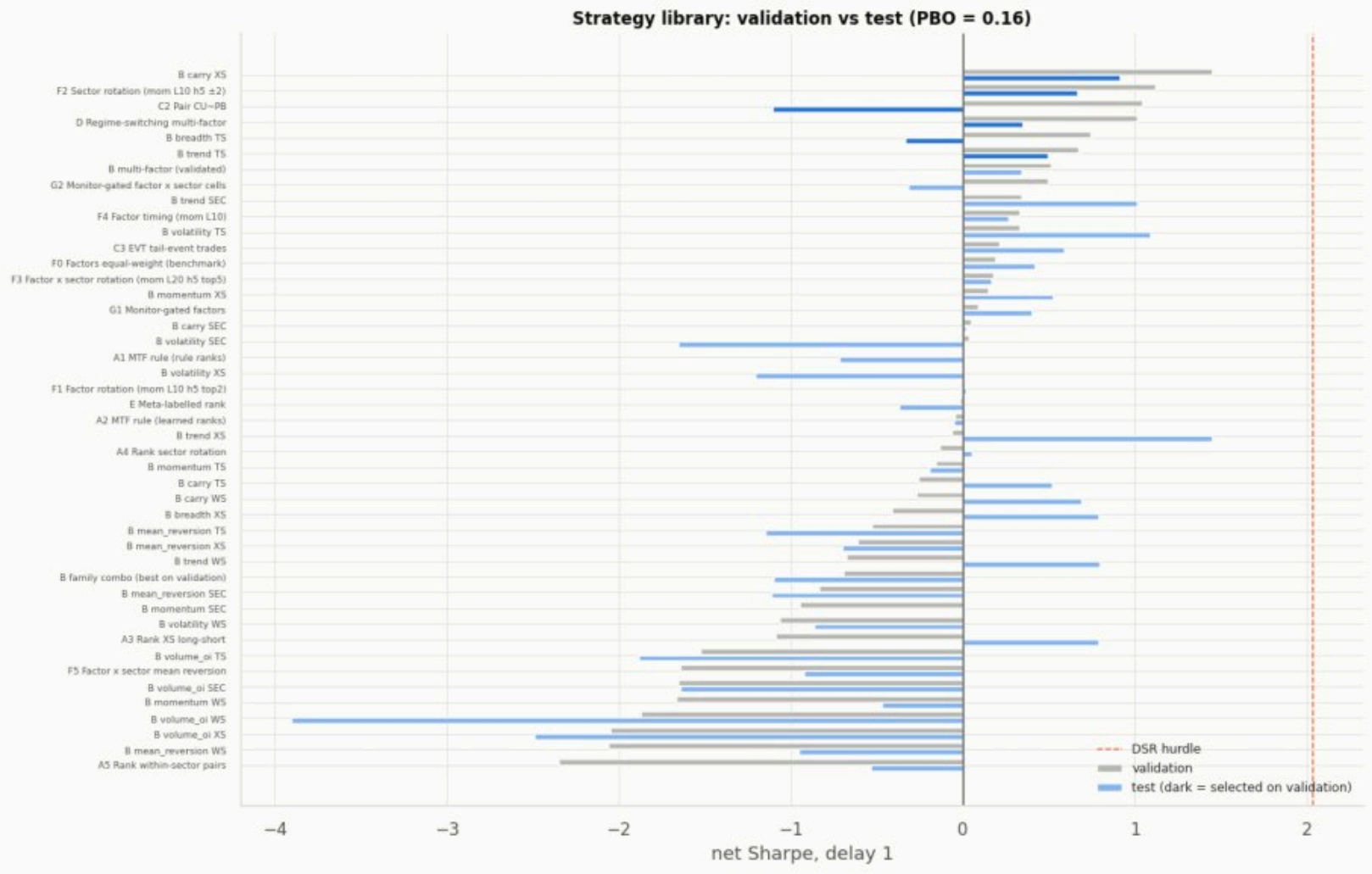
WHAT IT IS

Execution model: spread, passive fills and impact.

WHAT IT EXPLAINS

Median half-spread is 1.84 bp against flat slippage of 2.56 bp. Costs are applied to every strategy.

Strategy library Sharpe ratios



WHAT IT IS

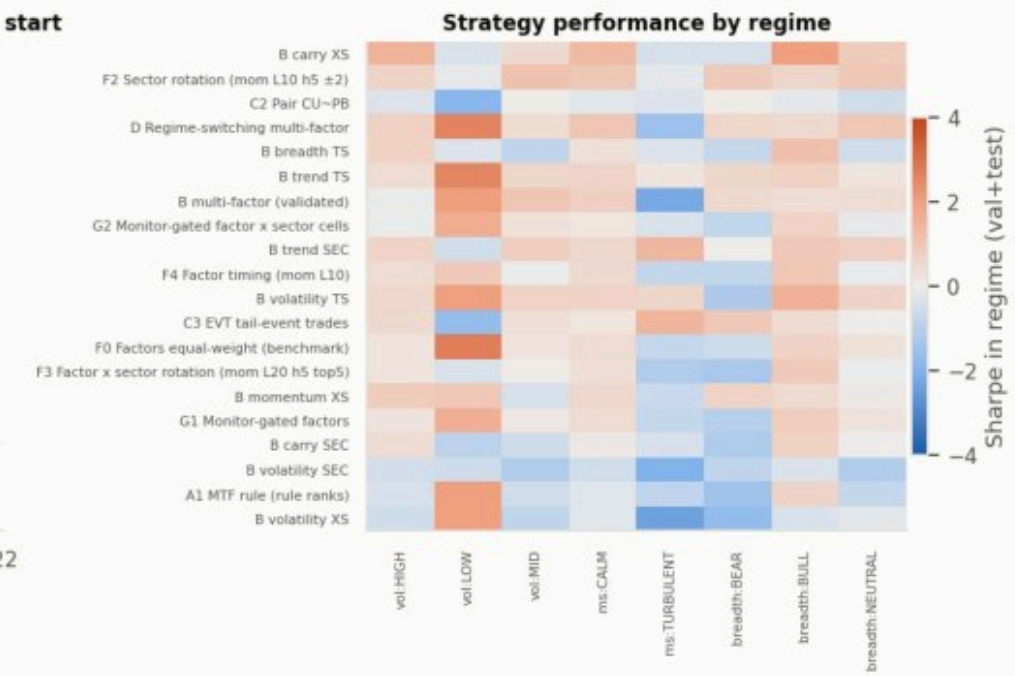
Strategy library Sharpe ratios.

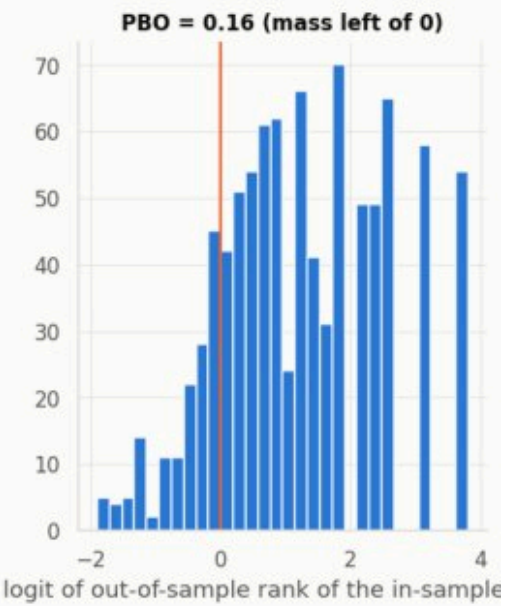
WHAT IT EXPLAINS

None clears every threshold. B volatility TS, B trend SEC, B carry XS and C3 EVT trades are on watch.

Books, regime dependence, PBO







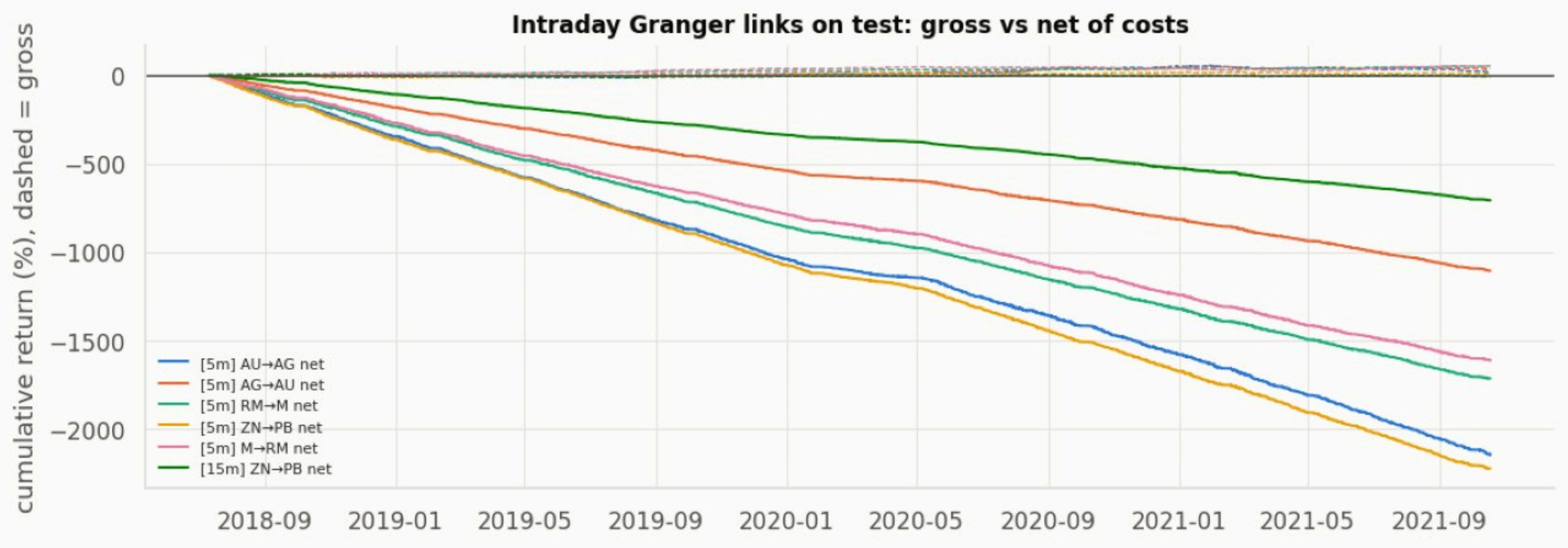
WHAT IT IS

Books, regime dependence and probability of backtest overfitting (PBO).

WHAT IT EXPLAINS

PBO is 0.16. Strategies depend on the regime.

Intraday Granger strategies



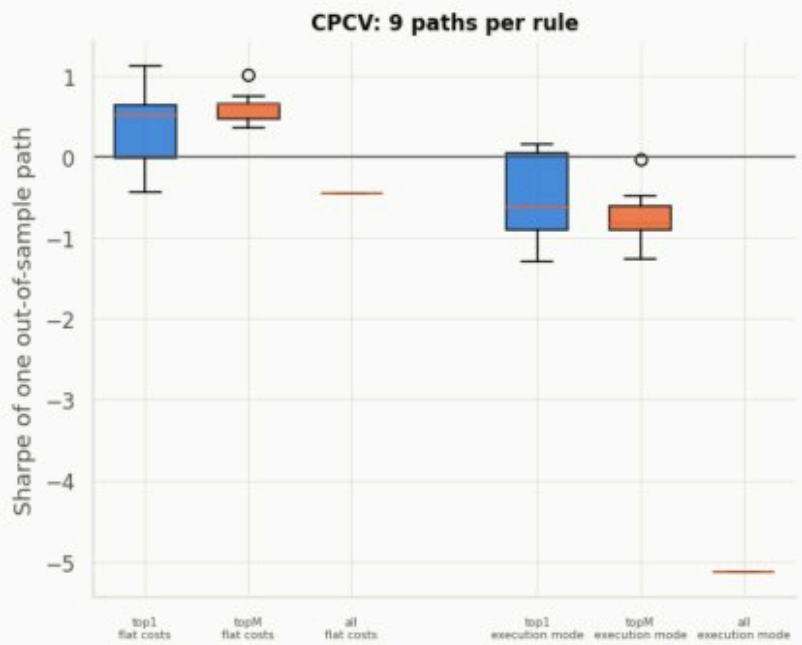
WHAT IT IS

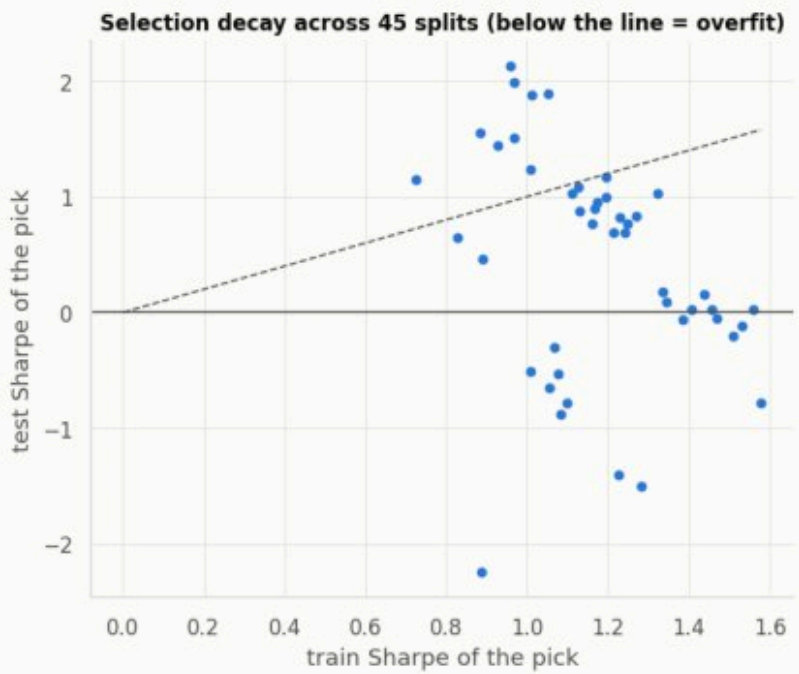
Intraday Granger strategies.

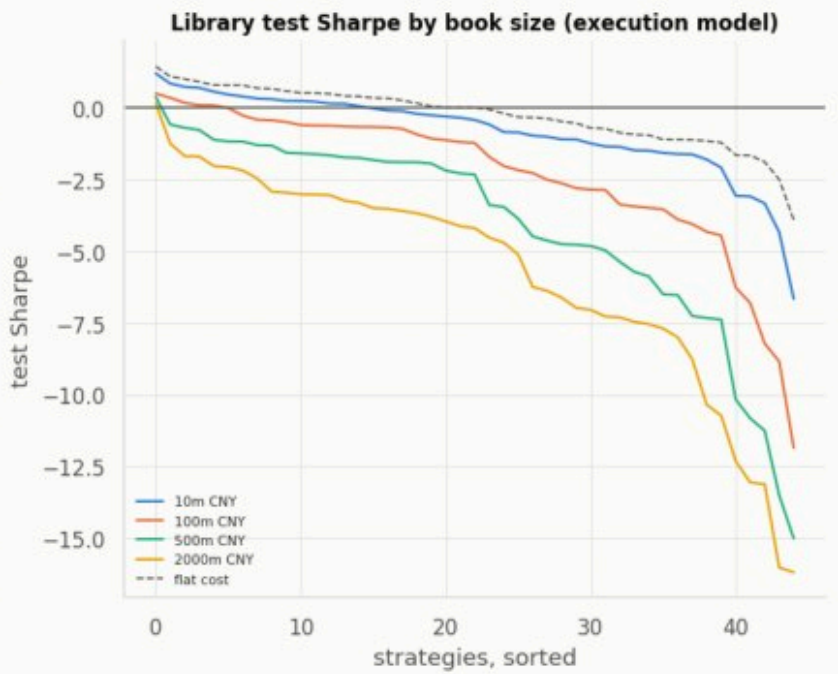
WHAT IT EXPLAINS

Trading the replicated intraday links: they do not survive cost.

CPCV paths and execution-adjusted library







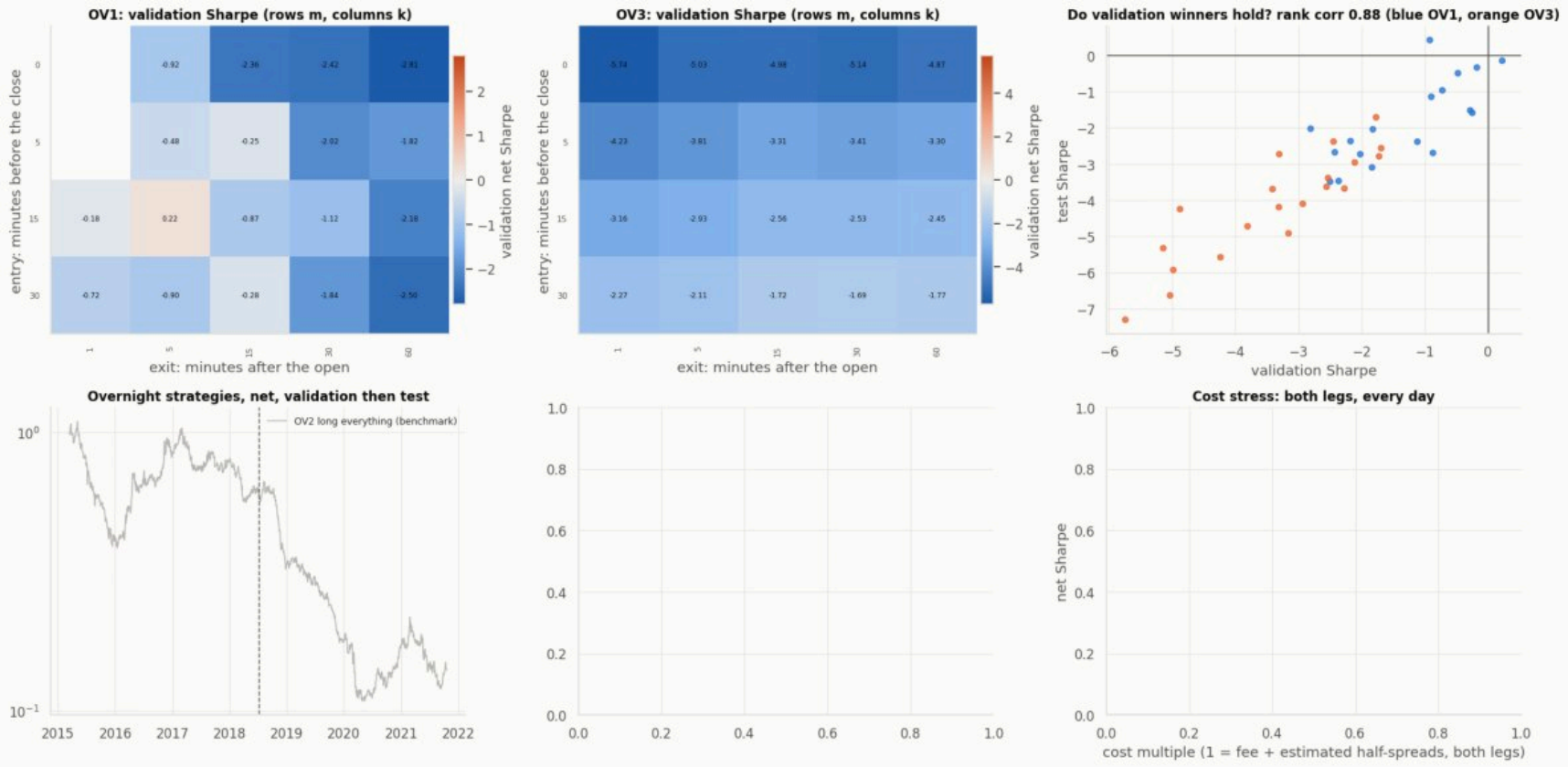
WHAT IT IS

CPCV paths and the execution-adjusted library.

WHAT IT EXPLAINS

Selecting top strategies on past performance gives mean path Sharpe 0.62 with no negative paths.

Overnight strategy: grid, selection, sectors, cost



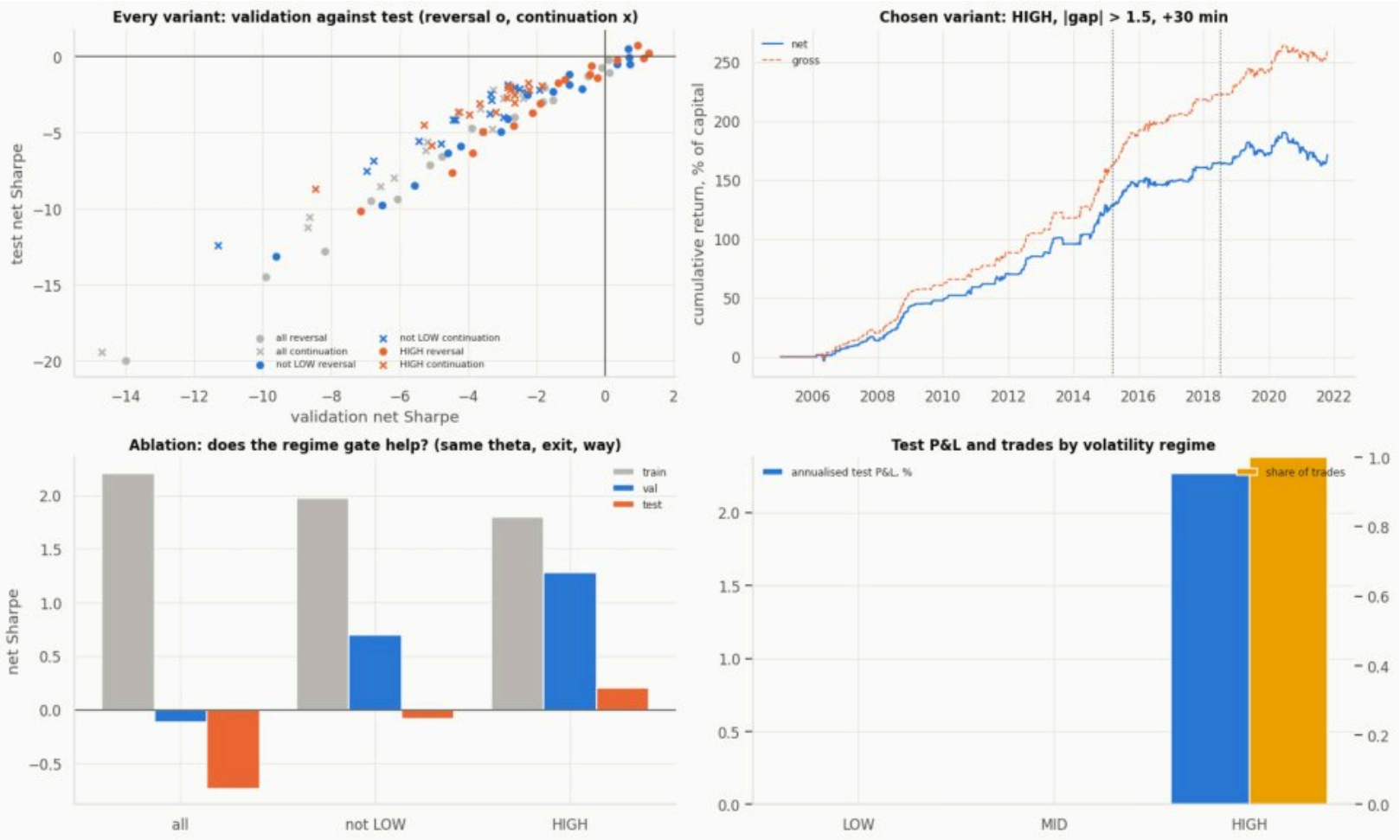
WHAT IT IS

Overnight strategy: grid, selection, sectors and cost.

WHAT IT EXPLAINS

No variant had a positive Sharpe on train and validation, so nothing was added.

Opening reversal gated by the volatility regime



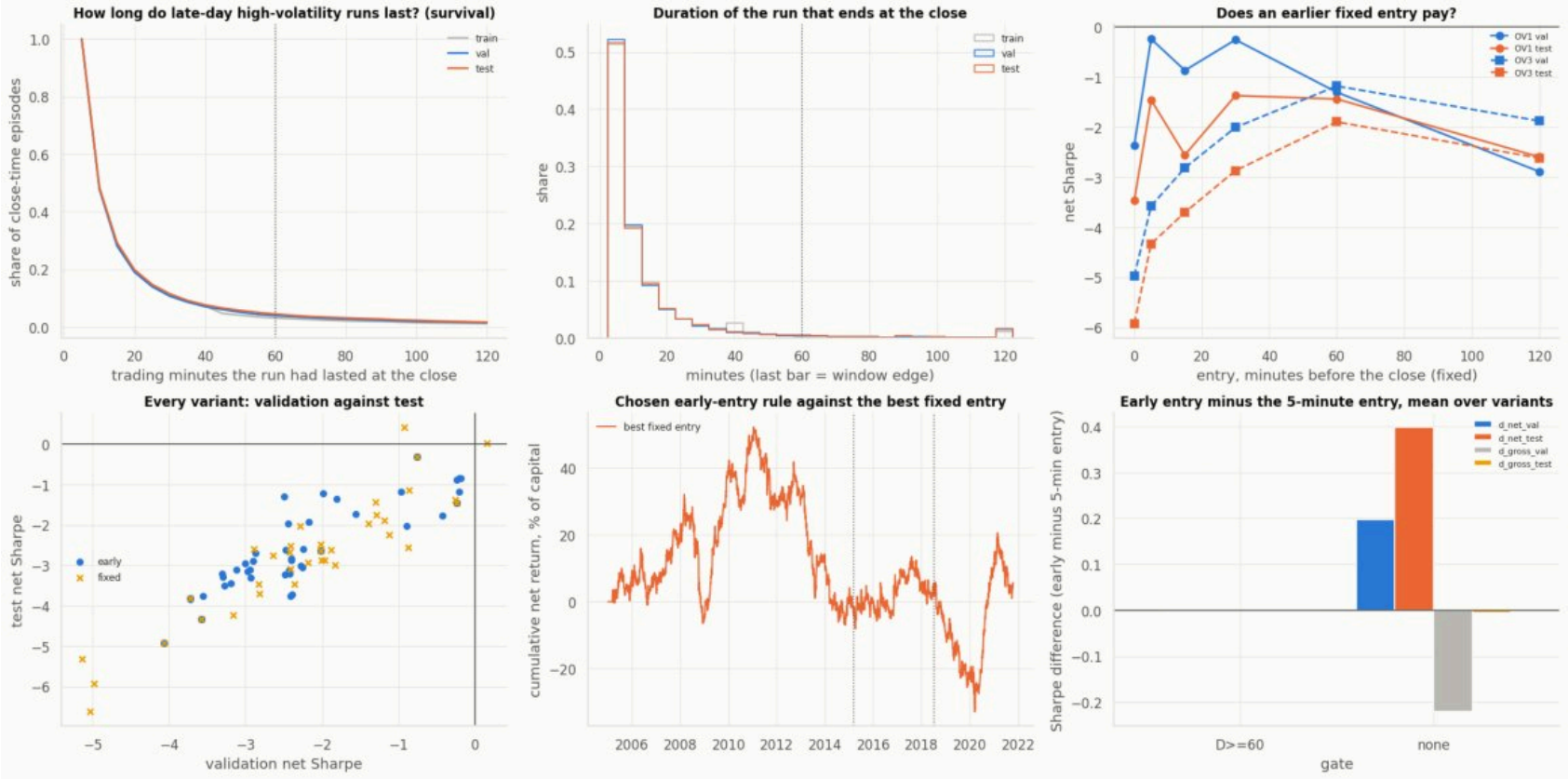
WHAT IT IS

Opening reversal gated by the volatility regime.

WHAT IT EXPLAINS

The HIGH-volatility gate leaves Sharpe +0.21 on test after cost, down from +1.80 in train.

Late-day high-volatility runs and the early-entry rule



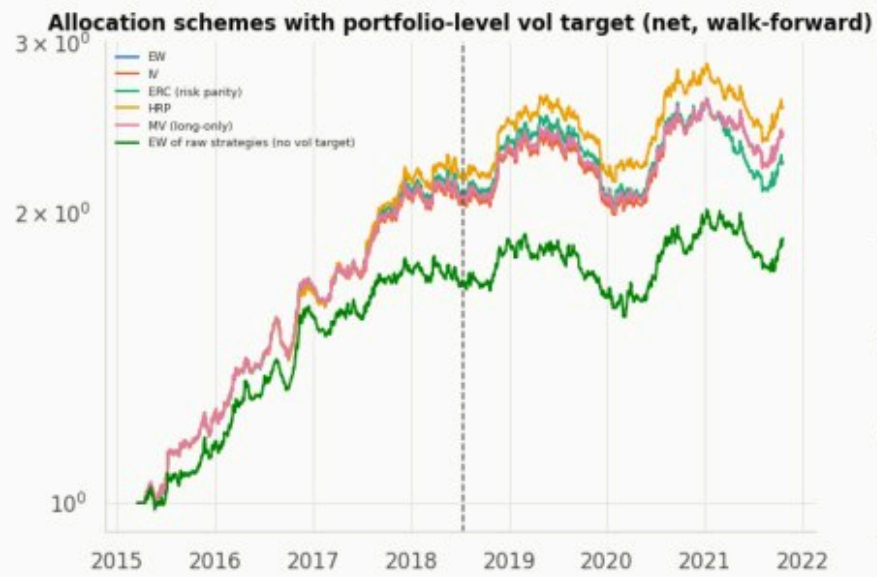
WHAT IT IS

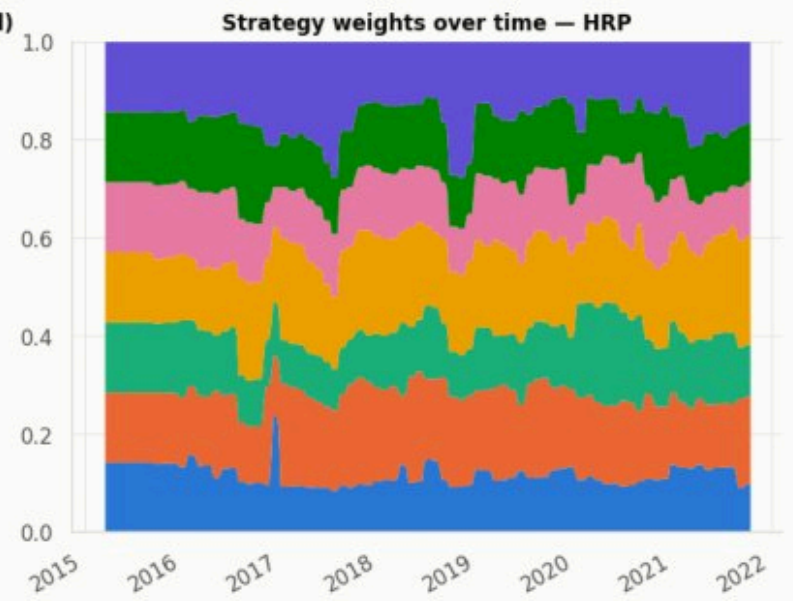
Late-day high-volatility runs and the early-entry rule.

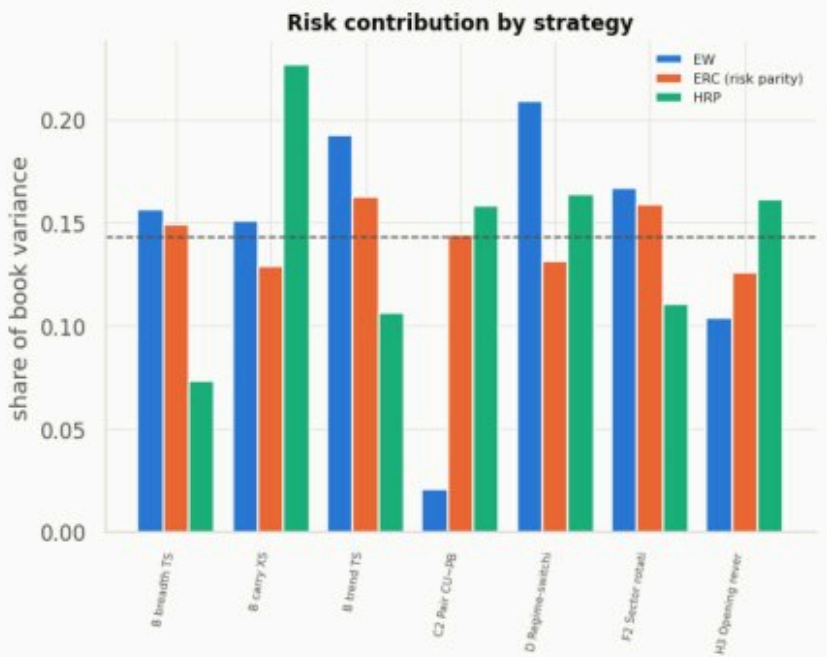
WHAT IT EXPLAINS

Runs last 15 trading minutes on average; no early-entry variant had a positive Sharpe on train and validation.

Portfolio construction: schemes, weights, risk contributions







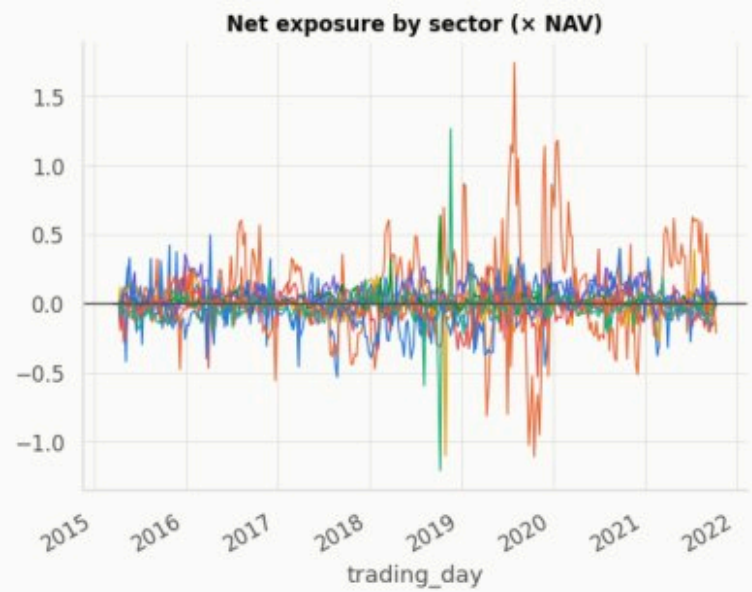
WHAT IT IS

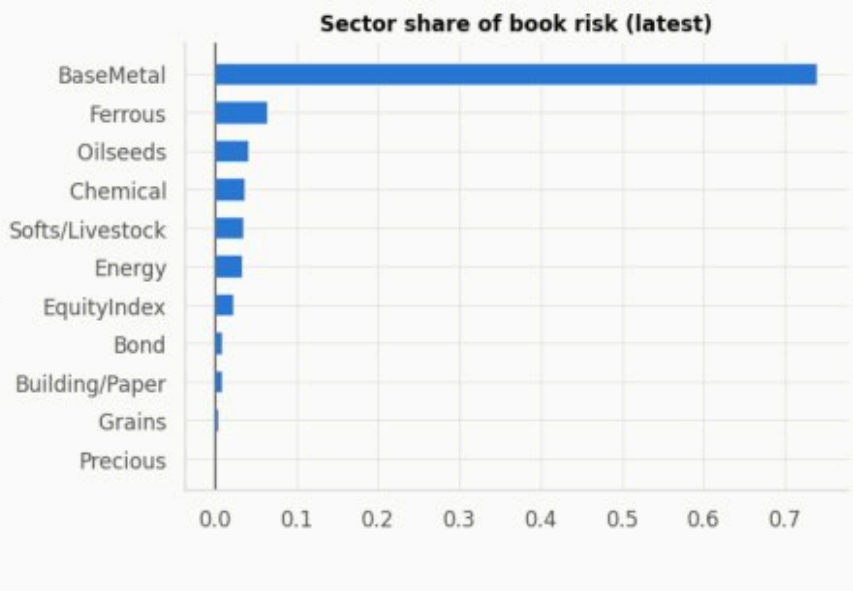
Portfolio construction: schemes, weights and risk contributions.

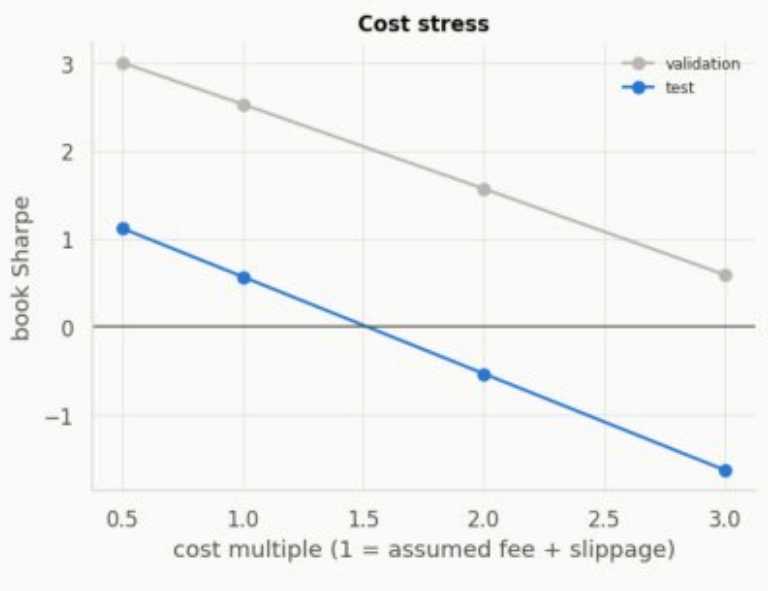
WHAT IT EXPLAINS

Equal weight, inverse volatility, risk parity, HRP and long-only mean-variance compared on test Sharpe (about 0.29-0.57).

Exposure, sector risk, cost stress







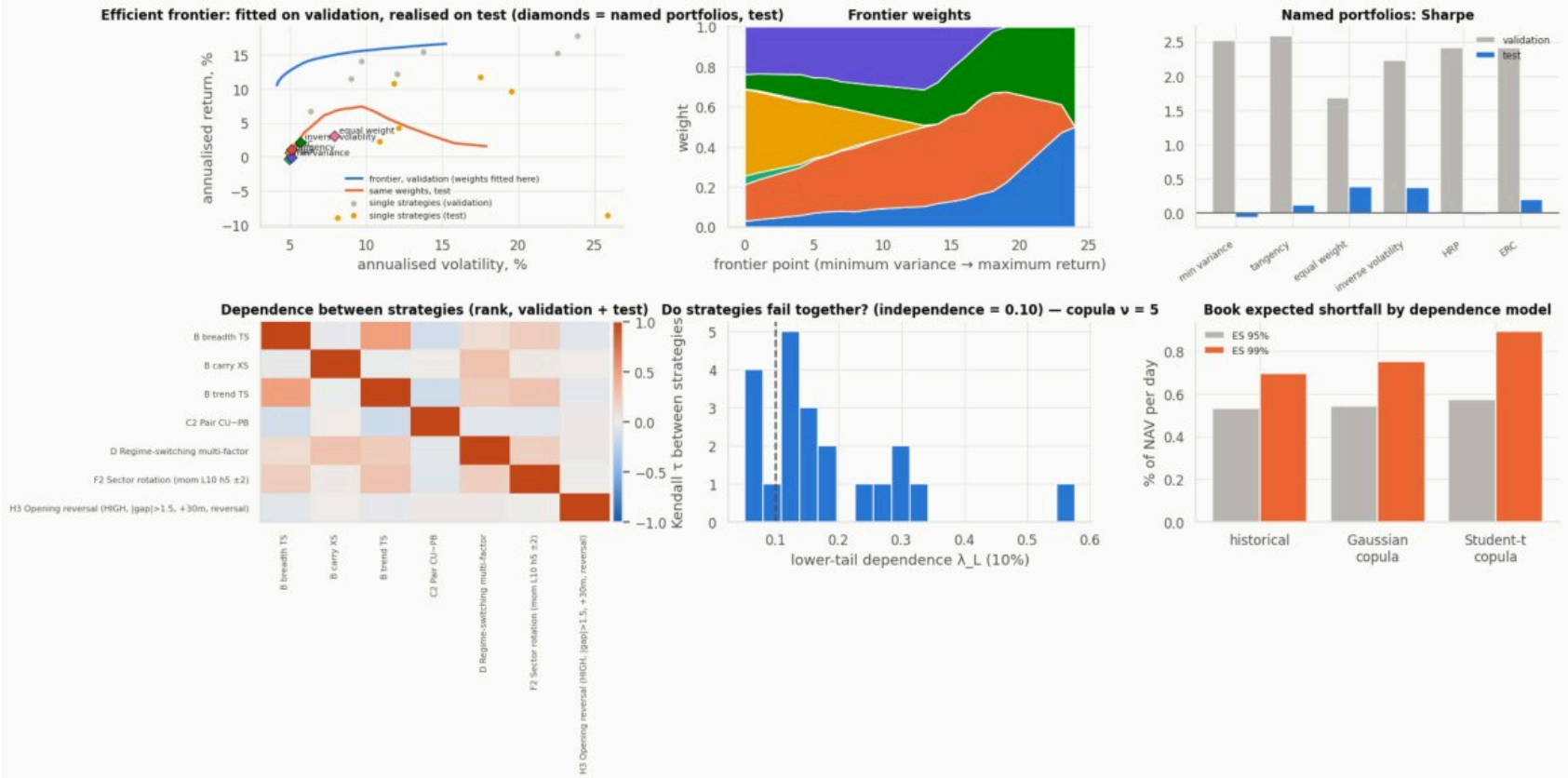
WHAT IT IS

Exposure, sector risk and cost stress.

WHAT IT EXPLAINS

How exposure splits by sector and how results degrade as cost rises.

Efficient frontier and portfolio copula



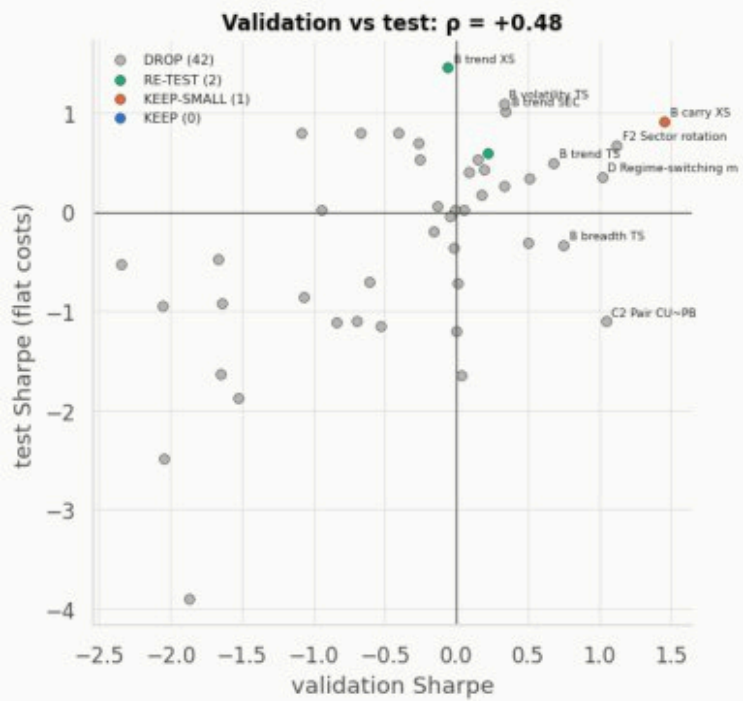
WHAT IT IS

Efficient frontier and portfolio copula.

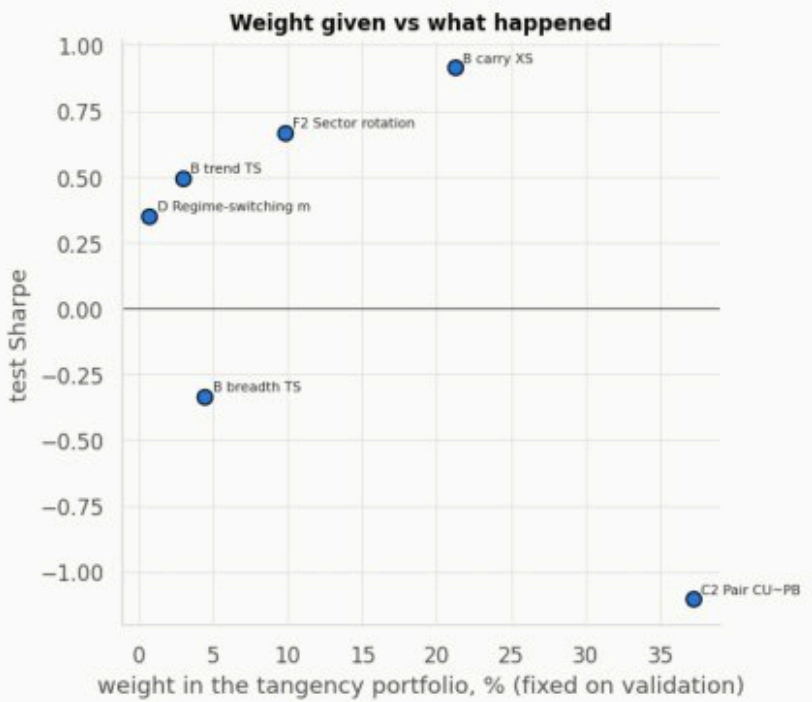
WHAT IT EXPLAINS

Tangency portfolio looks excellent on validation (2.59) but only 0.11 on test.

Selection audit and keep/drop ledger







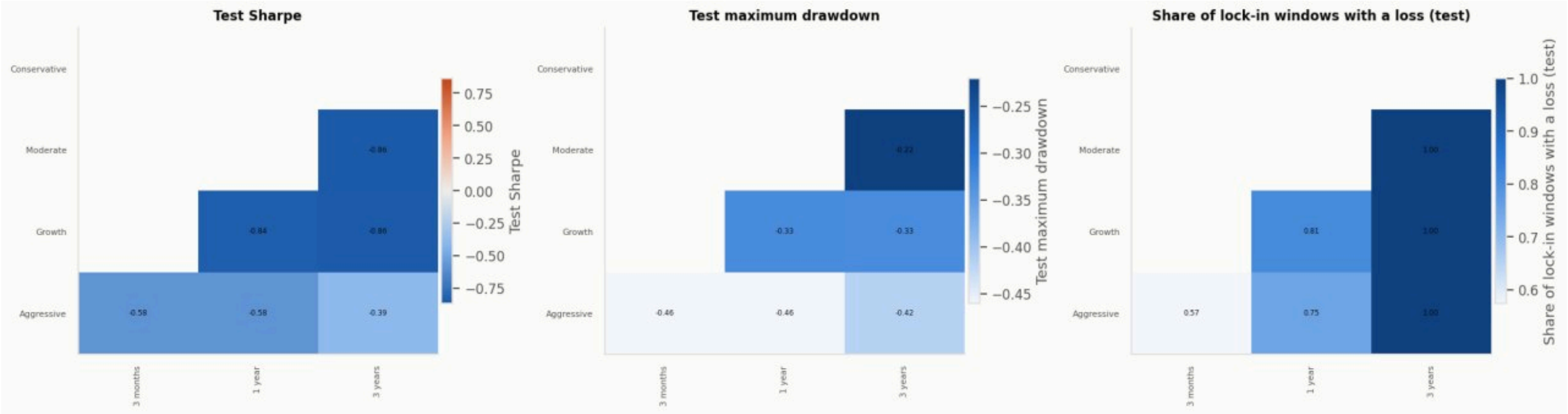
WHAT IT IS

Selection audit and keep/drop ledger.

WHAT IT EXPLAINS

Validation and test Sharpe rank-correlate +0.83 under the execution model.

Investor profiles: risk appetite x lock-in



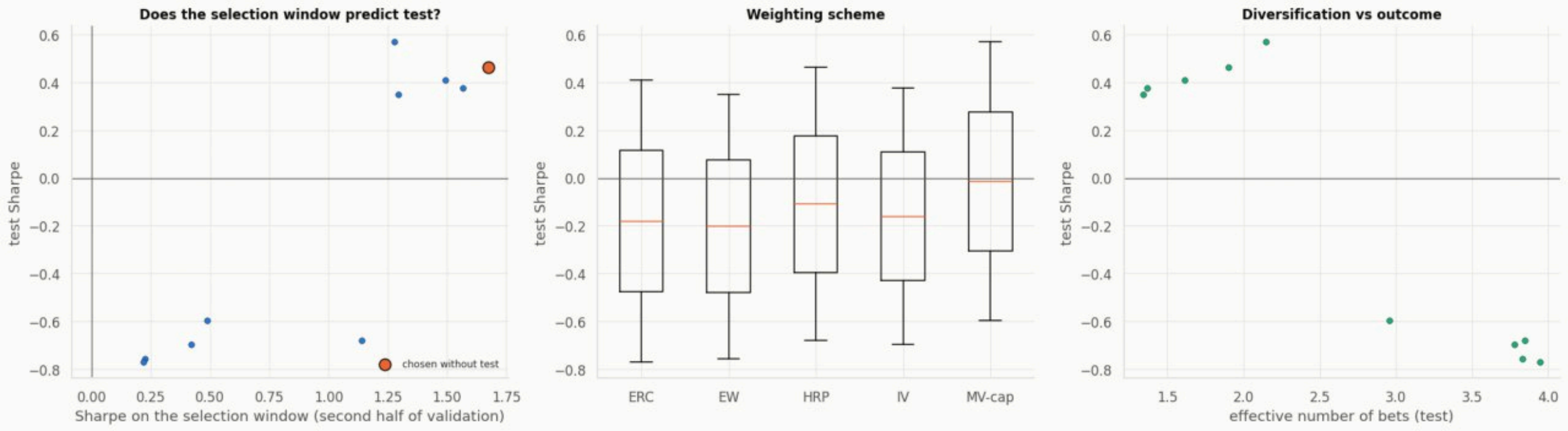
WHAT IT IS

Investor profiles: risk appetite x lock-in.

WHAT IT EXPLAINS

6 of 12 profiles pass admission on validation; none of the admitted were profitable on test.

Profitable portfolio search



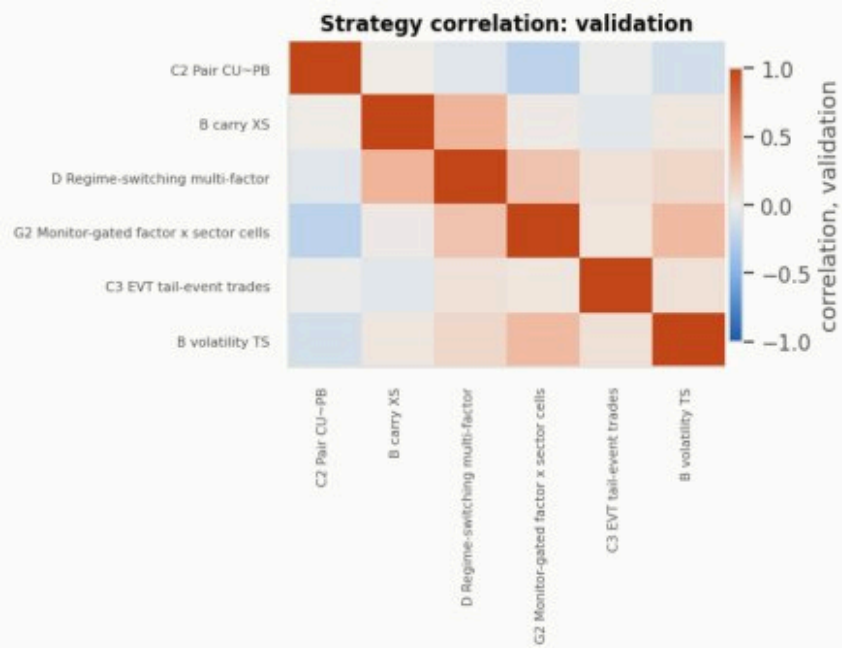
WHAT IT IS

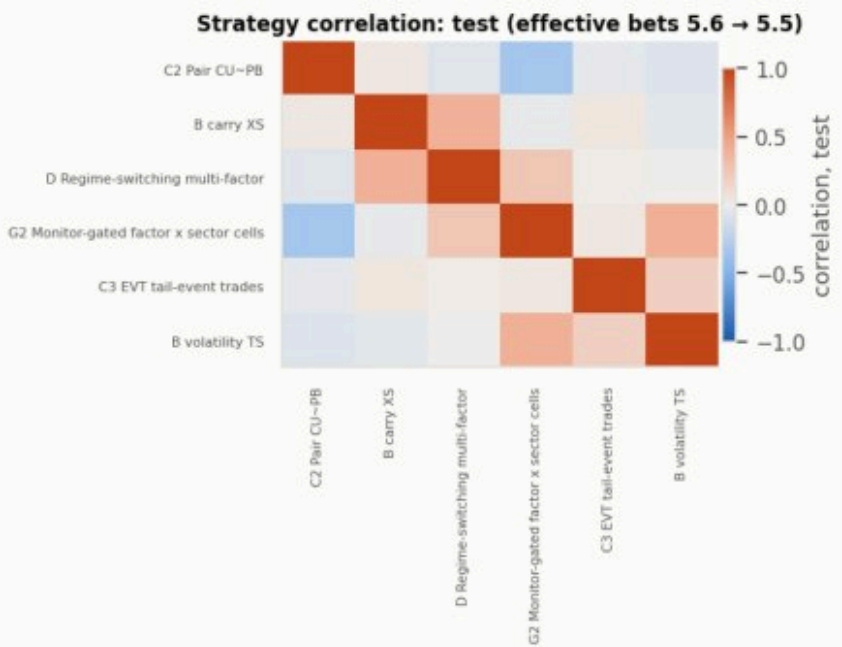
Profitable portfolio search.

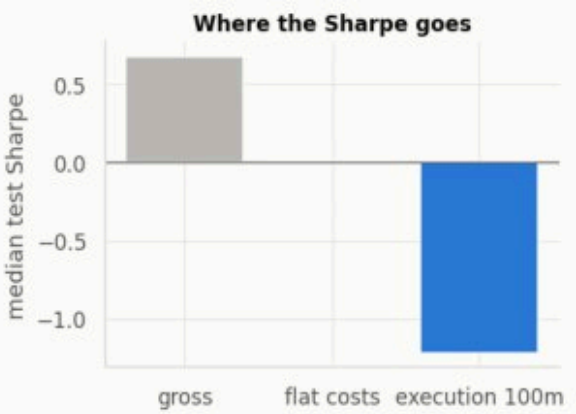
WHAT IT EXPLAINS

10 recipes tried; 5 had test Sharpe above zero, median -0.12.

Test-phase behaviour and levers







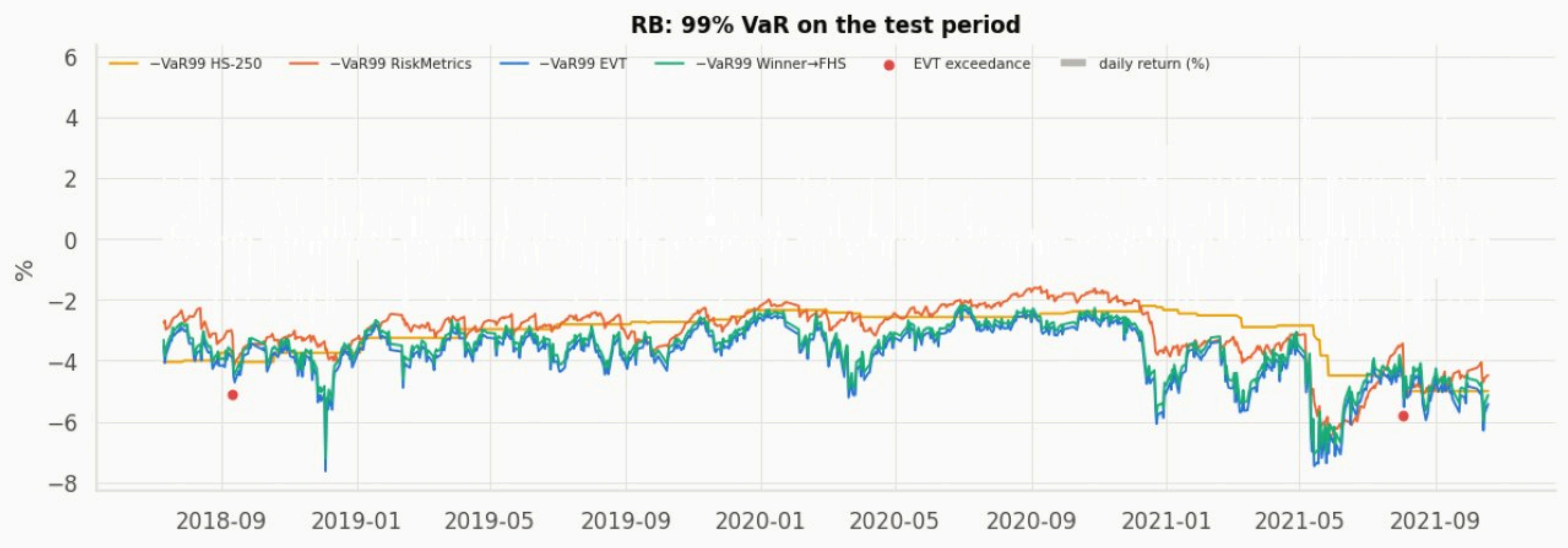
WHAT IT IS

Test-phase behaviour and levers.

WHAT IT EXPLAINS

The recipe chosen without the test (band 4.0, HRP) earns Sharpe +0.46 on test.

VaR band, showcase series



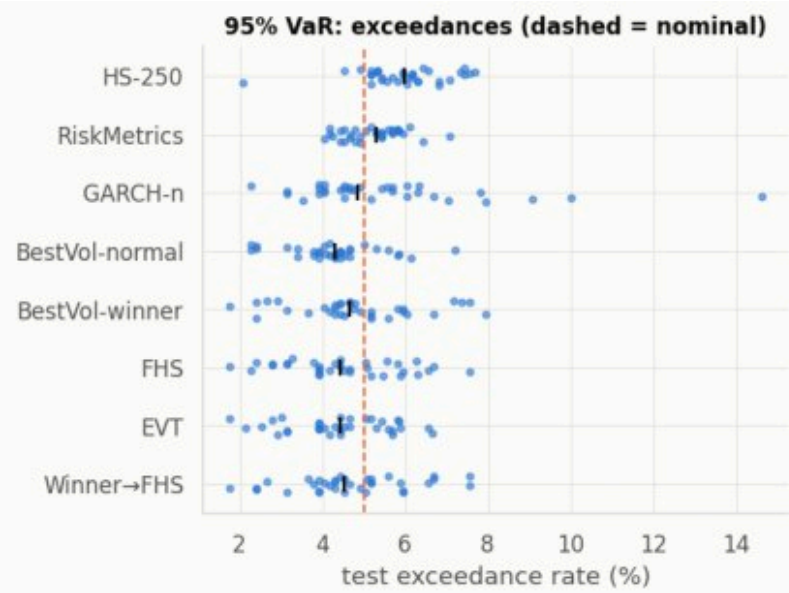
WHAT IT IS

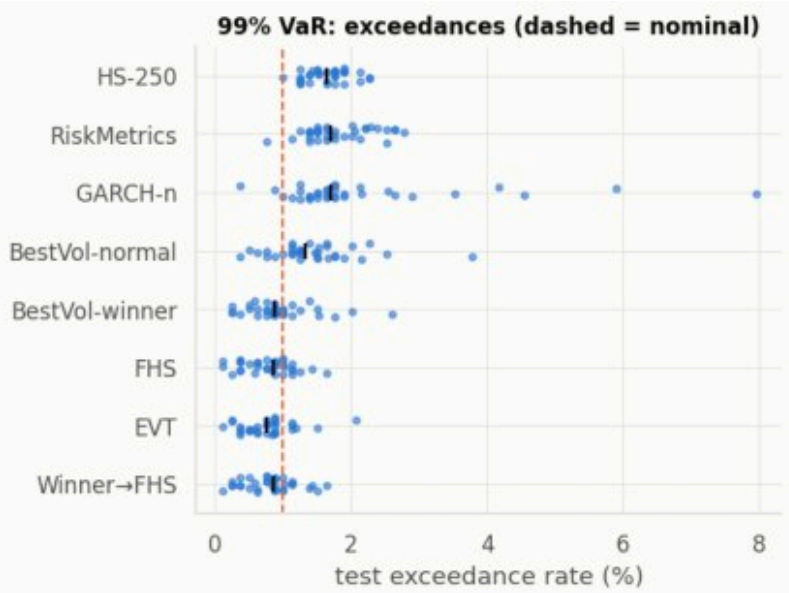
VaR band and showcase series.

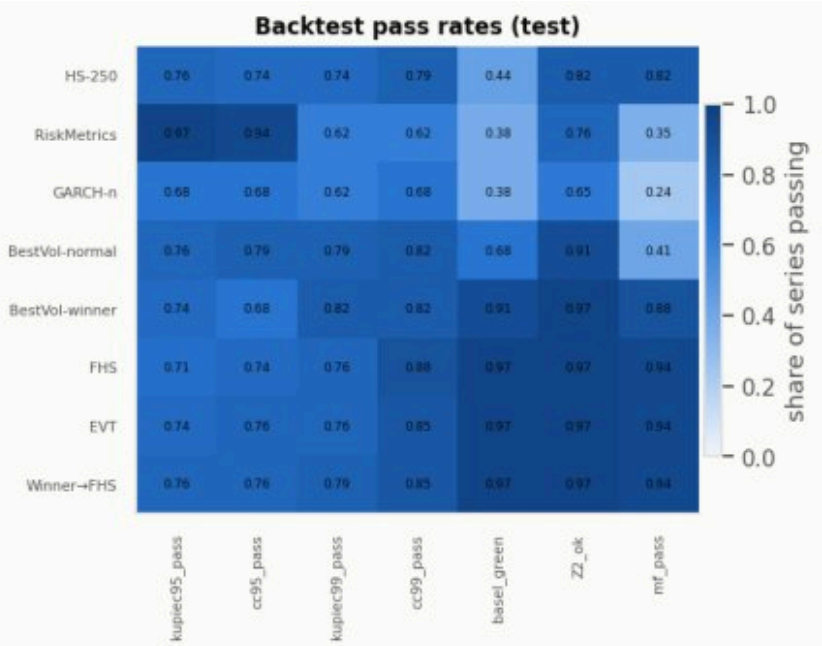
WHAT IT EXPLAINS

Value-at-risk band over time for sample series.

VaR/ES backtests







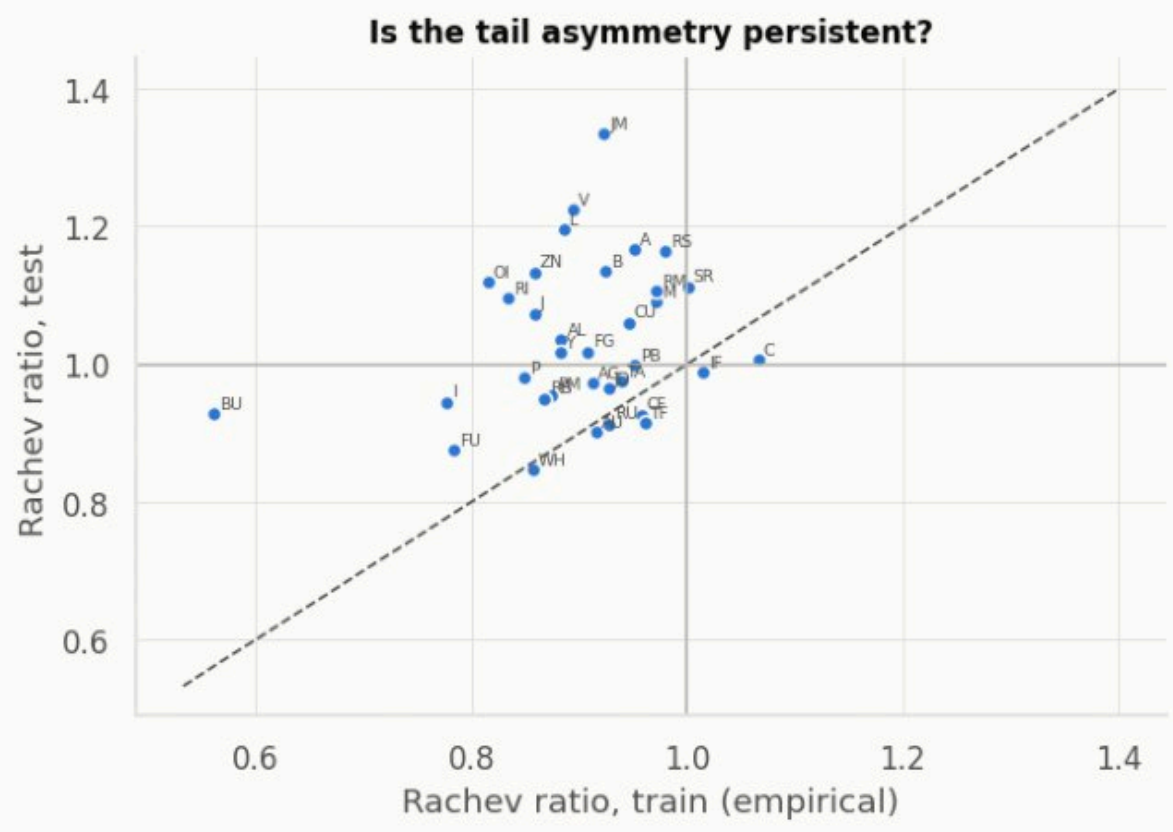
WHAT IT IS

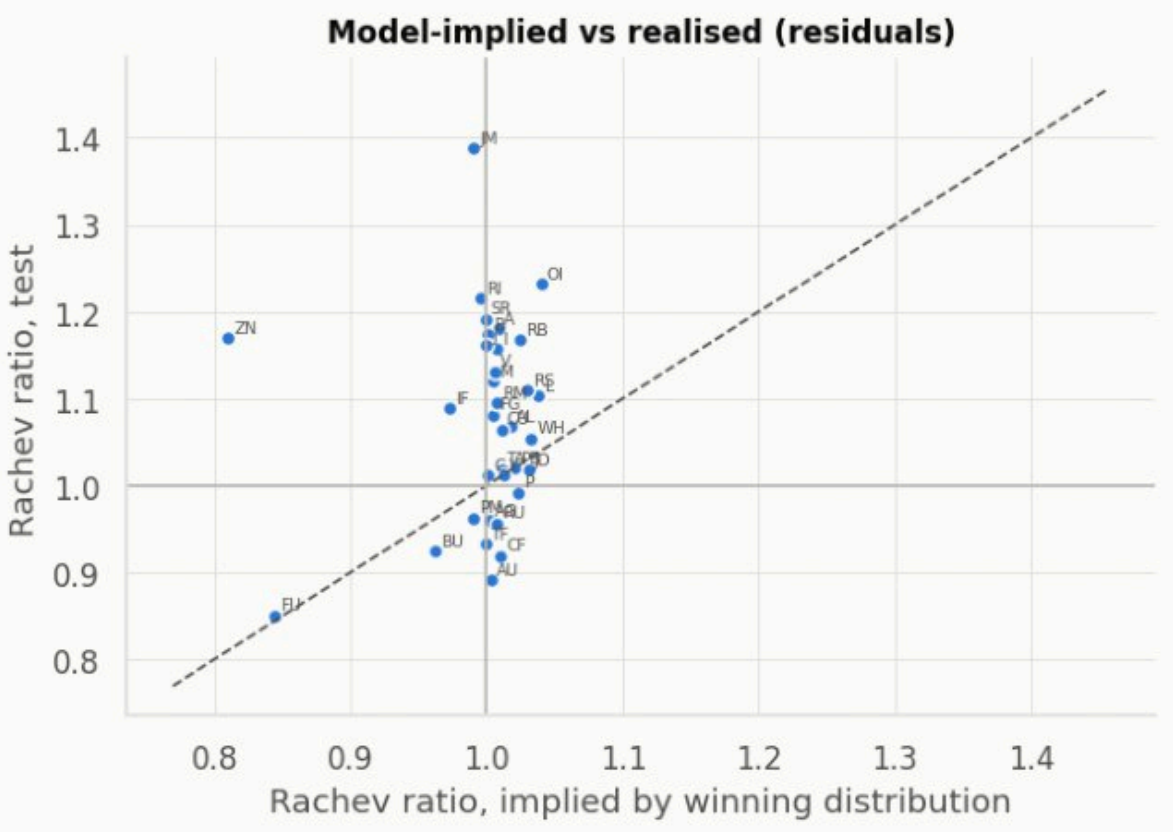
VaR/ES backtests.

WHAT IT EXPLAINS

BestVol-winner has the best median Kupiec p-value, exceedance 1.0% against 1.0% nominal.

Rachev ratios





WHAT IT IS

Rachev ratios.

WHAT IT EXPLAINS

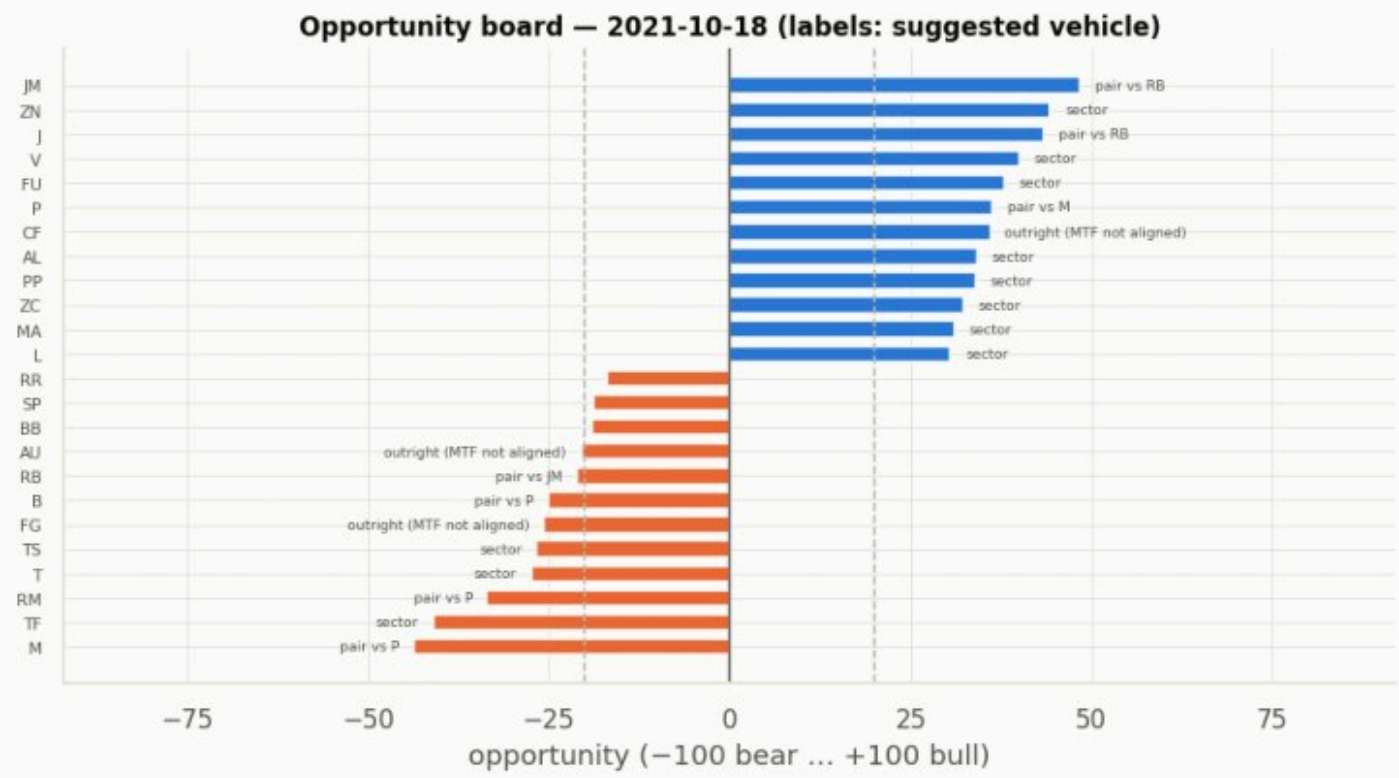
Tail reward-to-risk comparison of strategies.

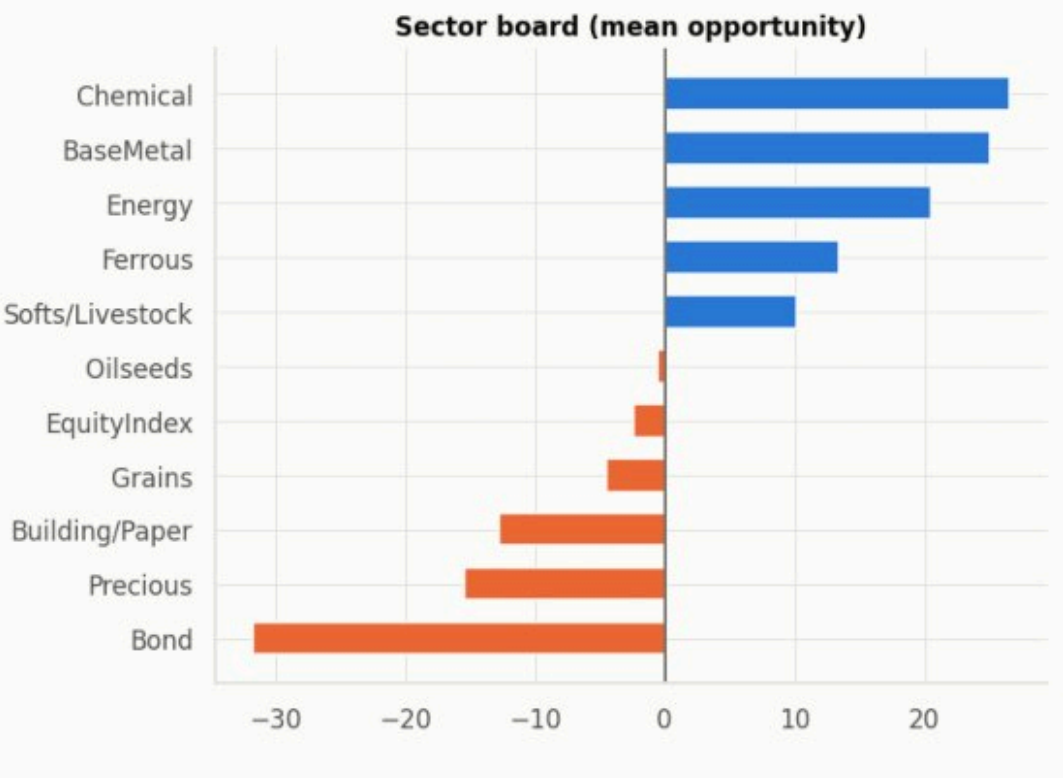
Part H

The opportunity board

Where things stand on 2021-10-18 and the cautions that apply.

Opportunity board





WHAT IT IS

Opportunity board: the current ranks, regime and volatility state.

WHAT IT EXPLAINS

As of 2021-10-18: HIGH volatility, CALM Markov state, BULL breadth. Treat it as a watch list, not an order list.

PART H · FINDINGS

The board today and cautions

- **Regime:** volatility HIGH, Markov state CALM, breadth BULL.
- **Suggested mix:** B carry XS, C2 pair CU~PB, B breadth TS, F2 sector rotation, D regime-switching multi-factor.
- **Bull candidates:** JM, ZN, J, V, FU, P, CF, AL, PP, ZC. **Bear candidates:** M, TF, RM, T, TS, FG, B, RB, AU.
- **Cautions:** hundreds of pairs are tested, only replicated links count. The test period is 3.2 years; a Sharpe on it has a standard error near 0.6, so smaller differences are noise. Costs use bar-based half-spreads and a square-root impact law; there is no order-book data, so capacity is approximate.

NEXT STEPS

What to do next

- Build the Stage 2 long, range-regime version as the base and test it with volume and squeeze as a size and entry gate (must beat ungated Stage 2 in train, validation and test; keep the tail).
- Learn the 18n component signs on train_R instead of using the pasted signs.
- Use range expansion with rising incremental volume, passive entry and calendar conditioning as timing tools for the daily signal, not as direction signals.
- Keep sub-30-minute work to volatility and range forecasting; it does not pay as direction.

Testing so far uses the real data on Kaggle for the findings; code changes were checked on synthetic data only.